



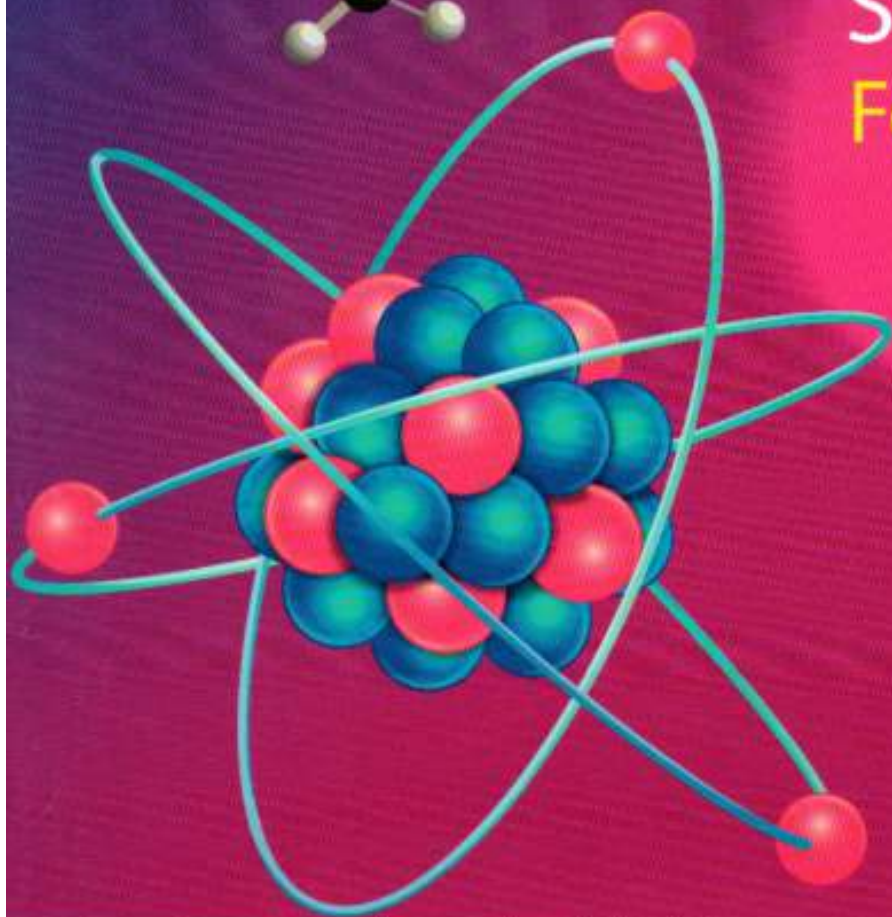
Federal Republic of Somalia
Ministry of Education, Culture and
Higher Education



Chemistry

Secondary

Form Four



Student Book

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Muqdisho -Soomaaliya





MAHADNAQ IYO ORAAHDA WASIIRKA

Alle ayey mahad u sugnaatay, Wasaaradda Waxbarashada Hiddaha iyo Tacsiinta sare (WWH&TS) waxa ay muddo dheer ku taameysay in ay soo saarto manhaj ka tarjumaya ummadda Soomaaliyeed, kaas oo la jaan qaadi kara casriga cusub, kuna dhisan aragti midayneysa ummadda Soomaaliyeed, una horseedaysa midnimo iyo horumar.

Waxaa sharaf u ah **WWH&TS** in ay soo saarto manhajkan cusub oo tayadiisu sareeyso, kana qeyb qaadanaya horumarinta maskaxda ubadkeena iyo garaadka fekerkooda si ku dhisan cilmi.

Manhajkan cusub waxa uu daaweyn doonaa fikradaha kala duwan ee ummadda Soomaaliyeed kala geeyay, oo ay sababtiisu tahay kala duwanaanshaha manaahijta dugsiyada kala duwan loo dhigo, waxa uuna keeni doonaa in la helo ummad Soomaaliyeed oo hal ujeeddo iyo hal fikir leh.

Manhajkan waxa uu ku dhisan yahay feker Soomaalinimo, si loo gaaro arrinkaas waxa lagu caddeeyey manhajkaan waxa la bari doono ardayga Soomaaliyeed, habka barashada, dhigista iyo ciddii dhigi leheydba, kaas oo lasiin doono tababar ku filan iyo bare kaaliye haga baraha.

Waxaana rajeyneynaa inuu noqodo kii ka saari lahaa ubadka Soomaaliyeed kala qoqobnaanta una horseedi lahaa midnimada iyo horumarka.

Ugu danbeyn, WWH&TS waxay mahad iyo bogaadin u jeedinaysaa Madbacadda Beder oo maalgelisay qorista iyo soo saarista manhajkan.

A. Goodax B.

Mudane, Xil. Cabdullaahi Goodax Barre
Wasiirka Wasaaradda



Mahadnaq

Marka hore waxa aan u mahad naqeynaa Alle (SW), oo noo fududeeyey howshan baaxadda leh, intaa kaddib, Wasaaradda Waxbarashada, Hiddaha iyo Tacliinta Sare iyo Xafiiska Horumarinta Manaahijta ee Wasaaraddu waxa ay mahad ballaaran u celinayaan dhammaan dhinacyadii iyo shaqsiyaadkii ka qayb qaatay manhajka cusub ee waxbarashada dalka, iyo soo saaridda buuggan, gaar ahaan:

- Madbacadda Beder oo gacan ka geysatay daabacaadda iyo maalgalinta mashruucan
- Guddiga dib u eegista ee Xafiiska Manaahijta oo si hufan u turxaan bixiyay buugga
- Qorayaasha buugga oo waqti badangeliyay
- Khubarada maadada oo dajiyay casharrada
- La taliyeyaasha oo soo jeediyay talooyin wax ku ool ah
- Khubarada kale ee xagga farsamada iyo isku xirka casharrada naga caawiyay
- Khubarada naqshadeynta oo xagga sawirrada iyo naqshadeynta naga caawiyay

Ugu dambayntii, waxaan u rajayneynaa in uu buuggani noqon doono mid faa'iido u leh ardayda iyo barayaashaba, uuna door muuqda ka ciyaari doono horumarinta waxbarashada dalka.

Mahadsanidiin

Xafiiska Horumarinta Manaahijta
WWH & TS
Mudisho - Soomaaliya
2020

Summary of the Somali National Curriculum

Education is the key to development. However it has to be quality education that delivers outcomes relevant to the society. It is important for a number of reasons – to help the country and the people to develop.

The new curriculum is intended to help reform the education system, modernize it and ensure that its graduates fit for this 21st century. Prosper, to support and encourage children and others to help them live a peaceful, happy and healthy life with a strong basis in Islamic values; to help them develop and achieve their ambitions; to prepare them for a creative and productive working life, and to prepare them to be active citizens in a changing world. We look forward to develop an education system which meets all of these needs, for all the Somalis.

The Ministry of Education intends that all students learn to acquire the following values:

Appreciate and apply Islamic values, develop a strong sense of patriotism and love of their country and community, an appreciation of their own cultural inheritance, ascetic values which appreciate beauty and its expressions in nature and the arts, self-discipline including appropriate behavior and time keeping, respect and value the rights of all people irrespective of gender, ability, value integrity in themselves and others including respect for property and life, social standing or geography, respect and value the environment, develop and value a healthy mind, body and spirit, develop creative and problem solving skills and become confident individuals with skills and knowledge that will enable them to adapt and contribute to a changing society, economically and socially.

Learning and Teaching Approaches

In delivering this new curriculum in the classroom, teachers shall use a variety of instructional strategies that actively engage all learners in the learning process. In this regard, teachers shall employ learning and teaching approaches that promote:

1. **Active learning:** Learners meaningfully engage in activities that contribute towards the attainment of learning outcomes.
2. **Experiential learning:** learning through action, learning by doing, learning through experience, and learning through discovery.
3. **Skills development:** Acquisition of a wide range of skills such as critical thinking, creativity and problem solving among other skills.

Assessment approaches

The school curriculum regards assessment as an integral part of the teaching and learning process. This curriculum places greater emphasis on assessment **for learning** rather than **of learning**. In this regard, the role of assessment will be to:

1. Provide feedback on learners' progress in demonstrating the anticipated learning outcomes to be used as guide when providing remedial support.
2. Enable learners to identify their own strengths and areas to improve for purposes of furthering their own learning.
3. Inform the teachers, schools, parents and ministry officials on the general performance of learners with respect to intended learning outcomes.



All assessment activities will focus on checking all learners' progress in achieving the intended outcomes prescribed in the teaching syllabus and lesson plan. In particular, assessment will focus on the following:

1. Learners' acquisition of knowledge, skills and values
2. Learners' ability to apply skills and knowledge acquired in class in their daily lives
3. Evidence of learning and behavioural change in learners

Teachers shall ensure that learners with special needs are given comparable assessment items without compromising the quality of assessment.

This curriculum promotes an inclusive assessment regime. It seeks to empower learners to participate in evaluating their own learning progress. Therefore, the following modes of assessment have been adopted:

1. Self-assessment: The learner is able to evaluate his or her own learning progress. This strategy should be encouraged in upper primary and secondary education
2. Peer assessment: Learners are empowered to comment on the learning progress of one another. The learner's work is assessed by fellow peers
3. Teacher assessment: Teachers will facilitate continuous and summative assessment in a more balanced manner.

Chemistry

The study of chemistry enables students to build an interpretive and predictive model for understanding the properties of matter in all its forms and the potential for change through chemical reactions. By the end of this level the following broad outcomes will be accomplished

Learners will be able to identify the halogens, alkali metals, noble gases, transition metals, to explain the structure of the periodic table in terms of the atomic number and chemical properties of the elements, to develop understanding of atomic structure, bonding and structure, moles and stoichiometry, develop understanding of enthalpy changes (including bond enthalpies & measuring and calculating enthalpy changes), to draw a diagram to show how the outer electrons form covalent bonds, draw diagrams to show the shapes of simple two element molecules, to explain in terms of solubility and essential elements why ammonium salts, potassium salts, nitrates & phosphates are useful synthetic fertilisers. Learners will be able to demonstrate appropriate safety techniques and proper use of protective equipment, to demonstrate skills in measuring and in recording data, to communicate results and data in clear and understandable forms, to demonstrate an ability to design, perform, and analyses a titration experiment involving the primary standards, standardized solutions, titration curves, appropriate indicators.

Learners will be able to give examples of the pollution problems associated with oil and coal and explain their occurrence, state that sulphur dioxide in the atmosphere reacts with water to produce acid rain, give examples of the damaging effects of acid rain on building, or other structures, soils and plant and animal life, explain that a surface barrier to air and water can provide physical protection against corrosion (e.g., painting, greasing, electroplating, galvanising, tin plating, coating with plastic), to state what is meant by monomer, polymer and polymerisation.

The learners will also be able to differentiate between organic and inorganic compounds, and have a basic knowledge of nuclear chemistry



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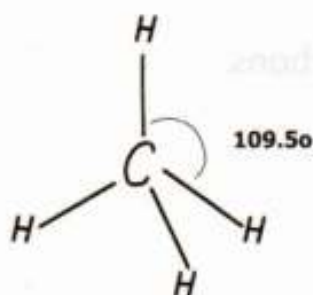
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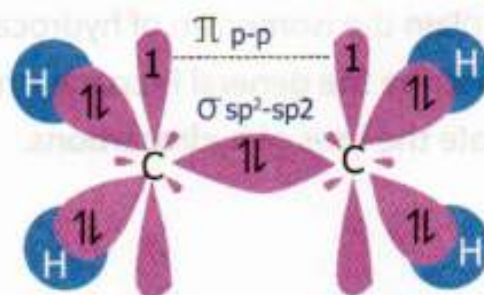


CHAPTER 1

HYDROCARBONS



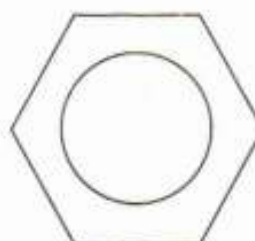
Alkane



Alkene



Alkyne



Aromatics

CHAPTER CONTENTS

- 1.1 Introduction to hydrocarbons
- 1.2 Classification of hydrocarbon
- 1.3 Alkanes
- 1.4 Alkenes
- 1.5 Alkynes
- 1.6 Aromatics
- Chapter Review Questions

CHAPTER OBJECTIVES

By the end of the chapter, the learner should be able to:

- Define and describe hydrocarbon.
- Classify hydrocarbons.
- Distinguish among the structures of alkanes, alkenes, alkynes and aromatics.
- Write structural formulas and names for Alkanes, alkenes, alkynes and aromatics.
- Relate the properties of different types of hydrocarbons to their structures.
- Explain the isomerism of hydrocarbons.
- Describe the general Preparation of hydrocarbons.
- State the uses of hydrocarbons.

1.1 INTRODUCTION TO HYDROCARBONS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define and describe hydrocarbon.
- Classify hydrocarbons.
- State the uses of hydrocarbons.

**ACTIVITY 1**

We look at these pictures, and find out which are considered organic Substances? We discuss with our partners.



Salt



Sugar



Maize



Water



Tomato

**ACTIVITY 2**

We will Identify and classify the following compounds into organic and inorganic, and then state the type of elements:

Compounds	Organic	Inorganic	Type of element
$C_2H_4O_2$			
H_2SO_4			
$CaCl_2$			
C_2HCl_3O			
Na_2SO_2			

Hydrocarbons are compounds that contain carbon and hydrogen only.

For examples;

- a) CH_4 methane
- b) C_3H_4 propene
- c) C_5H_8 pentyne

Hydrocarbons are mainly obtained from



Natural Gas



Coal



Petroleum

Hydrocarbons are commonly used as fuels. Candle wax, oil, gasoline, liquid propane (LP) gas and natural gas are all composed of hydrocarbons.

Hydrocarbons are also the raw materials in the synthesis of many different consumer products, including fabrics, soaps, dyes, cosmetics, drugs, plastic, and rubber.



EXERCISE

1. Define hydrocarbons
2. List at least three common uses of hydrocarbons
3. List at least three hydrocarbon compounds?

1.2 CLASSIFICATION OF HYDROCARBONS

Lesson objectives

By the end of the lesson, the Learner should be able to:

- classify hydrocarbons

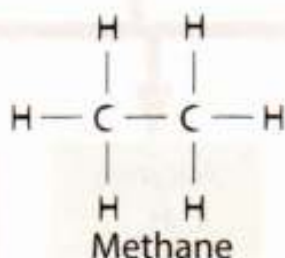
On the basis of bond, hydrocarbons are classified into:

1. Saturated
2. Unsaturated

1. Saturated hydrocarbons

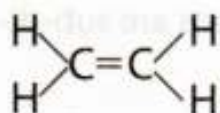
These are hydrocarbons that contain only C-C single bond. Saturated hydrocarbons are also **called alkanes**.

For example

**2. Unsaturated hydrocarbons**

These are hydrocarbons that contain carbon-carbon multiple bonds (double or triple bond)

For examples



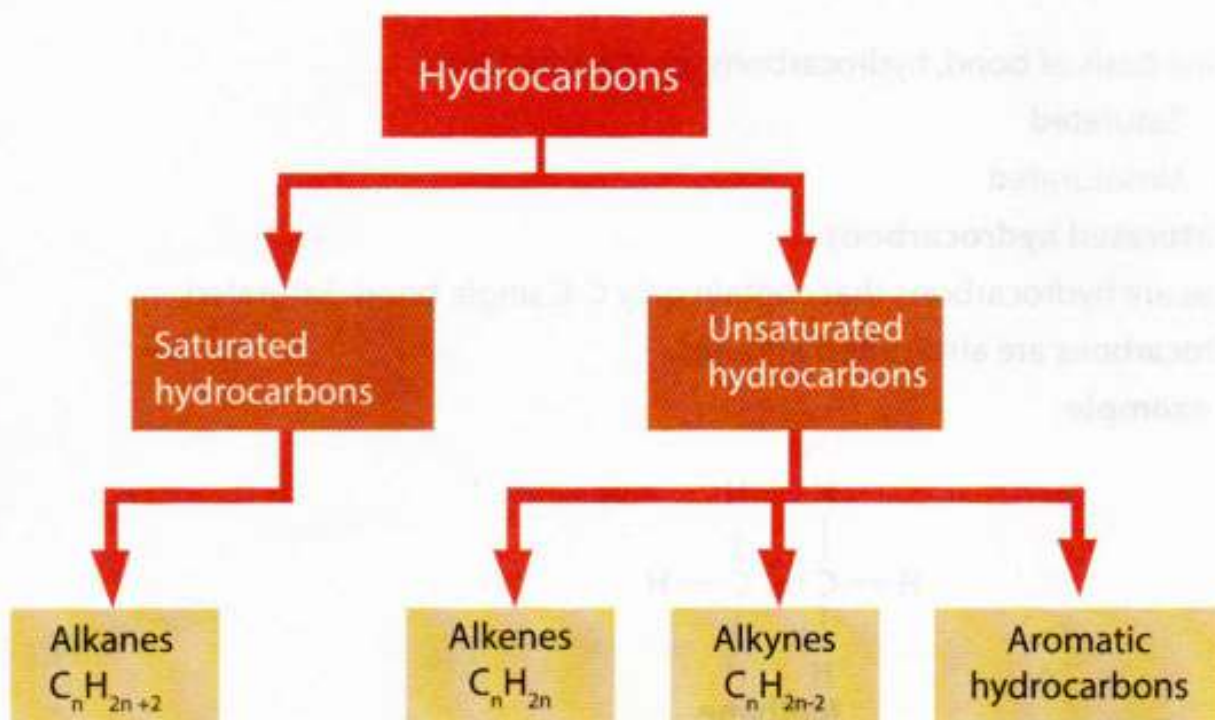
Ethene



Ethyne

Unsaturated hydrocarbons are further divided into:

- a) Alkenes
- b) Alkynes
- c) Aromatic hydrocarbons



On the basis of structure, hydrocarbons are classified into:

- 1. Cyclic hydrocarbons
- 2. Alicyclic hydrocarbons

1. Cyclic hydrocarbons.

These are closed or ring structure hydrocarbons. Cyclic hydrocarbons are sub-divide into:

- (i) Alicyclic hydrocarbons, which further divided into:
 - a) Cyclo-alkanes: closed structure alkanes
 - b) Cyclo-alkenes: closed structure alkenes
 - c) Cyclo-alkynes: closed structure alkynes.

(ii) Aromatic hydrocarbons, which is further divided into:

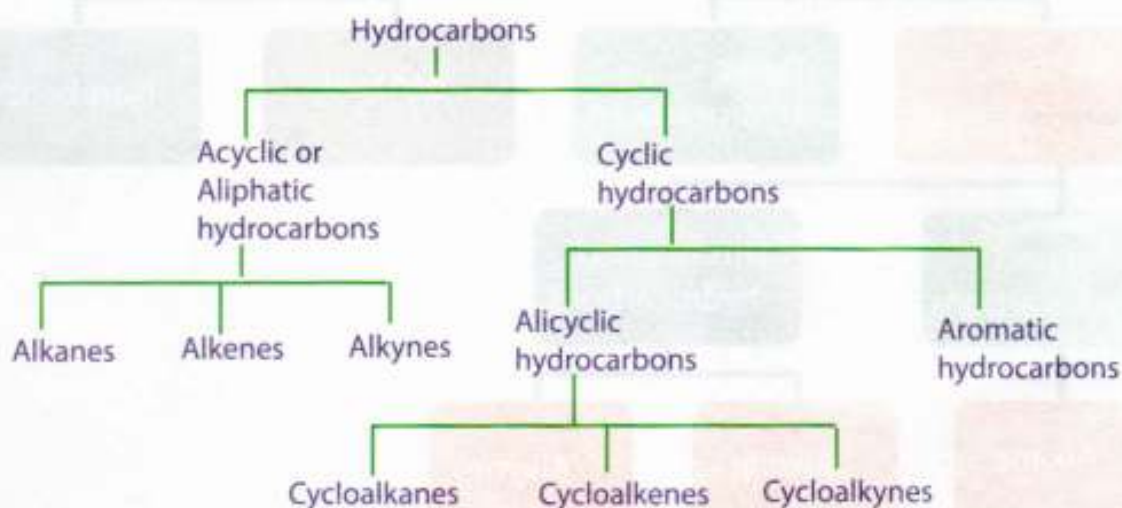
- a) **Benzenoid:** Aromatic hydrocarbons that contain benzene ring
- b) **Non-Benzenoid:** Aromatic hydrocarbons that do not contain benzene ring

2. Acyclic hydrocarbons

These are open chain structure hydrocarbons and they are also called aliphatic hydrocarbons.

They are sub-divided into;

- (i) Alkanes
- (ii) Alkenes
- (iii) Alkynes



Classification of hydrocarbon

On the basis of their sources and properties, hydrocarbons are classified into:

1. Aliphatic hydrocarbons
2. Aromatic hydrocarbons

1. Aliphatic hydrocarbons

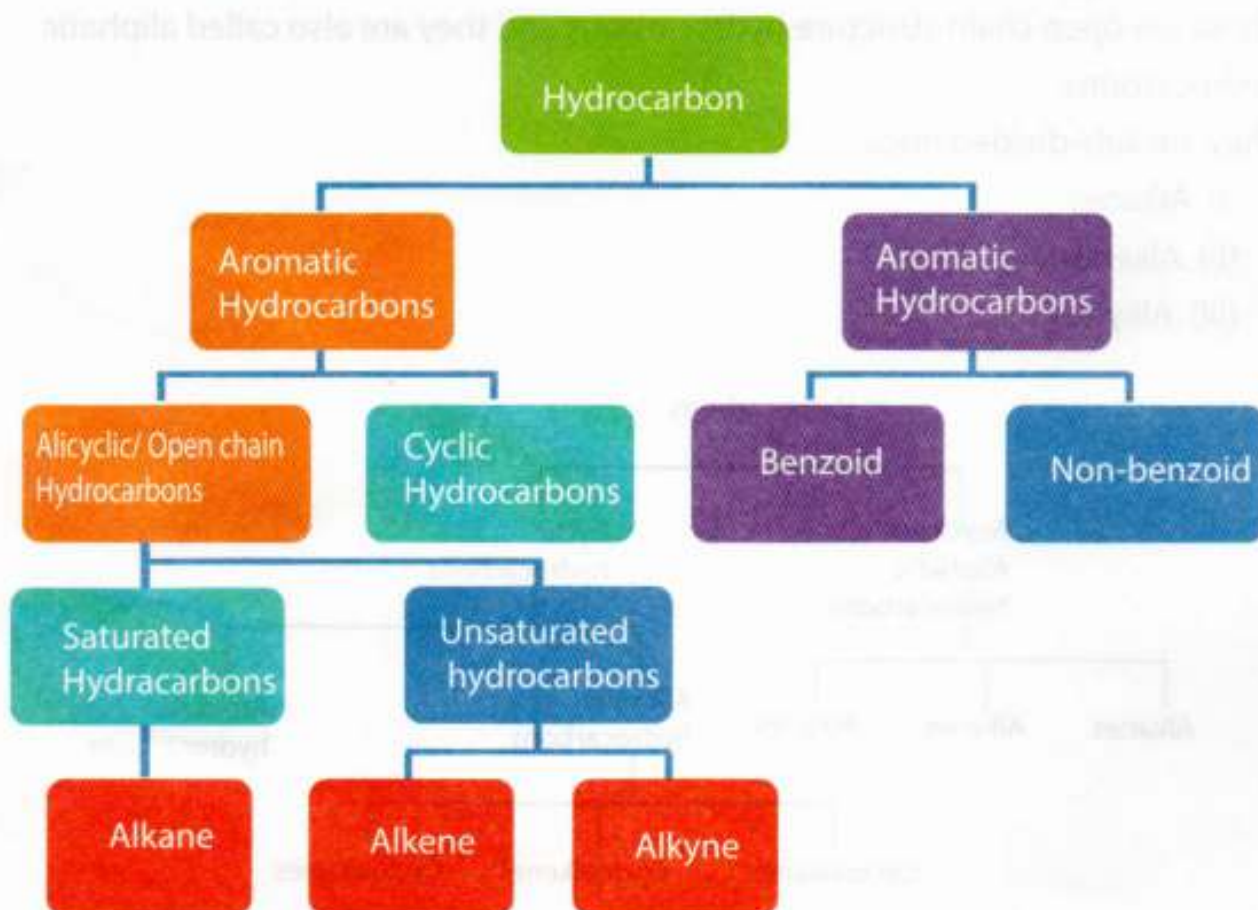
These are hydrocarbons derived by the chemical degradation of fats or oils. They sub-divided into:

- a) Alkane
- b) Alkene
- c) alkyne

2. Aromatic hydrocarbons

These are hydrocarbons obtained by chemical degradation of certain pleasant-smelling plant extracts. It is subdivided into:

- Benzenoid
- Non-benzenoid



Classification of hydrocarbons

Experiment 1:

The effect of bromine solution and potassium permanganate on saturated and unsaturated compounds

Materials

- Cyclohexane
- Cyclohexene
- Bromine water
- Potassium permanganate in acidic solution
- 4 beakers
- Two A4 sheets of paper
- 2 plastic pipettes

Procedure

1. Label one piece of paper A and the other piece B
2. Pour 20 mL of cyclohexane into a beaker and place the beaker on paper A.
3. Pour 20 mL of cyclohexane into a beaker and place the beaker on paper B
4. Repeat steps 2 and 3 with cyclohexene.
5. Take 12 mL of bromine water and add it to the beaker of cyclohexane on paper A.
6. Repeat step 5 with the beaker of cyclohexene on paper A.
7. Take 12 mL of KMnO_4 and add it to the beaker of cyclohexane on paper B.
8. Repeat step 7 with the beaker of cyclohexene on paper B.
9. Record your results in the table below:

Compound	Initial colour	Solution added	Final colour
Cyclohexane (A)		Bromine water	
Cyclohexane (B)		KMnO_4	
Cyclohexene (A)		Bromine water	
Cyclohexene (B)		KMnO_4	

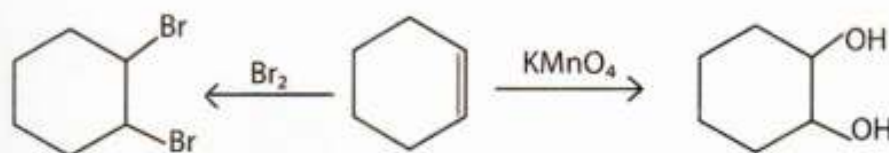
Observation

1. What did you observe?
2. Which of these compounds is saturated and which is unsaturated?
3. What colour change did you observe with the saturated compound?
4. What colour change did you observe with the unsaturated compound?
5. Can you suggest a reason for the difference?

Conclusion

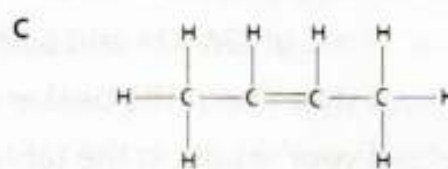
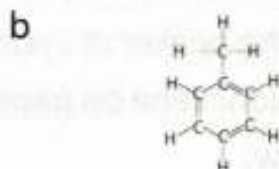
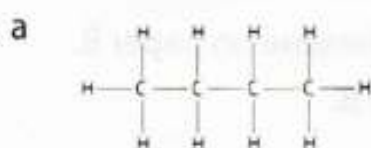
Cyclohexane is a saturated colourless liquid. When bromine water and KMnO_4 are added to the cyclohexane there is no reaction and the solution assumed the colour of bromine water or KMnO_4 .

Cyclohexene is unsaturated colourless liquid. Cyclohexene reacts with bromine water and KMnO_4 bromohexane. Bromohexane is a colourless liquid and the solution turned colourless



**EXERCISE**

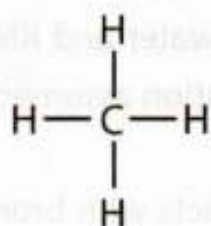
1. How many ways can hydrocarbons be classified?
2. What is the most common classification of hydrocarbons?
3. What are the three different types of aliphatic hydrocarbons? Define each type?
4. Indicate whether each molecule is an aliphatic or an aromatic hydrocarbon; if aliphatic, identify the molecule as saturated or unsaturated.

**1.3 ALKANES (PARAFFINS)****Lesson objectives**

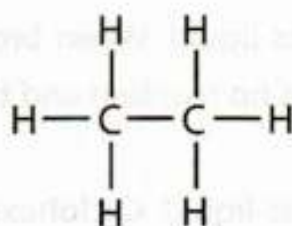
By the end of the lesson, the learner should be able to:

- Define alkane and give examples
- Classify

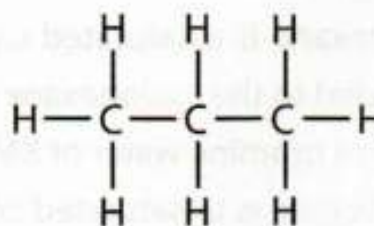
Alkanes are saturated hydrocarbons containing only carbon-carbon single bonds. Alkanes are also called saturated hydrocarbons because they are saturated with hydrogen.

For examples

Methane

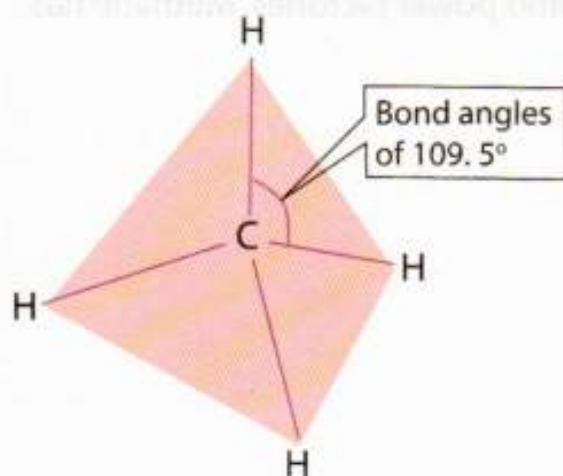


Ethane

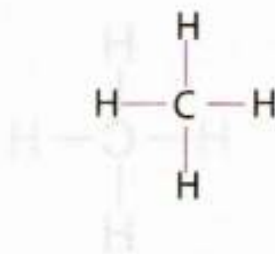


Propane

Each carbon in alkane is sp^3 hybridized. Therefore the geometric structure of alkane is tetrahedron with bond angle of 109.5° .



Tetrahedral structure



Planar representation



Space-filling model

Alkanes form homologous series with the general formula $C_n H_{2n+2}$ where "n" is the number of carbon atoms in the molecule.

The number of carbon and hydrogen atoms can be calculated by this formula:

$$H = 2n + 2$$

Example 1

Find the number of hydrogen atoms of an alkane with 5 carbon atoms.

Solution

Given $n = 5$ $H = ?$

$$H = 2n + 2$$

$$= 2(5) + 2$$

$$= 10 + 2$$

$$= 12$$

Example 2

Find the number of carbon atoms of an alkane with 10 hydrogen atoms.

Solution

Given $H = 10$ $n = ?$

$$2n + 2 = H$$

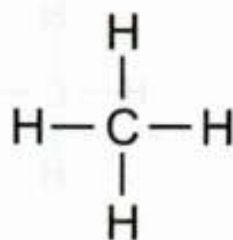
$$2n + 2 = 10$$

$$2n = 10 - 2$$

$$2n = 8$$

$$n = 4$$

Methane is the primary compound in natural gas. Tremendous quantities of natural gas are consumed worldwide because methane is an efficient, clean-burning fuel. It is used to heat homes, cook food, and power factories. Methane has the following structure.



METHANE

1.3.1 Classification of Alkanes

Alkanes are classified into two types:

1. Straight chain Alkanes
2. Branched chain Alkanes

1. Straight chain Alkanes

These are Alkanes in which each carbon atom is bonded to not more than two carbon atoms. Straight Chain Alkanes are also known as normal alkanes (n-alkane)

Examples

- a) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ butane
- b) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$ pentane

2.1 Branched chain Alkanes

These are Alkanes in which at least one carbon atom is bonded to more than two other carbon atoms.

Examples

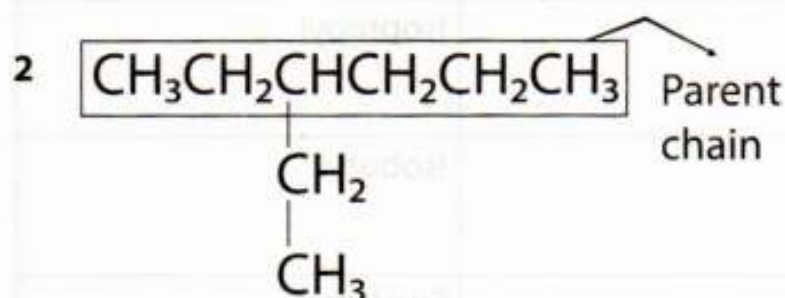
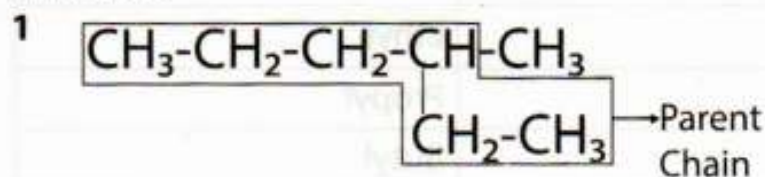
- c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{-CH}_2\text{-CH-CH}_3 \end{array}$ 2-Methylbutane
- d) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH-CH}_3 \\ | \\ \text{CH}_2\text{-CH}_3 \end{array}$ 3-Methylhexane

Structurally, branched chain Alkanes consist of two parts

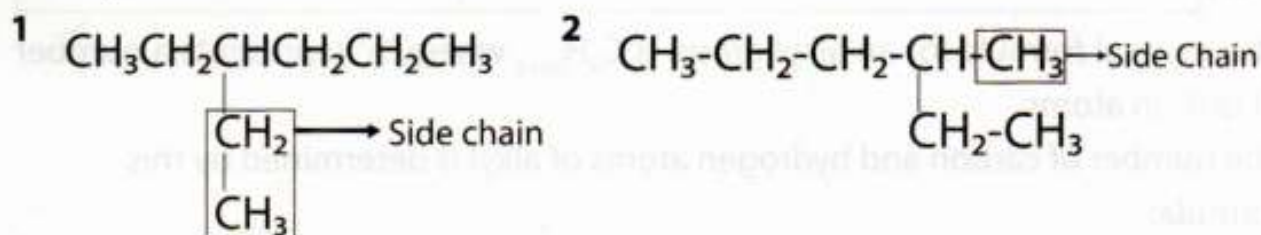
- a) Parent chain
- b) Side chain

a) Parent chain

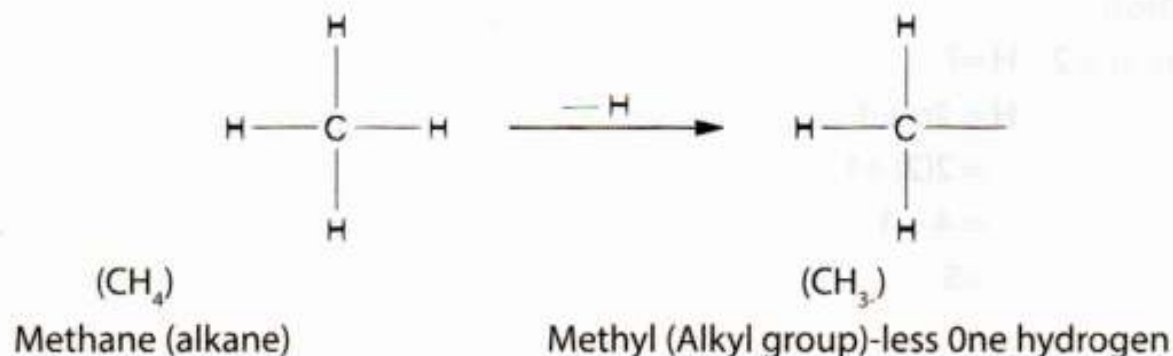
This is the longest straight carbon – carbon chain of carbon atoms.

Examples**b) Side chain**

These are short chains attached to the parent chain. The side chain is also known as substituent or branch. In hydrocarbons all the hydrocarbons side chains are known as alkyl group.

Examples**1.3.2 Alkyl groups**

In organic chemistry, an alkyl group is an alkane that has one hydrogen less.



Some common alkyl groups are represented the table below:

Table 1.1: Common alkyl Groups

Condensed structural formula	Name
$-\text{CH}_3$	Methyl
$-\text{CH}_2-\text{CH}_3$	Ethyl
$-\text{CH}_2-\text{CH}_2\text{CH}_3$	Propyl
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	Butyl
$\begin{array}{c} -\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array}$	Isopropyl
$\begin{array}{c} -\text{CH}_2\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array}$	Isobutyl
$\begin{array}{c} -\text{CHCH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	Sec-butyl
$\begin{array}{c} \text{CH}_3 \\ \\ -\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$	Tert-butyl

The general formula for an alkyl group is $\text{C}_n\text{H}_{2n+1}$ where "n" represents a number of carbon atoms.

The number of carbon and hydrogen atoms of alkyl is determined by this formula:

$$\text{H} = 2n + 1$$

Example 1

Find the number of hydrogen atoms of an alkyl group with 2 carbon atoms.

Solution

Given; $n = 2$ $\text{H} = ?$

$$\text{H} = 2n + 1$$

$$= 2(2) + 1$$

$$= 4 + 1$$

$$= 5$$

Example 2

Find the number of carbon atoms of an alkane with 7 hydrogen atoms.

Solution

Given: $H = 7$ $n = ?$

$$2n + 1 = H$$

$$2n + 1 = 7$$

$$2n = 7 - 1$$

$$2n = 6$$

$$n = 3$$

Alkyl groups are named by using a prefix which indicates the number of carbon atoms and suffix **-yl** which signifies that the group is alkyl group.

Examples

a) CH_3 methyl

b) CH_2-CH_3 or C_2H_5 ethyl

c) $CH_2-CH_2-CH_3$ or $-C_3H_7$ propyl

The first ten prefixes are presented the table below

Table 1.2: Prefixes for base names of alkanes

Number of carbon atoms	Prefixes
1	meth-
2	eth-
3	prop-
4	but-
5	pent-
6	hex-
7	hept-
8	oct-
9	non-
10	dec-



The first four Prefixes have historical origin while the other prefixes were derived from the Greek.

1.3.2 Nomenclature of Alkanes

Nomenclature is the naming of hydrocarbons. Generally the names of alkane contain prefix which indicates the number of carbon and suffix “-ane” which signifies the alkane family.

Examples

- a) CH_4 Methane
- b) $\text{CH}_3\text{-CH}_3$ ethane
- c) $\text{CH}_3\text{-CH}_2\text{-CH}_3$ propane
- d) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ butane

However Alkanes are named in two different systems:

1. Trivial system (common system)

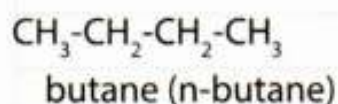
In this system, Alkanes are named using three terms:

- a) Normal (n)
- b) Iso-(i)
- c) Neo-(N)

a) Normal (n)

This system is used to name straight chain alkanes

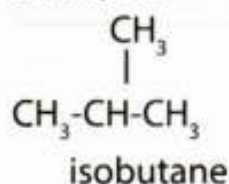
Examples



b) Iso-(i)

This system is used to name alkanes in which one methyl group is attached to the second carbon of the parent chain (iso-alkane).

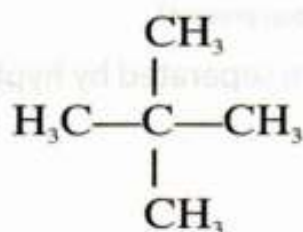
Examples



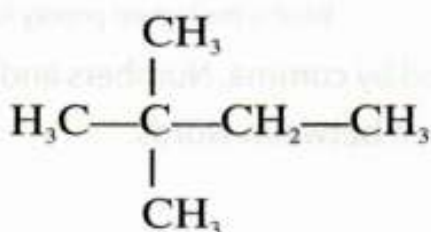
c) Neo- (N)

This system is used to name alkanes in which two methyl groups are attached to the second carbon of the parent chain (Neo-alkane).

Examples



neo pentane



neo hexane

NOTE: This Common system (Trivial system) cannot be used to name larger and more complicated molecules. Therefore another system was adopted.

2. IUPAC(International Union Pure and Applied Chemistry) system:

This is a systematic nomenclature modified by a group of scientist in 1958. The IUPAC rules for naming Alkanes are:

1. Determine the parent and side chains of the molecule.
2. Beginning with the end closest to the branching, number the carbon atoms in the parent chain and assign a number to each substituent. The parent chain should be the longest continuous carbon-carbon chain
3. Name the substituent as alkyl group prefixing the number of the carbon attached.
4. If there are two or more substituents, give each one a number and list them alphabetically with hyphens between words and numbers.
5. If a compound has two or more identical substituents, designate the number of identical substituents with the prefix di- (2), tri- (3), or tetra- (4) before the substituent's name.
6. Name the parent chain using appropriate prefix and suffix -ane
7. Then Write the name of the compound in the format:

How many carbons are in the longest chain?

Substituents

Parent Chain

Suffix

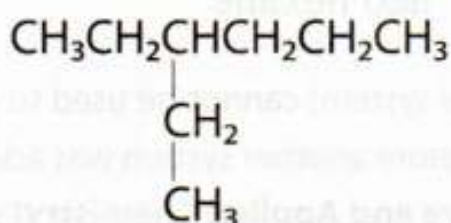
What is attached to the parent chain, and where?

What is the highest priority functional group present?

Numbers are separated by comma, Numbers and words are separated by hyphen (-) and there is no space between words.

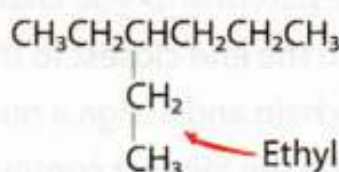
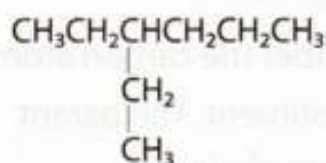
Example 1

Name the following alkane

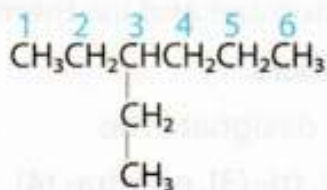


Solution

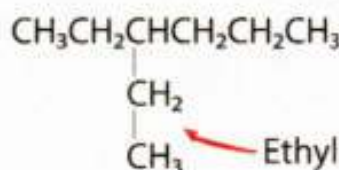
Step 1: Determine the parent and side chains of the molecule



Step 2: Number the carbon atoms from the end nearer a substituent to indicate the position of the substituent.



Step 3: Name the substituent as alkyl group prefixing the number of the carbon attached and ordering alphabetically.



Step 4: Name the parent chain using appropriate prefix and suffix -ane Hexane

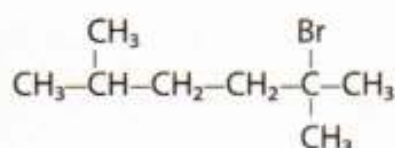
Step 5: Then Write the name of the compound in the format:

substituent number– substituent name Parent name

Therefore, the name of the compound is 3-ethylhexane

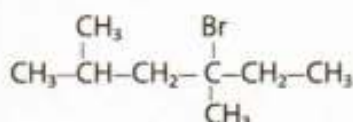
Example 2

Write the IUPAC name for the following alkane:

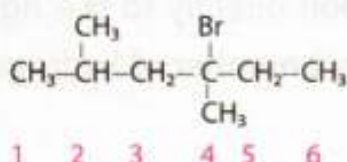


Solution

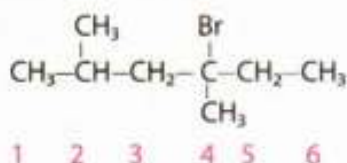
Step 1: Determine the parent and side chains of the molecule



Step 2: Number the carbon atoms from the end nearest a substituent to indicate the position of each substituent.



Step 3: Name the substituent as alkyl group prefixing the number of the carbon attached and ordering alphabetically. Note that Numbers are separated by comma, Numbers and words are separated by hyphen (-) and there is no space between words.



4-bromo-2,4-methyl



Step 4: Name the parent chain using appropriate prefix and suffix -ane

Hexane

Step 5: Then Write the name of the compound in the format:

substituent number– substituent name Parent name

Therefore, the name of the compound is 4,4-bromo-2,4-methylhexane.

Example 3

Write full structural and condensed structural formula for octane, C_8H_{18} .

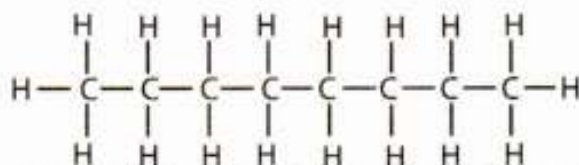
Solution

To write the full structural formula of octane

Step 1: write out the carbon atoms of the parent chain



Step 2: add H atoms so that all carbons have four bonds.



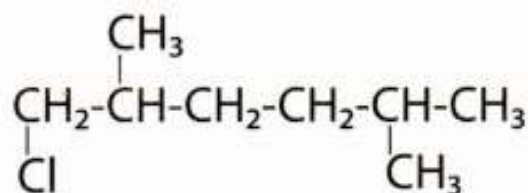
This is the structural formula of octane

Step3: write the hydrogen atoms bonded to each carbon directly to the right of the carbon atom. Use subscripts to indicate the correct number of hydrogen atoms.



EXERCISE

1. Write the structural and condensed structural formula for C_5H_{12} .
2. Write the IUPAC name for



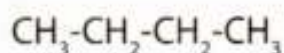
1.3.3 Isomerism in Alkanes

Isomers are defined as molecules with the same molecular formula but different structural formula. Because of their different structures, isomers have different properties; indeed, they are different compounds. Alkanes exhibit chain isomerism

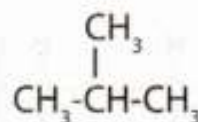
For example



Molecular formulas



n-butane



isobutane

The first three members of Alkanes have only one possible structure therefore the number of possible isomers of other alkanes is calculated by this formula: $2^{n-4} + 1$ where ($n \geq 4$) and it indicates the number of carbon atoms.

Note: This is only applicable up to 7 carbon atoms.

Example 1

Draw the five isomers for hexane

Solution

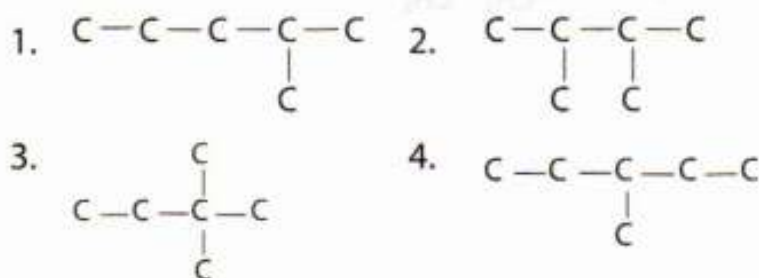
Step 1: find the number of possible structures (isomers) of hexane

$$\begin{aligned} n = 6 \qquad 2^{n-4} + 1 &= \\ &= 2^{6-4} + 1 \\ &= 2^2 + 1 \\ &= 4 + 1 \\ &= 5 \end{aligned}$$

Step 2: draw the carbon chain of the molecule to make the first isomer.

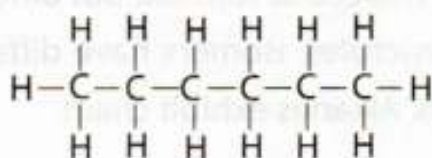


Step 2: determine the carbon chain structure of the other isomers by arranging the carbon atoms in four other unique ways:

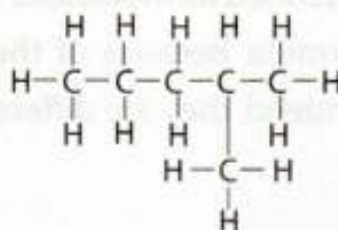


Step 3: fill in all the hydrogen atoms so that each carbon has four bonds.

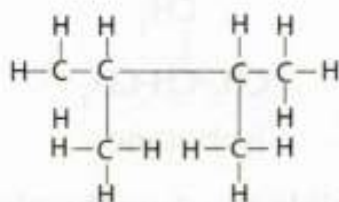
1.



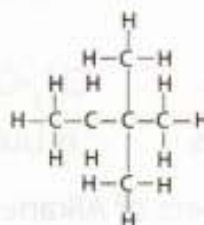
2.



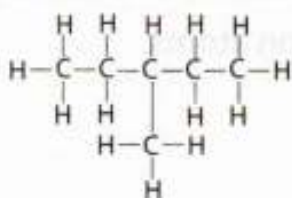
3.



4.



5.



EXERCISE

1. Draw the number of possible isomers of

a) Butane

b) Pentane

2. Which two molecules are isomers?

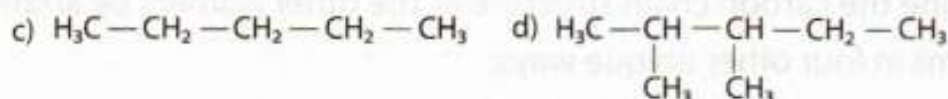
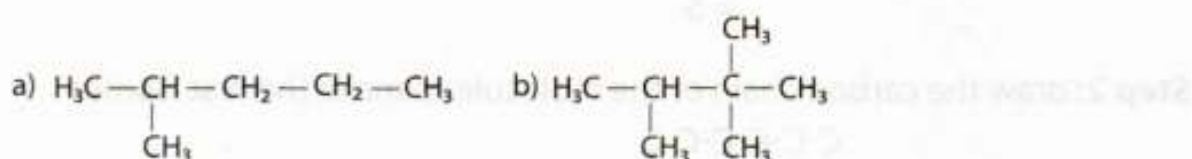


Table 1.3: the Number of Possible Isomers of the first ten alkanes

Name	Molecular Formula ($C_n H_{2n+2}$)	Condensed Structural Formula	Number of Possible Isomers
Methane	CH_4	CH_4	—
Ethane	C_2H_6	CH_3CH_3	—
Propane	C_3H_8	$CH_3CH_2CH_3$	—
Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$	2
Pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$	3
Hexane	C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$	5
Heptanes	C_7H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	9
Octane	C_8H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	18
Nonane	C_9H_{20}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	35
Decane	$C_{10}H_{22}$	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	75

1.3.4 Physical properties of Alkanes

1. Alkanes are non polar compounds therefore soluble in non polar solvent and not soluble in polar solvent
1. The first four members of straight chain alkanes — methane, ethane, propane and butane are gases. The next thirteen are liquids (C_5-C_{17}). Those containing eighteen carbons or more are waxy solids.
2. The boiling point of branched chain alkanes is lower than that of corresponding straight chain alkanes because the surface area of alkanes decreases with branching,
3. Alkanes are lighter than water.



Table 1.4: Physical Properties of Some Alkanes

Name	Molecular Formula	Melting Point (°C)	Boiling Point (°C)	Density (20°C)*	Physical State (at 20°C)
Methane	CH ₄	-182	-164	0.668 g/L	Gas
Ethane	C ₂ H ₆	-183	-89	1.265 g/L	Gas
Propane	C ₃ H ₈	-190	-42	1.867 g/L	Gas
Butane	C ₄ H ₁₀	-138	-1	2.493 g/L	Gas
Pentane	C ₅ H ₁₂	-130	36	0.626 g/mL	Liquid
Hexane	C ₆ H ₁₄	-95	69	0.659 g/mL	Liquid
Octane	C ₈ H ₁₈	-57	125	0.703 g/mL	Liquid
Decane	C ₁₀ H ₂₂	-30	174	0.730 g/mL	Liquid

*Note the change in units going from gases (grams per liter) to liquids (grams per milliliter). Gas densities are at 1 atm pressure.

1.3.5 chemical properties of alkane

Alkanes are generally inert towards acids, bases, oxidising and reducing agents. However, Alkanes are quite inert towards common reagents such as acids, alkalis and oxidizing agent. Alkanes do not undergo many reactions; they are called **paraffins**, from the Latin *parum affinis* meaning *little affinity*. However, alkanes undergo the following reactions under certain conditions:

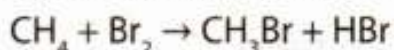
1. Substitution
2. Combustion

1. Substitution

In a substitution reaction, one atom is displaced by another atom. Generally Alkanes undergo substitution reaction with the halogens in the presence of light this type of substitution is known as halogenation.

For example

Methane reacts with chlorine and bromine.



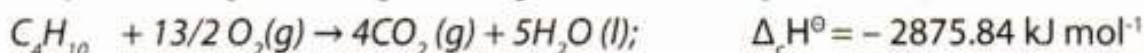
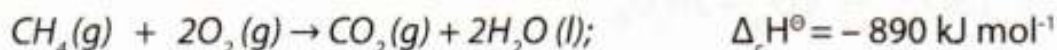
This reaction is a substitution reaction because one of the hydrogen atoms from methane is replaced by a bromine atom.



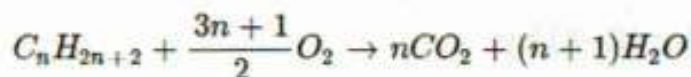
2. Combustion Reaction

Alkanes burnt in the presence of sufficient oxygen to form carbon dioxide and water and enormous amount of heat energy is released. This type of combustion is known as complete combustion.

For example



The general combustion reaction of alkanes can be generalized as follows:

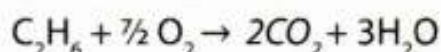
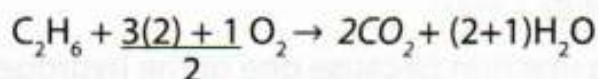
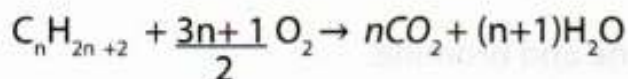


Example 1

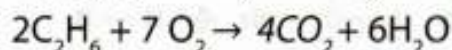
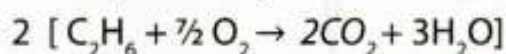
Write a balanced combustion equation of ethane (C_2H_6)

Solution

ethane (C_2H_6) $n = 2$



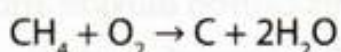
Multiply the whole equation by 2 to make the coefficient whole number.



Alkanes can also undergo incomplete combustion in insufficient amount of oxygen to form carbon monoxide and water.

For example

Methane + (insufficient) Oxygen $\rightarrow$ Carbon + Water



Methane + (insufficient) Oxygen $\rightarrow$ Carbon Monoxide + Water

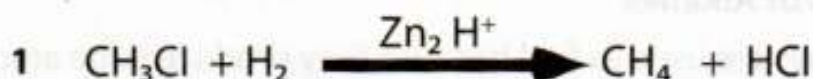
**1.3.6 General Methods of Preparation of Alkanes.**

The following methods are used to prepare alkanes

1. From Alkyl halides (R-X) by:**a) Reduction reaction**

All Alkyl halides (except alkyl fluorides) are reduced to Alkanes in the presence of zinc and dilute hydrochloric acid

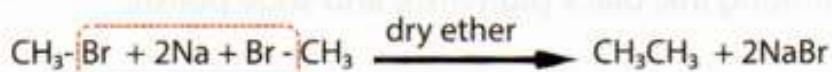
For examples



b) Wurtz reaction

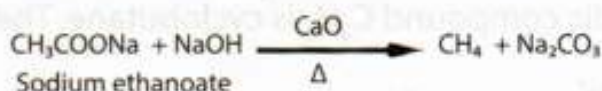
In this process Haloalkanes (Alkyl halides) react with metallic sodium in dry ether to form symmetrical alkanes with double the number of carbon atoms present in the alkyl halide. This reaction is known as Wurtz reaction.

For examples



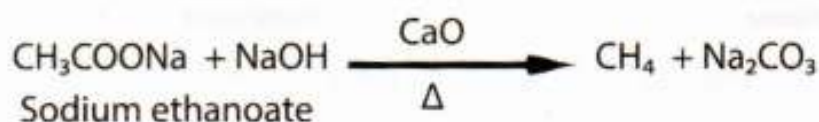
2. Decarboxylation

Sodium salts of carboxylic acids (RCOONa) on heating with soda lime ($\text{NaOH} + \text{CaO}$) form alkanes containing one carbon atom less than the salt.

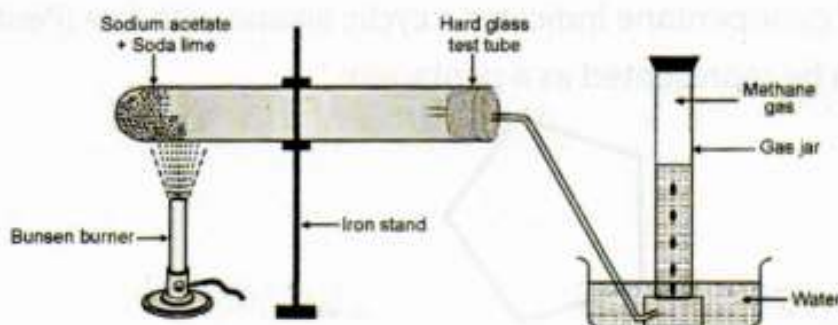


Laboratory preparation of methane

Sodium acetate (CH_3COONa) is mixed with soda lime (NaOH) and is heated as shown below. The evolved gas is collected by upward delivery and downward displacement of water and the reaction is:



The mechanism of the reaction is:

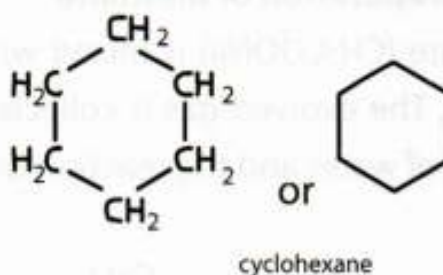
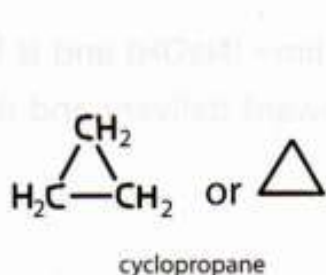


1.3.7 Uses of Alkanes

1. Alkanes are used as fuel because they produce large amount of heat in combustion.
2. Liquid alkanes are used as solvents
3. Higher alkanes that have 17 to 20 carbon atoms are viscous liquid and used as lubricants
4. Solid Alkanes with 21 to 30 carbon atoms are the constituents of paraffin wax used in making candles or base in cosmetics.
5. Carbon black obtained by incomplete combustion of alkane is used in the manufacture of printing ink, black pigments, and shoe polish.

1.3.8 Cycloalkanes

Cycloalkanes are cyclic hydrocarbons with only single carbon-carbon bonds. Cycloalkanes are named by adding the suffix *cyclo-* to the name of the open-chain compound having the same number of carbon atoms as there are in the ring. Thus the name for the cyclic compound C_4H_8 is cyclobutane. The general formula of cycloalkanes is C_nH_{2n} .



Example

Draw the structure of:

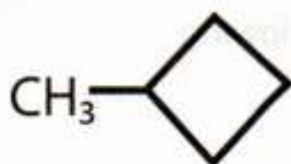
- a) Cyclopentane
- b) Methylcyclobutane

Solution

- a) The name of cyclopentane indicates a cyclic alkane with five (Pent-) carbon atoms. It can be represented as a pentagon:

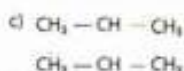
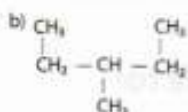
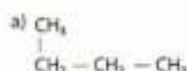


b) The name of methylcyclobutane indicates a cyclic alkane with four (but-) carbon atoms. It can be represented as:

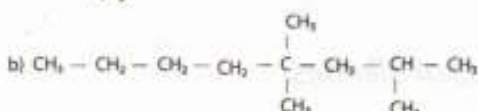
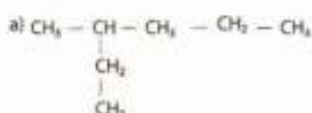


EXERCISE

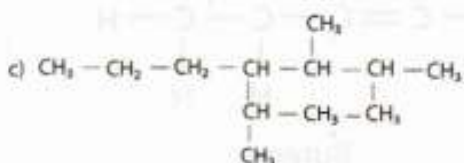
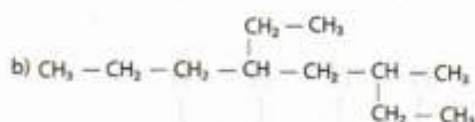
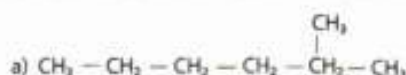
1. How does a branched hydrocarbon differ from a normal hydrocarbon?
2. Identify the longest carbon chain in the following.



3. Decide how to correctly number the longest chain in the following according to IUPAC rules.



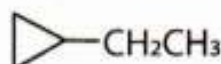
4. Give the correct IUPAC name to each of the following:



5. Without referring to a table, predict which has a higher boiling point between hexane and octane? Explain your answer.
6. If 25mL of hexane were added to 200mL of water in a beaker which of the following would you expect to happen?
 - a) Hexane would dissolve in the water.
 - b) Hexane would not dissolve in the water and would float on top.
 - c) Hexane would not dissolve in the water and would sink to the bottom.



7. Why are Alkanes sometimes called paraffins?
8. Name the most important reaction of alkane.
9. Draw the structure of each of the following
 - a) Cycloheptane
 - b) Ethylcyclohexane
10. What is the IUPAC name for this compound



1.4 ALKENES (OLEFINS)

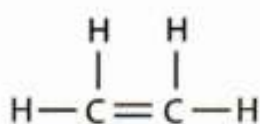
Lesson objectives

By the end of the lesson, the learner should be able to:

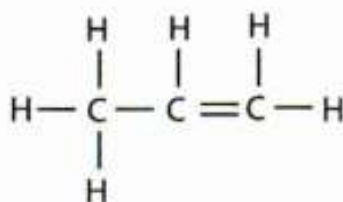
- Define alkenes and give examples.
- Classify alkenes.
- Name and draw alkenes.
- Identify different types of isomerism in alkenes.
- Explain general preparation of alkenes.
- Explain the physical and chemical properties of alkenes.
- State common uses of alkenes.

Alkenes are unsaturated hydrocarbons containing at least one carbon-carbon double bond between carbon atoms.

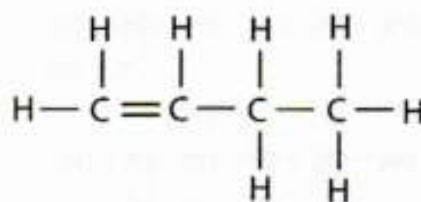
For examples



Ethene



Propene

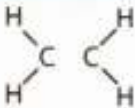
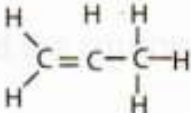
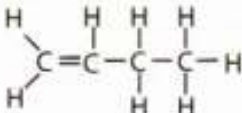
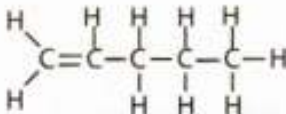
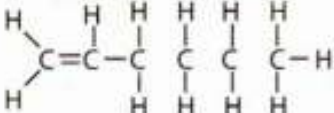


Butene

Because of the double bond, alkenes have fewer hydrogen atoms than the corresponding alkane and are therefore called unsaturated hydrocarbons. Alkenes are also called olefins because they form oily liquids.

Alkenes have molecular formula of C_nH_{2n} therefore Alkenes form homologous series.

Table 1.5: the first five homologous series of Alkenes

n	Name	Molecular Formula (C_nH_{2n})	Structural Formula	Condensed structural Formula
1	ethene	C_2H_4		$CH_2=CH_2$
2	propene	C_3H_6		$CH_2=CHCH_3$
3	1-butene	C_4H_8		$CH_2=CHCH_2CH_3$
4	1-pentene	C_5H_{10}		$CH_2=CHCH_2CH_2CH_3$
5	1-hexene	C_6H_{12}		$CH_2=CHCH_2CH_2CH_2CH_3$

The number of carbon and hydrogen atoms of alkene can be calculated by this formula **$H = 2n$**

Example 1

Find the number of hydrogen atoms in an alkene with 5 carbon atoms.

Solution

$$n = 5 \quad H = ?$$

$$H = 2n$$

$$= 2(5)$$

$$= 10$$

Example 2

Find the number of carbon atoms of an alkene with 4 hydrogen atoms.

Solution

$$n = ? \quad H = 4$$

$$2n + 2 = H$$

$$2n = 4$$

$$n = 2$$

1.4.1 Classification of alkenes

Alkenes are classified into two groups:

1. Symmetric alkenes
2. Asymmetric alkenes

1. Symmetric alkenes

These are alkenes in which two same groups are being attached to the carbon-carbon double bond.

For example

- a) $\text{CH}_3\text{-CH=CH-CH}_3$
- b) $\text{CH}_3\text{-CH}_2\text{-CH=CH-CH}_2\text{-CH}_3$

2. Asymmetric alkenes

These are alkenes in which two different groups are being attached to the carbon-carbon double bond.

For example

- a) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_3$
- b) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH=CH-CH}_2\text{-CH}_3$

1.4.2 Nomenclature of Alkenes

Generally the names of Alkanes contain prefix which indicates the number of carbon and suffix **-ane** which signifies the alkane family.

For examples

- | | |
|--|---------|
| a) CH_4 | Methane |
| b) $\text{CH}_3\text{-CH}_3$ | ethane |
| c) $\text{CH}_3\text{-CH}_2\text{-CH}_3$ | propane |
| d) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ | butane |

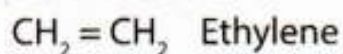
However, Alkenes are named in two different systems.

1. Common system
2. IUPAC system

1. Common system

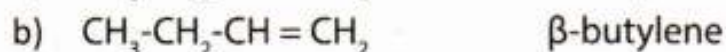
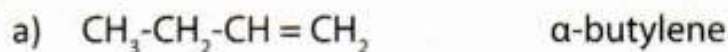
In this system alkenes are named using appropriate prefix and suffix **-ylene**.

For example,



The position of the double bond is indicated by the Greek letters α, β, γ etc.

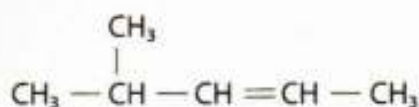
For example

**2. IUPAC system**

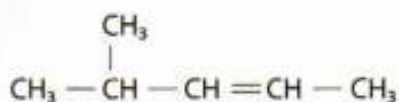
In this system, alkenes are named as Alkanes but the parent chain of alkene must contain the Carbon-Carbon double bond and the double bonds must be indicated.

Example 1

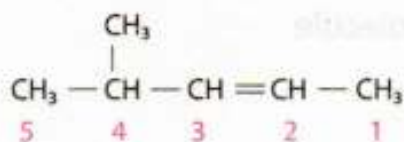
Write the IUPAC name for the following:

**Solution**

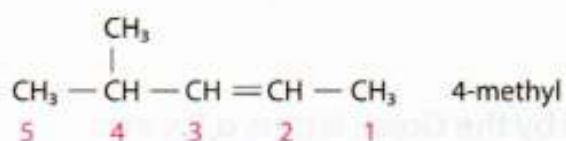
Step 1: Determine the parent and side chains of the molecule. The parent chain must contain the carbon-carbon double bond.



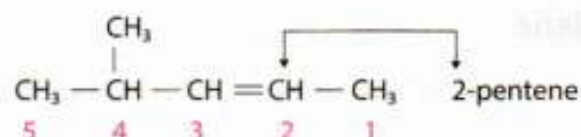
Step 2: Number the parent chain carbon atoms from the end nearer to the double bond. Alkenes with two or three carbons do not need numbers.



Step 3: Name the substituent as alkyl group prefixing the number of the carbon attached and ordering alphabetically.



Step 4: Name the parent chain using appropriate prefix and suffix -ene. Indicate the position of the double bond.

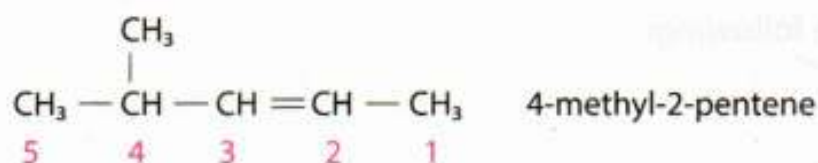


Step 5: Write the name of the compound in the format:

Substituents

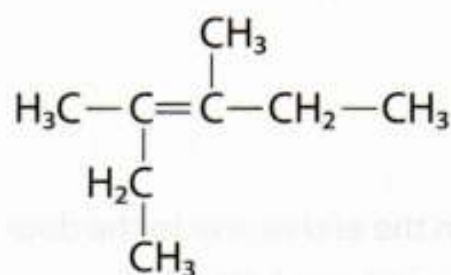
Parent Chain

Suffix



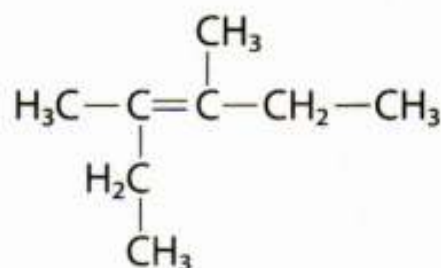
Example 3

Write the IUPAC name for the following:

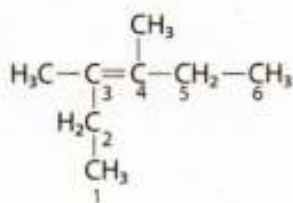


Solution

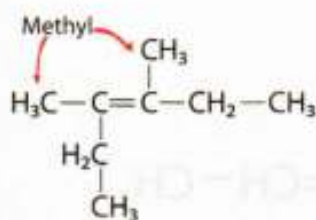
Step 1: Determine the parent and side chains of the molecule



Step 2: Number the carbon chain from the end nearer the double bond.

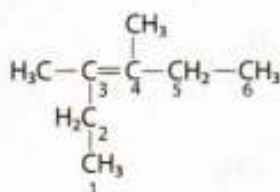


Step 3: Name the substituent as alkyl group prefixing the number of the carbon attached and ordering alphabetically.



3,4-dimethyl

Step 4: Name the parent chain using appropriate prefix and suffix -ene. Indicate the position of the double bond.



3-hexene

Step 5: Write the name of the compound in the format:

Substituents

Parent Chain

Suffix

3,4-dimethyl-3-hexene

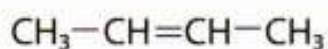
1.4.3 Isomerism in alkenes

Alkenes exhibit;

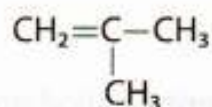
- Chain isomerism
- Position isomerism
- Geometrical isomerism

a) Chain isomerism**For example**

1-butene



and

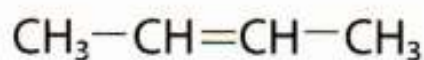


2-butene

2-methylpropene

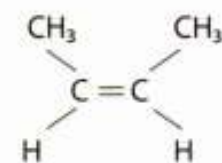
b) Position isomerism**For example**

and

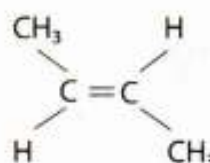


1-butene

2-butene

c) Geometrical isomerism (Cis-trans isomerism)**For example**

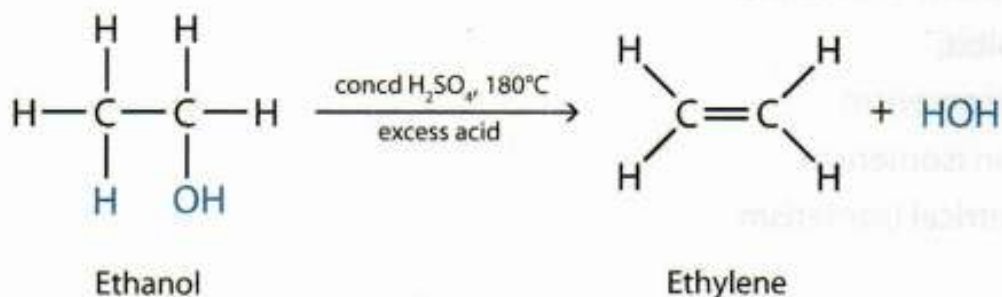
Cis-



Trans-

1.4.4 General methods of Preparation of alkenes**1. Dehydration of alcohol**

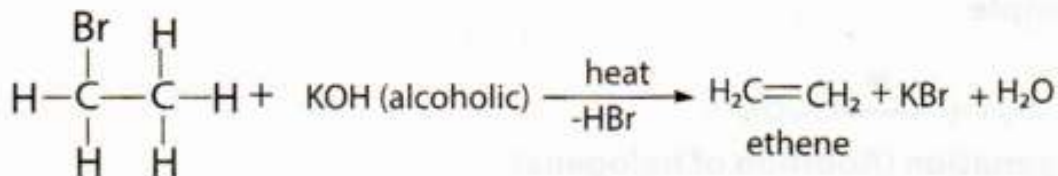
In this process alcohol is heated in the presence of sulfuric acid or phosphoric acid to produce alkane.

For example

2. Dehydrohalogenation of alkyl halides

In this process alkyl halide is heated in alcoholic solution of potassium hydroxide to produce alkene.

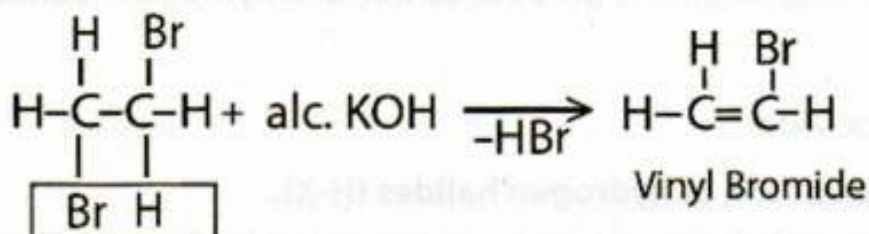
For example



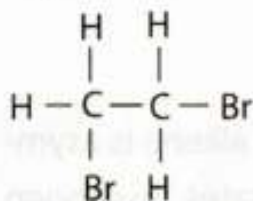
3. Dehalogenation of vicinal dihalide

In this process vicinal dihalide is heated in the presence of zinc in alcoholic solution to produce alkenes

For example



Vicinal dihalide is dihalide that the two halogen atoms are bonded to the adjacent carbons



Vicinal dihalide

1.4.5 Physical properties of alkenes

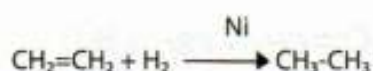
1. The first three alkenes are gases, next few members are liquids and higher members are solids.
2. Ethylene is colourless gas with sweet odour while others are colourless.
3. They are soluble in organic solvents like benzene but are insoluble in water.
4. Boiling points of alkene increases with the increase in molecular mass.
5. Alkenes with branches have lower boiling points.

1.4.6 Chemical properties of alkenes

a) Hydrogenation (Addition of H₂)

Alkenes react with hydrogen in the presence of a metal catalyst such as nickel to produce alkane.

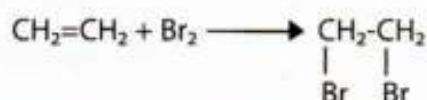
For example



b) Halogenation (Addition of halogens)

Alkenes react with halogens such as chlorine, bromine and iodine

For example



Ethene reacts explosively with fluorine to produce carbon and hydrogen fluoride gas.



c) Hydrohalogenation (addition of hydrogen halides (H-X)).

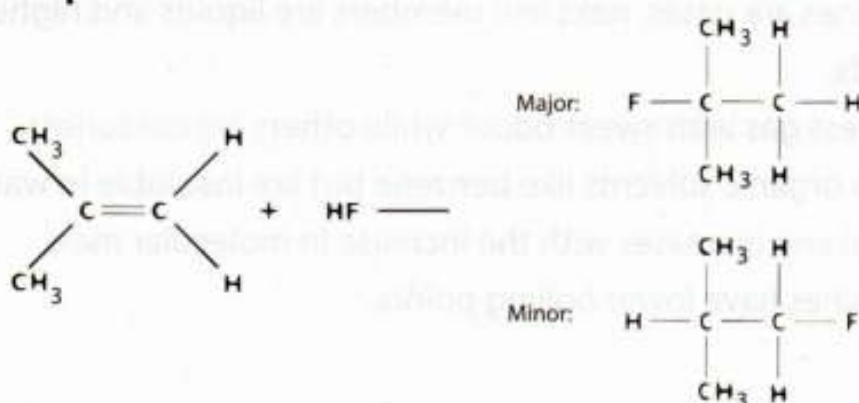
Alkenes react with hydrogen halides such as hydrogen chloride and hydrogen bromide to form monohaloalkanes.

For example



In this process we use a rule known as Markovnikov's rule if the alkene is asymmetric to rule predict the major products. Markovnikov's rule states 'hydrogen is added to the carbon atom of the double bond with the greatest number of carbon atoms.

For example



d) Hydration (Addition of H₂O)

Alkenes react with water to produce alcohol.

For example**e) Combustion**

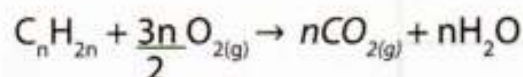
Alcohols react with sufficient or insufficient supply of oxygen to produce carbon monoxide or carbon dioxide respectively.

For example

$\text{C}_2\text{H}_{4(g)} + 3\text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)} + 2\text{H}_2\text{O}_{(l)}$ (Complete combustion of ethene-excess supply of oxygen)

$\text{C}_2\text{H}_{4(g)} + 2\text{O}_{2(g)} \rightarrow 2\text{CO}_{(g)} + 2\text{H}_2\text{O}_{(l)}$ (Incomplete combustion of ethene – insufficient supply of oxygen)

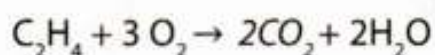
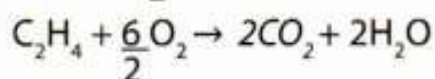
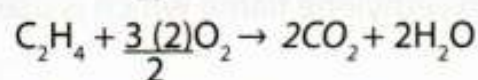
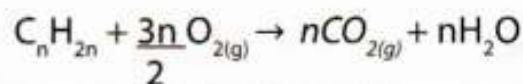
The general balanced equation for complete combustion of alkene is

**Example**

Write the balanced combustion equation of ethene (C₂H₄)

Solution

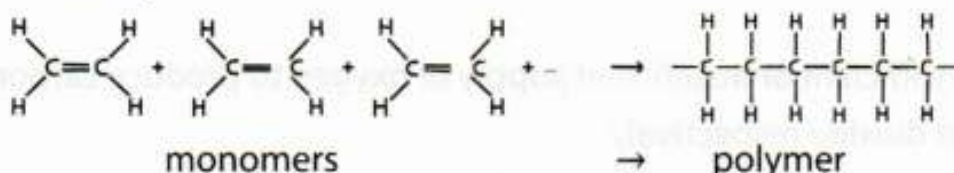
Given: ethane (C₂H₄) n = 2



f) Polymerization

This is the most useful commercial reactions of alkenes. During this reaction small molecules of alkenes called monomers are assembled together into large molecules called polymers.

For example



The following table illustrates some common addition polymers and their uses.

Table 1.7: common polymers and their uses

Monomer	Polymer	Polymer Name	Some Uses
$\text{CH}_2=\text{CH}_2$	$\sim\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\sim$	polyethylene	plastic bags, bottles, toys, electrical insulation
$\text{CH}_2=\text{CHCH}_3$	$\begin{array}{c} \sim\text{CH}_2\text{CHCH}_2\text{CHCH}_2\text{CH}\sim \\ \quad \quad \\ \text{CH}_3 \quad \text{CH}_3 \quad \text{CH}_3 \end{array}$	polypropylene	carpeting, bottles, luggage, exercise clothing
$\text{CH}_2=\text{CHCl}$	$\begin{array}{c} \sim\text{CH}_2\text{CHCH}_2\text{CHCH}_2\text{CH}\sim \\ \quad \quad \\ \text{Cl} \quad \text{Cl} \quad \text{Cl} \end{array}$	polyvinyl chloride	bags for intravenous solutions, pipes, tubing, floor coverings
$\text{CF}_2=\text{CF}_2$	$\sim\text{CF}_2\text{CF}_2\text{CF}_2\text{CF}_2\text{CF}_2\text{CF}_2\sim$	polytetrafluoroethylene	nonstick coatings, electrical insulation

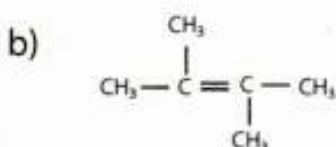
1.4.7 Uses of alkenes

1. Ethylene is used to manufacture polythene
2. In combustion with oxygen, ethene is used oxyethylene flame which is used for cutting and welding metals.
3. Ethene is also used as a ripening agent for foods.
4. Alkenes are used to prepare alkanes, alcohols etc.

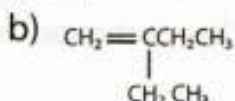
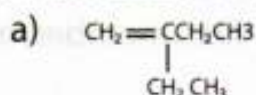


EXERCISE

1. Write the IUPAC name for each of the following



2. Name each of the following compounds



3. Briefly identify the important distinction between saturated and unsaturated hydrocarbons.

4. Draw the structure of each compound:

a) 2-methyl-2-pentene

b) 2,3-dimethyl-2-pentene

c) 5-methyl-1-hexene

d) 2-ethyl-3-heptene

5. What is the principal difference and similarity in properties between alkenes and alkanes?

6. If C_4H_8 reacts with HBr , what is the molecular formula of the product?

7. Define a monomer and a polymer

1.5 ALKYNES (ACETYLENE)

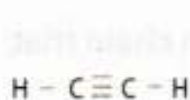
Lesson objectives

By the end of the lesson, the learner should be able to:

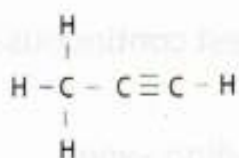
- Define alkynes and give examples.
- Classify alkynes.
- Name and draw alkynes.
- Identify different types of isomerism in alkynes.
- Explain general preparation of alkynes.
- Explain physical and chemical properties of alkynes.
- State common uses of alkynes

Alkynes are defined as unsaturated hydrocarbon compounds containing at least one carbon-carbon triple bond between carbon atoms.

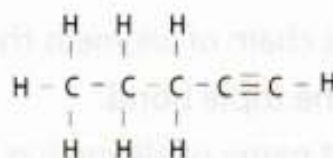
For Examples



Ethyne (acetylene gas)



Propyne



Pentyne



Because of the triple bond, alkynes are highly unsaturated hydrocarbons. Alkynes are also called acetylene because the common name of the first member of alkyne is known as acetylene.

Alkynes have a general molecular formula of C_nH_{2n-2} . Where n , is the number of carbon atom in alkynes, the simplest alkyne is ethyne (C_2H_2) and is also called acetylene. Alkynes form same homologous series.

Table 1.8: The homologous series of the first five Alkynes

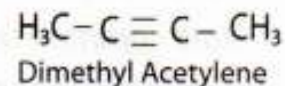
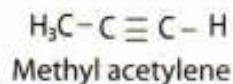
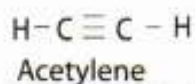
n	Name	Molecular formula C_nH_{2n-2}	Structural formula	Condensed structural formula
1	ethyne	C_2H_2	$H-C \equiv C-H$	$CHCH$
2	propyne	C_3H_4	$ \begin{array}{c} H \\ \\ H-C \equiv C-C-H \\ \\ H \end{array} $	$CH \equiv CCH_3$
3	1-butyne*	C_4H_6	$ \begin{array}{c} H \quad H \\ \quad \\ H-C \equiv C-C-C-H \\ \quad \\ H \quad H \end{array} $	$CH \equiv CCH_2CH_3$
4	1-pentyne*	C_5H_8	$ \begin{array}{c} H \quad H \quad H \\ \quad \quad \\ H-C \equiv C-C-C-C-H \\ \quad \quad \\ H \quad H \quad H \end{array} $	$CH \equiv CCH_2CH_2CH_3$
5	1-hexyne*	C_6H_{10}	$ \begin{array}{c} H \quad H \quad H \quad H \\ \quad \quad \quad \\ H-C \equiv C-C-C-C-C-H \\ \quad \quad \quad \\ H \quad H \quad H \quad H \end{array} $	$CH \equiv CCH_2CH_2CH_2CH_3$

1.6.1 Nomenclature of Alkynes

1. Common system

In this system the carbon-carbon triple bond is named as acetylene therefore the alkynes are named alkyl acetylene.

For example



2. IUPAC system

In this system, alkynes are named in the same way as alkenes with the following exceptions:

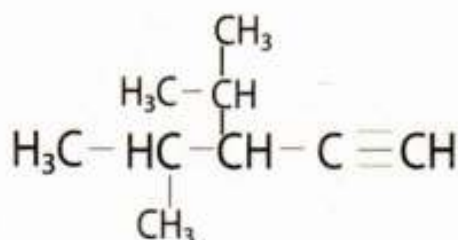
- The parent chain of alkyne is the longest continuous carbon chain that contains the triple bond.
- The parent name of alkyne has the ending -yne.
- The parent chain is counted starting from the end nearest to triple bond.

Example 1

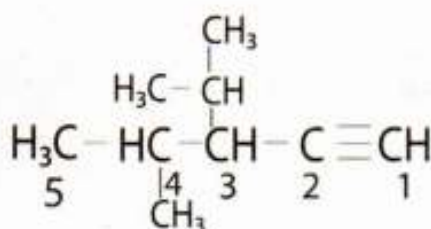
Name the following alkynes

Solution

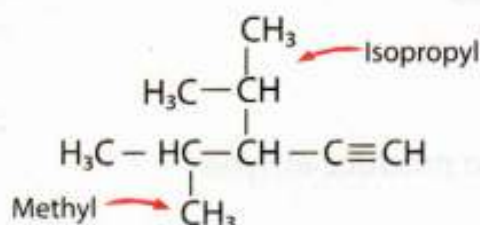
Step 1: Determine the parent and side chains of the molecule



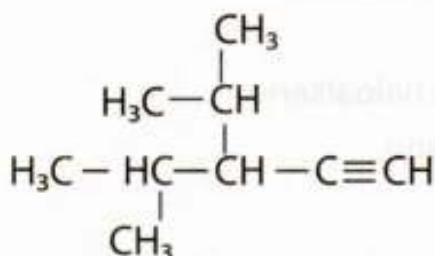
Step 2: Number the parent chain, giving the triple bond the lowest number, 1.



Step 3: Name the substituent as alkyl group prefixing the number of the carbon attached and ordering alphabetically.



Step 4: Name the parent chain using appropriate prefix and suffix -yne. Indicate the position of the triple bond.



Step 5: Then write the name of the compound in the format:

Substituents

Parent Chain

Suffix

4-methyl-3-isopropyl-1-pentyne



1.5.2 Isomerism in alkynes

Alkynes exhibit:

a) Chain isomerism

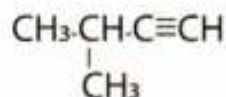
For example



pent-1-yne



and

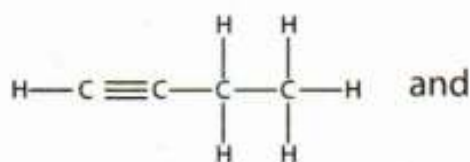


3-methyl-1-butyne



b) Position isomerism

For example



1-butyne



and



2-butyne

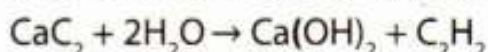


1.6.3 General methods of Preparation of alkynes

Alkynes are prepared by:

1. Action of water with Calcium carbide

In this process calcium carbide reacts with water to produce ethyne



Note: This process is used in preparing only ethyne.

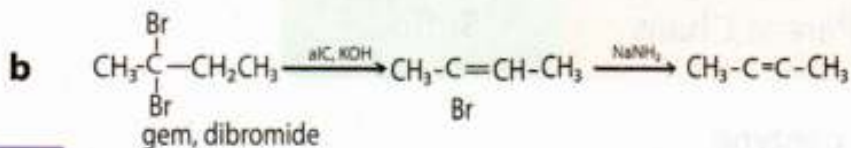
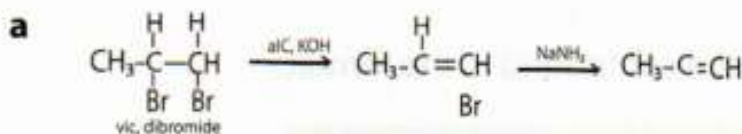
2. Dehydrohalogenation of Vicinal and germinal dihalides

This process undergoes two steps.

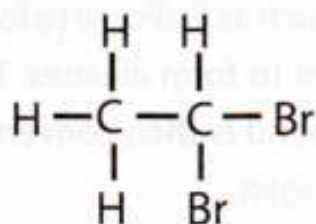
Step 1: Alcoholic KOH is used as catalyst to produce haloalkene.

Step 2: Sodamide is used as catalyst to produce alkyne.

For example



Geminal dihalide is dihalide that the two halogen atoms are bonded to the same carbon atom.

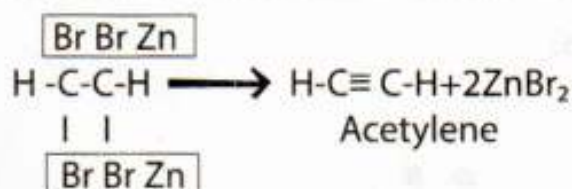


geminal dihalide

3. Dehalogenation of tetrahalides

In this process tetrahalides are heated with zinc dust to get alkynes.

For example



1.5.4 Physical properties of alkynes

1. The first three alkynes are gases, the next eight members are liquids and higher members are gases.
2. All alkynes except ethyne are odourless.
3. Alkynes have slightly higher boiling points than corresponding Alkanes and alkenes.
4. The melting and boiling points increases with increase in molecular mass or number of number of carbon atoms.
5. They are less dense in water and they are insoluble in water.

1.5.6 Uses of alkyne

1. Acetylene is used in oxy-acetylene flame which is used for cutting and welding metals
2. Acetylene is used in manufacture of westron



3. Alkynes are used in formation of aldehydes, ketones, alkanes etc.
4. Alkynes are also used for manufacturing plastics and synthetic rubber.



EXERCISE

1. Briefly describe the important differences between an alkene and an alkyne. How are they similar?
2. Do alkynes show cis-trans isomerism? Explain.
3. Draw the structure for each compound.
 - a) acetylene
 - b) 3-methyl-1-hexyne
 - c) 4-methyl-2-hexyne
 - d) 3-octyne
4. Name each alkyne.
 - a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
 - b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CCH}_3$

1.6 AROMATICS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define aromatic compounds.
- Describe bonding in benzene.
- Name aromatic compounds.
- Draw line-angle structural formula.

Aromatic hydrocarbons are hydrocarbons which have pleasant aromas. For example, the pleasant smells of cinnamon, vanilla, and jasmine are all caused by aromatic compounds.

Aromatic hydrocarbons are defined as cyclic organic compounds which contain at least one aromatic ring that resembles benzene.

Aromatic hydrocarbons are classified into two main types:

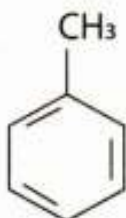
1. Benzenoid

These are aromatic hydrocarbons that contain benzene ring structure.

For example



benzene



toluene

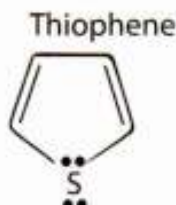
2. Non-benzenoid

These are aromatic hydrocarbons that do not contain benzene ring structure.

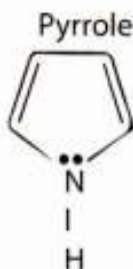
For example



Furan



Thiophene

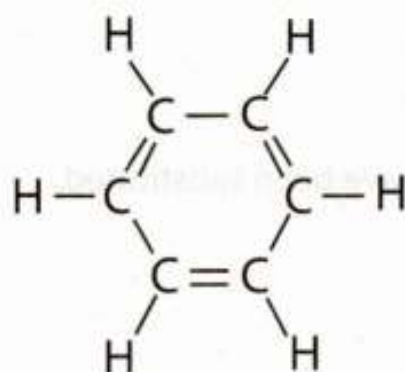


Pyrrole

In this level, we will study Benzenoid aromatic hydrocarbon-the basic building structure of benzenoid hydrocarbons benzene.

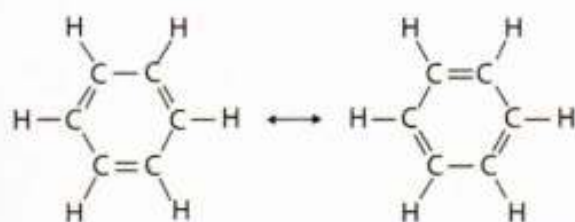
1.6.1 Benzene and its structure

Benzene was discovered by Michael Faraday in 1825 and given its structure by Friedrich August Kekulé in 1865.

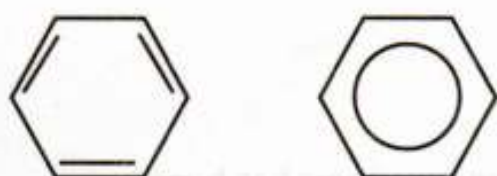


Benzene

The Kekulé's structure of benzene consists of alternating single and double carbon to carbon bonds. The structure of benzene is better represented by the following resonance structures.



Today, benzene is often represented with the shorthand notations:



Recognized representations of benzene

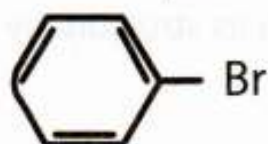
Each point in the hexagon represents a carbon atom with a hydrogen atom attached to it. The ring structure of benzene occurs in many organic compounds.

1.6.2 Classification of benzene

Benzenes are classified into:

1. Monosubstituted benzenes

These are benzenes in which only one of the hydrogen atoms has been substituted

For example

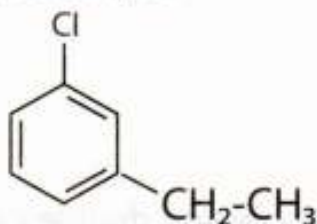
Bromobenzene



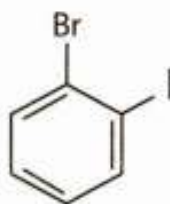
Ethylbenzene

2. Distributed benzenes.

These are benzenes, in which two hydrogen atoms have been substituted,

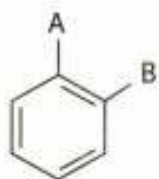
For example

1-Chloro-3-ethylbenzene



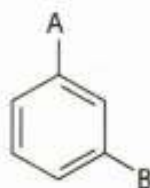
1-bromo-2-iodobenzene

Distributed benzenes exhibit three possible positional isomerism



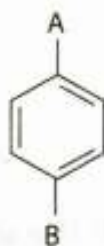
ortho-

1,2-



meta-

1,3-

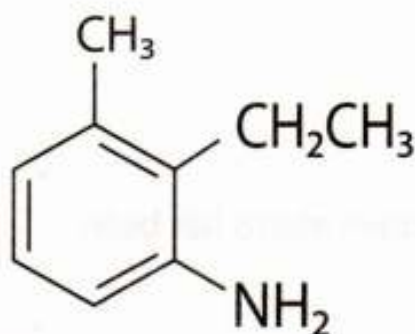


para-

1,4-

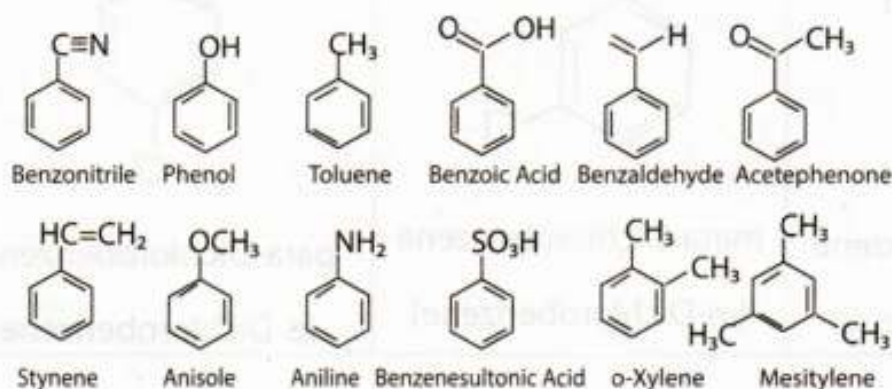
3. Polysubstituted benzene

These are benzenes, in which more than two hydrogen atoms has been substituted

For example

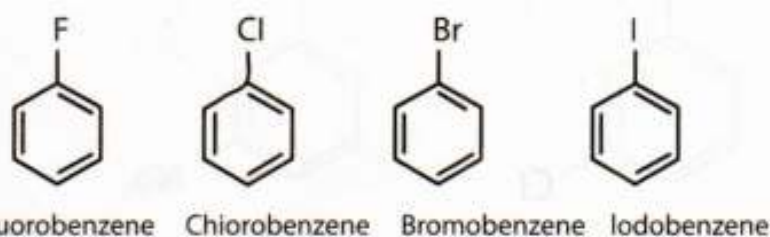
1.6.3 Nomenclature of benzene

Many monosubstituted benzenes have common names that can be learned only through familiarity.

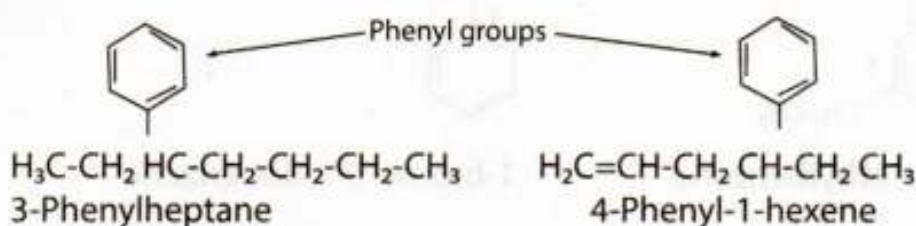


They can also be named by using benzene as parent chain and prefixing by the substituent.

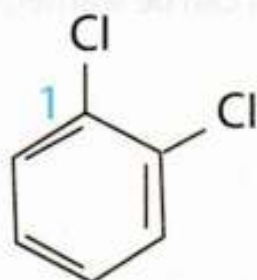
For examples



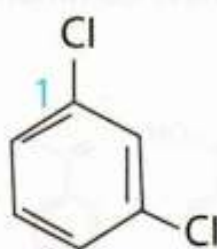
Some substituted benzenes, especially those with large substituents, are named by treating the benzene ring as the substituent. In these cases, the benzene substituent is called a phenyl group.



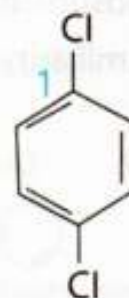
Disubstituted benzene with identical substituents, we use the prefix "di-".



ortho-Dichlorobenzene
(o-Dichlorobenzene)



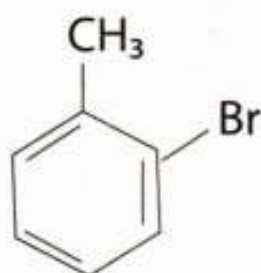
meta-Dichlorobenzene
(m-Dichlorobenzene)



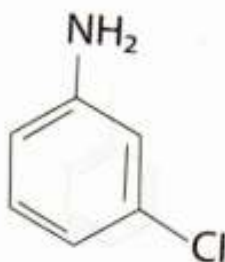
para-Dichlorobenzene
(p-Dichlorobenzene)

Disubstituted benzene where one group imparts a special name, name the compound as a derivative of that molecule:

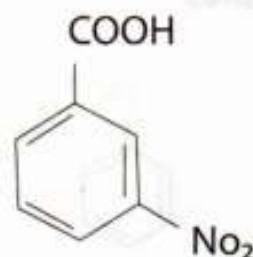
For examples



2-bromotoluene

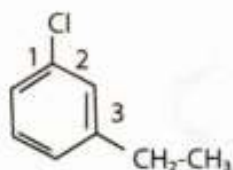


3-chloroaniline

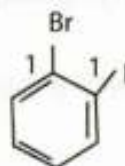


3-nitrobenzoic acid

Disubstituted benzene where neither group imparts a special name, locate the groups and alphabetize.



1-Chloro-3-ethylbenzene



1-bromo-2-iodobenzene

1.6.4 Chemical properties of benzene

Benzene resists addition reaction but readily undergoes:

1. Combustion reactions

In combustion reaction benzene burns with a smoke flame because it is highly flammable.

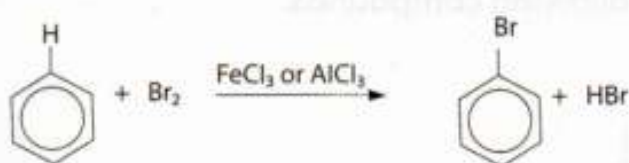


2. Substitution reaction

The common substitution reaction that benzene can undergo is:

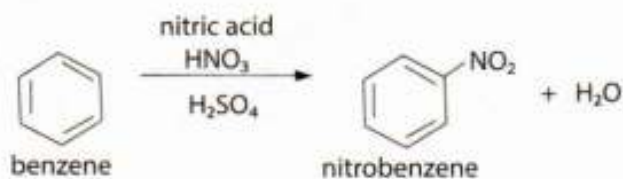
a) Halogenations

In this process halogen is added to benzene with presence of the catalysts such as ferric halide or aluminum halide



b) Nitration

In this process benzene is treated with a mixture of nitric acid and sulfuric acid.



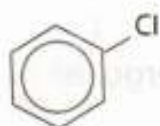
1.6.5 Uses of benzene

1. Benzene can be converted to styrene which is used in the manufacture of synthetic fibres like nylon
2. Benzene is an important source of chemicals which are used as pesticides and detergents
3. Derivatives of benzene such as toluene is used in making explosives
4. Derivatives of benzene aniline are used as dyes and drugs

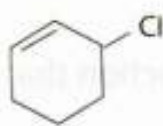
**EXERCISE**

1. Briefly describe the bonding in benzene.
2. How do typical reactions of benzene differ from those of the alkenes?
3. What does the circle mean in the chemist's representation of benzene?
4. Which of the following compounds are aromatic?

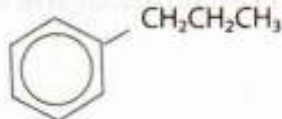
a)



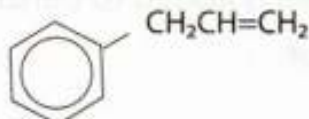
b)



c)



d)



5. Write the IUPAC name for each of the following compounds.

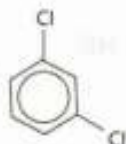
a)



b)



c)



6. Draw the structural formula for each of the following compounds.

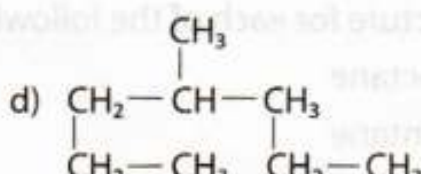
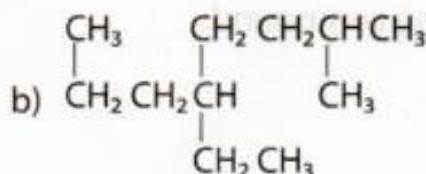
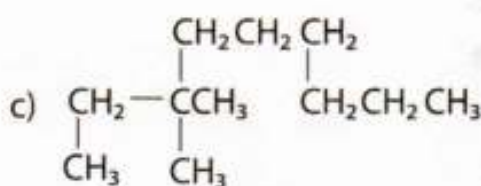
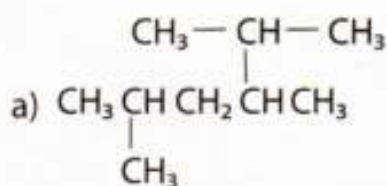
a) aniline

b) 1-chloro-4-fluorobenzene

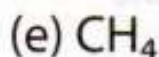
c) 4-ethyltoluene

Chapter Review questions

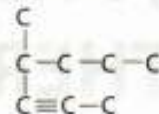
1. Name and draw the structural formulas for four smallest Alkanes.
2. What is the main component of natural gas?
3. What is meant by the term saturated, when applied to hydrocarbons?
4. What is meant by fractional distillation?
5. Where in the fractioning column will the longest hydrocarbon condense?
6. Name and draw the structural formula for the four smallest alkenes.
7. C_nH_{2n} can represent an alkane or an alkene. Explain.
8. What does the term aromatic imply about an organic molecule?
9. Which of the following could be the formula of a cyclic alkane?
 - a) C_8H_{18}
 - b) C_5H_8
 - c) C_6H_6
 - d) C_5H_{10}
10. How many carbons are in the longest chain of each of the following compounds? Write the name of the longest alkane chain for each.



11. Name the substituents attached to the longest chains in question 10 above and give the IUPAC names for the compounds.
12. What does the term normal imply when used in Alkanes?
13. Identify the following as an alkane, alkene, or alkyne. Assume that the compounds are not cyclic.

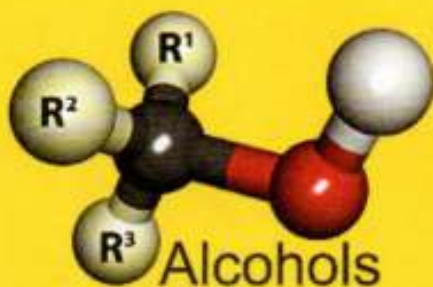


14. Explain why the name 1-propene is incorrect. What is the proper name for this molecule?
15. Explain why the name 3-butene is incorrect. What is the proper name for this molecule?
16. Name and draw the structural formula for each isomer of pentene.
17. Name and draw the structural formula for each isomer of hexyne.
18. Write a chemical equation for the reaction between methane and bromine.
19. Write a chemical equation for the reaction between ethane and chlorine.
20. Draw the structure of the product of the reaction of,
 - a) Bromine with propene.
 - b) Chlorine with 2-butene.
 - c) Hydrogen with 1-butene.
 - d) Hydrogen with 2-pentene.
21. Write the balanced chemical equation for the combustion of,
 - a) Heptane.
 - b) Benzene
 - c) Ethyne
 - d) Hexene
22. Draw the structure for each of the following compound.
 - a) 3,4-diethyloctane
 - b) 3-methylpentane
 - c) 4-ethyl-2-methylheptane
 - d) n -hexane
 - e) 3-isopropylhexane
 - f) 2,4,5-trimethyloctane
 - g) dimethyl-4-t -butylnonane
 - h) 2,2-dimethyl-4-propylnonane
 - i) 3-ethyl-4-methyl-3-heptene
 - j) 3,3-diethyl-1-pentyne
 - k) 4-ethyl-4-propyl-2-octyne
 - l) 5-butyl-2,2-dimethyldecane
 - m) 3,4-diethyl-1-hexyne
 - n) 4-propyl-3-ethyl-2-methyloctane

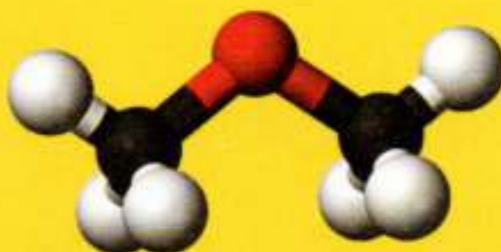
23. The name 2-ethylhexane is incorrect. Draw the structure and write the correct name for this molecule.
24. The name 3-butyl-7-methyloctane is incorrect. Draw the structure and write the correct name for this molecule.
25. The following skeletal structures represent alkenes or alkynes. Fill in the proper number of hydrogens on each carbon.
- a) $\text{C}-\text{C}\equiv\text{C}-\text{C}-\text{C}$
- b) 
- c) $\text{C}-\text{C}-\text{C}-\text{C}\equiv\text{C}$
- d) 
26. Write the condensed structures of a compound with seven carbon atoms in each of the four hydrocarbon classes.
27. Explain the process of polymerization

CHAPTER 2

ALCOHOLS, PHENOLS AND ETHERS



Phenols



Ethers

CHAPTER CONTENTS:**2.1 Introduction****2.2 Alcohols**

2.2.1 Classification of alcohols

2.2.2 Nomenclature of alcohols

2.2.3 Drawing the structure of alcohols

2.2.4 Physical properties of alcohols

2.2.5 Preparation of alcohols

2.2.6 Reactions of alcohols

2.2.7 Uses of alcohols

2.3 Phenols

2.3.1 Nomenclature of phenols

2.3.2 Physical properties of phenols

2.3.3 Sources of phenols

2.3.4 Reactions of phenols

2.3.5 Uses of phenols

2.4 Ethers

2.4.1 Nomenclature of ethers

2.4.2 Physical Properties of ethers

2.4.3 Preparation of ethers

2.4.4 Uses of ethers

Chapter review questions

CHAPTER OBJECTIVES:

By the end of the chapter, the learner should be able to:

- Define functional group.
- Explain why functional groups are important.
- Name alcohols, phenols & ethers using IUPAC system.
- State the common names of simple alcohols, phenols and ethers.
- Draw the structures of some common alcohols, phenols and ethers.
- Explain the physical properties of alcohols, phenols and ethers.
- State method of preparations for some common alcohols, phenols and ethers.
- Explain the chemical properties of alcohols, phenols and ethers.
- Describe the uses of some common alcohols, phenols and ethers.

2.1 INTRODUCTION TO FUNCTIONAL GROUP**Lesson Objectives:**

By the end of this lesson the learner should be able to:

- Define functional groups.
- Explain why functional groups are important.

**ACTIVITY 1**

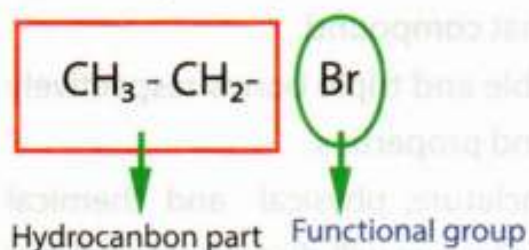
In groups, differentiate these functional groups of the chapter picture into their structures of different families of organic compounds.

Chemists have learned through experience that organic compounds can be classified into families according to their structural features known as functional groups.

Generally, an organic compound is composed of two parts;

- i. Hydrocarbon part which is chemically inert .
- ii. Active part which is called functional group.

For example:



A functional group is an atom or a group of atoms within a molecule that give a certain characteristic to the organic compounds.

Table 1.6: the first seven homologous series of Alkenes

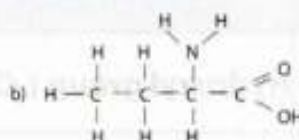
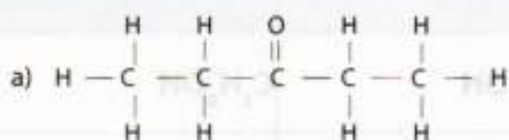
Homologous series	Functional group	General Formula	Example
Alcohols	Hydroxyl group (-OH)	R-OH	C ₂ H ₅ OH
Aldehydes	Aldehyde group (-CHO)	R-CHO	CH ₃ -CHO
Ketones	Carbonyl group (-CO-)	R-CO-R	CH ₃ -CO-CH ₃
Ethers	Oxy group (-O-)	R-O-R	CH ₃ -O-CH ₃
Carboxylic acids	Carboxyl group (-COOH)	R-COOH	CH ₃ COOH
Esters	Ester group (-COOR)	R-COOR	CH ₃ COOC ₂ H ₅
Amines	Amino group(-NH ₂)	R-NH ₂	C ₂ H ₅ NH ₂

Functional group can also be defined as active part of an organic compound which is responsible for the characteristics of that compound.

For example, alkenes and alkynes contain double and triple bonds respectively which are responsible for their nomenclature and properties.

The Functional group determines the nomenclature, physical and chemical properties of the organic molecule. In this chapter, we will discuss more about alcohols, phenols and ethers, the other functional groups will be discussed the next chapter. Exercise:

1. Define a functional group.
2. State three properties of molecules which are determined by a functional group.
3. Identify the functional group(s) in the following compounds;



2.2 ALCOHOLS

Lesson Objectives:

By the end of this lesson the learner should be able to:

- Define alcohols and phenols.
- Describe the classification of alcohols.
- Name alcohols using IUPAC system.
- Explain the physical and chemical properties of alcohols and phenols.
- Describe the uses of some common alcohols and phenols.

Alcohols are organic compounds which contain hydroxyl (-OH) as a functional group. Alcohols can also be defined as hydroxyl derivatives of alkane in which one or more hydrogen is replaced by a hydroxyl group (-OH).

For examples:

- a) $\text{CH}_3\text{-OH}$ methanol
- b) $\text{CH}_3\text{-CH}_2\text{-OH}$ ethanol

The general structural formula of alcohols is R-OH where R is an alkyl group such as -CH_3 (methyl), $\text{-C}_2\text{H}_5$ (ethyl) etc. Therefore alcohols contain two parts:

1. Hydrocarbon (alkyl group) part which is hydrophobic
2. Alcohol part (hydroxyl group) which is hydrophilic.

The simplest of alcohols is methanol, $\text{CH}_3\text{-OH}$, was once known as wood alcohol because it was prepared by heating wood in the absence of air. Methanol is toxic to humans, causing blindness in low doses (15 mL) and death in larger amounts (100–200 mL).

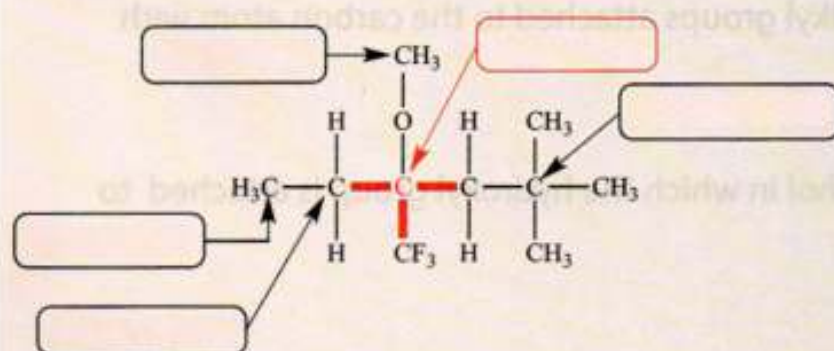
Ethanol ($\text{CH}_3\text{CH}_2\text{-OH}$) is one of the oldest known pure organic chemicals. Its production is by the fermentation of grains and sugars. Sometimes called grain alcohol, ethanol is the alcohol present in all wines.

2.2.1 Classification of alcohols



ACTIVITY 2

Identify each of the following molecules into primary, secondary and tertiary carbon



Alcohols are classified into four categories, according to the number of hydroxyl group attached to the chain:

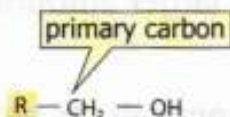
1. Monohydroxy Alcohols (Monohydric Alcohols): are alcohols that contain one-hydroxyl (-OH) group.

For examples

- | | |
|---|---|
| a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
(Butanol) | b) $\begin{array}{c} \text{CH}_3 \\ \\ \text{CHCH}_2\text{OH} \\ \\ \text{CH}_3 \end{array}$
(2-methyl propanol) |
| c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
(Propan-1-ol) | d) $\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{CH}_3 \\ \\ \text{OH} \end{array}$
(Propan-2-ol) |

According to the carbon attached the hydroxyl group, Monohydric alcohols are further classified into three classes:

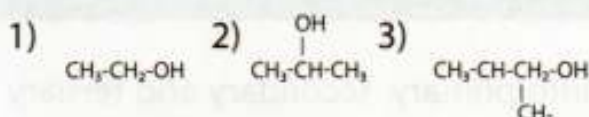
a) Primary (1°): is an alcohol in which the hydroxyl group is attached to primary Carbon.



a primary alcohol

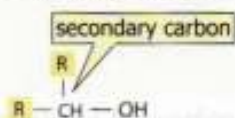
In a primary alcohol, the carbon with the hydroxyl group is only attached to one alkyl group.

For example,



There is an exception to this. Methanol, CH_3OH , is counted as a primary alcohol even though there are no alkyl groups attached to the carbon atom with hydroxyl group.

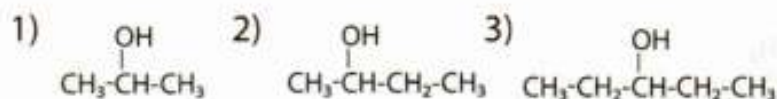
b) Secondary (2°): is an alcohol in which the hydroxyl group is attached to secondary carbon.



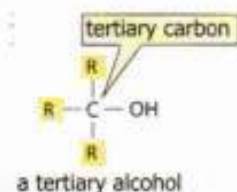
a secondary alcohol

In a secondary alcohol, the carbon with the hydroxyl group is attached to two alkyl groups.

For examples

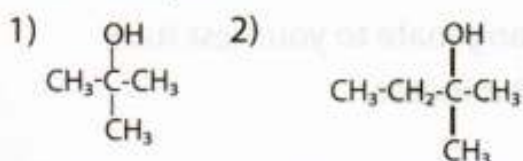


c) Tertiary (3°), is an alcohol in which the hydroxyl group is attached to secondary carbon



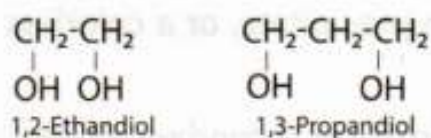
In a tertiary alcohol, the carbon with the hydroxyl group is attached to two alkyl groups.

For examples



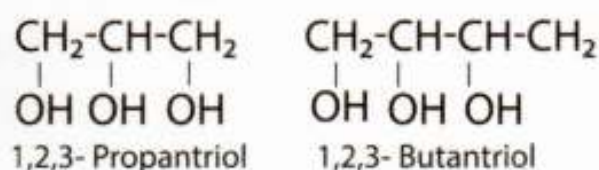
2. Dihydroxy Alcohols (Dihydric Alcohols): are alcohols contain two hydroxyl groups (-OH) groups. alcohols with two hydroxyl groups are also called diols.

For examples



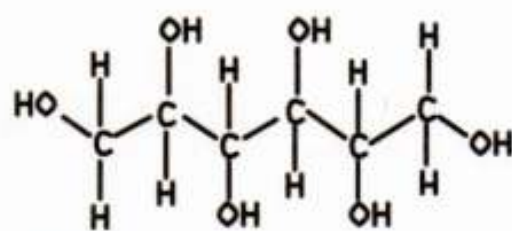
3. Trihydroxy Alcohols (Trihydric Alcohols): are alcohols contain three hydroxyl (-OH) groups. Alcohols with three hydroxyl groups are also called triols

For examples



4. Polyhydroxyalcohols: are alcohols contain more than three hydroxyl (-OH) groups.

For example



Sorbitol

Experiment 1:

Test for Primary and Secondary Alcohols.

Materials

- 2 test tubes
- Test tube rack
- Disposable pipettes
- A hot water bath
- Potassium permanganate KMnO_4 (0.1% solution in 0.1M H_2SO_4)

Procedure

1. Add 1 drop of the acidified potassium permanganate to your test tube
2. Warm the mixture in a hot water bath for 1-2 minutes.

Observation

Observe if the purple color of potassium permanganate changes or a precipitate is formed.

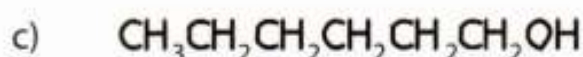
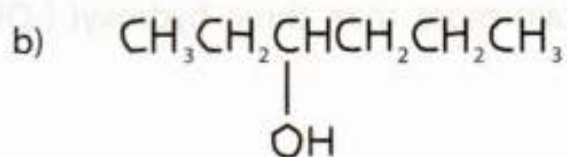
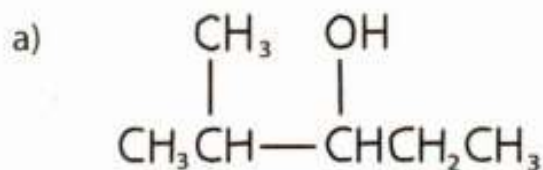
Conclusion

If the purple color is removed (you may see a brown precipitate, or a colorless mixture), then you have an alcohol.

If the purple color remains, then you do not have a primary or secondary alcohol.

**EXERCISE**

Classify each of the following alcohols as monohydroxy or dihydroxy:



2.2.2 Nomenclature of alcohols:

Alcohols are named by both IUPAC and common or trivial systems.

In common or trivial system, the alcohols are named by combining the name of the alkyl group with the word *alcohol*. As "Alkyl Alcohol" If the alkyl group is complex, the IUPAC name should be used.

Examples :

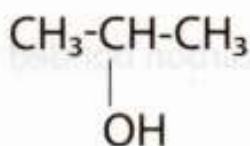
Alcohol	Common name
a) $\text{CH}_3\text{-OH}$	Methyl alcohol
b) $\text{CH}_3\text{-CH}_2\text{-OH}$	Ethyl alcohol
c) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$	Propyl alcohol
d) $\begin{array}{c} \text{CH}_3\text{-CH-CH}_2\text{-OH} \\ \\ \text{CH}_3 \end{array}$	Isobutyl alcohol

In the IUPAC system, alcohols are named as follows :

1. Find the longest carbon chain containing the carbon bonded to the OH group.
2. Number the carbon chain to give the OH group the lowest possible number.
3. Indicate the position of the hydroxyl immediately before the parent alcohol name by using the number of the carbon atom to which the -OH is bonded.
4. Change the name of the alkane corresponding to this chain by dropping the final -e and adding the suffix -ol.
5. If alcohol contains substituents, name each alkyl side chain or other group, and designate its position by number.

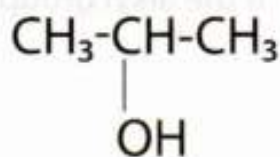
Example: 1

Give the IUPAC name of the following alcohol.



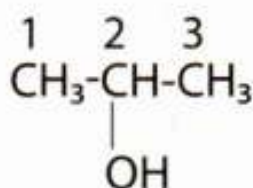
Solution:

Step 1: Find the longest carbon chain containing the carbon bonded to the OH group

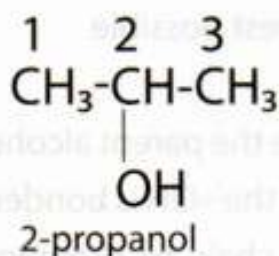


3 Carbons are in the longest chain.

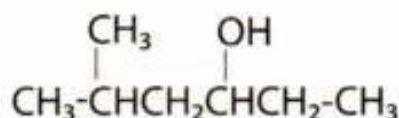
Step 2: Number the carbon chain to give the OH group the lowest possible number.



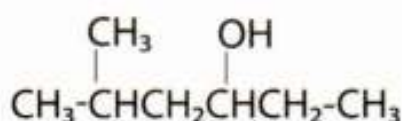
Step 3: Indicate the position of the hydroxyl and change the name of the alkane corresponding to this chain by dropping the final -e- and adding the suffix -ol.

**Example: 2**

Give the IUPAC name of the following alcohol.

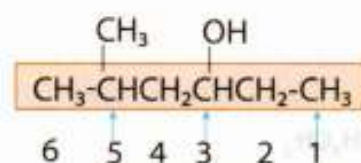


Step 1: Find the longest carbon chain containing the carbon bonded to the OH group.



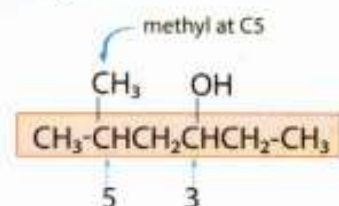
6 carbon atoms are in the longest chain.

Step 2: Number the carbon chain to give the OH group the lowest possible number.

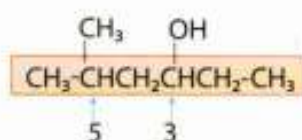


Number the chain to put the OH group at C3, not C4.

Step 3: Name and number the substituents.



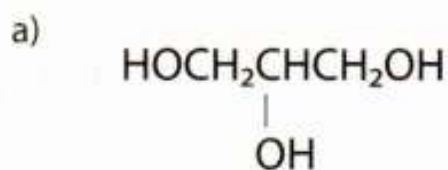
Step 4: Form the parent alcohol name by replacing the -e ending of the parent alkane to the suffix -ol.



Answer: 5-methyl-3-hexanol

In the naming of an alcohols which contain two hydroxyl groups, the suffix -diol is added to the name of the parent alkane and numbers are used in the prefix to indicate the location of the two OH groups.

For example,



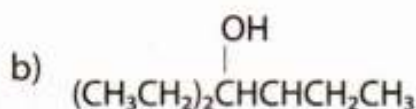
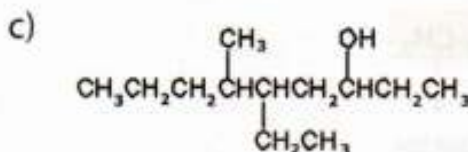
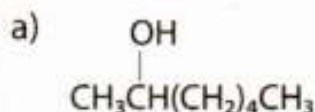
IUPAC name: 1,2,3-propanetriol

Or ethane-1,2-diol

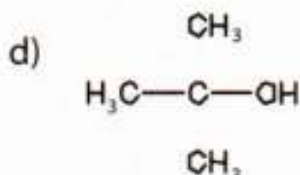
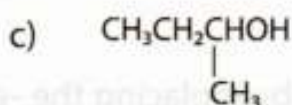


EXERCISE

1. Give the IUPAC name of each of the following alcohols;



2. Give the common name of each of the following alcohols;



2.2.3 Drawing the structure of alcohols:

The structural formula of a molecule is drawn from the IUPAC or common name.

Example

Draw the condensed structural formula of 2-methyl-1-pentanol

Solution

Draw the condensed formula of 2-methyl-1-pentanol

1. Identify the parent chain and side chain

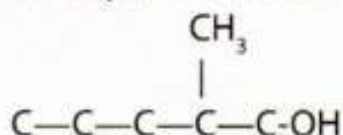
2. Draw the main chain of 5 carbon atoms.



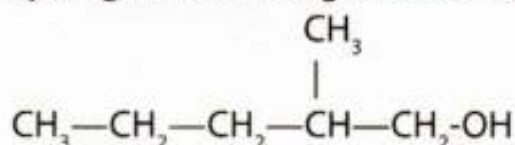
3. Attach hydroxyl group (-OH) to the end of carbon chain



4. Attach methyl (-CH₃) to C-2 on the main chain counting from right side



5. Add hydrogen atoms to give 4 bonds to each C.



**EXERCISE**

Draw the condensed structural formula for each of the following:

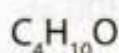
- 3-pentanol
- 2-methyl-2-butanol
- 3-bromo-3-hexanol
- 2-methyl-1-propanol

2.2.4 Isomerism in Alcohols

Alcohols show the following types of Isomerism:

1. Chain Isomerism: Alcohols containing four or more carbon atoms exhibit this type of Isomerism.

For example

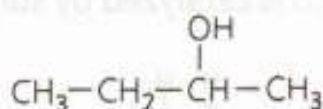


a)



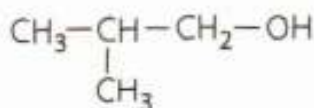
1-butanol

b)



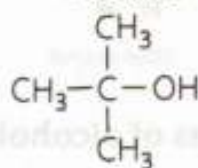
1-butanol

c)



2-methyl-1-propanol

d)

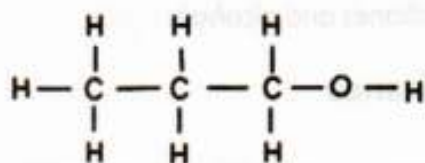


2-methyl-1-propanol

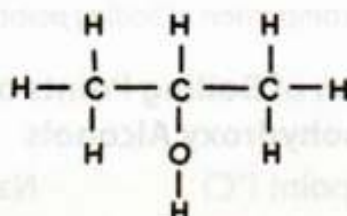
2. Position isomerism: alcohols containing three or more carbon atoms exhibit this type of isomerism

For example

Isomers of Propanol: C_3H_7OH



propan-1-ol

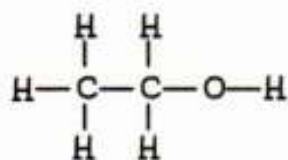


propan-2-ol

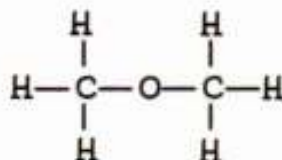
3. Functional Isomerism: Monohydric alcohols containing two or more carbon atoms show functional isomerism with corresponding ethers.

For example

Isomerism of Ethanol and dimethyl ether



Ethanol (Alcohol)

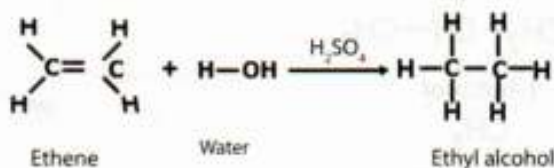


Dimethylether

2.2.5 Preparation of Alcohols

Alcohols can be prepared by hydration of alkenes:

Commercially, ethanol (or ethyl alcohol) is prepared by the addition of water across an alkene double bond (a hydration reaction) at about 280°C and 300 atmosphere. This reaction is catalyzed by sulfuric acid:



Ethene

Water

Ethyl alcohol

2.2.4 Physical properties of alcohols:

The physical properties of alcohols are responsible to the hydroxyl group they contain. Some of these properties are:

1. The physical state of lower alcohols are liquids while the higher members with than more than 12 carbon atoms are waxy solids.
2. The boiling points of alcohols are higher than those of corresponding hydrocarbons.

The following table shows the comparison of boiling points of alkanes and alcohols:

Table 2.1 Comparison of Boiling Points of Alkanes and Monohydroxy Alcohols

Name	Boiling point (°C)	Name	Boiling point (°C)
Pentane	36	1-Pentanol	138
Hexane	69	1-Hexanol	157
Octane	126	1-Octanol	195
Decane	174	1-Decanol	233

The boiling point of alcohols increases with the increases in the molecular mass or number of carbon atoms.

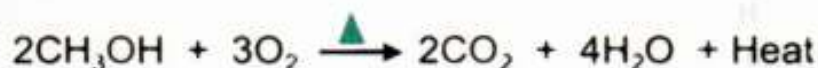
3. The Lower molecular weight alcohols (those having less than six carbons) are completely miscible in water while higher molecular weight alcohols (those having six carbons or more) are not soluble in water. The solubility of alcohol increases with the increase in temperature.

2.2.6 Reactions of alcohols:

Some common reactions that alcohols undergo are:

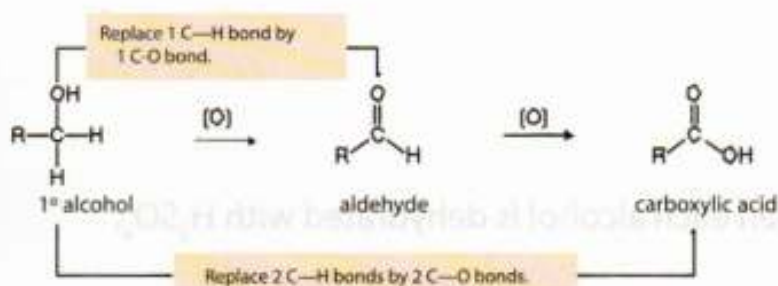
1. Combustion of alcohols:

Alcohols burn in air, producing carbon dioxide, vapour water and heat, that can be used as fuels:

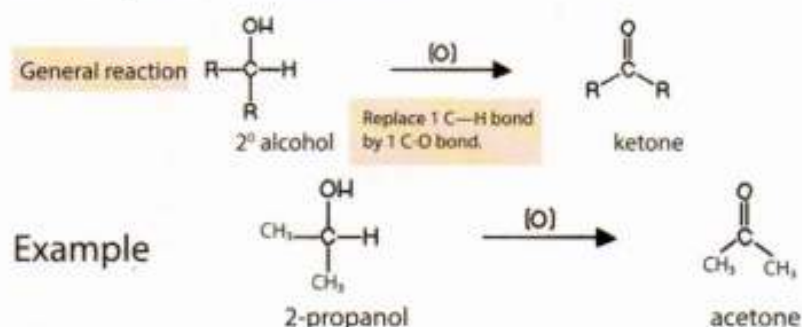


2. Oxidation of alcohols:

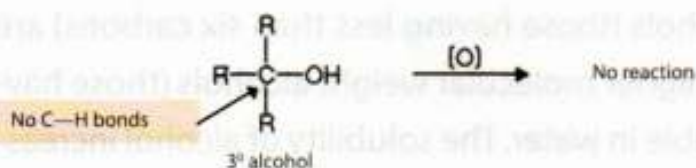
Primary (1°) alcohols are first oxidized to aldehydes (RCHO), which are further oxidized to carboxylic acids (RCOOH) by replacing one and then two C—H bonds by C—O bonds.



Secondary (2°) alcohols are oxidized to ketones (R_2CO), by replacing one C—H bond by one C—O bond.



Tertiary 3° alcohols have no H atoms on the carbon with the OH group, so they are not oxidized.

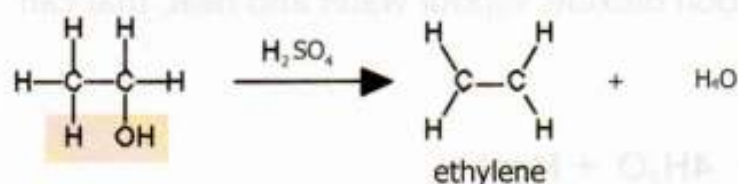


3. Dehydration of alcohols:

Alcohols are dehydrated to form alkenes when heated with sulfuric acid.

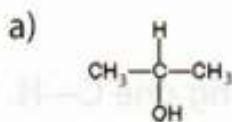
During dehydration, the H and OH from two adjacent carbon atoms are removed and a carbon-carbon double bond is formed.

For example



EXERCISE

- Write the combustion reaction for ethane and balance?
- Draw the carbonyl products formed when each alcohol is oxidized with oxidizing agent:
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- Draw the products formed when each alcohol is dehydrated with H_2SO_4 .



2.2.7 Use of Alcohols:

Some common uses of alcohols are:

1) **Ethanol ($\text{CH}_3\text{-CH}_2\text{-OH}$)**

Ethanol has countless applications as a solvent for organic chemicals and as a starting compound for the manufacture of dyes, synthetic drugs, cosmetics. It is also a constituent of alcoholic beverages. Ethanol is used as a gasoline additive since it readily combusts to form CO_2 and H_2O with the release of energy.

2) **Methanol (CH_3OH)** is used as a solvent and as antifreeze for car radiators and also can be used as fuel for cars. Methanol is a useful solvent and starting material for the synthesis of plastics.

2-Propanol [$(\text{CH}_3)_2\text{CHOH}$, isopropyl alcohol] is the major component of rubbing alcohol. When rubbed on the skin it evaporates readily, producing a pleasant cooling sensation. 2-Propanol is used to clean skin before medical procedures and to sterilize medical instruments.

3) **Ethylene glycol ($\text{HOCH}_2\text{CH}_2\text{OH}$)** is the major component of antifreeze. It is sweet tasting but extremely toxic. Ethylene glycol is a starting material in the synthesis of synthetic fibers such as Dacron.

4) **Glycerol [$(\text{HOCH}_2)_2\text{CHOH}$]** is a triol used in lotions, liquid soap, and shaving cream. It is sweet tasting and nontoxic. Its three OH groups form readily hydrogen bond with water, so it helps to retain moisture in these products.

2.3 PHENOLS

Lesson Objectives:

By the end of this lesson the learner should be able to:

- Define phenols.
- Name phenols using IUPAC system.
- Explain the physical and chemical properties of phenols.
- State the method of preparation of Phenol.
- Describe the uses of some common phenols.

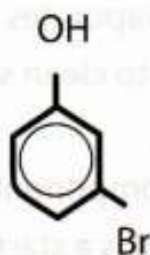
Phenols are compounds that have a hydroxyl group bonded directly to Benzene.

Phenols can also be defined as Hydroxyl derivatives of Benzene

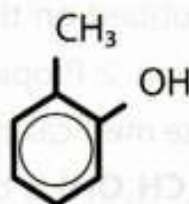
For example;



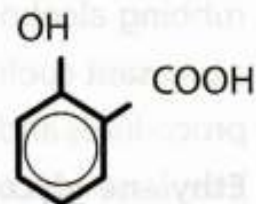
phenol



m-bromophenol



o-cresol



salicylic acid

The general structural formula of phenol is ArOH where Ar is Aryl or Benzene derivatives. Phenols are an important industrial chemical.

Experiment 2

Test for Phenols.

Materials

- Test tubes
- Test tube rack
- Disposable pipettes
- A hot water bath
- Iron(III) sulfate $\text{Fe}_2(\text{SO}_4)_3$ (0.1% in 0.1 M H_2SO_4)

Procedure:

1. Add 2-3 drops of the acidified iron(III) sulfate solution to your test tube

Observation:

1. Observe if the color of the acidified iron(III) sulfate solution changes.

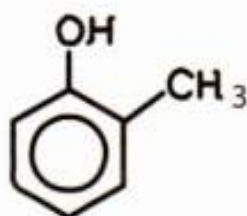
Conclusion:

- If you see an intense purple or blue color, you have a phenol.
- If you see no obvious change to purple or blue color, you do not have a phenol

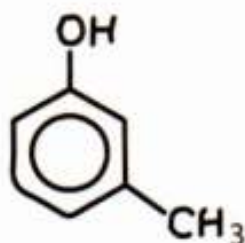
3.3.1 Classification of Phenols

Depending on the number of hydroxyl groups attached, phenols can be classified into three types:

1. Monohydric phenols: are phenols that contain one hydroxyl (-OH) group.

For examples

2-methyl phenol

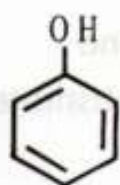


3-methyl phenol

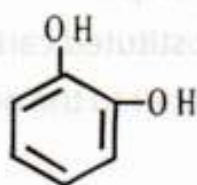


4-methyl phenol

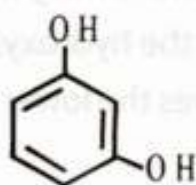
2. Dihydric phenols: are phenols that contain two hydroxyl (-OH) groups.

For examples

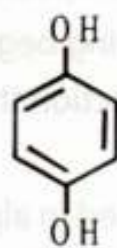
Phenol



Catechol



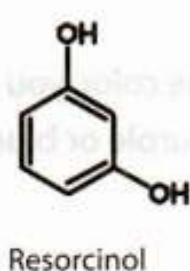
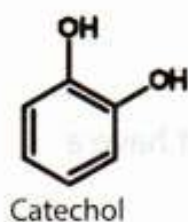
Resorcinol



Quinol

They may be ortho-, meta- or para- derivative.

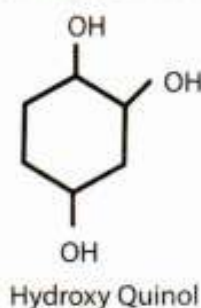
For examples



1,2-dihydroxybenzene 1,3-dihydroxybenzene 1,4-dihydroxybenzene

3. Trihydric phenols: are phenols that contain three hydroxyl (-OH) groups

For example:

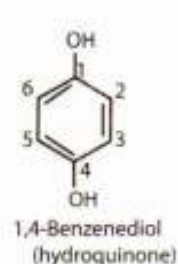
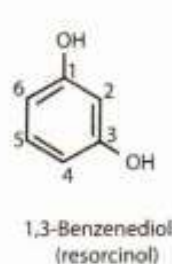
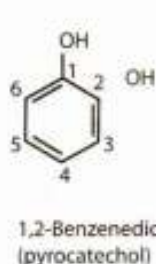
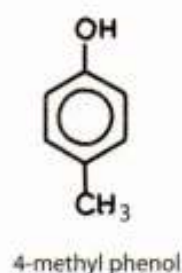
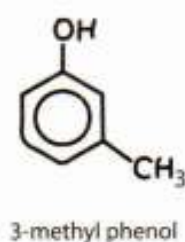
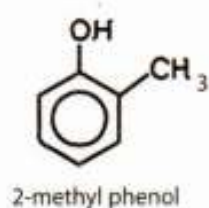


2.3.2 Nomenclature of phenols:

Although benzenol is the systematic name for C_6H_5OH , Phenol is the preferred as IUPAC name. Therefore, the substituted derivatives are named on the basis of phenol as the parent compound and following these steps:

- Numbering of the ring begins at the hydroxyl-substituted carbon and proceeds in the direction that gives the lower number to the next substituted carbon.
- Substituents are cited in alphabetical order.

For examples



2.3.3 Physical properties of phenols:

Many of the properties of phenols are analogous to those of alcohols. The physical properties of phenols are strongly influenced by the hydroxyl group, which permits phenols to form hydrogen bonds with other phenol molecules and water. Thus, phenols have higher melting points and boiling points and are more soluble in water than Arenes and Aryl Halides of comparable molecular weight.

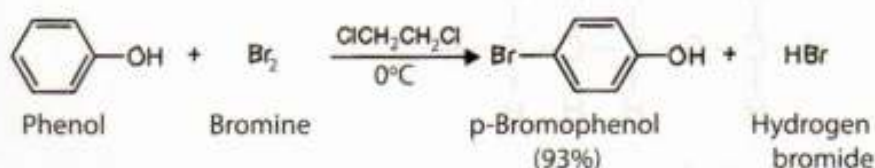
TABLE 2.2 Comparison of Physical Properties of an Arene, a Phenol, and an Halide

Physical property	Compound		
	Toluene, $C_6H_5CH_3$	Phenol, C_6H_5OH	Fluorobenzene, C_6H_5F
Molecular weight	92	94	96
Melting point	-95°C	43°C	-41°C
Boiling point (1 atm)	111°C	182°C	85°C
Solubility in water (25°C)	0.05 g/100 mL	8.2 g/100 mL	0.2 g/100 mL

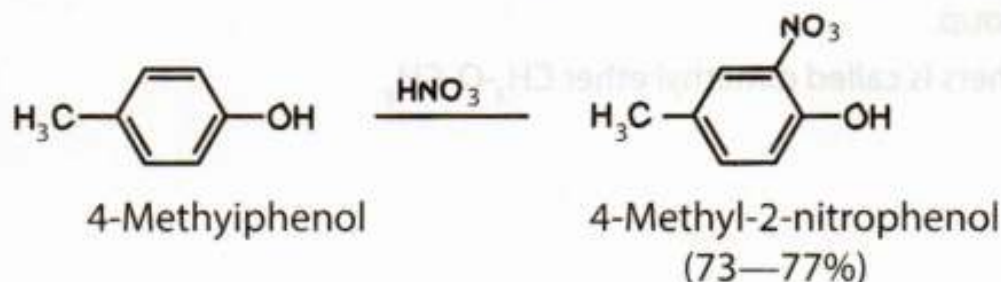
2.3.4 Reactions of Phenols:

Some common substitution reactions that phenols undergo are:

1. Halogenation: Bromination (addition of bromine) and chlorination (addition of chlorine) of phenols occur readily even in the absence of a catalyst. Substitution occurs primarily at the position para to the hydroxyl group. When the para position is blocked, ortho substitution is observed.



2. Nitration: Phenols are nitrated on treatment with a dilute solution of nitric acid in either water or acetic acid. It is not necessary to use mixtures of nitric and sulfuric acids as benzenes because of the high reactivity of phenols.



2.3.5 Use of phenols:

Phenols have a wide range of applications in industries. Some of the important uses of phenols are:

1. Phenols are used to prepare aspirin and dyes, and plastics.
2. Phenols are used as antiseptics for washing surgical instruments.
3. Phenols are also used in perfumery.

2.4 ETHERS

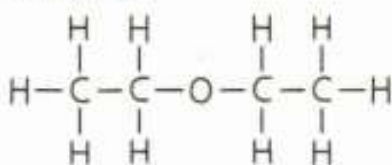
Lesson Objectives:

By the end of the lesson, the learner should be able to:

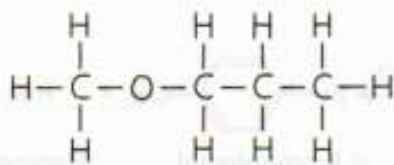
- Define ethers.
- Name ethers using IUPAC system.
- State the common names of ethers.
- Explain the physical and chemical properties of ethers.
- State the method preparation of ethers.
- Describe the uses of some common ethers.

Ethers are organic compounds which contain oxy (-O-) as a functional group. Ethers also can be defined as derivatives of water by replacing both hydrogen into Alkyl group or Aryl group.

Examples



diethyl ether



methyl propyl ether

The general formula of ether is R-O-R or R-O-Ar or Ar-O-Ar

Where;

R=Alkyl group

Ar= Aryl group.

The simplest of ethers is called dimethyl ether $\text{CH}_3\text{-O-CH}_3$

2.4.1 Classification of Ether

Depending on the type of the alkyl or aryl groups attached to the oxygen atom in ether, it can be classified into two types.

1. **Symmetrical ether:** Also known as the simple ether, the alkyl or the aryl group attached to either side of the oxygen atoms are the same.

Examples

a) $\text{CH}_3\text{-O-CH}_3$ dimethyl ether

b) $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$ diethyl ether

2. **Unsymmetrical ether:** Also known as the mixed ether, the alkyl or the aryl group attached to either side of the oxygen atoms, are not the same.

Examples

a) $\text{CH}_3\text{-O-C}_2\text{H}_5$ ethyl methyl ether

b) $\text{C}_2\text{H}_5\text{-O-C}_3\text{H}_7$ ethyl propyl ether

2.4.2 Nomenclature of Ethers

Ethers can be named two ways:

1. Common system
2. IUPAC system

1. Common system

Ethers are named by the combining the names of alkyl groups on the both sides of the ether group with the word ether. The names of alkyl groups are listed in alphabetical order followed by the word ether and a prefix is used if both alkyl groups are the same.

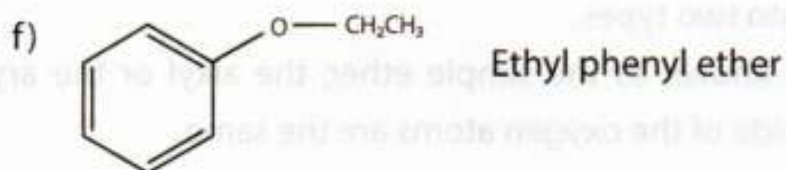
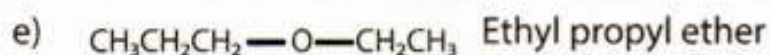
For examples

a) $\text{CH}_3\text{-O-CH}_3$ Dimethyl ether

b) $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$ Diethyl ether

c) $\text{CH}_3\text{---}\boxed{\text{O}}\text{---CH}_2\text{CH}_3$
 ↑ ↑ ↑
 methyl ether ethyl
 ethyl methyl ether

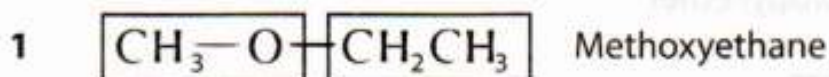
d) $\text{CH}_3\text{---}\boxed{\text{O}}\text{---CH}_3$
 ↑ ↑ ↑
 methyl ether ethyl
 ethyl methyl ether



2. IUPAC system:

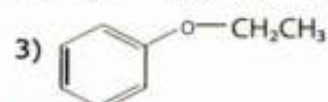
In the IUPAC System, ethers are named by choosing larger alkyl group attached to the ether group as parent and named as alkane while the smaller alkyl group attached to the ether group is named with oxygen as alkoxy. Therefore, the complete name for ether will be Alkoxy alkane.

Examples



EXERCISE

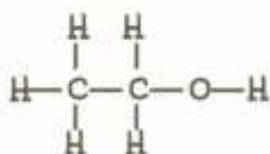
Write the common name for each of the following:



2.4.3 Isomerism in ethers

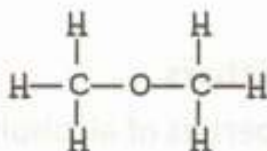
Ethers exhibit functional isomerism with alcohols having the same number of carbon atoms.

For example



Ethanol (Alcohol)
 $\text{CH}_3\text{H}_6\text{O}$

and



Dimethylether
 $\text{CH}_3\text{H}_6\text{O}$

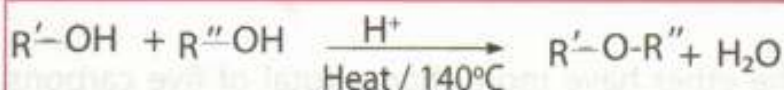
2.4.4 Preparation of Ethers

Ethers are prepared by:

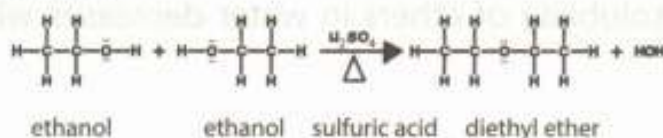
1. Dehydration of alcohols

In this process two molecules of alcohols are heated with concentrated H_2SO_4 at 140°C then, the alcohols undergo dehydration to give symmetrical ether by heating them in the presence of an acid as the following

General Equation



For example



This process is suitable for primary alcohols because secondary alcohols give a mixture of alkene and ether while tertiary alcohols give only alkene.

2. Dehalogenation of Haloalkanes: In this, Ethers are prepared from alkyl halides and sodium alkoxides via a substitution reaction known as the Williamson synthesis. This synthesis is often used to prepare symmetric and asymmetric ethers.

General Equation

General Equation



Example:

Sodium
ethoxide

bromomethane ethyl methyl ether

2.4.5 Physical Properties of Ethers

Some important physical properties of alcohols are:

1. Dimethyl ether and Ethyl methyl ether are gases at ordinary temperature while other ethers are liquids.
2. **The boiling point of Ethers**

The boiling point of Ethers is higher than hydrocarbons of comparable size and shape and lower than alcohols of comparable size and shape.

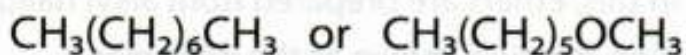
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	$\text{CH}_3\text{OCH}_2\text{CH}_3$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
butane	ethyl methyl ether	1-propanol
boiling point -0.5°C	boiling point 12°C	boiling point 97°C

Increasing boiling point 

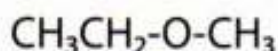
3. When the alkyl groups of the ether have more than a total of five carbons, the ether is insoluble in water but dissolve readily in organic solvents such as benzene, chloroform etc. The solubility of ethers in water decreases with the increase in the molecular mass.

**EXERCISE**

1. Which compound in each pair has the higher boiling point?



2. Label each of the following ethers as water soluble or water insoluble.



**EXERCISE**

1. Write the chemical equation for the preparation of dimethyl ether by dehydration of alcohols at 96% H_2SO_4 & 140°C .
2. Write the chemical equation for the preparation of Ethyl propyl ether by the Williamson synthesis reaction.

2.4.6 Chemical properties of alcohols

Ethers are not very reactive, however under suitable condition ethers undergo:

1. Combustion reaction:

In this process ethers burn in air to produce carbon dioxide and water because they are volatile and highly flammable.

For example**2. Dehydration:**

In this process, ether vapors are passed over Alumina (Al_2O_3) heated at 633K to produce alkene.

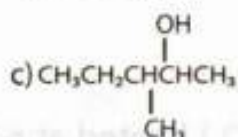
For example**2.4.7 Uses of Ethers**

1. **Ethers** are widely used as solvents for oils, gums, resins etc.
2. Diethyl ether is used as a general anesthetic in surgery but its usage for anesthetic is diminishing as better anesthetics are now available,

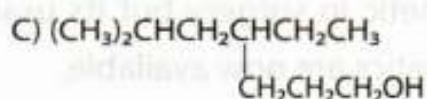
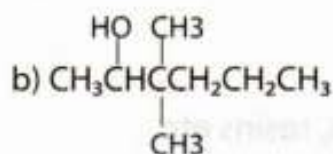
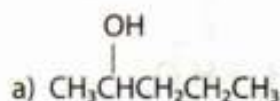


Chapter review questions:

1. What are the characteristics of alcohols and ethers?
2. How are alcohols classified?
3. What are the properties of alcohols and ethers?
4. State the IUPAC name for diethyl ether
5. What products are formed when an alcohol undergoes dehydration?
6. What products are formed when an alcohol is oxidized?
7. Draw structures for the four constitutional isomers of molecular formula $C_4H_{10}O$ that contain an OH group.
8. Classify each alcohol as 1° , 2° , or 3° .



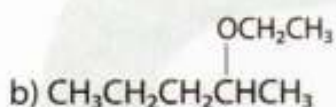
9. Give the IUPAC name for each alcohol.



10. Give the structure corresponding to each IUPAC name.

- a) 3-methyl-3-pentanol
- b) 4-methyl-2-pentanol
- c) 2,4-dimethyl-2-hexanol
- d) 1,3-propanediol
- e) 3,5-dimethylcyclohexanol
- f) 6,6-diethyl-4-nonanol

11. Give an acceptable name IUPAC or common for each below.



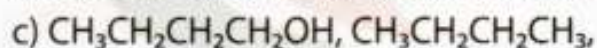
12. Give the structure corresponding to each IUPAC name.

a) 2-methoxypropane

b) cyclobutyl ethyl ether

c) 1-ethoxy-2-ethylcyclohexane

13. Rank the compounds in each group in order of increasing boiling point.



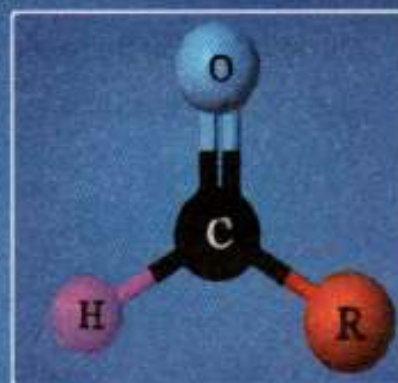
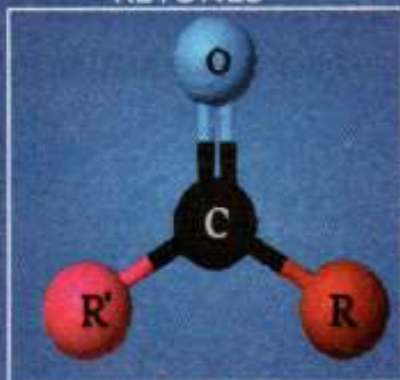
14. Explain the following observation. Dimethyl ether $[(\text{CH}_3)_2\text{O}]$ and ethanol $(\text{CH}_3\text{CH}_2\text{OH})$ are both water soluble compounds. The boiling point of ethanol is 78°C , however, is much higher than the boiling point of dimethyl ether (-24°C) .

15. Methanol (CH_3OH) is used as a fuel for some types of race cars, because it readily combusts in air to form CO_2 and H_2O . Write a balanced equation for the complete combustion of methanol.

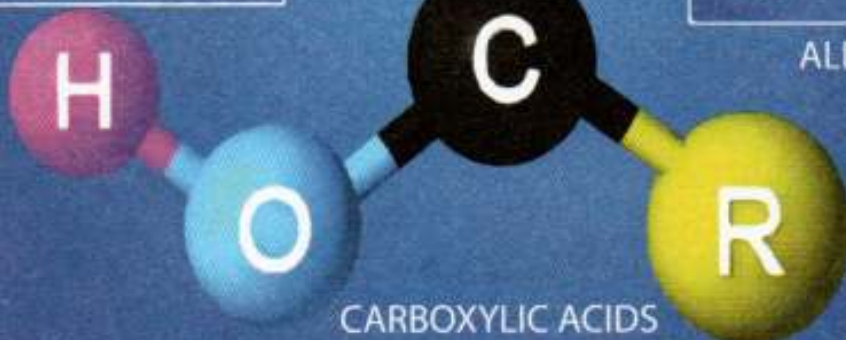
CHAPTER 3

ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

KETONES



ALDEHYDES



CARBOXYLIC ACIDS

CHAPTER CONTENTS:**3.1 Aldehydes and ketones**

3.1.1 Aldehyde

3.1.2 Ketones

3.1.3 Physical properties of aldehydes and ketones

3.1.4 Preparation of aldehydes and ketones

3.1.5 Reactions of aldehydes and ketones

3.1.6 Uses of aldehydes and ketones

3.2 Carboxylic acids

3.2.1 Classification of carboxylic acids

3.2.2 Nomenclature of carboxylic acids

3.2.3 Physical properties of carboxylic acids

3.2.4 Synthesis of carboxylic acids

3.2.5 Reactions of carboxylic acids

3.2.6 Uses of carboxylic acids

Chapter review questions

CHAPTER OBJECTIVES:**By the end of the chapter, the learner should be able to:**

- Define aldehydes, ketones and carboxylic acids.
- Name aldehydes, ketones and carboxylic acids using IUPAC system.
- State the common names of aldehydes, ketones and carboxylic acids.
- Explain the physical and chemical properties of aldehydes, ketones and carboxylic acids.
- Describe the uses of some common carbonyl compounds and amines.

3.1 ALDEHYDES AND KETONES

Lesson Objectives:

By the end of the lesson, the learner should be able to:

- Define aldehydes and ketones
- Name aldehydes and ketones using IUPAC system
- State the common names of some aldehydes and ketones
- Explain the physical and chemical properties of aldehydes and ketones
- Describe the uses of some common aldehydes and ketones

Experiment 1

- Test for Aldehydes (Benedict's test)

Materials and reagents:

- 2 test tubes
- Test tube rack
- Disposable pipettes
- A hot water bath
- Benedict's solution

Procedure:

1. Add about 1 mL of Benedict's solution to the test tube with a reagent.
2. Shake thoroughly by holding the test tube at the top and flicking the bottom with the fingers of your other hand.
3. Heat the solution for about one minute at 95°C in a water bath.

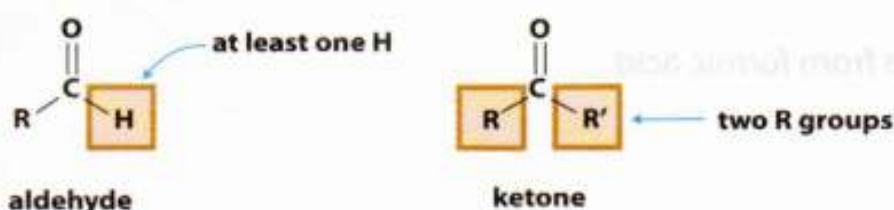
Observation:

- Observe if the blue color of Benedict's solution changes or precipitate formed.

Conclusion:

If you see the formation of an orange-red precipitate, you have an aldehyde. If the solution remains blue, and you don't see an orange-red precipitate, you don't have an aldehyde.

Aldehydes and Ketones constitute an important class of organic compounds. They possess the functional group ($>\text{C}=\text{O}$) commonly known as the carbonyl group. This is why aldehydes and ketones are collectively known as carbonyl compounds. Aldehydes and ketones occur widely in nature and they also serve as useful starting materials and solvents in industrial processes. All simple carbohydrates contain a carbonyl group. A carbonyl carbon of aldehydes and ketones is bonded to hydrogen or carbon atoms.



An aldehyde has at least one H atom bonded to the carbonyl group or if the carbonyl group is primary carbon the molecule is an Aldehyde while **A ketone** has two alkyl groups bonded to the carbonyl group or if the carbonyl group is Secondary carbon the molecule is Ketone.

Aldehydes are organic compounds which contain aldehydic carbonyl ($-\text{CHO}$) as a functional group. The general formula of aldehydes is $\text{R}-\text{CHO}$ or $\text{Ar}-\text{CHO}$.

Where;

R = Alkyl group

Ar = Aryl group or hydrogen

Ketones are organic compounds which contain carbonyl ($>\text{C}=\text{O}$) as a functional group.

The general formula of Ketones is;

Where R = Alkyl group or replace AR = Aryl.

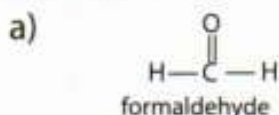


3.1.1 Nomenclature of Aldehydes

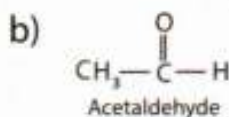
a) Common names of aldehydes:

An aldehyde has a common name for the simple aldehydes which are derived from corresponding common names of carboxylic acid by changing -ic acid- into aldehyde.

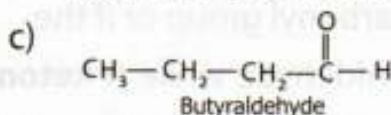
For examples



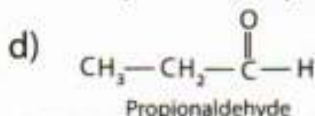
Formaldehyde from formic acid



Acetaldehyde from acetic acid



Propionaldehyde from propionic acid



Butyraldehyde from butyric acid

b) IUPAC nomenclature of aldehydes:

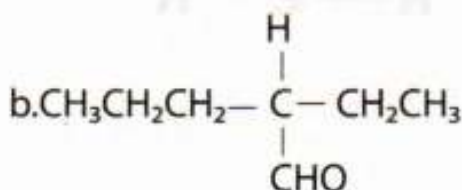
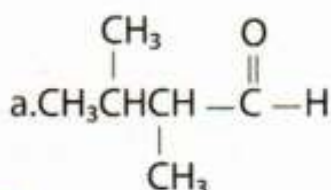
The IUPAC names of aldehydes are derived from the names of alkanes having the same number of carbon atoms by changing last "-e" into "-al".

Rules of IUPAC system

1. Find the longest chain containing the CHO group.
2. Number the chain or ring to put the CHO group at C1, but omit this number from the name.
3. The branched groups are indicated by using numbers.
4. Change the "-e" ending of the parent alkane to the suffix "-al".

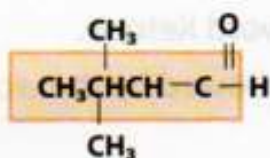
Example:

Give the IUPAC name for each of the following aldehydes:



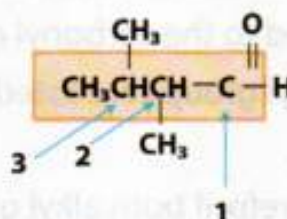
Solution:

- a. [1] Find and name the longest chain containing the CHO.



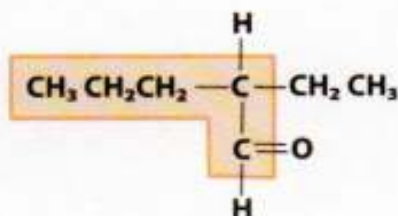
butane ————— butanal
(4 C's)

- [2] Number and name substituents, making sure the CHO group is at C1.



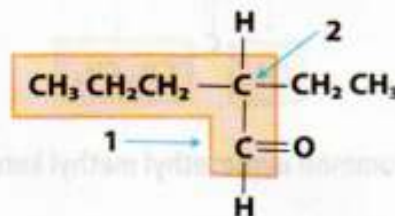
Answer: 2, 3- dimethylbutanal

- b. [1] Find and name the longest chain containing the CHO.



Pentane -----> Pentanal
(5 C's)

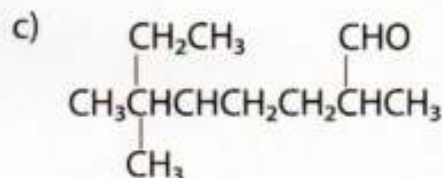
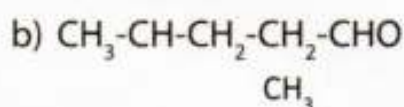
- [2] Number and name substituents, making sure the CHO group is at C1.



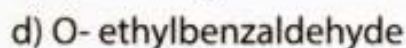
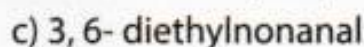
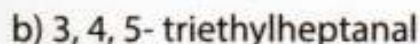
Pentane -----> Pentanal
(5 C's)

**EXERCISE**

1. Give the IUPAC name for each of the following aldehyde.



2. Give the structure corresponding to each IUPAC name of these aldehydes.



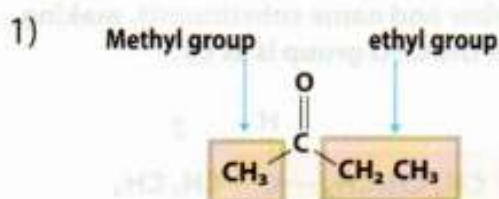
3.1.2 Nomenclature of Ketones:

a) Common names of Ketones

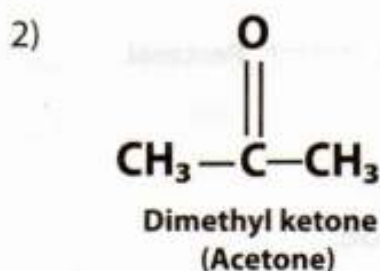
- i. Common names of Ketones are formed from the names of the two groups attached to the carbonyl group, followed by the word Ketone.
- ii. The alkyl groups are listed in alphabetical order followed by the word Ketone.
- iii. Use a prefix if both alkyl groups are the same.

Example

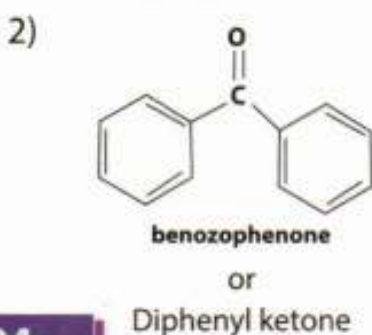
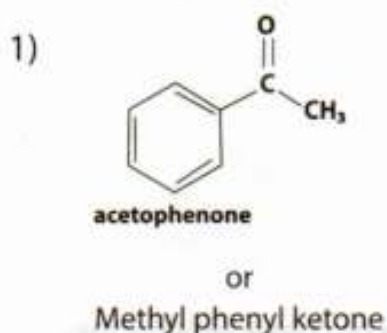
Give common name for each of the following Ketones:



Common name: ethyl methyl ketone



Common name for some aromatic ketones:



b) IUPAC nomenclature of Ketones:

The IUPAC names of Ketones are derived from the names of alkanes having the same number of carbon atoms;

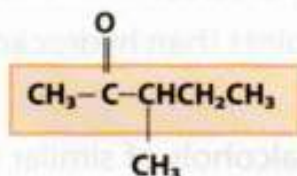
- Find the longest chain containing the group - >C=O
- Number the carbon chain to give the carbonyl carbon the lower number.
- Use a number to indicate the position of the branched groups.
- Indicate the position of carbonyl carbon using a number.
- Change the **-e** ending of the parent alkane to the suffix **-one**.

Example:

Give the IUPAC name for each of the following Ketones:

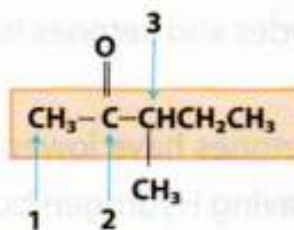
**Solution:**

- a) [1] Find and name the longest chain containing the CHO.



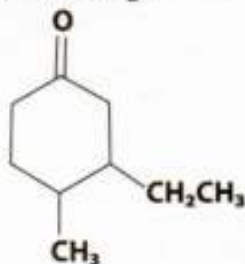
Pentane -----> Pentanone
(5 C's)

- [2] Number and name substituents, making sure the carbonyl carbon has the lowest possible number.



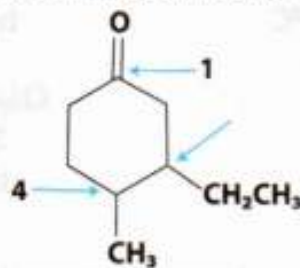
Answer: 3-methyl-2 pentanone

- b) [1] Name the ring.



Cyclohexane -----> cyclohexanone

- [2] Number and name substituents, making sure the carbonyl carbon is at C1.

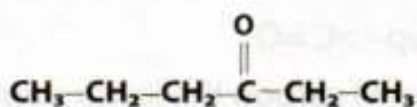


Answer: 3-ethyl-4-methylcyclohexanone

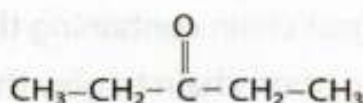
**EXERCISE**

1. Give common name for each of the following Ketones:

a)

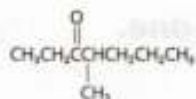


b)

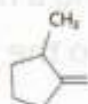


2. Give the IUPAC name for each of the following Ketones

a)



b)



c)



3. Give the structure corresponding to each name of the following ketones:

a)

butyl ethyl ketone

c)

p-ethylacetophenone

b)

2-methyl-3-pentanone

d)

2-propylcyclobutanone

3.1.3 Physical properties of aldehydes & ketones:**Boiling point:**

Aldehydes and ketones have a polar carbonyl group, therefore, they are polar molecules with stronger intermolecular forces than the hydrocarbons. Since they have no O-H bond, two molecules of RCHO or RCOR are incapable of intermolecular hydrogen bonding, giving them weaker intermolecular forces than alcohols. As a result: Aldehydes and ketones have higher boiling points than hydrocarbons of similar mass.

Aldehydes and ketones have lower boiling points than alcohols of similar mass because of not having Hydrogen bond between them.



pentane

bp 36 °C



butanal

bp 76 °C



1-butanol

bp 118 °C

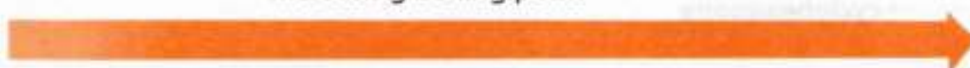


2-butanone

bp 80 °C

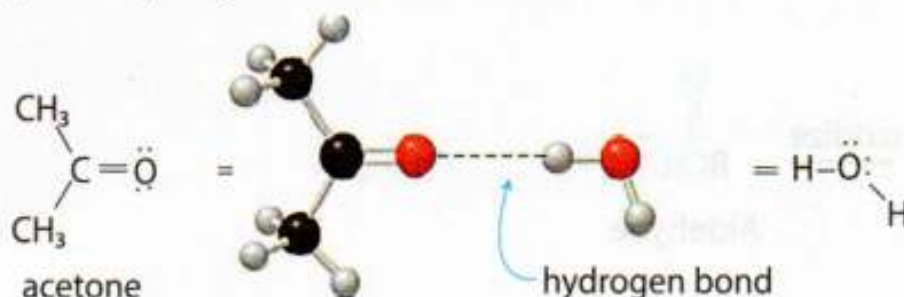
Increasing strength of intermolecular forces

Increasing boiling point



Solubility

Based on the general rule governing solubility i.e. like dissolves like, aldehydes and ketones are soluble in organic solvents. Moreover, because aldehydes and ketones contain an oxygen atom with an available lone pair, they can intermolecularly form hydrogen bond with water.

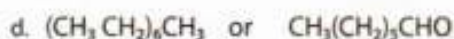
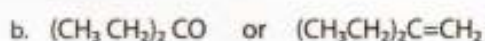
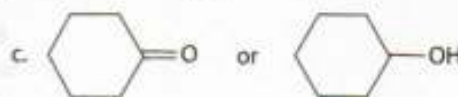
**As a result:**

Low molecular weight aldehydes and ketones, those having less than six carbons, are soluble in both organic solvents and water.

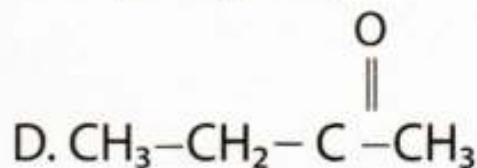
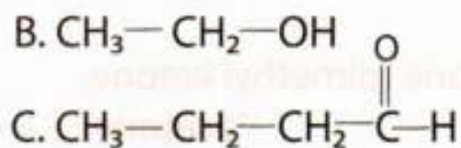
Higher molecular weight aldehydes and ketones, those having six carbons or more, are soluble in organic solvents, but insoluble in water.

**EXERCISE**

1. Which compound in each pair has the highest boiling point?



2. Indicate if each is soluble or insoluble in water:

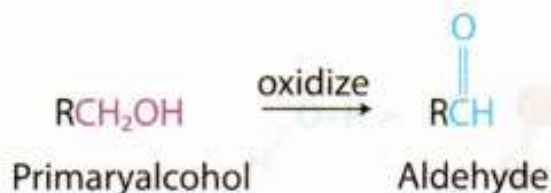


3.1.4 Preparation of aldehydes and ketones:

a) Oxidation of primary alcohols:

Primary alcohols are oxidized to produce aldehydes by a number of reagents especially pyridinium dichromate (PDC)

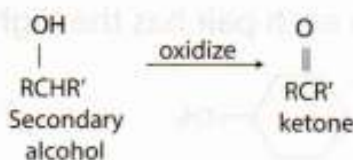
General Equation:



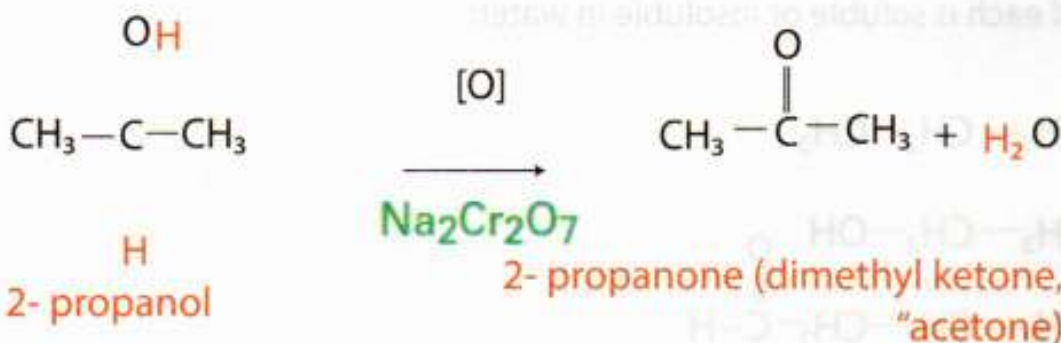
b) Oxidation of secondary alcohols:

Secondary alcohols are oxidized to produce ketones. For large-scale oxidation however, an inexpensive reagent such as chromium trioxide (CrO_3) or sodium dichromate ($\text{Na}_2\text{Cr}_2\text{O}_7$) in aqueous acetic acid is more economical.

General Equation:



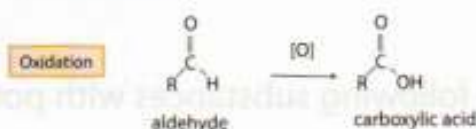
Example:



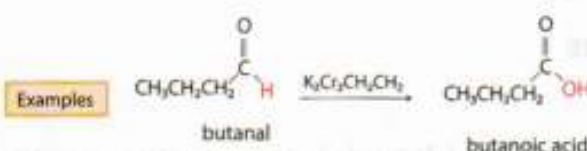
3.1.5 Reactions of aldehydes and ketones:

1. Oxidation of aldehydes and ketones:

Aldehydes can be oxidized to carboxylic acids this reaction is used to distinguish Aldehydes from Ketone because Aldehydes are easily oxidized while Ketones are not easily oxidized.



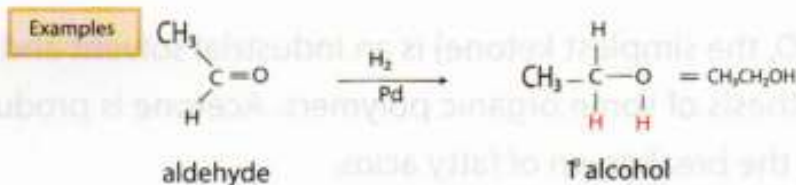
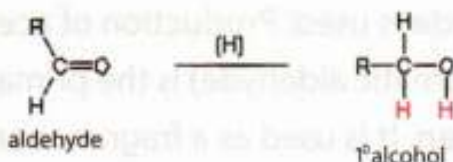
A common reagent for this oxidation is potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$, a red-orange solid that is converted to a green Cr^{3+} product during oxidation. Ketones (R_2CO) are not oxidized with $\text{K}_2\text{Cr}_2\text{O}_7$.



2. Reduction of aldehydes and ketones:

Aldehydes (RCHO) are reduced to 1° alcohols (RCH_2OH) by using the process of catalytic hydrogenation which aldehydes react with hydrogen in the presence of a metal catalyst like Pd, Pt or Ni.

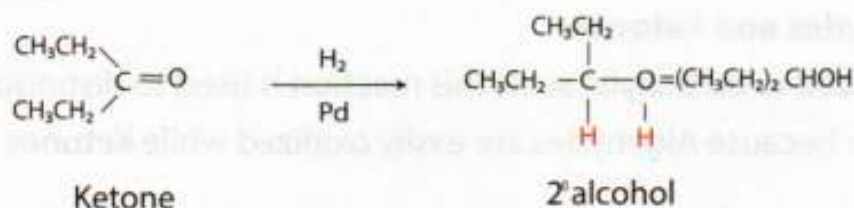
General Equation:



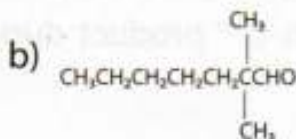
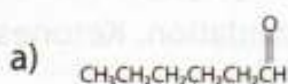
Ketones (RCOR) are reduced to 2° alcohols (R_2CHOH) by using the process of catalytic hydrogenation which ketones react with hydrogen in the presence of a metal catalyst like Pd, Pt or Ni.

General Equation:



Example:**EXERCISE**

Predict the products of the reaction of the following substances with potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$,

**3.1.6 Uses of aldehydes and ketones****a) Aldehydes:**

- **Formal Aldehyde** ($\text{CH}_2=\text{O}$, the simplest aldehyde) is a starting material for the synthesis of many resins like urea-formaldehyde and plastics such as Bakelite.
- **Formalin** (a 37% aqueous solution of formaldehyde) is used a disinfectant and preservative for biological specimens.
- **Acetaldehyde** is used: Production of acetic acid.
- **Vanillin** (aromatic aldehyde) is the primary component of the extract of the vanilla bean. It is used as a fragrance and also in food flavoring

b) Ketones:

Acetone [$(\text{CH}_3)_2\text{C}=\text{O}$, the simplest ketone] is an industrial solvent and a starting material in the synthesis of some organic polymers. Acetone is produced naturally in cells during the breakdown of fatty acids.

3.2 CARBOXYLIC ACIDS

Lesson Objectives:

By the end of the lesson, the learner should be able to:

- Define carboxylic acids.
- Name carboxylic acids using IUPAC system.
- State the common names of some carboxylic acids.
- Explain the physical and chemical properties of carboxylic acids.
- Describe the uses of some common carboxylic acids.

Carboxylic acids constitute of one the most frequently encountered classes of organic compounds. Countless natural products are carboxylic acids or are derived from them. Some carboxylic acids, such as acetic acid, have been known for centuries. The importance of carboxylic acids is magnified when we realize that they are the parent compounds of a large group of derivatives that includes acyl chlorides, acid anhydrides, esters, and amides.

Carboxylic acids are organic compounds which contain carboxyl ($-\text{COOH}$) as a functional group.

The general formula for carboxylic acids is: $\text{R}-\text{COOH}$ or $\text{Ar}-\text{COOH}$

where R =Alkyl group or hydrogen.

Ar =Aryl group.

The simplest of carboxylic acids is HCOOH where $\text{R} = \text{H}$ (hydrogen).

The word carboxyl (for a COOH group) is derived from carbonyl ($\text{C}=\text{O}$) + hydroxyl (OH).

Experiment 1

Test for carboxylic acids

Materials

- 2 test tubes
- Test tube rack
- Disposable pipettes
- Sodium bicarbonate, saturated solution (8.4% or 1.0 M)

Procedure:

Add 3-5 drops of the saturated sodium bicarbonate solution to the test tube a reagent.

Observation:

Observe if there is fizzing or bubbling a gas.

Conclusion:

If you observe fizzing or bubbling gas, you have a carboxylic acid.

If it doesn't fizz, you do not have a carboxylic acid.

3.2.1 Classification of carboxylic acids

Depending on the number of carboxyl group, carboxylic acids are classified in to :

1. Mono-carboxylic acids are carboxylic acids containing one carboxyl group in their molecules.

Forexamples

a) HCOOH Formic acid

b) CH_3COOH Acetic acid

2. Di-carboxylic acids: are carboxylic acids containing two carboxyl group in their molecules.

Forexamples

a) HOOC-COOH Oxalic acid

b) $\text{HOOC-CH}_2\text{-COOH}$ Malonic acid

3. Tri-carboxylic acids: are carboxylic acids containing three carboxyl group in their molecules.

Forexamples

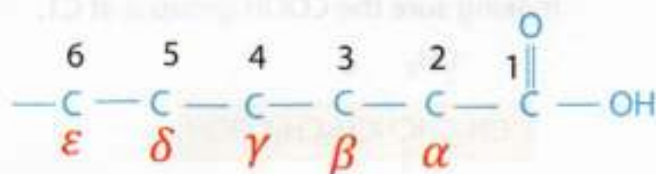
a) $\text{C}_6\text{H}_8\text{O}_7$ Citric acid

3.2.2 Nomenclature of carboxylic acids:**a) Common names of carboxylic acids:**

Many carboxylic acids have historical names obtained from their natural source.

Carboxylic acid	Common name	Latin & English
a) HCOOH	formic acid	formica (ant)
b) CH_3COOH	acetic acid	acetum (vinegar)
c) $\text{CH}_3\text{CH}_2\text{COOH}$	propionic acid	propionate (first salt)
d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	butyric acid	butyrum (butter)

Greek letters are sometimes used to designate the carbons near carbonyl groups. These terms can be used for any carbonyl group.



The carbon adjacent to the COOH is called the α (alpha) carbon.

The carbon bonded to the α carbon is the β (beta) carbon.

b) IUPAC nomenclature of carboxylic acids:

The IUPAC names of carboxylic acids are derived from the names of alkanes having the same

number of carbon atoms by changing the last **-e** with the suffix **-oic acid**:

- Find the longest chain containing the (-COOH) group.
- Number the chain or ring to put the (-COOH) group at C1, but omit this number from the name.
- The branched groups are indicated by using numbers.
- Change the **-e** ending of the parent alkane to the suffix **-oic acid**.

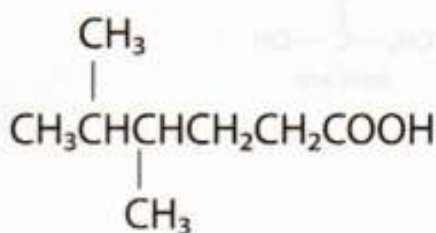
Example 1

Give the IUPAC name of each of the following carboxylic acids:

Carboxylic acid	IUPAC Name
a) HCOOH	methanoic acid
b) CH_3COOH	ethanoic acid
c) $\text{CH}_3\text{CH}_2\text{COOH}$	propanoic acid
d) $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CHCOOH} \end{array}$	2-methylpropanoic acid

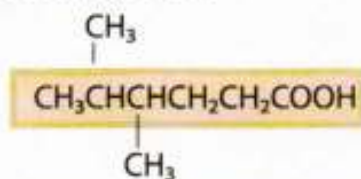
Example 2

Give the IUPAC name of the following carboxylic acid.



Solution:

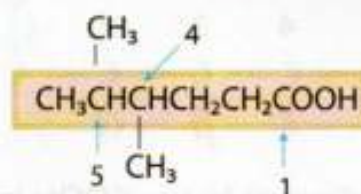
- [1] Find and name the longest chain containing COOH.



hexane.....→ hexanoic acid
(6 C's)

The COOH contributes
one C to the longest chain.

- [1] Find and name the substituents, making sure the COOH group is at C1.

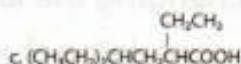
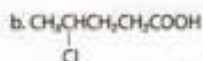
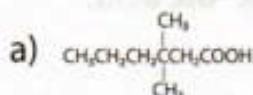


two methyl substituents on C4 and C5

Answer: 4,5- dimethylhexanoic acid

**EXERCISE**

1. Give the IUPAC name for each of the following compound.



2. Give the structure corresponding to each IUPAC name.

a) 2-bromobutanoic acid

c) 2-ethyl-5,5-dimethyloctanoic acid

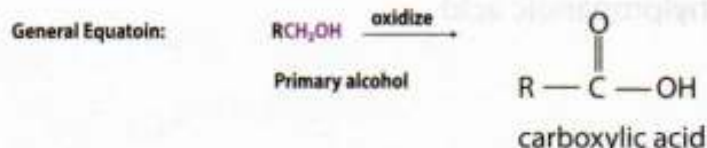
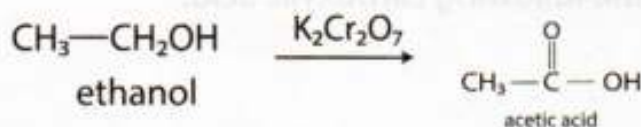
b) 2, 3-dimethylpentanoic acid

d) 3,4,5,6-tetraethyldecanoic acid

3.2.3 Preparation of carboxylic acids

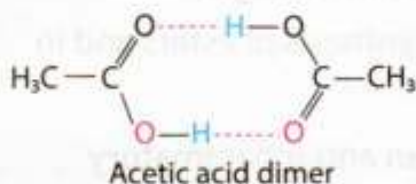
Oxidation of primary alcohols:

Primary alcohols are oxidized to produce carboxylic acids by a number of oxidizing agents, such as chromium trioxide (CrO_3) and potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) in an aqueous acid solution:

**Example:**

3.2.4 Physical properties of carboxylic acids

Carboxylic acids also exhibit intermolecular hydrogen bonding since they possess a hydrogen atom bonded to an electronegative oxygen atom. Carboxylic acids often exist as dimers, held together by two intermolecular hydrogen bonds, as shown below.



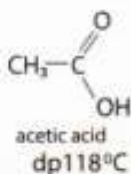
The carbonyl oxygen atom of one molecule forms hydrogen bonding with the hydrogen atom of another molecule.

As a result:

Carboxylic acids have higher boiling points and melting points than alcohols of similar mass.



1-propanol
dp 97°C



more hydrogen-bonding interactions possible
higher boiling point

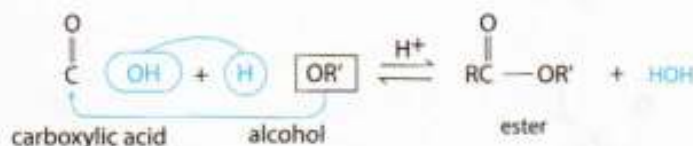
3.2.5 Reactions of carboxylic acids:

Esterification:

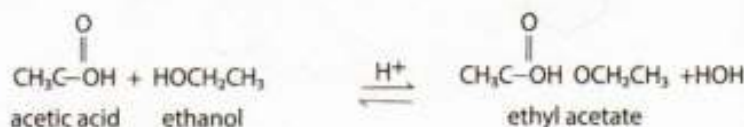
Alcohols react with carboxylic acids in the presence of an acid catalyst to form esters in a reaction known as esterification.

General

Equation:



Example:





EXERCISE

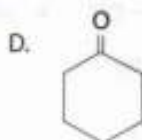
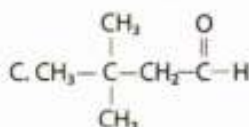
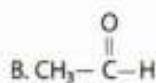
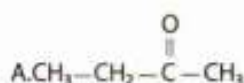
What product is formed when 1-propanol is oxidized by using potassium dichromate ($K_2Cr_2O_7$) in an aqueous acid solution?

3.2.4 Uses of carboxylic acids

1. **Acetic acid (CH_3COOH)** is the sour-tasting component of vinegar.
2. Acetic acid is also used industrially as a reagent for synthesis of esters and in the manufacture of plastics (polyvinyl acetate).
3. **Aspirin (acetylsalicylic acid)** is a pain reliever and an anti-inflammatory agent.
4. **Adipic acid** is used to make nylon 6.6.
5. **Fatty acids** are main components of fats and oils.

Chapter review questions:

1. Define each of the following:
 - a) Aldehydes
 - b) Ketones
 - c) Carboxylic acids
2. How are aldehydes and ketones named?
3. What are the characteristics of aldehydes and ketones?
4. Give examples of useful aldehydes and ketones.
5. What products are formed when aldehydes and ketones are reduced?
6. Sketch the hydrogen bonding between formaldehyde and water.
7. Classify each of the following as an aldehyde or ketone:



CHAPTER 4

BIOCHEMISTRY



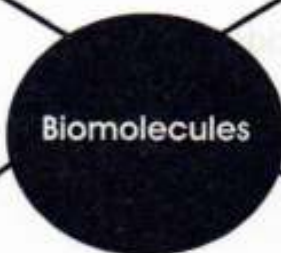
Carbohydrates



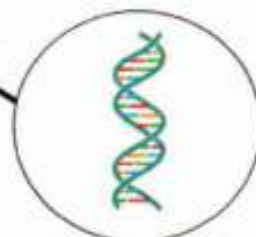
Proteins



Lipids



Biomolecules



Nucleic Acids

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Chapter review questions

Nucleic Acids



Carbohydrates



Lipids

CHAPTER OBJECTIVES

By the end of the chapter, the learner should be able to:

- Define biochemistry, carbohydrate, proteins, amino acids, lipid and nucleic acids
- Describe different types of carbohydrates, proteins, amino acids, lipids and nucleic acids
- Explain the structure and configuration of carbohydrates
- State some biological importance of carbohydrates, proteins, lipids and nucleic acids
- Compare and contrast the structure of DNA and RNA

4.1 INTRODUCTION TO BIOCHEMISTRY

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define biochemistry and biomolecules



ACTIVITY 1

In groups we are going to bring different food labels, read them and then discuss with the class on how these different ingredients affect our diet and health.

Biochemistry is an integrated science of biology and chemistry. It studies the chemical constituents and chemical processes in the living organism. It is divided into three fields:

1. Structural biology: it is the study of the structure of biomolecules.
2. Enzymology: this is a branch of biochemistry concerned with enzymes.
3. Metabolism: this is the sum of all chemical reactions in the organism.

The chemical processes that occur in the living system are collectively known as metabolism and the chemical constituents of the living system are known as biomolecules. They are mainly classified into:

1. carbohydrates
2. proteins
3. Lipids
4. Nucleic acids

In this chapter, we will study more about these biomolecules.



EXERCISE

1. Define
 - a) Biochemistry
 - b) Biomolecules
2. What are the major types of biomolecules?
3. What are the fields of biochemistry?
4. Why is biochemistry studied under the field of chemistry?

4.2 CARBOHYDRATES

Lesson objectives

By the end of the lesson, the learner should be able to:

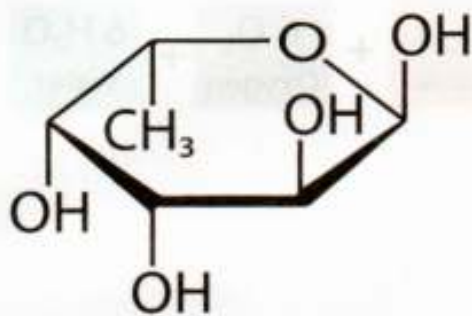
- Describe different types of carbohydrates
- Explain the structure and configuration of carbohydrates
- State functions of carbohydrates

Carbohydrates are the most abundant organic compounds in nature.

Generally, carbohydrates are substances with the general formula $C_x(H_2O)_y$. They are also called hydrates of carbon because they contain hydrogen and oxygen in the same proportion as in water (2:1). However, a number of compounds have been found, which are carbohydrates by chemical behaviour but, do not conform to the formula $C_x(H_2O)_y$.

For example

Rhamnose $C_6H_{12}O_5$ is carbohydrate but it can't be represented as $C_x(H_2O)_y$



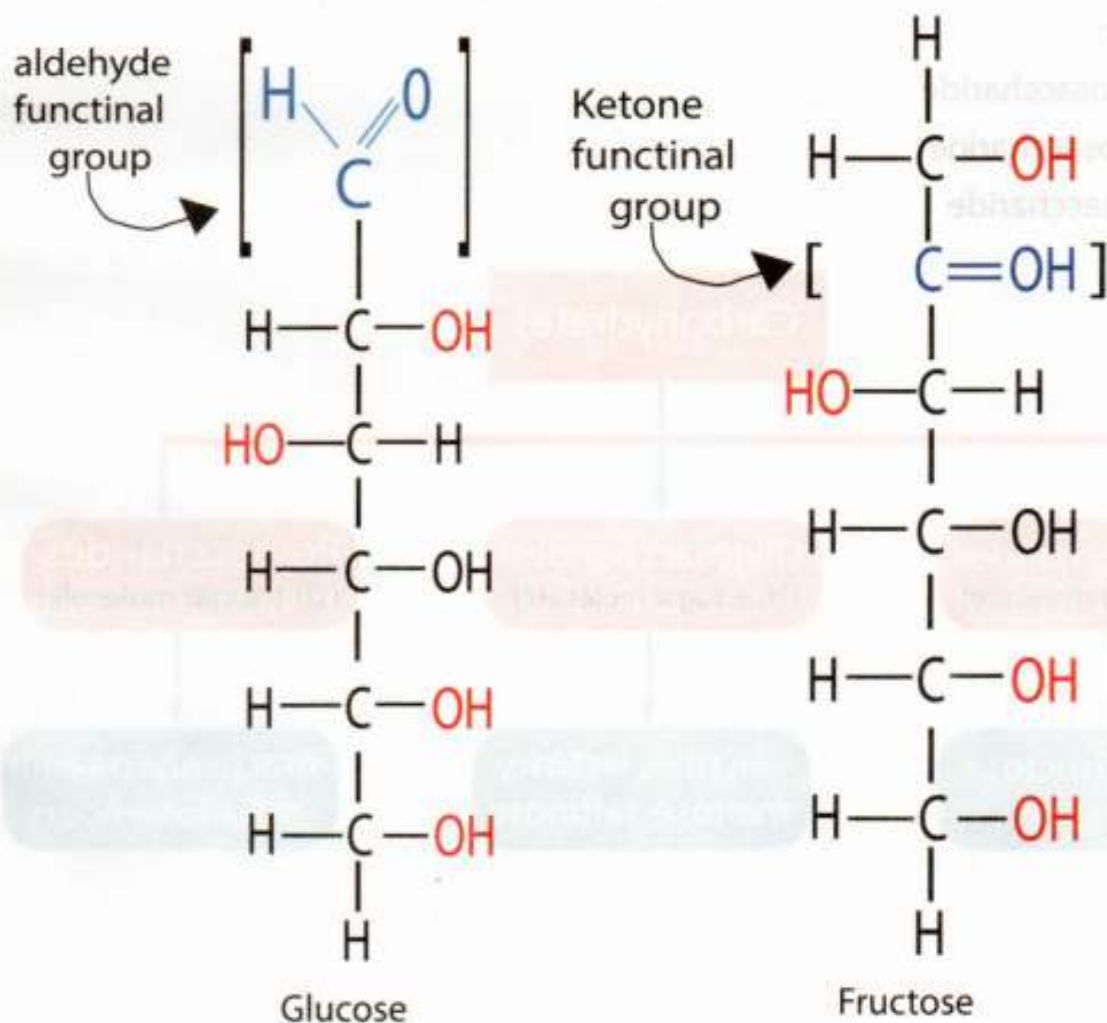
It was also observed that some substances other than carbohydrates can be represented as hydrates of carbons

Examples

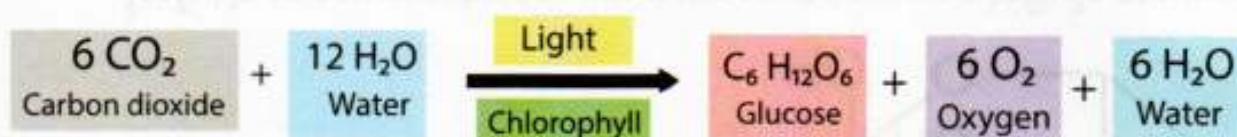
a) Acetic acid CH_3COOH can be represented as $C_2(H_2O)_2$

b) Lactic acid $C_3H_6O_3$ can be represented as $C_3(H_2O)_3$

Therefore, Carbohydrates can be defined as Polyhydroxy Aldehydes or Ketones.

For example

Plants synthesize carbohydrates through photosynthesis:



EXERCISE

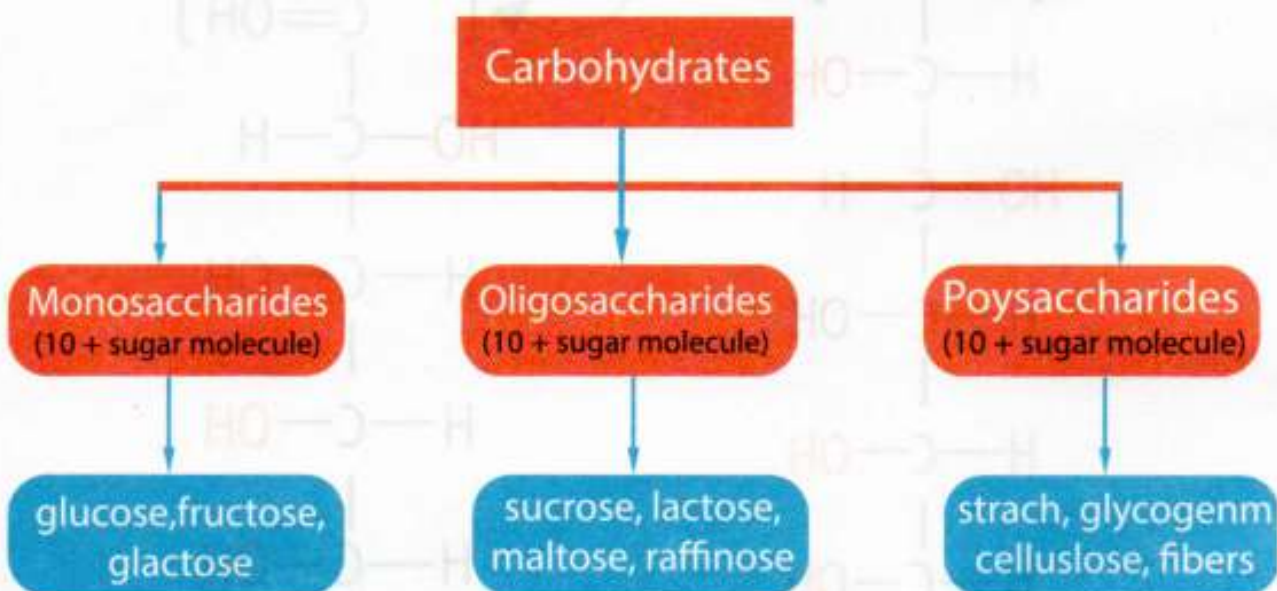
1. What are carbohydrates?
2. What is the general formula of carbohydrates?
3. What foods are found in glucose and fructose?

4.2.1 Classification of carbohydrates

Carbohydrates can be classified based on their test, behavior of hydrolysis and reducing.

On the bases of the behavior of hydrolysis, carbohydrates are classified into three categories:

1. Monosaccharide
2. Oligosaccharide
3. Polysaccharide

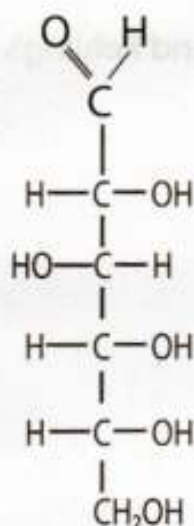


On the basis of taste, carbohydrates are classified into two categories:

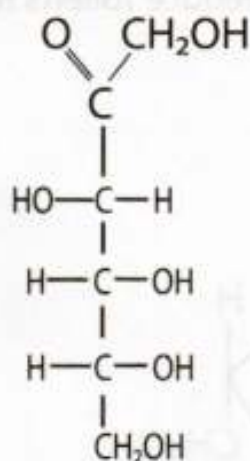
1. Sugar

These are carbohydrates that are sweet in taste. They are usually crystalline in nature and are soluble in water. All monosaccharides and disaccharides belong to this group.

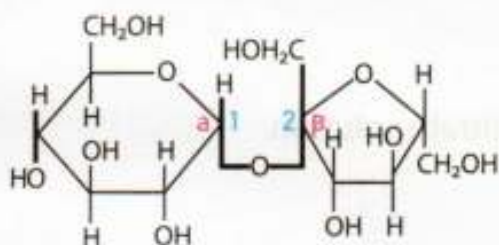
For examples



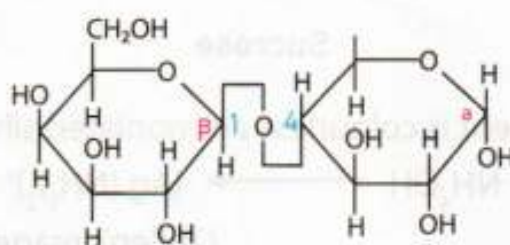
glucose



fructose



Sucrose



Lactose

2. Non sugars

These are carbohydrates which are not sweet in taste.

They are usually amorphous in nature and are insoluble or much less soluble in water.

For examples

- Starch
- cellulose

On the basis of reducing behavior carbohydrates are classified into two categories:

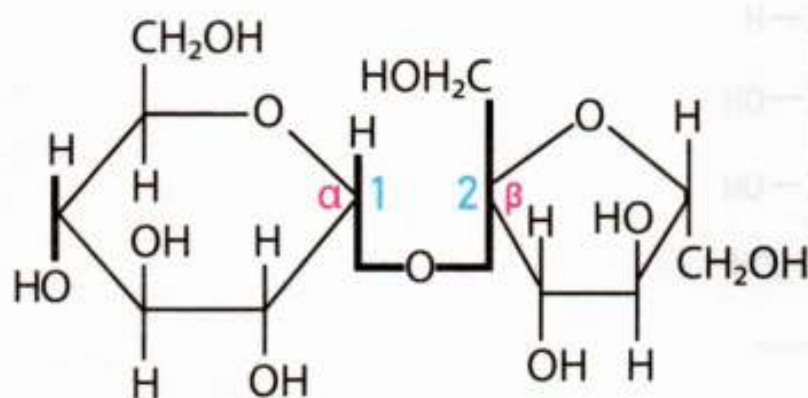
1. Reducing sugar

These are carbohydrate which is capable of reducing Tollen's reagent and Fehling's solution. All monosaccharide and disaccharides are reducing sugars except sucrose.

2. Non reducing sugars

These are carbohydrates that are unable to reduce Tollen's reagent and Fehling's solution.

For example

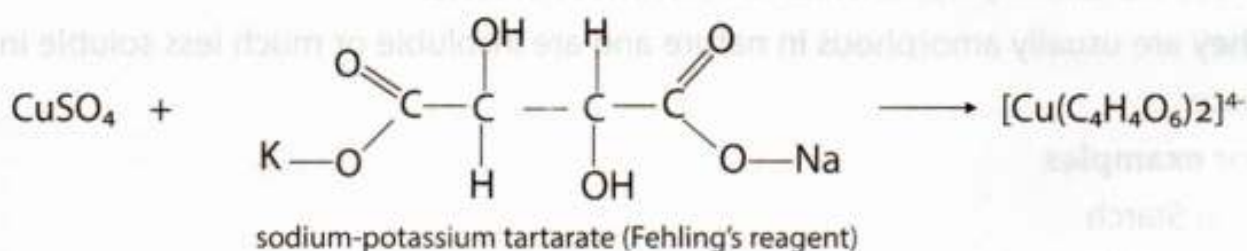


Sucrose

Tollen's reagent is colourless ammoniated silver nitrate solution.

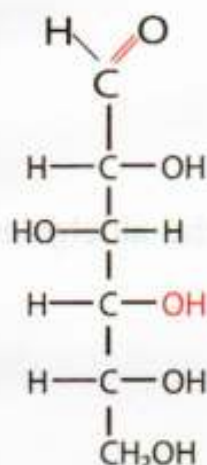


Fehling's reagent is a blue mixture of copper sulfate and sodium-potassium tartrate.

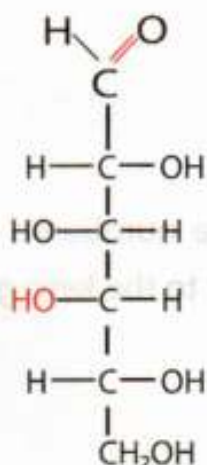


4.2.2 Monosaccharide

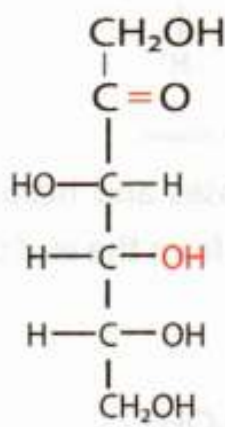
Monosaccharides are simplest carbohydrates which cannot be hydrolyzed. They have a general formula of $(CH_2O)_n$ where **n** represent the number of carbon atoms and it can be greater than 3 and less than 7. ($3 \leq n \leq 7$) Some common examples of monosaccharide are:



Glucose



Galactose



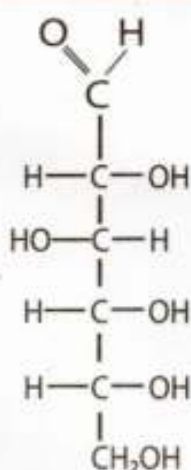
Fructose

4.2.3 Classification of Monosaccharide

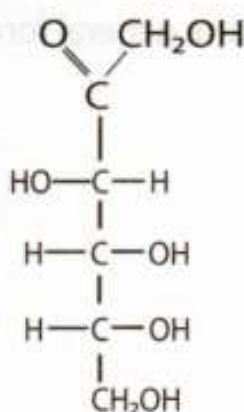


ACTIVITY 2

We study the following molecules to identify the functional groups present in each molecule.

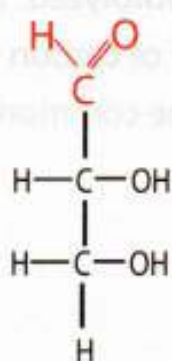


glucose

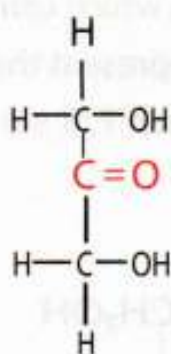


fructose

On the basis of the functional group, monosaccharide can be broadly classified into Aldoses and ketoses. Aldoses are monosaccharides that contain aldehyde group ($-\text{CHO}$) while Ketoses are monosaccharides that contain keto group. ($\text{C}=\text{O}$)

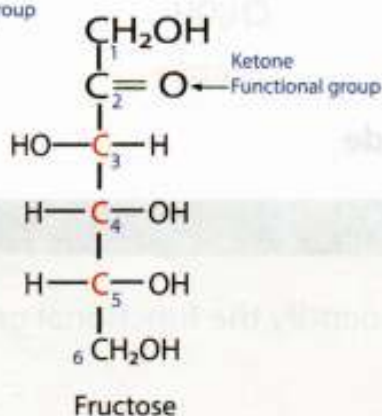
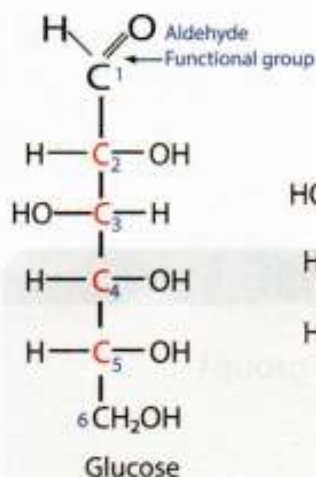
For example

An Aldose

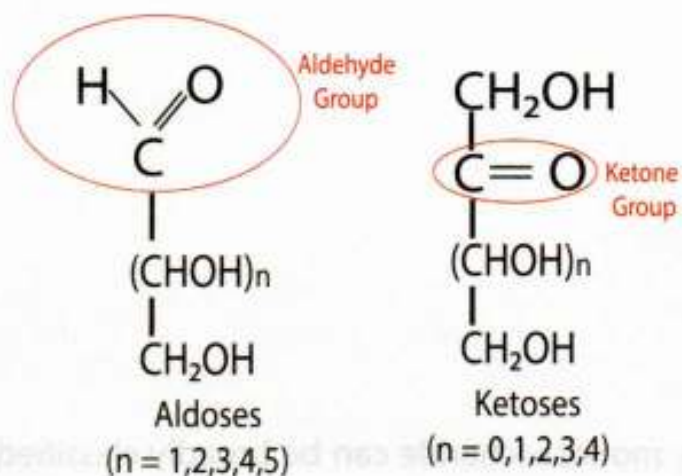


A Ketose

The carbon atoms of aldoses and hexoses are numbered from the aldehyde group and that of a ketose from the end closest to the keto group.

For example

Aldoses and ketoses can be represented by the general formula of



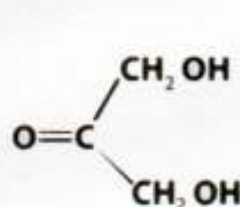
"n" represents the number of carbon atoms.

On the basis of the number of carbon, Monosaccharides can be further classified into:

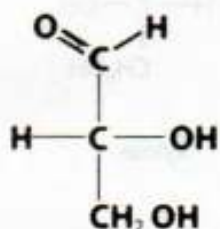
1. Trioses

These monosaccharide having three carbon atoms and they have the general molecular formula of $C_3H_6O_3$.

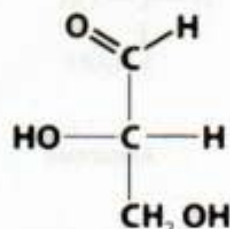
For example



Dihydroxyacetone



D-Glyceraldehyde

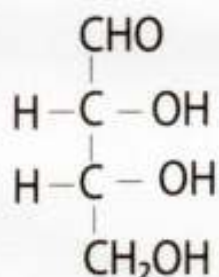


L-Glyceraldehyde

2. Tetroses

These Monosaccharides contain four carbon atoms and they have the general formula of $C_4H_8O_4$

For example

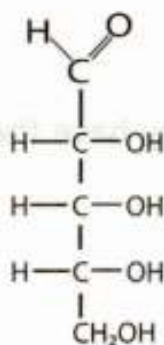


Erythrose

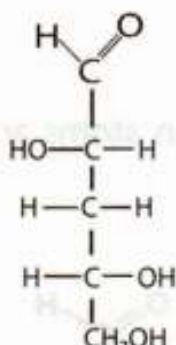
Erythrose is found in blood, as well as in human cartilage tissue.

3. Pentoses

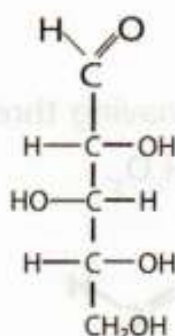
These are monosaccharide that contains five carbon atoms and they have the general formula of $C_5H_{10}O_5$

For example

Ribose



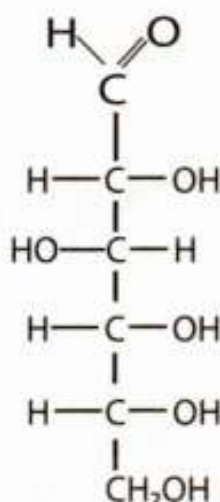
Arabinose



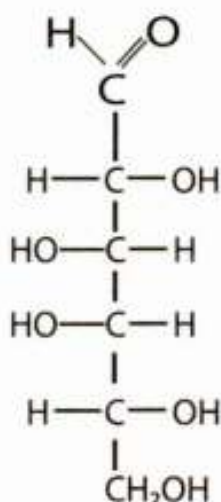
Xylose

4. Hexoses

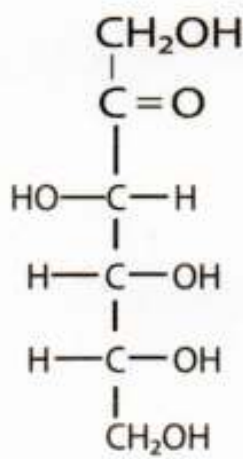
These are Monosaccharides that contain six carbon atoms and they have general formula $\text{C}_6\text{H}_{12}\text{O}_6$

For example

Glucose



Galactose



Fructose

Hexoses are the most common monosaccharide.

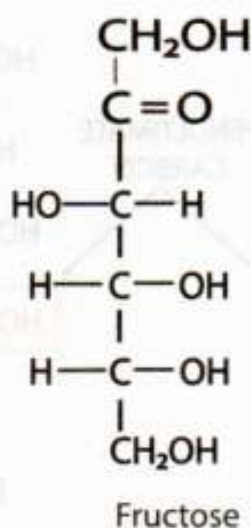
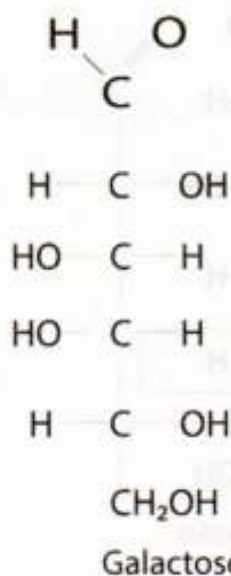
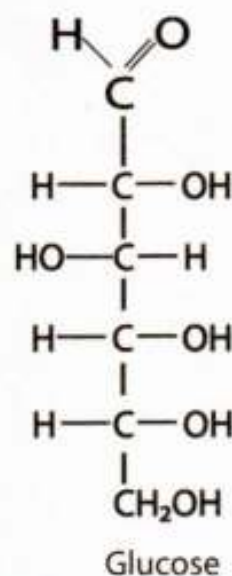
4.2.4 Structure of Monosaccharide

The structure of carbohydrates can be represented as:

1. Full Structural formula

This is the formula that represents the atoms and bond of the molecules.

For example

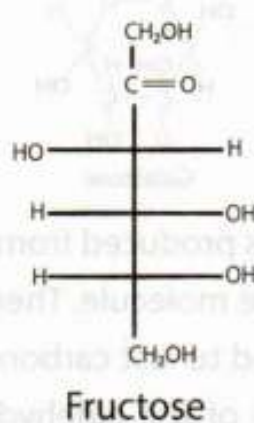
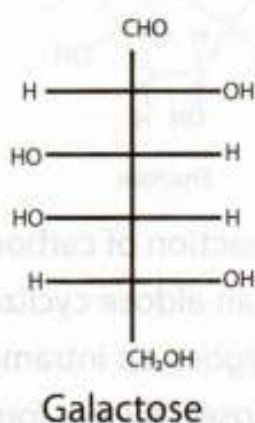
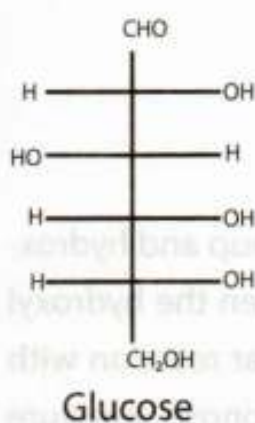


In the full structural formula, the central carbon atoms of the molecule which are attached to four different groups are referred to as chiral carbon or a symmetric carbon.

2. Fisher projection

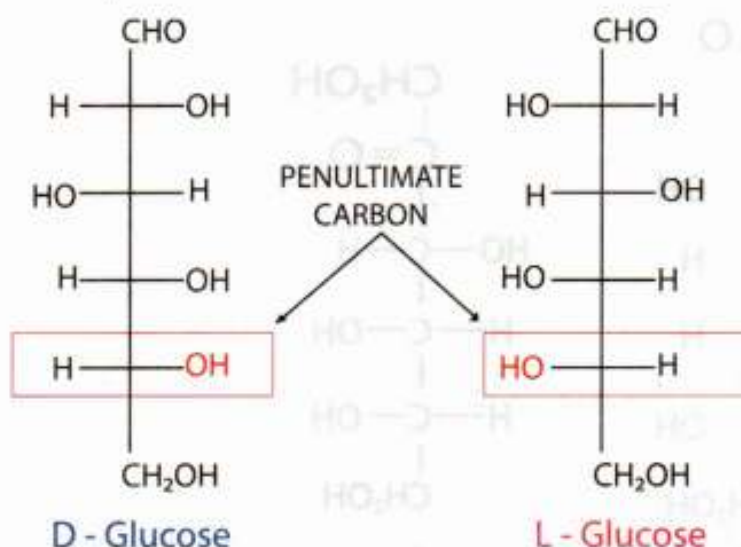
This is a formula that represents the straight-chain structures of monosaccharides. The chiral carbons are represented as intersection of horizontal and vertical lines.

For example



If the hydroxyl group (-OH) on the chiral carbon (penultimate carbon) that is farthest from the ketone or aldehyde functional group is on the right, the sugar is referred to as D-sugar but if the hydroxyl group is on the left the sugar is referred to as L-sugar. All natural carbohydrates have D-configuration.

For example

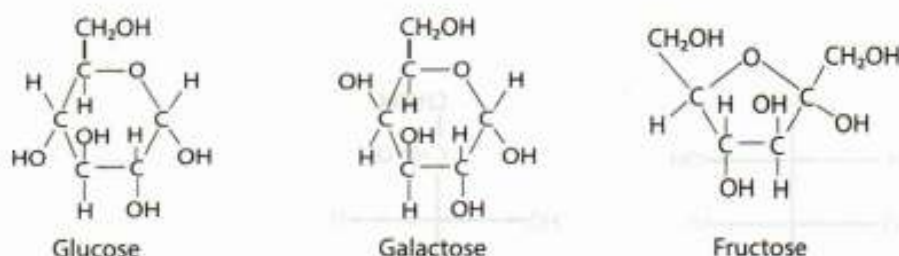


The letter "D" derived from Latin word "Dexter" means right while the letter "L" derived from Latin word "laevus" means left.

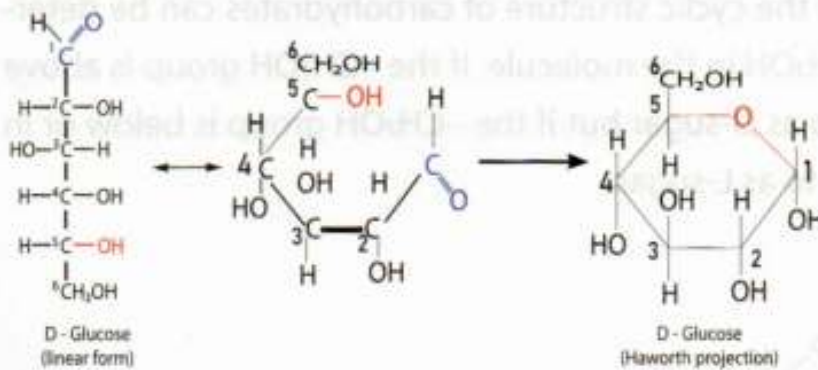
3. Haworth projection

This is a formula that represents the cyclic structure of monosaccharide.

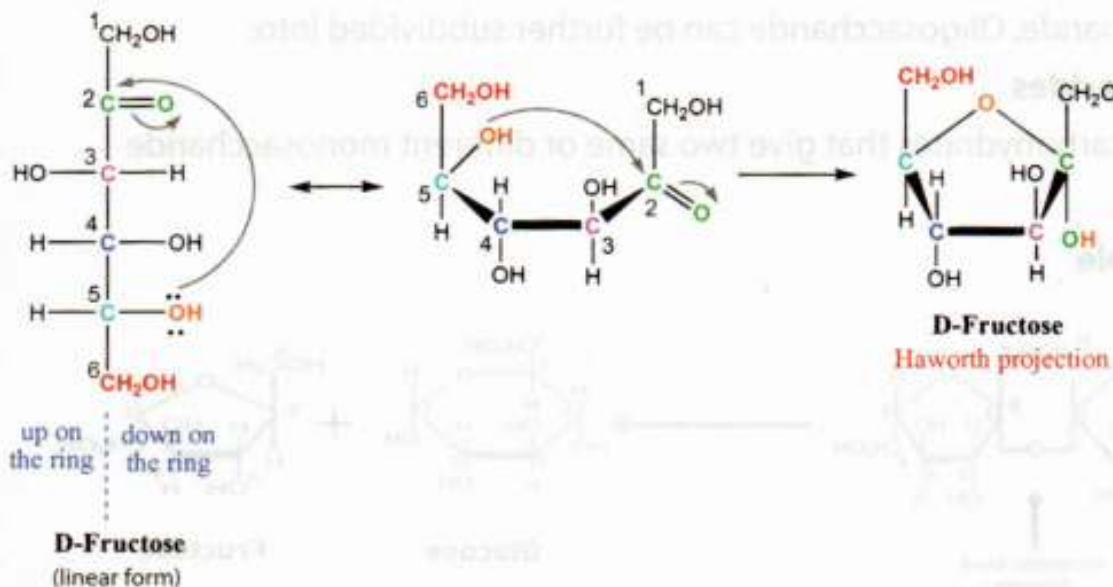
For example



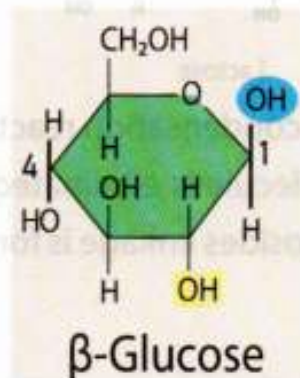
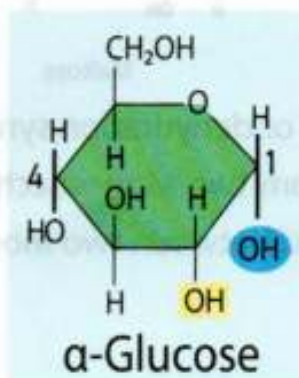
Haworth structure is produced from the reaction of carbonyl group and hydroxyl group in the same molecule. Therefore an aldose cyclizes, when the hydroxyl group on the second to last carbon undergoes an intramolecular reaction with the carbonyl group of the aldehyde. Aldoses always form pyranose structure which is a six membered rings containing an oxygen atom.



Similarly, a ketose cyclizes, when the hydroxyl group on the second to last carbon undergoes an intramolecular reaction with the carbonyl group. Ketoses and pentoses always form furanose structure which is a five member ring containing an oxygen atom.

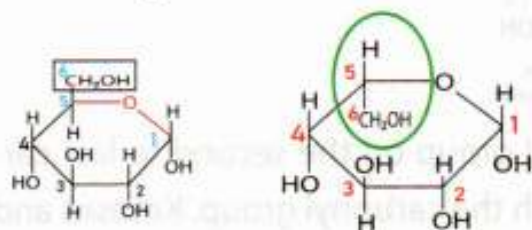


The carbon that formed the circle is known as anomeric carbon. If the $-OH$ group attached to the anomeric carbon is below the ring the sugar is referred to as α -sugar but, if the hydroxyl group on the C-atom is above the "ring" the sugar is referred to as β -sugar.



The L and D configuration of the cyclic structure of carbohydrates can be determined by the position of $-\text{CH}_2\text{OH}$ in the molecule. If the $-\text{CH}_2\text{OH}$ group is above the ring the sugar referred to as D-sugar but if the $-\text{CH}_2\text{OH}$ group is below or in the ring the sugar is referred to as L-sugar.

For example



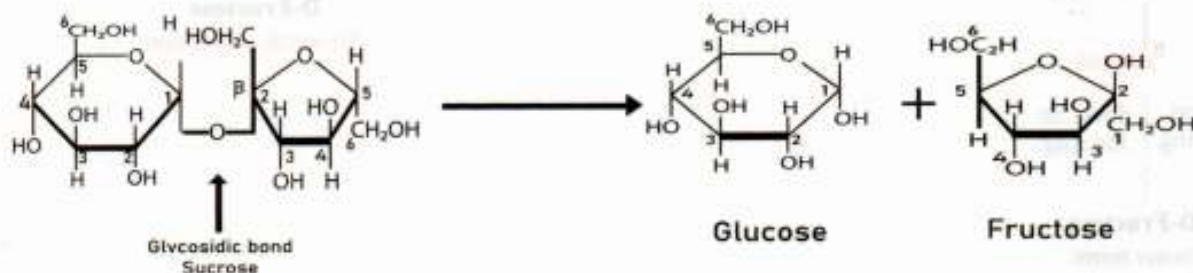
4.2.5 Oligosaccharides

These are carbohydrates which undergo hydrolysis into 2 to 10 molecule of monosaccharide. Oligosaccharide can be further subdivided into:

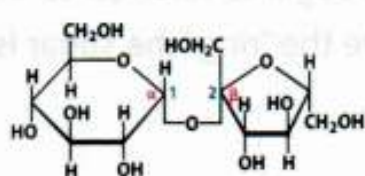
1. Disaccharides

These are carbohydrates that give two same or different monosaccharide molecules.

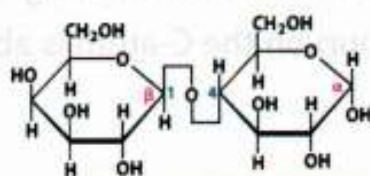
For example



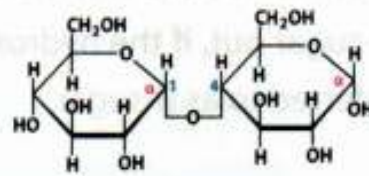
Some common examples of disaccharide are:



Sucrose



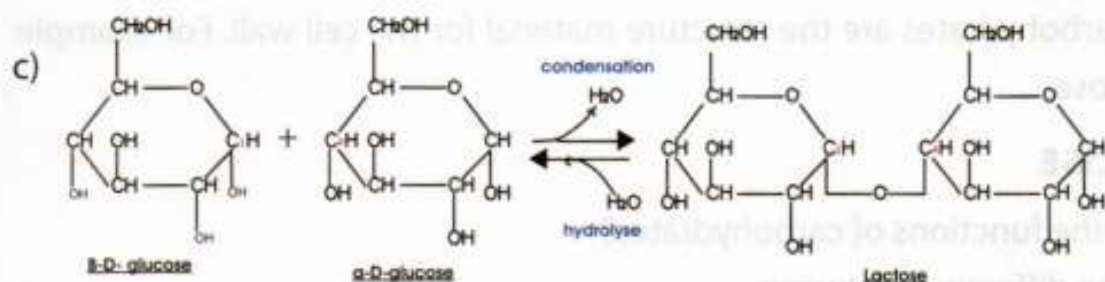
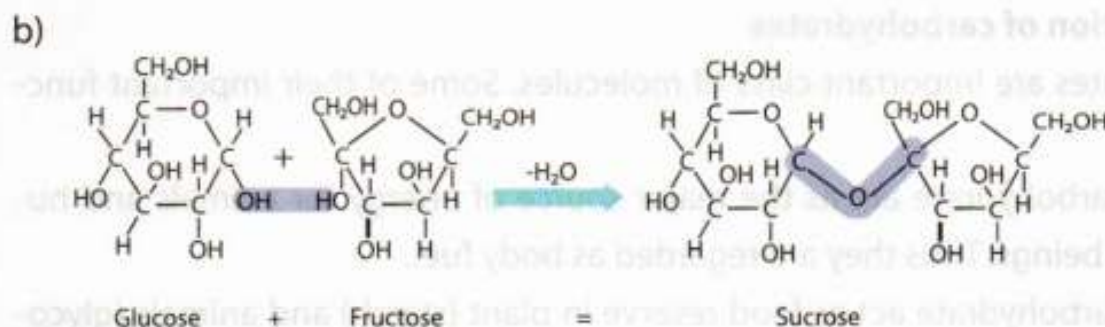
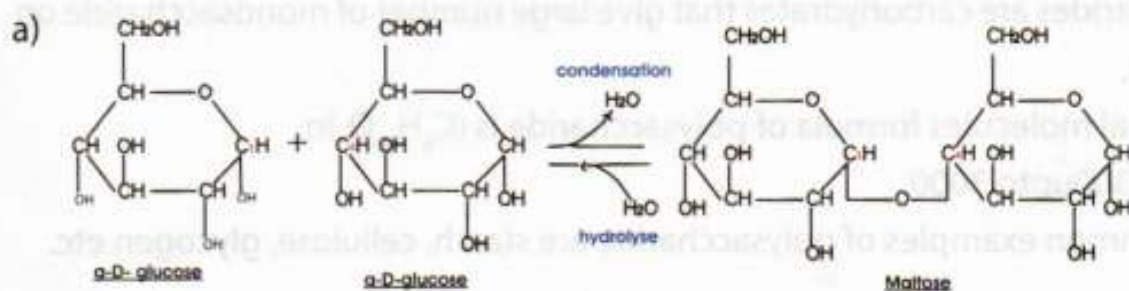
Lactose



Maltose

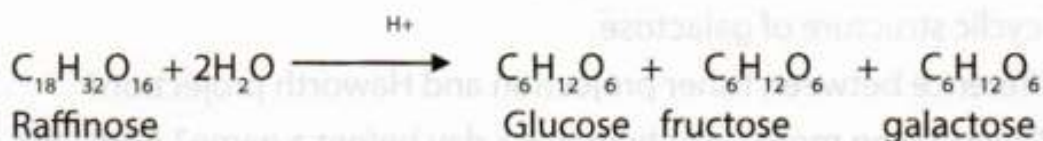
Disaccharides are formed by the condensation reaction or dehydration synthesis. During this reaction a water molecule is eliminated from two Monosaccharides then a new linkage known as glycosides linkage is formed between two Monosaccharides.

For examples



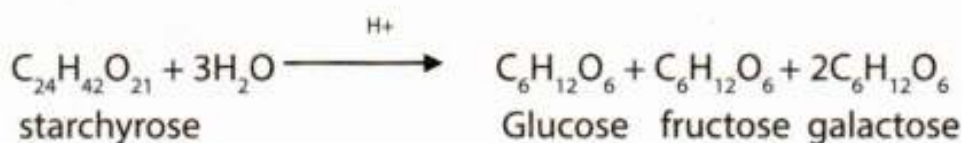
2. Trisaccharides

These are carbohydrates that can be hydrolyzed into three monosaccharide molecules. The typical example of Trisaccharides is Raffinose.



3. Tetrasaccharide

These are carbohydrates that furnish three Monosaccharides on hydrolysis. The typical example of Tetrasaccharide is starchyrose



4.2.6 Polysaccharides

Polysaccharides are carbohydrates that give large number of monosaccharide on hydrolysis.

The general molecules formula of polysaccharide is $(C_6H_{10}O_5)_n$.

Where $n=300$ upto 3000.

Some common examples of polysaccharide are starch, cellulose, glycogen etc.

4.2.7 Function of carbohydrates

Carbohydrates are important class of molecules. Some of their important functions are:

1. The carbohydrate act as the major source of energy for animals and humans beings. Thus they are regarded as body fuel.
2. The carbohydrate act as food reserve in plant (starch) and animals (glycogen).
3. The carbohydrates are the structure material for the cell wall. For example Cellulose.



EXERCISE

1. What are the functions of carbohydrates?
2. What is the difference between
 - a) Sugar and non-sugar?
 - b) Reducing sugar and non-reducing sugar?
 - c) Monosaccharide, disaccharide and polysaccharides?
3. Write the cyclic structure of galactose
4. What is difference between fisher projection and Haworth projection?
5. Why are Players given more carbohydrates a day before a game? Apply the concept from the lesson
6. What is the chemical name of the table sugar found at home?
7. Bring examples of fruits in your area and identify the type of sugar present.

4.3 PROTEINS

Lesson objectives:

By the end of the Lesson, the learner should be able to:

- Define amino acids and proteins
- Classify amino acids
- Explain the order of protein structure
- State biological importance of proteins

Proteins are polymers of amino acids with very high molar masses. Proteins are the most abundant biomolecules of the living system. Chief sources of proteins are milk, cheese, peanuts, fish, meat, etc.

Food items rich in proteins



Egg



Milk



Milk-preparations



Fish

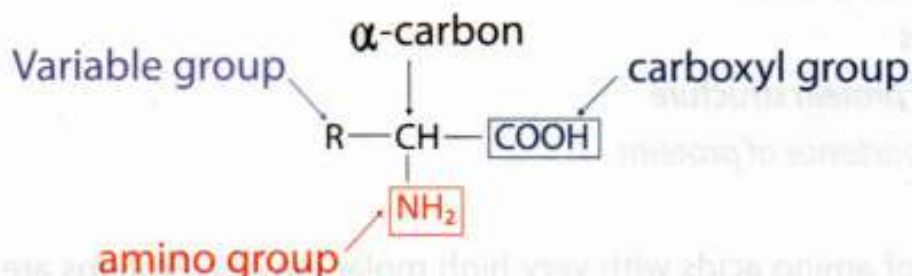


Meat

They occur in every part of the body and form the fundamental basis of structure and functions of life. They are also required for growth and maintenance of body. Proteins must be continually supplied to the body in the diet.

4.3.1 Amino acids

Amino acids are monomers of proteins. Amino acids can also be defined as organic compounds which contain amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) as a functional groups. The general formula of amino acids is:



The simplest of amino acids is called glycine, the $\text{R} = (-\text{H})$. The next one is called alanine whereas $\text{R} = (-\text{CH}_3)$.

Amino acids are generally represented by a three letter symbol or sometimes one letter symbol is also used. Structures of some commonly occurring amino acids along with their 3-letter and 1-letter symbols are given in Table 4.1.

Table 4.1: The 20 common naturally occurring amino acids

Neutral Amino Acids					
Name	Structure	Abbreviations	Name	Structure	Abbreviations
Alanine		Ala A	Phenylalanine		Phe F
Asparagine		Asn N	Proline		Pro p
Cysteine		Cys C	Serine		Ser S
Glutamine		Gln Q	Threonine		Thr T

Table 4.1: (Continued..)

Glycine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{H} \end{array}$	Gly G	Tryptophan	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{Indole ring} \end{array}$	Trp W
Isoleucine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3 \end{array}$	Ile I	Tyrosine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{Benzene ring with OH} \end{array}$	Try Y
leucine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2\text{CH}(\text{CH}_3)_2 \end{array}$	Leu L	Valine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}(\text{CH}_3)_2 \end{array}$	Val V

Acidic Amino Acids			Basic Amino Acids		
Name	Structure	Abbreviations	Name	Structure	Abbreviations
Aspartic acid	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2\text{COO}^- \end{array}$	Ala A	Arginine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ (\text{CH}_2)_3-\text{N}^+ \\ \\ \text{C}-\text{NH}_2 \end{array}$	Arg R
Glutamic acid	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2\text{CH}_2\text{COO}^- \end{array}$	Asn N	Histidine	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{Imidazole ring} \end{array}$	His H

4.3.2 Classification of amino acids



ACTIVITY 3

We discuss the following questions in groups and answer the following questions using table 4.1

1. What are the common functional groups in each molecule?
2. Which one is the smallest molecule?
3. Which one is the largest molecule?
4. What are the elements present in each molecule?
5. What do these letters represent?

a) Gln b) Gly c) Glu d) Ile e) Met

Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule.

- equal number of amino and carboxyl groups makes it neutral.
- More number of amino than carboxyl groups makes it basic.
- More carboxyl groups as compared to amino groups make it acidic.

See table 4.1

Amino acids are also classified as essential or non essential. The amino acids, which can be synthesized in the body, are known as nonessential amino acids. On the other hand, those which cannot be synthesized in the body and must be obtained through diet are known as essential amino acids as shown below.

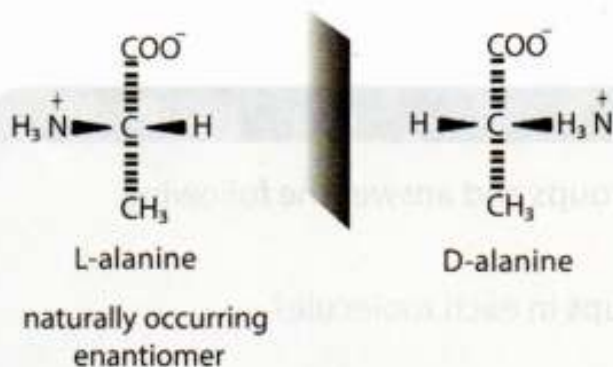
Essential amino acids

Arginine	Lysine
Histidine	Methionine
Isoleucine	Threonine
Leucine	Phenylalanine
Valine	Tryptophan

Non-essential amino acids

Alanine	Glycine
Asparagine	Proline
Aspartate	Serine
Glutamate	Cysteine
Glutamine	Tyrosine

Except glycine, all other naturally occurring α -amino acids are optically active, since the α -carbon atom is asymmetric. These exist in both 'D' and 'L' forms. Most naturally occurring amino acids have L-configuration. L-Amino acids are represented by writing the $-\text{NH}_3^+$ group on left hand side.



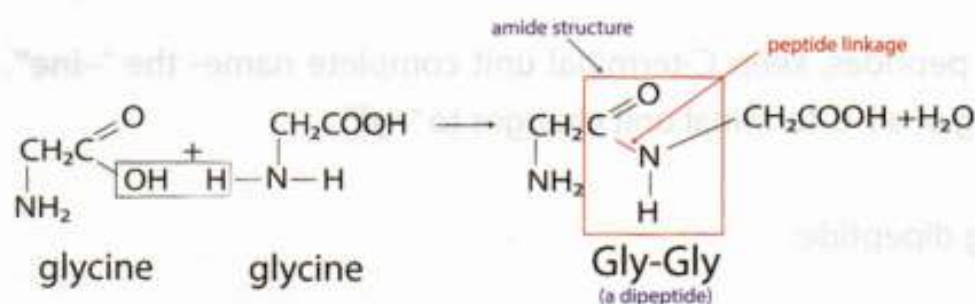
4.2.3 Structure of proteins

Proteins are the polymers of α -amino acids and they are connected to each other by peptide bond or peptide linkage. Chemically, peptide linkage is an amide formed between ($-\text{COOH}$) group and ($-\text{NH}_2$) group. A condensation reaction between two molecules of similar or different amino acids proceeds by the combination of the amino group of one molecule with the carboxyl group of the other, water molecule is removed and a peptide bond is formed. The product of the reaction is called a dipeptide because it is made up of two amino acids.

Example 1

Combine two glycine molecules to form dipeptide:

Solution:

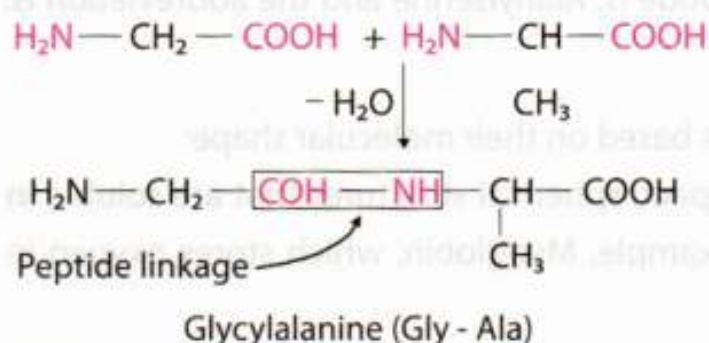


Name: Glycylglycine

Example: 2

Combine glycine with alanine to form dipeptide:

Solution:



If a third amino acid combines to a dipeptide, the product is called a tripeptide. A tripeptide contains three amino acids linked by two peptide linkages. Similarly when four or five amino acids are linked, the respective products are known as tetrapeptide or pentapeptide.

When the number of such amino acids is more than ten, then the products are called polypeptides. A polypeptide with more than hundred amino acid residues is called a protein.

The amino acid units in a peptide are called amino acid residues. The amino group end is called the N-terminal residue, the other end is called C-terminal residue.

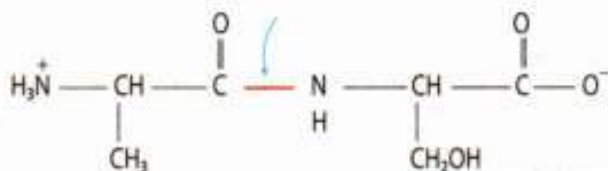
Example: Ala-Pro-Tyr-Met-Gly-Lys-Gly

The above peptide alanine is N-terminal residue whereas glycine is C-terminal residue.

When naming the peptides, keep C-terminal unit complete name- the “-ine”, whereas, the ending of all C-terminal unit changes to “-yl”.

Example:

Name the following dipeptide:



Answer: The N-terminal residue is alanine whereas the C-terminal residue is serine therefore the name of the dipeptide is: Alanylserine and the abbreviation is: Ala-ser

Proteins are classified into two types based on their molecular shape:

- Globular proteins:** are compact, spherical structures that are soluble in an aqueous environment. Example, Myoglobin, which stores oxygen in muscle.
- Fibrous proteins:** are long, threadlike structures that have high helical content. Examples are; Keratins, found in hair, nails, the scales of reptiles, and collagen.

Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary, each level being more complex than the previous one. Each level of protein structure is a result of interactions between the amino acids of the protein.

i) Primary structure of proteins: Is the sequence of amino acids that is joined together by peptide bonds. Any change in this primary structure i.e., the sequence of amino acids creates a different protein. The order or sequence of amino acids in a protein chain is important in determining its structure and function.

ii) Secondary structure of proteins: The three-dimensional arrangement of localized regions of a protein is called its secondary structure.

The two most common types of secondary structures are:

- Alpha helix (α helix)
- Beta-pleated sheet (β -pleated sheet)

α -Helix:

The α -helix forms when a peptide chain twists into a right-handed or clockwise spiral, as shown in figure 4.1. This coil is stabilized by hydrogen bonds between the carbonyl oxygen of one amino acid and the N—H hydrogen atom of another amino acid located four amino acids from it in the primary structure.

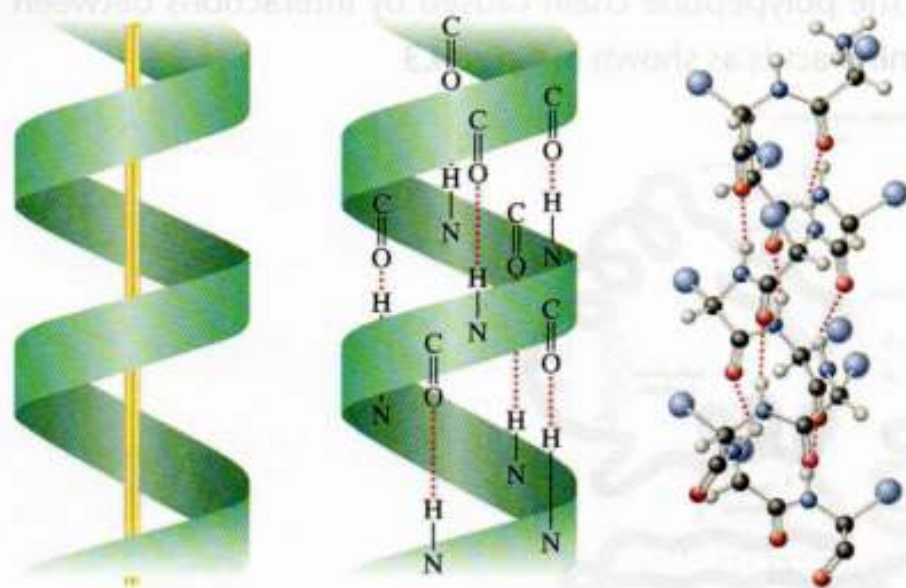


Figure 4.1: Alpha-helix of protein structure

β -Pleated Sheet:

The β -pleated sheet is an extended structure in which segments of the protein chain align to form a zigzag structure. Strands called *beta strands* are held together through hydrogen bonding interactions between the backbones as shown in figure 4.2

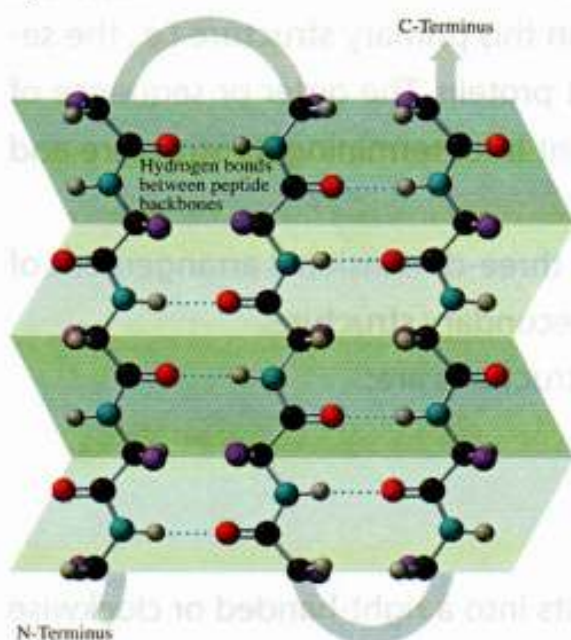


Figure 4.2: β -Pleated Sheet of protein structure

iii. Tertiary structure of proteins:

The tertiary structure is the three-dimensional structure of the protein. It involves twisting and folding of the polypeptide chain caused by interactions between the side chains of the amino acids as shown in figure 4.3

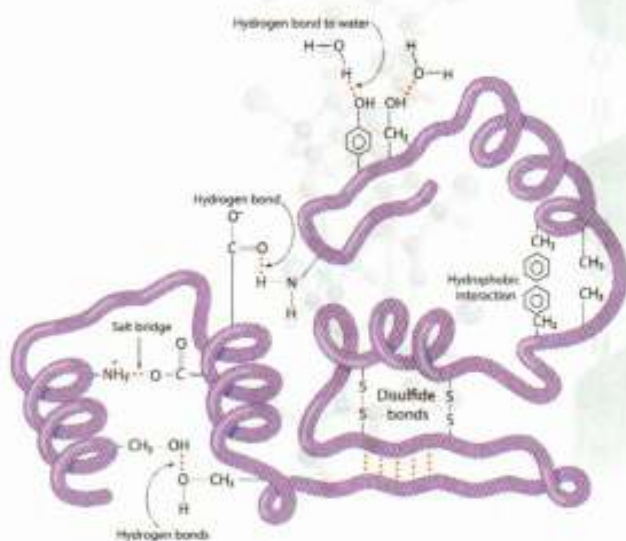


Figure 4.3: Tertiary structure of protein

iv. Quaternary structure of proteins:

The quaternary structure consists of two or more polypeptide chains interacting to form a biologically active protein.

Hemoglobin, an oxygen transport protein, is an example of a protein with a quaternary structure.

- It consists of four polypeptide chains or subunits.
- It has two identical alpha subunits and two identical beta subunits.

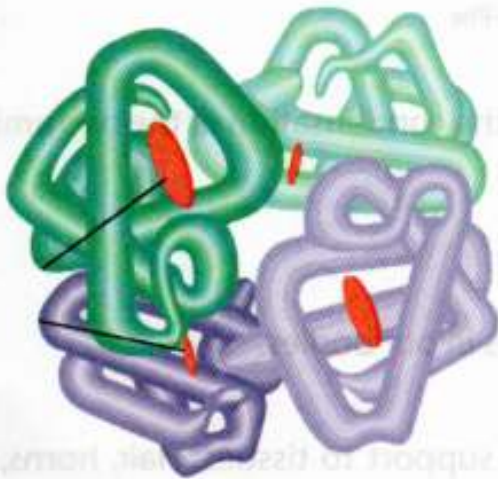


Figure 4.4: Quaternary structure of protein

Denaturation of proteins:

Denaturation is a process that disrupts secondary, tertiary, and quaternary structures.

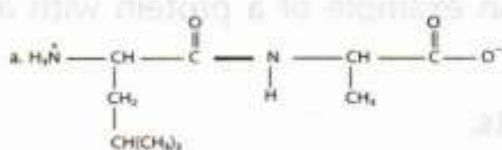
When the secondary, tertiary, or quaternary structure of a protein is disturbed, the properties of a protein are also altered and the biological activity is often lost. The primary structure is not destroyed during denaturation.

Denaturing agents that cause protein denaturation are:

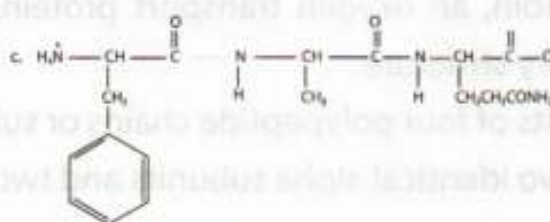
- a) Heat
- b) Mechanical agitation
- c) Organic compounds
- d) Changes in pH
- e) Heavy metals

**EXERCISE**

1. Identify the N-terminal and C-terminal amino acid in each peptide.



b. Arg-His-Asn-Tyr



d. Val-Thr-Pro-Phe

2. Draw the structure of the dipeptide formed by the condensation of the two amino acids Valine and glycine (Val-Gly)

4.3.3 Biological importance of proteins:

Proteins have structural and functional properties in our bodies.

Keratin:

Form long insoluble fibres, giving strength and support to tissues. Hair, horns, hooves, and fingernails are all made up of keratin.

Collagen:

It is found in bone, connective tissue, tendons, and cartilage.

Membrane proteins:

Transport small organic molecules and ions across cell membranes.

Insulin:

The hormone that regulates blood glucose levels, and

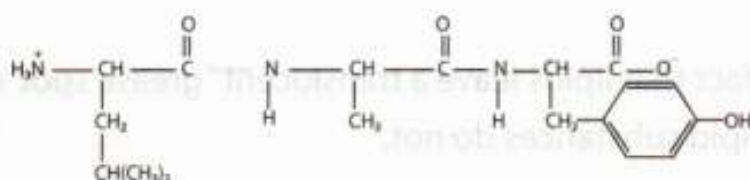
Hemoglobin:

Transport oxygen and carbon dioxide from lungs to tissues and tissues to the lungs respectively.



EXERCISE

- Define each of the following:
 - Amino acids
 - Proteins
 - Primary structure of proteins
 - Globular proteins
- State the two most common types of secondary structure of proteins.
- Classify amino acids based on the relative number of amino and carboxyl groups in their molecules?
- Explain briefly the tertiary structure of proteins.



4.4 LIPIDS

Lesson Objectives

By the end of the lesson, the learner should be able to:

- Describe lipids and state its importance.
- Classify and identify lipids.
- Explain basic structure of lipids.

Lipids are organic chemical compounds that are insoluble in water but soluble in nonpolar solvents. They are defined on the basis of physical property not by a particular functional group, thus they have a variety of structures and functions.

Experiment 1

Test for the presence of lipids

Materials

- piece of paper
- food item e.g. fries, piece of cooked meat, etc
- 10 ml of cooking oil
- 10 ml water

Procedure

1. Add cooking oil to brown paper bag until it is soaked up. The part of the paper that soaks up oil should be translucent compared to the part that does not.
2. Wet the paper with water. The paper may become wet and soggy, but should not become translucent.
3. Stain the brown paper bag with the food item to be tested and hold it up to the light.
4. Record your observations, noting any key differences between the controls and the experimental sample.

Observations

What did you observe?

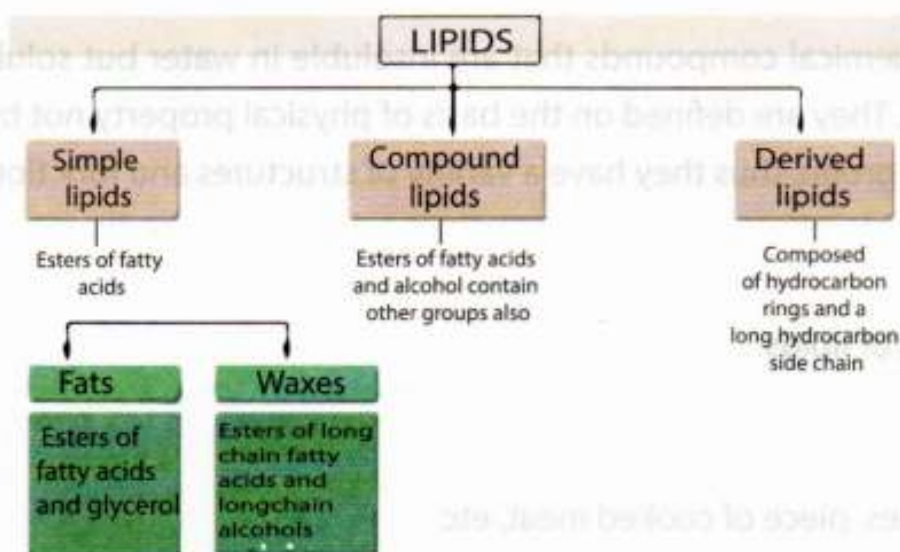
Conclusion

The test for lipids relies on the fact that lipids leave a translucent 'grease spot' on brown paper bags, while non-lipid substances do not.

4.4.1 Classification of lipids

In 1943, Bloor classified lipids on the basis of their chemical composition into three categories:

- a) Simple lipids
- b) Compound lipids
- c) Derived lipids



4.4.2 Simple lipids

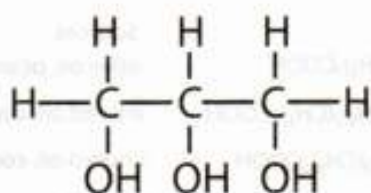
Simple lipids are esters of fatty acids and alcohol. On the basis of nature, simple lipids are divided into two categories:

1. Triglycerides

They are esters of fatty acids and glycerol.

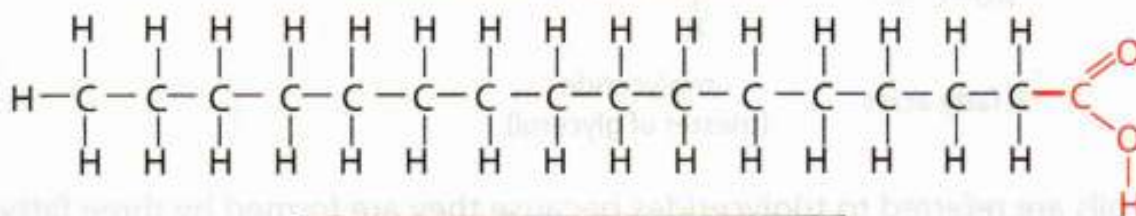
They are the most abundant lipids in nature and the storage form of lipids and concentrated source of energy. It is composed of fatty acids and glycerol.

Glycerol is an alcohol that contains three hydroxyl groups.



Fatty acids are aliphatic organic acids which usually contain an even number of carbon atom. Fatty acids can be saturated or unsaturated. If they have no double bond, they are referred to saturated fatty acids but if they have one or more double bonds they referred to as unsaturated fatty acids.

SATURATED FATTY ACID



UNSATURATED FATTY ACID

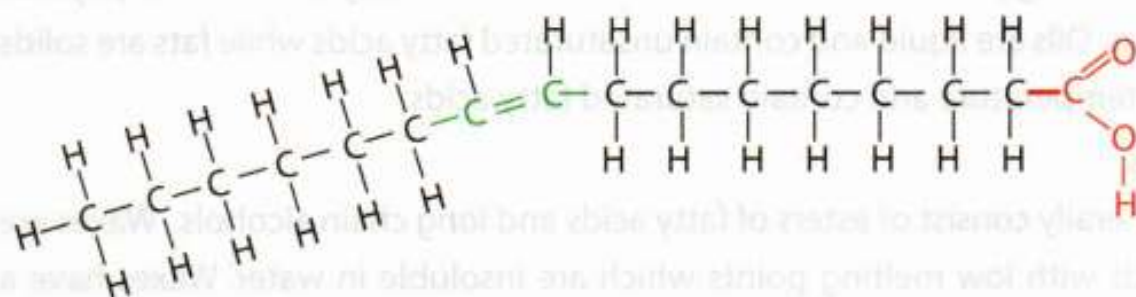
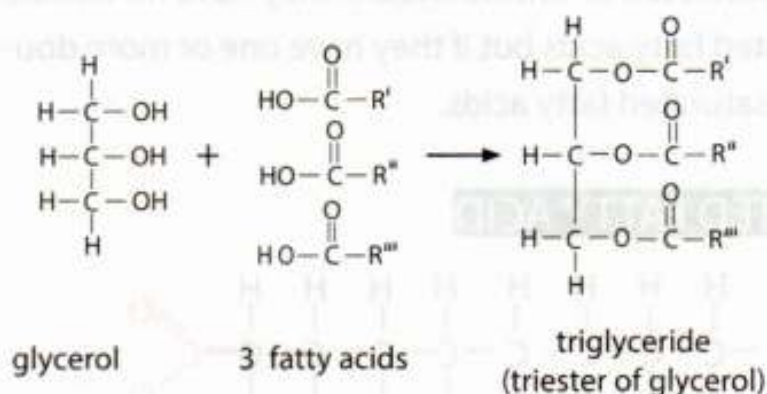


Table 4.2: Saturated and unsaturated fatty acids and some common sources for each

Sources		Saturated Fatty Acids		
Name	Number of Carbon Atoms	Structure	Sources	
butyric acid	4	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	milk fat	
capric acid	10	$\text{CH}_3(\text{CH}_2)_8\text{COOH}$	milk fat, whale oil	
myristic acid	14	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	butterfat, coconut oil	
palmitic acid	16	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	beef fat, butterfat	
stearic acid	18	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	beef fat, butterfat	

Unsaturated Fatty Acids				
Name	Number of Carbon Atoms	Number of Double Bonds	Structure	Sources
oleic acid	18	1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	olive oil, peanut oil
linoleic acid	18	2	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_2(\text{CH}_2)_6\text{COOH}$	linseed oil, corn oil
linolenic acid	18	3	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2)_3(\text{CH}_2)_6\text{COOH}$	linseed oil, corn oil

Triglycerides are formed when the three hydroxide groups of glycerol molecules combine with the carboxyl group of the fatty acid.



Fats and oils are referred to triglycerides because they are formed by three fatty acids and one glycerol. Fats and oils are similar chemically but differ in physical properties. Oils are liquid and contain unsaturated fatty acids while fats are solids at room temperature and contain saturated fatty acids.

2. Waxes

They generally consist of esters of fatty acids and long chain alcohols. Waxes are soft solids with low melting points which are insoluble in water. Waxes have a variety of applications in medicine, cosmetic, and other industries.

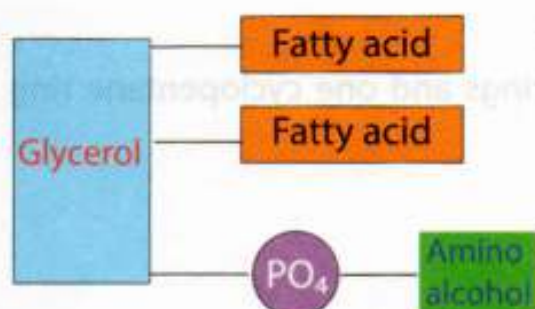
4.4.3 Compound lipids

Compound lipids are esters of fatty acids and with glycerol and additional group. According to the nature of the additional group, compound lipids are classified into:

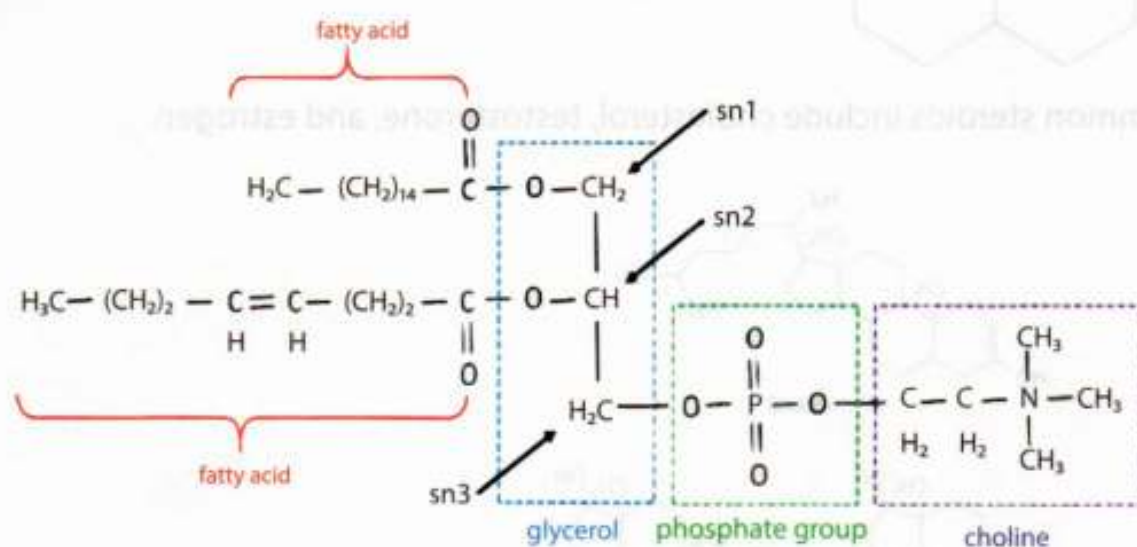
1. **Glycolipids:** is a molecule that is a combination of lipid and carbohydrate.
2. **Lipoproteins:** is a molecule that is a combination of lipid and proteins.
3. **Phospholipids:** They are lipids that contain phosphate group and amino alcohol. Phospholipids are amphibatic molecules having both hydrophilic part which is the hydrocarbon chain of the fatty acids and hydrophilic part which are the phosphate group and additional nitrogenous bases like choline etc therefore Phospholipids are classified into two types according to the type of alcohol:

a) Glycerophospholipids

They are phospholipids in which its alcohol is glycerol



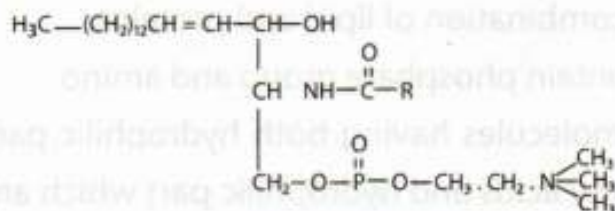
For example



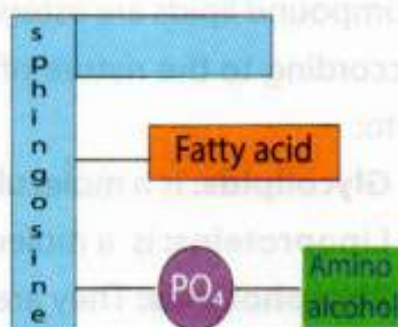
b) Sphingolipids

These are phospholipids in which its fatty alcohol is sphingosine (amino alcohol).

For example



Sphingomyeline



Note: glycolipids are lipids that contain carbohydrates, for example cerobroside.

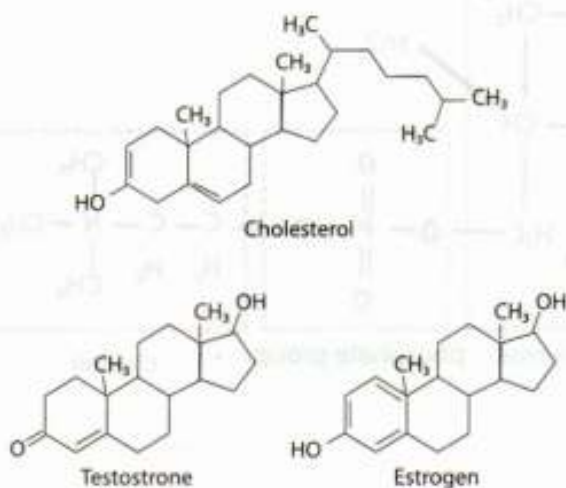
4.4.3 Derived lipids

Derived lipids are substances that are derived by the hydrolysis from the compound and simple lipids. They also include substances related to lipids. Derived lipids include; Fatty acids, glycerol, steroids etc.

Steroids are lipids having three cyclohexane rings and one cyclopentane ring fused as shown below



Some common steroids include cholesterol, testosterone, and estrogen.



Cholesterol is the most common steroid in the body. It is a needed structural molecule found in all normal animal cells. Although cholesterol is needed to synthesize other steroids, too much cholesterol can lead to blocked blood vessels and high blood pressure, then possibly heart attack and stroke. We can't live without cholesterol, and we can't live with too much of it, either.

4.4.4 Importance of lipids

1. Lipids act as a source of energy. They are the natural solvent for fat-soluble vitamins
2. They contain essential fatty acids.
3. Lipids in adipose tissue serve as energy store.
4. Lipids have a role in protection and fixation of internal organs
5. Lipids have myelin sheath of nerve fibres which serve as electrical insulator.
6. Lipids under the skin serve as thermal insulator.



EXERCISE

1. What are the general characteristics of lipids?
2. What are waxes?
3. What are triacylglycerols, and how do the triacylglycerols in fat and oil differ?
4. What are the major types of phospholipids?
5. What is the difference between a saturated and monounsaturated fatty acid? Give an example of each having the same number of carbons.

4.5 NUCLEIC ACIDS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Describe nucleic acid.
- State types and functions of nucleic acids.
- Explain the structure of nucleic acid.
- Compare and contrast the structure of DNA and RNA.

Nucleic acids are polynucleotides that occur in the nuclei of the living cells. There are two types of nucleic acids:

1. deoxyribonucleic acids (DNA)
2. Ribonucleic acid (RNA)



4.5.1 Structure of Nucleic acids

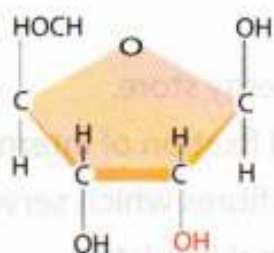
Nucleic acids are composed of polynucleotide. Each polynucleotide consists of three chemical constituents:

a) Sugar:

There are two types of pentose sugars namely β -D-ribose and β -D-2-deoxyribose. The nucleotide of DNA contains β -D-2-deoxyribose while that of nucleotides present in RNA contain β -D-ribose while that of RNA contains B-D-ribose.



2 - Deoxyribose



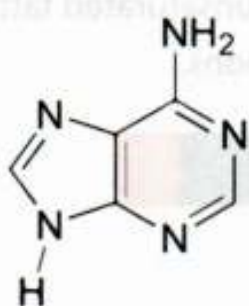
Ribose

Ribose has the molecular formula of $C_5H_{10}O_5$ while the molecular formula of deoxyribose is $C_5H_{10}O_4$.

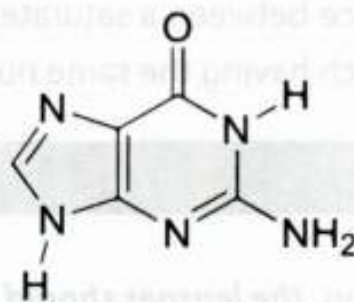
b) Nitrogen bases

The nitrogen bases present in DNA are:

i) Adenine and Guanine which are collectively known as **Purines**



adenine



guanine

ii) Thymine (T), Cytosine (C) and Uracil which are collectively known as Pyrimidine.

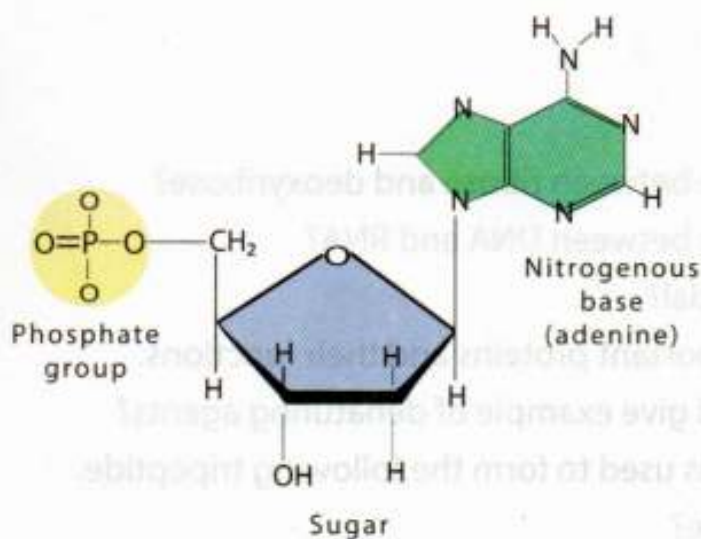


Pyrimidines

The nitrogen bases of DNA are adenine (A), guanine (G), cytosine(C) and thymine (T) while those in RNA are adenine (A), guanine (G), cytosine(C) and Uracil (U).

c) Phosphate group

This group is responsible for linking nucleotides together in a nucleic acid molecule. The complete structure of nucleotide is represented as



The nucleotides are joined together by phosphodiester bond to form the backbone of the DNA and RNA.

The structural difference between DNA and RNA are summarized as:

1. The sugar present in RNA is ribose while in DNA it is deoxy ribonucleic acid.
2. The nucleic basis in RNA are adenine(A), guanine(G), cytosine(G) and Uralic(U) while those present in DNA adenine(A), Guanine(G), cytosine(C) and type.
3. DNA has a double stranded helix structure where as RNA has a single strand-ed helix structure.

4.5.2 Functions of Nucleic acids

Nucleic acids are essential for life and play a variety of roles:

1. They control protein synthesis.
2. To store and transmit genetic information from one generation to the next.



EXERCISE

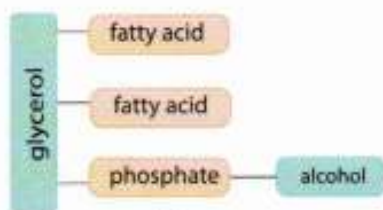
1. What is the true structure of DNA?
2. What nucleotide base does RNA contain but DNA does not?
3. What molecules do both DNA and RNA contain?
4. What is the function of nucleic acids?

Chapter Review Questions

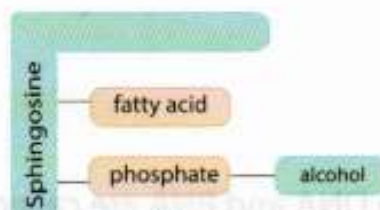
Answer the following questions

1. Define
 - a) Carbohydrates
 - b) Nucleic acid
 - c) Proteins
 - d) Lipids
2. What are the structural difference between ribose and deoxyribose?
3. What are the structural difference between DNA and RNA?
4. Why some amino acids are essential?
5. State at least two biologically important proteins and their functions.
6. What is protein denaturation, and give example of denaturing agents?
7. Identify the individual amino acids used to form the following tripeptide.
8. What is the name of the tripeptide?
9. Draw a triacylglycerol formed from two molecules of myristic acid and one molecule of palmitic acid?
10. Give an example of each type of lipid.
 - a) a monounsaturated fatty acid
 - b) saturated triacylglycerol

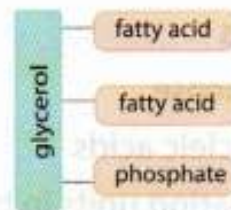
11. What type of lipids is represented by each of the following block diagrams:



A



B



C

Choose the Correct Answer

1. Which of the following is a polysaccharide?

- a) cellulose
- b) fructose
- c) glucose
- d) sucrose

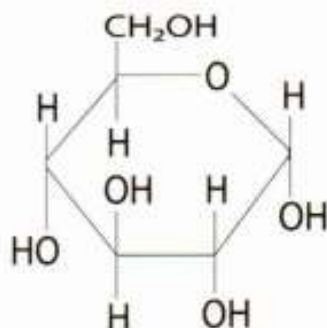
2. What elements make up carbohydrates?

- a) Carbon, hydrogen, nitrogen, oxygen
- b) sulfur, carbon, hydrogen
- c) glucose and fructose
- d) carbon, hydrogen, oxygen

3. $C_6H_{12}O_6$ is the molecular formula for what carbohydrate?

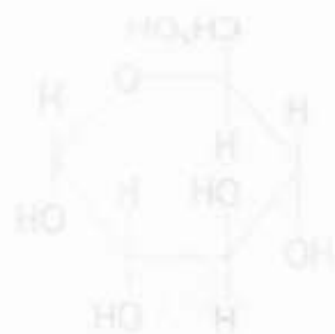
- a) Sucrose
- b) Glucose
- c) Fructose
- d) Both glucose and fructose

4. Which biomolecule is this a picture of?



- a) Protein
- b) Nucleic Acid
- c) Lipid
- d) Carbohydrate

5. The information for protein synthesis is stored in:
- amino acids
 - fats
 - glucose
 - nucleic acids
6. The repeating units in both DNA and RNA are called:
- amino acids
 - hydrogen bonds
 - nucleic acids
 - nucleotides
7. The sugar that is in the backbone of DNA is called:
- Delta glucose
 - deoxyribose
 - fructose
 - ribose
8. The sugar that is in the backbone of RNA is called:
- deoxyribose
 - fructose
 - ribose
 - sucrose
9. The RNA molecule is a:
- double helix
 - double-stranded molecule
 - none of the choices
 - single-stranded molecule
10. Adenine bonds with _____.
- Guanine
 - Thymine
 - Cytosine
 - Adenine



11. Guanine bonds with _____.
a) Adenine
b) Guanine
c) Cytosine
d) Thymine
12. Nucleotides are made up of a sugar, phosphate, and a _____.
a) nitrogenous base
b) nucleic acid
c) cholesterol
d) fatty acid
13. Which one of the following is a monosaccharide?
a) fructose
b) glycogen
c) lactose
d) sucrose
14. Which component is found in all sphingolipids?
a) a carbohydrate
b) a negative charge
c) a phosphate group
d) an amino alcohol
15. Which of these is not a lipid?
a) Fats
b) Oils
c) Proteins
d) Waxes
16. What are the functions of lipids?
a) Provide energy to the body
b) Form part of cellular membrane
c) Control metabolism
d) All options given are correct

CHAPTER 5

P- BLOCK
ELEMENTS

P-Block elements

P-Block elements																											
1 IA H Hydrogen 1.008																		2 IIA He Helium 4.003									
3 IIIA Li Lithium 6.941	4 IIA Be Beryllium 9.012																	10 VIIIA Ar Argon 39.95									
5 IIIA Na Sodium 22.99	6 IIA Mg Magnesium 24.31																	18 VIIIA Ne Neon 20.18									
19 IIIA K Potassium 39.10	20 IIA Ca Calcium 40.08																	36 VIIIA Kr Krypton 83.80									
37 IIIA Rb Rubidium 85.47	38 IIA Sr Strontium 87.62																	54 VIIIA Xe Xenon 131.29									
55 IIIA Cs Cesium 132.91	56 IIA Ba Barium 137.33																	72 VIIIA Rn Radon 222									
87 IIIA Fr Francium 223	88 IIA Ra Radium 226																	104 VIIIA Og Oganesson 294									

CHAPTER CONTENTS

5.1 Introduction to p-block elements

5.2 Group 13: Boron family

5.3 Group 14: Carbon family

5.4 Group 15: Nitrogen family

5.5 Group 16: Oxygen family

5.6 Group 17: Halogens

5.7 Group 18: Noble gases

Chapter review Questions

CHAPTER OBJECTIVES

By the end of the chapter, the learner should be able to:

- Identify groups and elements of p-block elements
- Write electronic configuration of p-block elements
- Explain physical and chemical properties of p-block elements
- State uses of p-block elements

P-Block elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H	He											B	C	N	O	F	Ne
Li	Be											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

5.1 INTRODUCTION TO P-BLOCK ELEMENTS

Lesson objectives

By the end of the lesson, learners should be able to:

- Define p-block element
- Identify p-block elements and their position in the periodic table
- Classify p-block elements into metals, non-metals and metalloids

**ACTIVITY 1**

We will study the periodic table and classify the elements of the periodic table into blocks. We will find and mane the number of groups that are in p-block elements.

The p-block elements in the periodic table are those in which the last electron enters the outermost p-orbital. The elements of this group include metals, non- metals and metalloids.

P-Block elements

13	14	15	16	17	18
IIIA	IVA	VSA	VIA	VIIA	VIIIA
B Boron 10.811	C Carbon 12.011	N Nitrogen 14.007	O Oxygen 15.999	F Fluorine 18.998	He Helium 4.003
Al Aluminum 26.982	Si Silicon 28.086	P Phosphorus 30.974	S Sulfur 32.06	Cl Chlorine 35.45	Ne Neon 20.180
Ga Gallium 69.723	Ge Germanium 72.630	As Arsenic 74.922	Se Selenium 78.96	Br Bromine 79.904	Kr Krypton 83.80
In Indium 114.82	Sn Tin 118.71	Sb Antimony 121.76	Te Tellurium 127.6	I Iodine 126.91	Xe Xenon 131.29
Tl Thallium 204.38	Pb Lead 207.2	Bi Bismuth 208.98	Po Polonium 209	At Astatine 210	Rn Radon 222
Nh Nihonium 286	Fl Flerovium 289	Mc Moscovium 288	Lv Livermorium 293	Ts Tennessine 289	Og Oganesson 294

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
La Lanthanum 138.91	Ce Cerium 140.12	Pr Praseodymium 140.91	Nd Neodymium 144.24	Pm Promethium 144.91	Sm Samarium 150.36	Eu Europium 151.96	Gd Gadolinium 157.25	Tb Terbium 158.93	Dy Dysprosium 162.50	Ho Holmium 164.93	Er Erbium 167.26	Tm Thulium 168.93	Yb Ytterbium 173.05	Lu Lutetium 174.97	
Ac Actinium 227	Th Thorium 232.04	Pa Protactinium 231.04	U Uranium 238.03	Np Neptunium 237.05	Pu Plutonium 244.06	Am Americium 243.06	Cm Curium 247.07	Bk Berkelium 247.07	Cf Californium 251.08	Es Einsteinium 252.08	Fm Fermium 257.10	Md Mendelevium 258.11	No Nobelium 259.10	Lr Lawrencium 262.11	

As the p-orbital can accommodate only 6 electrons, there are six groups of p-block elements in the Periodic Table, numbered from 13 to 18 as follows:

1. Group 13 or group IIIA (Boron family).
2. Group 14 or group IVA (Carbon family)
3. Group 15 or group VA (nitrogen family)
4. Group 16 or group VIA (Oxygen family)
5. Group 17 or group VIIA (Halogens)
6. Group 18 or group VIIIA (Noble gases)



EXERCISE

1. Why do you think we have metals, non metals and metalloids within the p-block groups?
2. What is the total number of p-block elements?
3. Why are they called p-block elements?
4. Classify the p-block elements into metals, non-metals and metalloids
5. Choose one p-block elements and write about it.

5.2 GROUP 13 BORON FAMILY

Lesson objectives

By the end of the lesson, the learner should be able to:

- Identify Boron Family
- Write electronic configuration of Boron Family
- Explain physical and chemical properties of Boron Family
- State the uses of Boron Family



ACTIVITY 2

We will use periodic table to identify Boron Family, and then we find out the total number of elements in Boron family and finally we name them.

Group 13 or group IIIA elements of the periodic table are called Boron family. The group 13 elements consist of five elements, namely;

1. Boron (B),
2. Aluminum (Al)
3. Gallium (Ga)
4. Indium (In)
5. Thallium (Tl)

Boron Family

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Boron is the lightest of the elements in this group. It is a metalloid. The others in the group are metals. Aluminum is the most abundant metal in this group and it is the third most abundant element in the earth's crust.

5.2.1 Electronic configuration of boron family



ACTIVITY 3

We will write the electronic configuration of Boron family, and then we compare and contrast their electronic configuration.

The electronic configurations of Boron family using noble gas core notation are shown in the **table 5.1**

Table 5.1: Electronic configuration of Boron Family:

Element	Atomic number	Electronic configuration
Boron	5	[He] $2s^2 2p^1$
Aluminum	13	[Ne] $3s^2 3p^1$
Gallium	31	[Ar] $3d^{10} 4s^2 4p^1$
Indium	49	[Kr] $4d^{10} 5s^2 5p^1$
Thallium	81	[Xe] $4f^{14} 5d^{10} 6s^2 6p^1$

These elements have three electrons in their outermost shell. Therefore their valence shell electronic configuration is $ns^2 np^1$

5.2.2 Physical properties of boron Family

Boron family shows a wide variation in properties. Some characteristics of Boron Family include:

1. In nature, boron is metalloids while the rest in the group are metals.
2. At room temperature, the physical state of boron family is solid.
3. Boron has a high melting point while gallium has the lowest melting point.
4. As we move down the group, +1 oxidation state turns out to be steadier than +3 states.
5. Aluminum is amphoteric. Therefore, it can react with both acids and bases.
6. Thallium prefers +1 oxidation state to form stable compounds while others prefer +3 oxidation states.

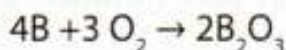
5.2.3 Chemical properties of boron Family

Some common reactions that boron family undergoes are:

1. Reaction with oxygen

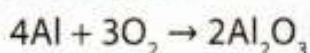
Pure Boron does not react with oxygen at all while impure boron reacts with oxygen to form Boron oxide (diboron trioxide).

For example

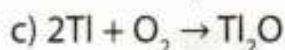
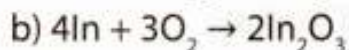
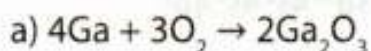


Aluminum reacts with oxygen at normal temperature to form a protective layer of its oxide on the surface. This layer makes it non-reactive.

For example



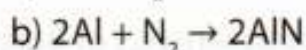
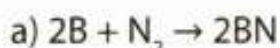
Similarly Gallium, Indium and Thallium react with oxygen to form Gallium oxide, Indium oxide and Thallium oxide respectively.



However, the Indium and Thallium are less affected by oxygen.

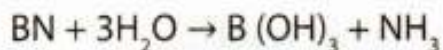
2. Reaction with Nitrogen

Boron and Aluminum react with nitrogen to form Boron nitride and Aluminum nitride



On hydrolysis of boron nitride produces boron hydroxide and ammonia

For example

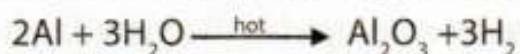


Gallium, Indium and Thallium do not react with nitrogen.

3. Reaction with water

Boron, Gallium, Indium and Thallium neither react with cold water nor with hot water.

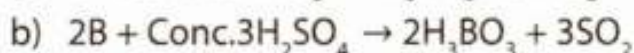
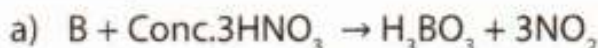
Only Aluminum reacts with hot water to form Aluminum oxide and hydrogen gas



4. Reaction with acids and bases

Boron doesn't react with normal acids and bases, but reacts with concentrated acids

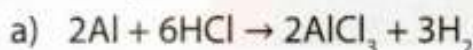
For examples



Here concentrated acids act as oxidizing agent.

Aluminum reacts with acid and base because it is amphoteric in nature.

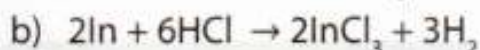
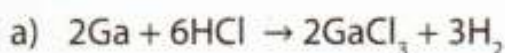
For examples



Sodium Aluminate

Gallium, Indium and Thallium are metals therefore, they react with acids and not bases to produce hydrogen gas.

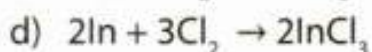
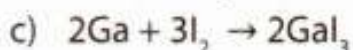
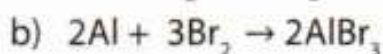
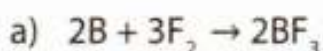
For examples



5. Reaction with Halogens

All boron family reacts with halogens to form tri-halides but Thallium forms mono-halides because Thallium prefers +1 oxidation state due to its stability.

For examples



5.2.4 Uses of boron family

1. Boron is used in manufacturing of Pyrex and making bullet proof vest.
2. Aluminum is used for making cooking utensils because it is light, cheap and conducts heat. It is also used as window frame.
3. Gallium is a semi-conductor material employed in the manufacture of compact-disc players and fibre optic communication devices.
4. Indium is also used in making semi-conductor devices such as transistors and electrical resistance thermometers.
5. Thallium is extremely toxic and has no commercial use.



EXERCISE

1. Define boron family.
2. Identify the position of boron family in the periodic table.
3. Write at least three physical properties of boron family?
4. State at least three common uses of boron family.

5.3 GROUP 14 CARBON FAMILY

Lesson objectives

By the end of the lesson, the learner should be able to:

- Identify Carbon Family
- Write electronic configuration of Carbon Family
- Explain physical and chemical properties of Carbon Family
- State uses of Carbon Family



ACTIVITY 4

We will use periodic table to identify Carbon Family and then, find out and name the total number of elements in Carbon Family?

Group 14 or IVA elements of the periodic table are called carbon family. Members of this group include:

1. Carbon
2. Silicon
3. Germanium
4. Tin (stannum)
5. Lead (plumbum)

Carbon Family

Carbon Family

1 IA H	2 IIA Li	3 IIIA Be	4 IVA B	5 VA C	6 VIA N	7 VIIA O	8 VIIIA F	9 VIIIA Ne									
11 IA Na	12 IIA Mg	13 IIIA Al	14 IVA Si	15 VA P	16 VIA S	17 VIIA Cl	18 VIIIA Ar										
19 IA K	20 IIA Ca	21 IIIB Sc	22 IVB Ti	23 VB V	24 VIB Cr	25 VIB Mn	26 VIB Fe	27 VIB Co	28 VIB Ni	29 VIB Cu	30 VIB Zn	31 IIIB Ga	32 IVB Ge	33 VB As	34 VIB Se	35 VIIA Br	36 VIIIA Kr
37 IA Rb	38 IIA Sr	39 IIIB Y	40 IVB Zr	41 VB Nb	42 VIB Mo	43 VIB Tc	44 VIB Ru	45 VIB Rh	46 VIB Pd	47 VIB Ag	48 VIB Cd	49 IIIB In	50 IVB Sn	51 VB Sb	52 VIB Te	53 VIIA I	54 VIIIA Xe
55 IA Cs	56 IIA Ba	57 IIIB La	58 IVB Hf	59 VB Ta	60 VIB W	61 VIB Re	62 VIB Os	63 VIB Ir	64 VIB Pt	65 VIB Au	66 VIB Hg	67 IIIB Tl	68 IVB Pb	69 VB Bi	70 VIB Po	71 VIIA At	72 VIIIA Rn
87 IA Fr	88 IIA Ra	89 IIIB Ac	90 IVB Th	91 VB Pa	92 VIB U	93 VIB Np	94 VIB Pu	95 VIB Am	96 VIB Cm	97 VIB Bk	98 VIB Cf	99 VIB Es	100 VIB Fm	101 VIB Md	102 VIB No	103 VIB Lr	

57 IIIB La	58 IVB Ce	59 VB Pr	60 VIB Nd	61 VIB Pm	62 VIB Sm	63 VIB Eu	64 VIB Gd	65 VIB Tb	66 VIB Dy	67 VIB Ho	68 VIB Er	69 VIB Tm	70 VIB Yb	71 VIB Lu
89 IIIB Ac	90 IVB Th	91 VB Pa	92 VIB U	93 VIB Np	94 VIB Pu	95 VIB Am	96 VIB Cm	97 VIB Bk	98 VIB Cf	99 VIB Es	100 VIB Fm	101 VIB Md	102 VIB No	103 VIB Lr

Carbon is the fourth most abundant element on earth. It is the second most common element in the earth's crust after oxygen. Germanium is a rare element. Tin (stannum) and lead (plumbum) are the two heaviest elements in group 14 (carbon family).

5.3.1 Electronic configuration of carbon family



ACTIVITY 5

We write the electronic configuration of carbon family, then we compare and contrast their electronic configuration.

The general valence shell electronic configuration of elements of group 14 is ns^2np^2 where n is the number outermost shell.

The electronic configurations of carbon family using noble gas core notation are shown in the table 5.2:

Table 5.2: Electronic configuration of carbon family

Element	Atomic number	Electronic configuration
Carbon (C)	6	(He) $2s^2 2p^2$
Silicon (Si)	14	(Ne) $3s^2 3p^2$
Germanium (Ge)	32	(Ar) $3d^{10} 4s^2 4p^2$
Tin (Sn)	50	(Kr) $4d^{10} 5s^2 5p^2$
Lead (Pb)	82	(Xe) $4f^{14} 5d^{10} 6s^2 6p^2$

As all the elements in group 14 have 4 electrons in the outermost shell, the valency of Group-14 elements is 4. They use these electrons in the bond formation in order to obtain octet configuration.

5.3.2 Physical Properties of carbon family

Some physical properties of carbon family are:

1. In nature carbon is non-metal, silicon and germanium are metalloids, tin and lead are metals.
2. At room temperature, the physical state of all members of carbon family is solids.
3. The melting and boiling points of group 14 elements are much higher than those of corresponding elements of group 13.

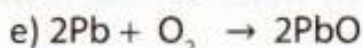
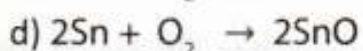
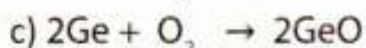
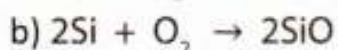
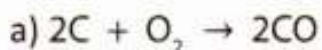
5.3.3 Chemical Properties of carbon family

Some common reactions that carbon family undergoes are:

1. Reaction with Oxygen

When carbon family reacts in limited supply of air they produce monoxides.

For examples:



Among monoxides, CO is neutral, GeO is acidic where SnO and PbO are amphoteric in nature. Note that SiO is insoluble in water so, we cannot determine whether it's acidic or basic.

But when carbon family reacts with excess or sufficient air they produce dioxides.

For examples:

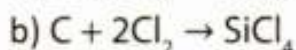


The oxides CO_2 and SiO_2 are acidic whereas GeO_2 , SnO_2 and PbO_2 are amphoteric (they show both basic and acidic properties)

2. Reaction with halogen

Carbon and silicon react with halogen to form tetrehalides only

For examples



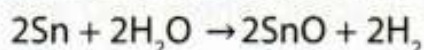
Germanium, tin and lead react with halogen to form both dihalides and tetrahalides

- a) $\text{Ge} + \text{Cl}_2 \rightarrow \text{GeCl}_2$
- b) $\text{Ge} + 2\text{Cl}_2 \rightarrow \text{GeCl}_4$
- c) $\text{Sn} + \text{Cl}_2 \rightarrow \text{SnCl}_2$
- d) $\text{Sn} + 2\text{Cl}_2 \rightarrow \text{SnCl}_4$

Note: Lead does not react with iodine.

3. Reaction with water

Carbon, Silicon, Germanium and Lead are not affected by water. Tin reacts with water to form oxide and hydrogen gas.



5.3.4 Uses of carbon family

1. Carbon is used in pencils and electrochemical cells in the form of graphite.
2. Since silicon and germanium are metalloids, they are used in electronics as semiconductors.
3. Tin is electroplated on steel cans to resist corrosion
4. Lead is used to protect your **torso** during dental x-rays.
5. Lead is also used in car batteries.



EXERCISE

1. Define carbon family
2. Where is the position of carbon family in the periodic table?
3. Write at least three physical properties of carbon family.
4. State at least three common uses of carbon family.
5. Choose two elements from carbon family and write notes about them.

5.4 GROUP 15 NITROGEN FAMILY

Lesson objectives

By the end of the lesson, learners should be able to:

- Identify nitrogen Family.
- Write electronic configuration of nitrogen family.
- Explain physical and chemical properties of nitrogen family.
- State the uses of nitrogen family.



ACTIVITY 6

We will use periodic table to identify nitrogen family and we find out the total number of Nitrogen Family then we name them.

Group 15 or VA elements of the periodic table are called **Nitrogen family**. This group is also known as **pnictogens** because all elements in this group are suffocating in nature. The term **pnictogens** is a greek work that means suffocation or choke. The members of this group include:

1. Nitrogen
2. Phosphorous
3. Arsenic
4. Antimony (stibium)
5. Bismuth

Nitrogen Family

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H	He													N	O	F	Ne
Li	Be												C	P	S	Cl	Ar
Na	Mg												Si	As	Se	Br	Kr
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	Sb	Te	I	Xe
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Bi	Po	At	Rn
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu			
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr			

5.4.1 Electronic configuration of Nitrogen family



ACTIVITY 7

We will write electronic configuration of nitrogen family, and then we compare and contrast their electronic configuration.

All Group 15 elements have the electron configuration;

ns^2np^3 where n is the principal quantum number.

The electronic configurations of nitrogen family using noble gas core notation are shown in the *table 5.3*

Table 5.3: Electronic configuration of nitrogen family

Element	Atomic number	Electronic configuration
Nitrogen (N)	7	(He) $2s^2 2p^3$
Phosphorus (P)	15	(Ne) $3s^2 3p^3$
Arsenic (As)	33	(Ar) $3d^{10} 4s^2 4p^3$
Antimony (Sb)	51	(Kr) $4d^{10} 5s^2 5p^3$
Bismuth (Bi)	83	(Xe) $4f^{14} 5d^{10} 6s^2 6p^3$

5.4.2 Physical Properties of Nitrogen family

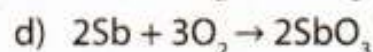
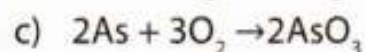
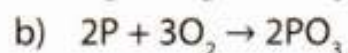
1. In nature nitrogen and phosphorus are non-metals, arsenic, bismuth, and antimony are metals.
2. At room temperature, the physical state of nitrogen is gas while others in the group are solids
3. They all show allotropes except nitrogen and bismuth
4. They all have isotopes for example nitrogen has two isotopes, nitrogen-14 and nitrogen-15
5. They all show a range of oxidation state of -3 to +5 except bismuth which shows +3 oxidation states.
6. Nitrogen exist as diatomic gas while others exist as tetra-atomic solids
7. Their boiling point increases down the group while their melting points increases up to arsenic and the decreases.

5.4.3 Chemical Properties of Nitrogen family

1. Reaction with oxygen

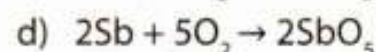
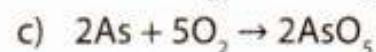
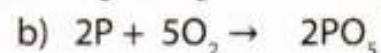
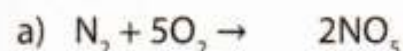
Nitrogen family reacts with oxygen to form their trioxides.

For examples



All Nitrogen family except bismuth also produces penta-oxides when they react with water.

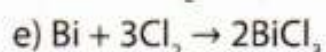
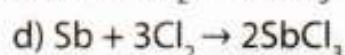
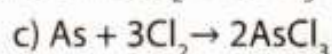
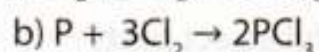
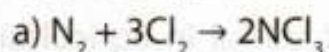
For examples:



2. Reaction with halogen

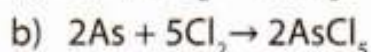
Nitrogen family react with halogen to form trihalides.

For examples



Phosphorus and Arsenic also form their pentahalides when they react with halogens.

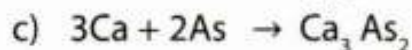
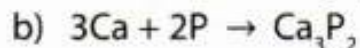
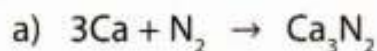
For examples



3. React with metals

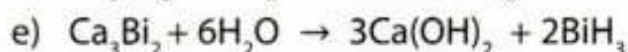
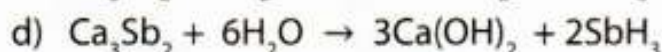
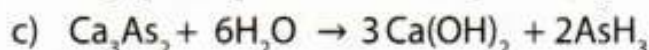
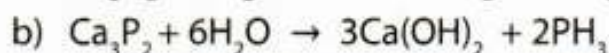
Nitrogen family reacts with metals to form nitride, phosphide, arsenide, etc.

For example



When these compounds are treated with water they produce their hydrides.

For example



5.4.4 Uses of Nitrogen family

- 1) Two of the elements, nitrogen and phosphorus, are used for fertilizers and essential for life. Phosphorus also is used in matches, fireworks, and to make phosphoric acid.
- 2) Arsenic, which is notorious for its toxicity, is most useful in agriculture, where its compounds are an aid in controlling harmful insect pests
- 3) Antimony and bismuth are used chiefly in metal alloys,.
- 4) Bismuth is also used in medications, paint, and as a catalyst.



EXERCISE

1. Define nitrogen family
2. Where is the position of nitrogen family in the periodic table?
3. Write three physical properties of nitrogen family?
4. State the common uses of nitrogen family.
5. Choose two elements from the group and write notes about them.

5.5. GROUP 16 OXYGEN FAMILY

Lesson objectives

By the end of the lesson, learners should be able to:

- Identify oxygen family
- Write electronic configuration of oxygen family
- Explain the physical and chemical properties of oxygen family
- State the uses of oxygen family

**ACTIVITY 8**

We will use periodic table to identify oxygen family and then, find out the total number of oxygen family and name them.

Group 16 or VIA elements of the periodic table are called **Oxygen family**. This group is also known as **Chalcogen** (Ore forming) because most of the Ores in the earth's crust are either oxides or sulfides. Ores are minerals from which metals are extracted. The members of this group include:

1. Oxygen
2. Sulphur
3. Selenium
4. Tellurium
5. Polonium

Oxygen Family

1	2											10	11	12	13	14	15	16	17	18
H	He											B	C	N	O	F	Ne			
Li	Be											Al	Si	P	S	Cl	Ar			
Na	Mg	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr			
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr			
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe			
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn			
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og			
La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu																				
Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr																				

Oxygen is the most abundant element by mass in both the Earth's crust and the human body. It is second most abundant element in the atmosphere.

Sulfur is the sixteenth most abundant element in Earth's crust. It exists naturally in a variety of forms, including elemental sulfur, sulfides, sulfates, and organo-sulfur compounds. Selenium is rare to find in its elemental form in nature; It is often found in soils and in plant tissues that have bio-accumulated the element.

Tellurium is an extremely rare element, and is commonly found as a telluride of gold. Polonium is a very rare, radioactive metal.

5.5.1 Electronic configuration of oxygen family



ACTIVITY 9

We write electronic configuration of oxygen family, then we compare and contrast their electronic configuration.

The electronic configurations of oxygen family using noble gas core notation are shown in the **table 5.4**

Table 5.4: Electronic configuration of oxygen family:

Element	Atomic number	Electronic configuration
Oxygen (O)	8	(He) $2s^2 2p^4$
Suiphur (S)	16	(Ne) $3s^2 3p^4$
Selenium (Se)	34	(Ar) $3d^{10} 4s^2 4p^4$
Tellurium (Te)	52	(Kr) $4d^{10} 5s^2 5p^4$
Polonium (Po)	84	(Xe) $4f^{14} 5d^{10} 6s^2 6p^4$

Group 6 elements have 6 valence electrons, two short of the desired 8. Group 6 elements gain 2 electrons to reach 8 valence electrons.

5.5.2 Physical Properties of oxygen family

1. In nature oxygen, sulfur, and selenium are non metals, tellurian and Po ions are metalloid.
2. At room temperature oxygen is gas while all other members of the group are solids.
3. Group 16 elements show negative oxidation state of -2 and positive oxidation state of +2, +3, +4 and +6.

Note: All oxygen family except oxygen show positive oxidation state. Oxygen exhibit, +2 oxidation state only in OF_2 .

5.5.3 Chemical Properties of oxygen family

Oxygen is the reactive of these elements. However the oxygen family undergoes:

1. Reaction with hydrogen

Oxygen family reacts with hydrogen to form hydrides.

For examples

- a) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- b) $\text{H}_2 + \text{S} \rightarrow \text{H}_2\text{S}$
- c) $\text{H}_2 + \text{Se} \rightarrow \text{H}_2\text{Se}$
- d) $\text{H}_2 + \text{Te} \rightarrow \text{H}_2\text{Te}$
- e) $\text{H}_2 + \text{Po} \rightarrow \text{H}_2\text{Po}$

2. React with Oxygen

Sulfur reacts with oxygen to form its dioxides and trioxides

For examples

- a) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
- b) $2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_3$

Similarly Selenium and tellurian reacts with oxygen to form their dioxides

For examples

- a) $\text{Se} + \text{O}_2 \rightarrow \text{SeO}_2$
- b) $\text{Te} + \text{O}_2 \rightarrow \text{TeO}_2$

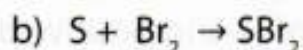
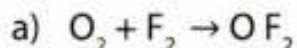


3. Reaction with Halogens

Oxygen family reacts with halogens to form halides.

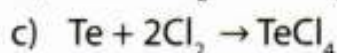
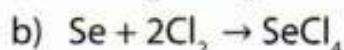
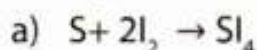
Note: Only oxygen and sulfur form di-halides.

For examples



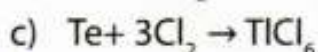
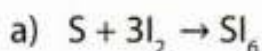
Sulfur, selenium and tellurium form tetra-halides

For examples



However, when Sulfur, selenium and tellurium react with fluorine they form hexa-halides:

For example



5.5.4 Uses of Oxygen group

1. Oxygen is essential for respiration in animals and humans. During respiration Carbon dioxide is produced and taken in by plants releasing Oxygen to atmosphere.
2. Ozone (O_3) protects the atmosphere from the harmful UV,
3. Sulphur is widely used in industries to produce Sulphuric acid,
4. Selenium is mainly used in electronic equipment, in pigments and in production of Glass,
5. Tellurium has no much use, Polonium has some scientific application as source of Alpha.



EXERCISE

1. Define oxygen family
2. Where is the position of oxygen family in the periodic table?
3. Write at least three physical properties of oxygen family?
4. State at least three uses of oxygen family.
5. Choose two elements from the group and write notes about them.

5.6 GROUP 17: HALOGENS

Lesson objectives

By the end of the lesson, learners should be able to:

- Identify Halogens
- Write electronic configuration of Halogens
- Explain physical and chemical properties of Halogens
- State the uses of Halogens



ACTIVITY 10

We will use periodic table to identify halogens and then we find out the total number of halogens then name them

Group 7 elements are also called **halogens** (salt makers) because they make salt with metals. Members of the halogens are:

1. Fluorine
2. Chlorine
3. Bromine
4. Iodine
5. Astatine

Halogen

The image shows a standard periodic table of elements. A black arrow points to the element Fluorine (F), which is located in the top right corner of the main body of the table. The table is color-coded by groups: Group 1 (red), Group 2 (orange), Groups 3-10 (blue), Groups 11-18 (purple), and Groups 19-20 (pink). The Lanthanide and Actinide series are shown at the bottom in green and blue respectively.

The last element of the group is astatine which is radioactive.

5.6.1 Electronic configuration of Halogens



ACTIVITY 11

We write electronic configuration of halogens, then we compare and contrast their electronic configuration.

The electronic configurations of halogens using noble gas core notation are shown in the **table 5.5**:

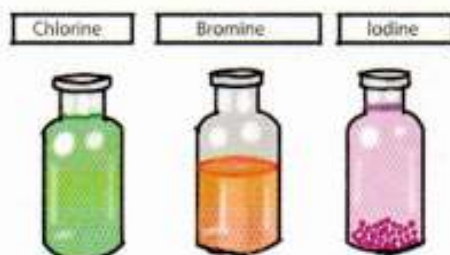
Table 5.5: Electronic configuration of alkali metals:

Element	Atomic number	Electronic configuration
Fluorine (F)	9	(He) $2s^2 2p^5$
Chlorine (Cl)	17	(Ne) $3s^2 3p^5$
Bromine (Br)	35	(Ar) $3d^{10} 4s^2 4p^5$
Iodine (I)	53	(Kr) $4d^{10} 5s^2 5p^5$
Astatine (At)	85	(Xe) $4f^{14} 5d^{10} 6s^2 6p^5$

All Halogens have 7 valence electrons, meaning they only require 1 more electron to reach the desired 8(octet) in the valence. Since they only require 1 more electron, the halogens are quite reactive.

5.6.2 Physical properties of Halogens

1. Group 17 elements are poor conductor of electricity.
2. They are poor conductor of heat because they are made up of covalent bonds.
3. Halogens have low melting and boiling points because their molecules are attached to each other by London dispersion force.
4. The physical state of halogens changes gradually from gas (chlorine) to liquid (Bromine) and finally to solid (iodine).
5. The halogens are di-atomic and exist as F_2 , Cl_2 , Br_2 , and I_2 .
6. All group 17 elements are non metals therefore they are insulators of heat and electricity.
7. The density of group 17 elements increases down the group due to increase in their atomic masses.
8. They are coloured; chlorine (Cl_2) is yellow-green gas, bromine (Br_2) is reddish-brown liquid and Iodine (I_2) is purple solid.



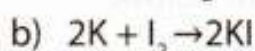
5.6.3 Chemical Properties of Halogens

Halogens show different chemical properties.

1. Reaction with metals

Group 17 elements react with the group 1 metals to form salts

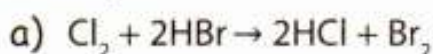
For example



2. Displacement reaction

The more reactive halogens displaces the ions of less reactive halogen from solution. The reactivity of halogen decreases down the group $\text{F} > \text{Cl} > \text{Br} > \text{I}$. Therefore, Fluorine is the most reactive halogen thus, can displace all the other halide ions (Cl, Br, and I) from solution. None of the other halogens can displace Fluorine ions (F) from solution. Chlorine displaces bromine and iodine ions from solution but not fluorine ions.

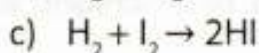
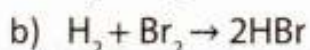
For examples



3. Reactive with hydrogen

Halogens form hydrides with hydrogen.

For examples



4. Reaction with water

We will study in the following three experiments

Experiment 1:

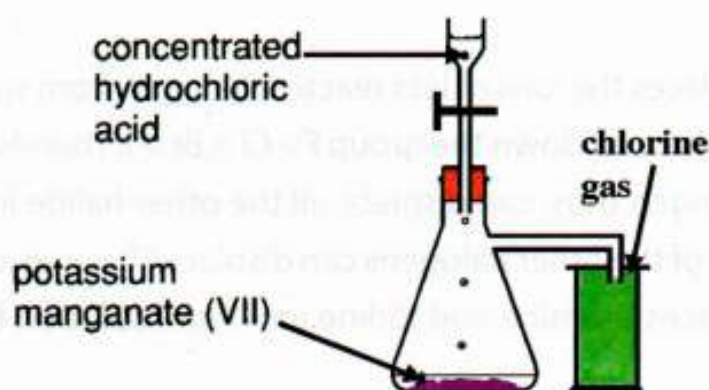
Reaction of chlorine with water

Materials

- Distilled water
- hydrochloric acid (HCl)
- potassium permanganate (VII)
- blue litmus paper
- flask
- Gas collecting jar

Procedure

1. Prepare chlorine gas by reacting concentrated hydrochloric acid (HCl) with potassium permanganate (VII) as shown in the experiment below.



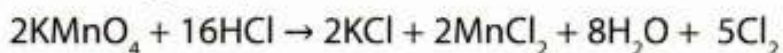
2. Add 5cm³ of distilled water to a test tube.
3. Pass chlorine gas into water in the test tube.
4. Dip a piece of blue litmus paper into the solution formed.

Observation

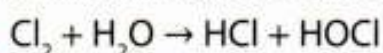
1. What did you observe?
2. What happened to the litmus paper, when it comes in contact with the solution formed?

Conclusion

When potassium permanganate reacted with concentrated hydrochloric acid chlorine gas is produced



Then the chlorine gas dissolves in water to form a pale yellow solution.



The solution formed is acidic and contains bleaching agent. Therefore it turns the blue litmus paper to red before it's bleached.

Experiment 2:

Reaction of bromine with water

Materials

- distilled water
- test tube
- blue litmus paper
- Bromine liquid
- Dropper

Procedure

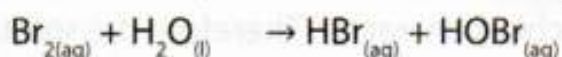
1. Add 5cm³ of distilled water to test tube
2. Add two drops of liquid bromine into water in the test tube.
3. Dip a piece of blue litmus paper into the solution formed.

Observation

1. What did you observe?
2. What happened to the litmus paper, when it comes in contact with the solution formed?

Conclusion

Bromine liquid dissolves slowly in water to form a yellowish-brown solution which contains Hydrobromic acid and Hypobromous acid.



The hypobromous acid is a weak bleaching agent. Therefore the solution turns the blue litmus paper to red before it's bleached slowly.

Experiment 3:

Reaction of iodine with water

Materials

- Test tube
- Dropper
- Solid iodine
- Distilled water
- Blue litmus paper

Procedure

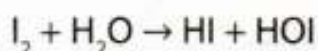
1. Add 5cm³ of distilled water to test tube
2. Add a small piece of solid iodine into distilled water in the test tube.
3. Close the test tube with a cotton wool.
4. Dip a piece of blue litmus paper into the solution formed.

Observation

1. What did you observe?
2. What happened to the litmus paper, when it comes in contact with the solution formed?

Conclusion

Only little iodine dissolves in water to form yellowish solution of hydroiodic acid and hypoiodous acid.



Hypoiodous acid has a very weak bleaching character. Therefore the solution turns the blue litmus paper to red but does not have a bleaching effect.

5.6.4 Uses of halogen

1. Chlorine is used as a bleaching agent in paper industry and textile.
2. It is also used in the manufacture of plastics and insecticides.
3. Halogens like chlorine are used in water purification, sanitation, pharmaceutical, plastics, papers, bleaching, pesticides, antiseptic and oil industries.
4. Sodium chloride and Sodium iodide are essential dietary minerals for human being.

**EXERCISE**

1. Define halogens
2. Where is the position of halogens in the periodic table?
3. Write at least three physical properties of halogens.
4. State three common uses of halogens.
5. Choose three elements from the group and write notes about them.

5.7 GROUP 18: NOBLE GASES**Lesson objectives**

By the end of the lesson, learners should be able to:

- Identify noble gases
- Write electronic configuration of noble gases
- Explain the physical and chemical properties of noble gases
- State the uses of noble gases

**ACTIVITY 12**

We will use periodic table to identify noble gases and then, we find out the total number of noble gases then name them.

Group 18 or VIIIA elements of the periodic table are called **noble gasses**. The six noble gases that occur naturally are:

1. helium (He)
2. Neon (Ne)
3. Argon (Ar)
4. Krypton (Kr)
5. Xenon (Xe)
6. Radon (Rn)

Noble gases

1 IA H 1.008																	10 VIIIA Ne 20.18	11 VIIIA Ar 39.95	12 VIIIA Kr 83.80	13 VIIIA Xe 131.3	14 VIIIA Rn 222																											
3 IIIA Li 6.941	4 IIA Be 9.012																	5 VA B 10.81	6 IVA C 12.01	7 VA N 14.01	8 VIA O 16.00	9 VIIA F 18.998	10 VIIIA Ne 20.18																									
11 IA Na 22.99	12 IIA Mg 24.31																	13 IIIA Al 26.98	14 IVA Si 28.09	15 VA P 30.97	16 VIA S 32.06	17 VIIA Cl 35.45	18 VIIIA Ar 39.95																									
19 IA K 39.10	20 IIA Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc 98.91	44 Ru 101.07	45 Rh 102.91	46 Pd 106.36	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.91	54 Xe 131.3													
55 IA Rb 85.47	56 IIA Sr 87.62	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm 144.91	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po 209	85 At 210	86 Rn 222	87 Fr 223	88 Ra 226	89 Ac 227	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu 239.05	95 Am 243.06	96 Cm 247.07	97 Bk 247.07	98 Cf 251.08	99 Es 252.08	100 Fm 257.10	101 Md 258.10	102 No 259.10	103 Lr 262.10

Note: Helium (He) belongs to this group but it does not belong to p-block elements. It belongs to this group because the valence shell is full but, the last valency electron is in s-orbital therefore belongs to s-block group.

5.7.1 Electronic configuration of Noble gases



ACTIVITY 13

We write electronic configuration of noble gases, then we compare and contrast their electronic configuration.

The general configuration of the noble gas family is given as ns^2np^6 (except helium which has $1s^2$). the electronic configurations of noble gases are shown in the table 5.6

Table 5.6: Electronic configuration of noble gases:

Element	Atomic number	Electronic configuration
Helium (He)	2	$1s^2$
Neon (Ne)	10	$1s^2 2s^2 2p^6$
Argon (Ar)	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Krypton (Kr)	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
Xenon (Xe)	54	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6$
Radon (Rn)	86	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 6s^2 6p^6$

These elements already have a full valence shell with 8 valence electrons.

This leaves the elements non-reactive as they are already stable and at low energy state.

5.7.2 Physical properties of noble gases

1. Noble gases exist as gaseous state in the atmosphere.
2. Noble gases are non-metals
3. They are monatomic in nature.
4. Noble gases are slightly soluble in water. Their solubility in water increases down the group
5. They have low melting and boiling points. Their melting and boiling points increase down the group.
6. They are easily liquefied due to their low melting point.

5.7.3 Chemical properties of noble gases

Noble gases are chemically stable because they have 8 valence electrons except helium. Due to this, noble gases are also called inert **gases or rare gases**. However in 1962 Neil Baitlett first observed that xenon forms compound known as xenon-hexafluoroplatinate (XePtF_6) This compound was the first true compound of noble gases. Later many other compounds of xenon and krypton with highly electronegativity elements such as fluorine and oxygen were formed as well.



For examples

- a) XeF_2 xenon difluoride
- b) XeO_3 xenon trioxide
- c) KrF_2 krypton difluoride

Note: krypton forms only KrF_2 krypton difluoride and helium, neon and argon do not form true compounds because of their high ionization energy.

5.7.4 Uses of noble gases

- 1. Helium is used to fill aero-plane tyres and balloon for Meteorological observations because it is much less denser than air.
- 2. Neon emits light when high voltage electricity is passed through it because of this; it is used in glowing *neon* advertising signs and fluorine lights.
- 3. Argon is used in filament bulbs because the metal filament will not burn in Argon.
- 4. Krypton and Xenon are used in fluorescent bulbs flash bulbs and laser beams.
- 5. Redon is used in radiotherapy for treatment of cancer.

**EXERCISE**

- 1. Define noble gases
- 2. What is the position of noble gases in the periodic table?
- 3. Write at least three the physical properties of noble gases?
- 4. State three common uses of noble gases.
- 5. Choose two elements from the group and write notes about them.

Chapter Review Question

- 1. Define p-block elements?
- 2. What are the types of elements in p-block elements?
- 3. What is the valence shell electronic configuration of
 - a) Boron family b) Carbon family
 - c) nitrogen family d) oxygen family
 - e) Halogens f) Noble gases
- 4. What elements of boron family do not react with nitrogen?
- 5. List the element of
 - a) Carbon family b) oxygen family
 - c) Halogens d) Noble gases
- 6. Why are noble gases chemically stable?
- 7. What happens to melting points, boiling points and density of noble gases

Choose and circle the correct answer

1. Group 13 on the periodic table is the
 - a) Boron Group
 - b) Noble Gases
 - c) Transition Metals
 - d) Inner transition metals
2. Group 13 on the periodic table has how many valence electrons?
 - a) 3
 - b) 8
 - c) 18
 - d) 13
3. What is the valance electron configuration of group 13
 - a) ns^2np^2
 - b) ns^2np^1
 - c) ns^2np^4
 - d) ns^2np^5
4. Which one is a group 13 element?
 - a) Sn
 - b) In
 - c) Cd
 - d) Ge
5. Which element is metalloid in group 13 element
 - a) B
 - b) Gn
 - c) Al
 - d) Ti
6. Group 18 on the periodic table are also called____.
 - a) alkali metals
 - b) alkaline earth metals
 - c) transition metals
 - d) noble gases



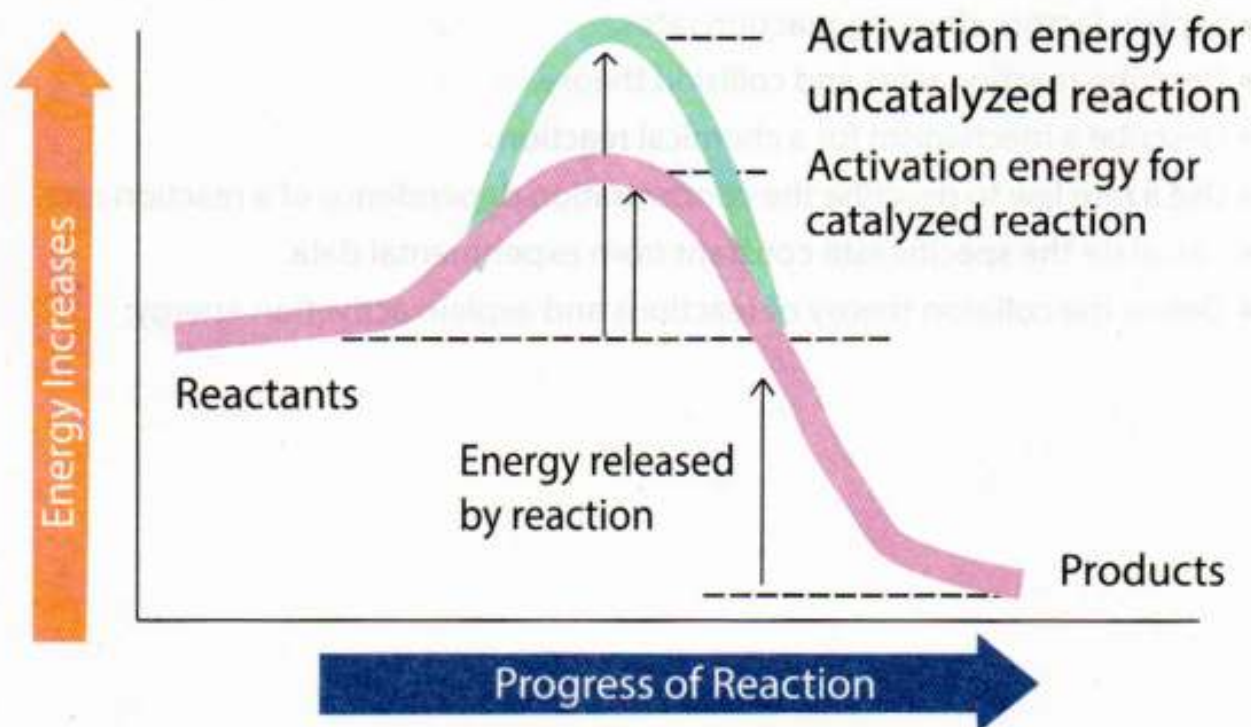
7. Name the group that contains inert (nonreactive) elements.
- noble gases
 - halogens
 - alkali metals
 - alkaline earth metals
8. Group 17 on the periodic table are called ____.
- transition metals
 - metalloids
 - halogens
 - alkali metals
9. Which group has the most reactive nonmetals?
- alkali metals
 - noble gases
 - metalloids
 - halogens
10. Some groups of the periodic table are more likely to contain elements that are gases than other groups. Which of the following groups contains the greatest number of gaseous elements?
- IIA
 - IVA
 - VIA
 - VIIIA
11. Which of the following is not true for the halogens?
- They are nonmetals.
 - They show the -1 oxidation number in most of their compounds.
 - The electronic configuration of their outermost electrons is ns^2np^6 .
 - Their compounds with metals are generally ionic in nature.
 - Elemental halogens exist as diatomic molecules.

Research and presentation

Pick one element from the P-block elements and research about it, then present your findings to the class using poster or power point presentation.

CHAPTER 6

CHEMICAL KINETICS



CHAPTER CONTENTS

- 6.1 Introduction to Chemical Kinetics
- 6.2 Rates of chemical reactions
- 6.3 Reaction rate and Stoichiometry
- 6.4 Factors affecting reaction rates.
 - 6.4.1 Nature of reactants
 - 6.4.2 Temperature
 - 6.4.3 Concentration
 - 6.4.4 Surface Area
 - 6.4.5 Catalyst
- 6.5 Reaction rate law and order of reactions
- 6.6 Collision theory.
- 6.7 Mechanism of reaction.
- Chapter review questions

CHAPTER OBJECTIVES

By the end of the chapter the learner should be able to:

- Define reaction rates.
- Relate the concentrations of reactants, products and reaction rate.
- Explain factors affecting reaction rates.
- Describe reaction rates and collision theory.
- Describe a mechanism for a chemical reaction.
- Use a rate law to describe the concentration dependence of a reaction rate.
- Calculate the specific rate constant from experimental data.
- Define the collision theory of reactions and explain activation energy.

6.1 INTRODUCTION

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define chemical kinetics
- State the importance of chemical kinetics

**ACTIVITY 1**

In groups we discuss the following different types of reactions and how quickly or slowly they occur.

- Explosion
- Corrosion (Rusting of iron).
- Photosynthesis
- Combustion

Chemical reactions convert substances into new a substance which has new properties. However, it is very important to know how quickly chemical reactions occur.

The branch of chemistry that deals with the study of the rate of the reactions, the factors affecting these rates and the mechanisms by which the reaction occurs is called **chemical kinetics**.

Chemical kinetics has broad importance because it explains the rate of chemical reactions such as: how quickly a medicine is able to work, how rapidly food spoils etc.

**EXERCISE**

1. Define chemical kinetics
2. Why do we study chemical kinetics?
3. If you put your colored shirt on the sun for a month, would its color change?

Explain Why?

What happens if it stays there for 3, 6, and 12 months? Give a reason to your answer.

6.2 RATES OF CHEMICAL REACTIONS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Describe the rates of chemical reaction
- Calculate the rate of chemical reactions

Reaction rates vary greatly for everyday processes. For example, banana ripens in few days while silver tarnishes in several months.



Figure 6.1; A banana ripens in a few days



Figure 6.2; Silver tarnishes in several months

Experiment 1:

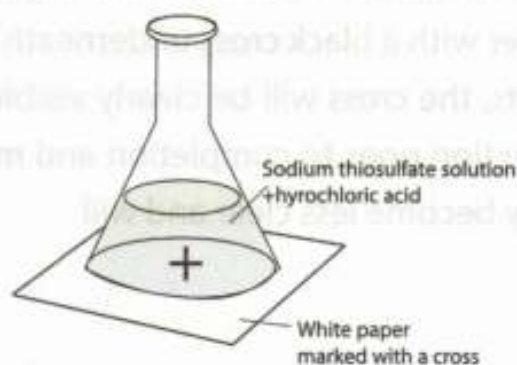
To estimate the average rate of reaction

Materials

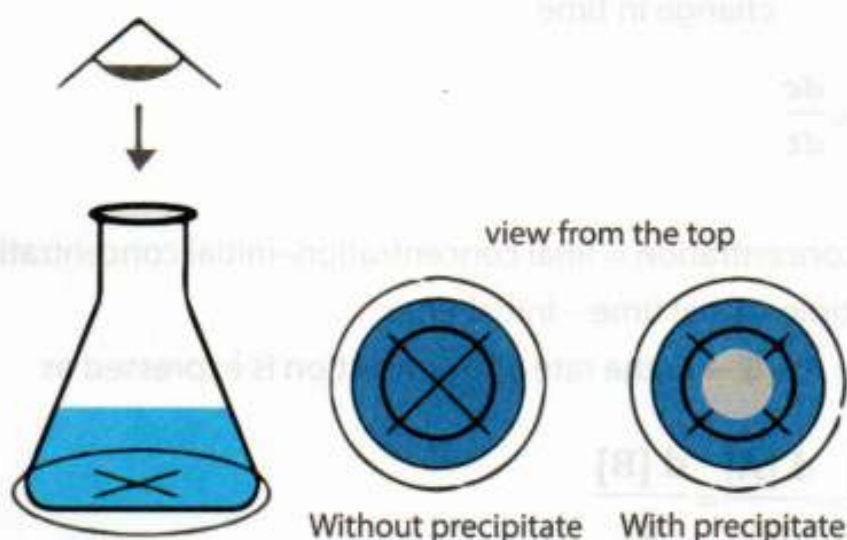
- Conical flask
- Sodium thiosulfate solution ($\text{Na}_2\text{S}_2\text{O}_3$)
- Hydrochloric acid (HCl).
- Pencil
- Filter paper
- Stop watch
- Data sheet

Procedure

1. Using a pencil, draw a cross in the middle of the filter paper.
2. Place 20 mL of the 0.1 M sodium thiosulfate solution into a conical flask.
3. Add 20 mL of 1 M hydrochloric acid to the flask, while starting the stop watch at the same time.
4. Swirl the flask and place it on a paper marked with a cross.



5. Record the time taken for the cross to disappear.
6. Repeat the experiment for a second time using different volumes of 10, 12, 14, 16 and 18 mL of sodium thiosulfate solution respectively. In each step, add water to increase the volume up to 20 mL and mix before adding HCl.
7. Record the results in data sheet.



Observation

1. What did you observe?
2. How long did the cross take to disappear?
3. Explain: how measuring the time that it takes for this to happen will give an idea of the reaction rate.

Conclusion

One way to estimate the average rate of reaction is to carry out the investigation in a conical flask and to place a piece of paper with a black cross underneath the bottom of the flask. When the reaction starts, the cross will be clearly visible as you look into the flask. However, as the reaction goes to completion and more precipitate is formed, the cross will gradually become less clear and will eventually disappear entirely.

In chemical reactions, reactants and products are involved. As the chemical reaction proceeds, the concentration of the reactants decreases while the concentration of products increases. The speed of the reaction at which the reactants disappear or products appear is called **reaction rate**. The reaction rate is expressed as change in concentration of reactants or products per unit time.

$$\text{Rate of reaction} = \frac{\text{change in concentration of reactant or product}}{\text{change in time}}$$

$$\text{rate} = \frac{dc}{dt}$$

Where:

dc is change in concentration = final concentration – initial concentration

dt is change of time = final time – initial time

For the general equation $A + B \rightarrow C$, the rate of the reaction is expressed as

$$\text{rate} = -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{d[C]}{dt}$$

The concentration of the reactant is always expressed using negative sign (-ve) because the concentration of reactant always decreases while the concentration of product is always expressed using positive sign (+ve) because the concentration of product always increases.

The unit of the rate of the reaction is **mol L⁻¹s⁻¹** or **atm time⁻¹**

The rate or speed of reaction is determined by measuring the amount of a reactant used up or the amount of a product formed in a certain period of time.

Example 1

The hypothetical reaction of $A + B \rightarrow C$.

The initial concentration of A is 0.054 M. After 1.5 minutes $[A] = 0.032$ M. Calculate the rate of the reaction in M/sec.

Solution

Initial concentration (C_1) = 0.032 M

Final concentration (C_2) = 0.054M

Time (t) = 1.5 x 60 seconds = 90sec

$$\begin{aligned}\text{rate} &= -\frac{d[C]}{dt} \\ \text{rate} &= -\frac{C_2 - C_1}{T_2 - T_1} \\ &= -\frac{(0.032 - 0.054)}{90} \\ &= 0.00024 \text{ m/s}\end{aligned}$$

Example 2

The following reaction takes place



After two minutes, 4 g of lithium has been used. Calculate the rate of the reaction.

Solution

Step 1: Calculate the number of moles of lithium that are used in the reaction

$$\begin{aligned}\text{Number of moles (n)} &= \frac{\text{Mass (m)}}{\text{Molar mass (M)}} \\ &= \frac{4\text{g}}{6.94\text{g.mol}^{-1}} \\ &= 0.576\text{mol}\end{aligned}$$

Step 2: Calculate the time (in seconds) for the reaction

$$t = 2 \text{ minutes} = 2 \times 60\text{s} = 120 \text{ seconds}$$

Step 3: Calculate the rate of the reaction

$$\text{Rate of reaction of Li used} = \frac{\text{Moles of Lithium used}}{\text{Time}}$$

$$= \frac{0.576\text{mol}}{120 \text{ s}}$$

$$= 0.00483 \text{ mol.s}^{-1}$$



EXERCISE

1. Explain what the reaction rate indicates about a particular chemical reaction.
2. The table below shows the number of moles of reactant used at a particular time for each reaction. Complete the table by calculating the average rate of each reaction.

Reaction	Reactants used (mol)	Time (s)	Reaction rate (mole.s ⁻¹)
1	2	30	
2	5	120	
3	1	90	
4	4	90	
5	6	30	

3. Iron reacts with oxygen according to the balanced reaction:



4g of iron and 1.04g of oxygen are used during the reaction. 5.2g of FeO is produced the reaction takes 60 minutes.

Calculate the average rate of reaction of:

- a) The use of Fe
- b) The use of O₂
- c) The formation of FeO

6.3 REACTION RATE STOICHIOMETRY

Lesson objectives

By the end of the lesson, the learner should be able to:

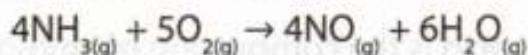
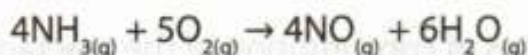
- Relate the concentration of reactants, products and reaction rate

Since the rate of a reaction can be expressed as the change in concentration of any reactant or product, the two rate expressions are all related to one another according to the reaction stoichiometry following this step:

1. Write the concentrations of - reactants in terms of $-\frac{d[R]}{dt}$ and products in terms of $+\frac{d[P]}{dt}$
2. Equalize the rate expressions of reactants and products.
3. The stoichiometric coefficient of the equation is always expressed in denominator.

Example

Write the rate expression for the following reaction in terms of the disappearance of the reactants and the appearance of the products:

**Solution**

$$\text{Rate} = -\frac{1}{4} \frac{d[\text{NH}_3]}{dt} = -\frac{1}{5} \frac{d[\text{O}_2]}{dt} = \frac{1}{4} \frac{d[\text{NO}]}{dt} = \frac{1}{6} \frac{d[\text{H}_2\text{O}]}{dt}$$

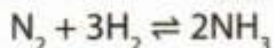
The individual rate expression in terms of various species of chemical reaction is called relative rates.

We can calculate the rate of reactants in relation to products or products in relation to reactants using the following formula:

$$\text{Rate required} = \frac{\text{Coefficient of the required}}{\text{coefficient of given}} \times \text{rate given}$$

Example 1

For the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ the rate of appearance of NH_3 is $4.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$. Calculate the rate of disappearance of H_2 and N_2 .

Solution

Rate of appearance of $\text{NH}_3 = 4.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$.

$$\text{Rate of disappearance of } \text{H}_2 = \frac{\text{Coefficient of the required}}{\text{coefficient of given}} \times \text{rate given}$$

$$= 3 \times 4.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$$

$$= 12.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$$

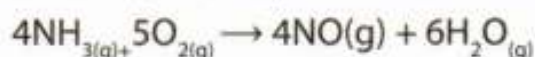
$$\text{Rate disappearance of } \text{N}_2 = \frac{\text{Coefficient of the required}}{\text{coefficient of given}} \times \text{rate given}$$

$$= 4.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$$

$$= 4.0 \times 10^{-3} \text{ mol. L}^{-1} \text{ sec}^{-1}$$

**EXERCISE**

1. The first step in the production of nitric acid is the combustion of ammonia:



Write the equations that relate the rates of consumption of the reactants and the rates of formation of the products.

2. The rate of formation of Br_2 is $6.0 \times 10^{-6} \text{ mol/L/s}$ in a reaction described by the following net ionic equation:



Write the equations that relate the rates of consumption of the reactants and the rates of formation of the products.

3. Consider the following reaction: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. At a certain point in the reaction, hydrogen gas is being used at a rate of 0.096 M/sec. What is the rate of formation of ammonia at this point?
- 4) The decomposition of N_2O_5 is described by the following reaction:
 $2\text{N}_2\text{O}_{5(g)} \rightarrow 4\text{NO}_{2(g)} + \text{O}_{2(g)}$. If O_2 is being formed at a rate of 2.40 mol/min, what is the rate at which N_2O_5 is being used up

6.4 FACTORS AFFECTING REACTION RATES.

Lesson objectives

By the end of the lesson, the learner should be able to:

- Name factors which affect reaction rates.
- Explain how different factors affect reaction rates.

There are several factors that affect the rate of a reaction. It is very important to study these factors so that reaction rates can be controlled. This is especially important when it comes to industrial reactions, where more productivity results to greater profits for companies. The following are some of the factors which affect the average rate of a reaction.

1. Nature of reactants
2. Temperature
3. Concentration of reactants
4. Surface area
5. The presence of a catalyst.

6.4.1 Nature of Reactants

Experiment 2

Determine the effect of the nature of reactants on the average rate of a reaction.

Materials:

- Oxalic acid ($\text{C}_2\text{O}_4\text{H}_2$)
- Iron (II) sulphate (FeSO_4)
- Potassium permanganate (KMnO_4).
- Sulphuric acid (H_2SO_4).
- Spatula
- 2 test tubes
- Medicine dropper
- Glass beaker
- Glass rod

Procedure

1. Use gloves and safety glasses.
2. Prepare iron (II) sulphate solution in a test tube by dissolving two spatula tips of FeSO_4 in 10cm^3 of water and label it 1.
3. Label the other test tube 2 and prepare a solution of oxalic acid ($\text{C}_2\text{O}_4\text{H}_2$) in the same way.
4. Prepare another solution of H_2SO_4 by adding 2cm^3 of concentrated H_2SO_4 to 10cm^3 of water. (always remember to add the acid to the water and not the other way round).
5. Add 2cm^3 of the H_2SO_4 to the FeSO_4 and $\text{C}_2\text{O}_4\text{H}_2$ solution.
6. Add a few drops of KMnO_4 to the two test tubes by using the medicine dropper.

Observation

1. What did you observe?
2. How quickly does KMnO_4 solution discolor in each solution?
3. Why does KMnO_4 discolor in the $\text{C}_2\text{O}_4\text{H}_2$ solution much more than in the FeSO_4 solution?

Conclusion

In a reaction between a solid and an aqueous/liquid/gas species, increase in the surface area of the solid-phase reactant increases the number of collisions per second and therefore increases the reaction rate. By increasing surface area, there are more collisions per unit of time.

Experiment 3:

The nature of reactants explains the different rates of different reactions but it cannot be used to make one reaction go fast or slow.

Materials:

- Two Test tubes
- Water
- Zinc (Zn)
- Sodium (Na)
- Glass rod

Procedure

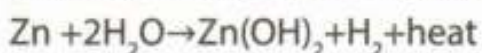
1. Put pieces of Zinc and Sodium into two test tubes.
2. Add few drops of water to each test tube

Observation

1. What did you observe?
2. Compare the reaction of Zinc and Sodium to water.

Conclusion

When Zinc mixes with water, it reacts to produce hydrogen gas, zinc hydroxide and heat. If you watch carefully you may be able to see bubbles of hydrogen gas forming and if you give it time you may be able to feel increase of temperature.



When sodium is mixed with water a similar reaction takes place producing hydrogen gas, Sodium hydroxide and heat.



In both cases, a single displacement reaction occurs between a metal and water. Both reactions produce hydrogen gas and heat, but the result appears very different due to the rate at which the hydrogen and heat are produced.

However, the reaction of sodium and water happens fast enough that you can see fizzing on the surface of the sodium and can hear the hydrogen bubbles being released and much more heat is released.

6.4.2 Temperature Effects

The rate of chemical reaction affects increase in temperature because increasing the temperature of the reactants increases the average kinetic energy of its constituent species. As the average kinetic energy increases, the particles move faster and collide more rapidly per unit and the reaction rate increases.

6.4.3 Concentration Effects

Experiment:

Effect of concentration on reaction rate

Materials

- Solid zinc granules
- Hydrochloric acid
- Two conical flasks
- Two beakers
- Two balloons
- Bunsen burner
- Splint of wood

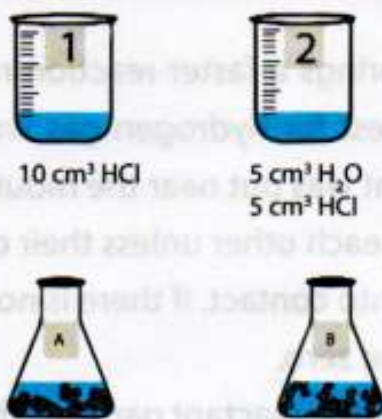
Procedure

Warning: Don't touch hydrochloric acid (HCl) with your bare hands, always use gloves. Also protect your eyes by wearing safety glasses.

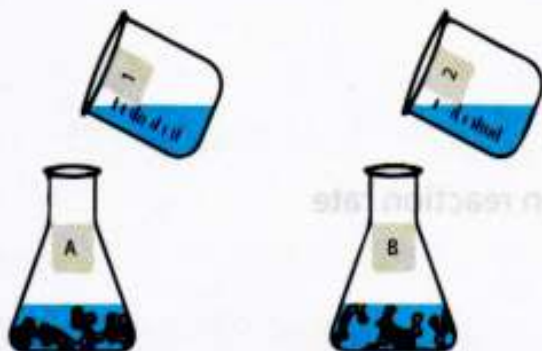
1. Label a conical flask A and put 5g of zinc granules into it.
2. Repeat the same with the second conical flask and label it B.



3. Label a beaker 1 and pour 10cm^3 of HCl into it.
4. Label the other beaker 2, pour 5cm^3 of water into it.
5. Add 5cm^3 of HCl into the second beaker.



6. Pour quickly the liquid in beaker 1 into the conical flask A and pour the liquid in beaker 2 into the conical flask B.



7. Attach one balloon firmly to each conical flask



8. Light the gas in the balloons

Observation

1. What did you observe?
2. Which of the two beakers contained more concentrated of HCl solution?
3. Which balloon filled more quickly?

Conclusion

A more concentrated solution brings a faster reaction rate - the balloon filling with H_2 gas more quickly. The test for hydrogen gas would have resulted in a loud *pop sound* when the lit splint was put near the mouth of a test tube.

Two reactants cannot react with each other unless their constituent particles: atoms, molecules or ions come into contact. If there is no contact of the reactants, the reaction rate will be zero.

Conversely, the more collisions of the reactant particles that collide per unit time, the more often a reaction between them can occur.

Consequently, the rate of a reaction usually increases as the concentration of the reactants increases.

6.4.4 Surface Area

Experiment 5:

To determine effect of surface area on reaction rate

Materials

- 2g of marble chips
- 2g of powdered marble
- Concentrated hydrochloric acid (HCl)
- A beaker
- 2 test tubes

Caution: Concentrated HCl can cause serious burns - Use gloves and goggles.

Procedure

1. Add 2cm^3 of concentrated HCl to 20cm^3 of H_2O
2. Put the marble chips into one of the test-tube
3. Put the powdered marble into the other test tube
4. Add 10cm^3 of diluted HCl to each of the test tubes.
5. Note down what you observe

Observation

1. What did you observe?
2. Which reaction proceeds faster?
3. Can you explain this?

Conclusion

The reaction with the powdered marble is faster because the powdered marble has a greater surface area than the marble chips.

The rate of a chemical reaction can be accelerated by increasing the surface area of a solid reactant.

The surface area, is increased by breaking down the substance into small pieces, or by grinding the pieces into a powder.

A decrease in the surface area is done by having pieces stay in a big clump.

If the surface area of a solid reactant is increased the result will be:

1. More particles are exposed to other reactants.
2. There will be more collisions.
3. The rate of reaction increases.

6.4.5 Catalyst**Experiment 6:**

Effect of a catalyst in a reaction rate

Materials

- 1 liter graduated cylinder
- Large watch glass
- Large tray
- Absorbent paper
- Gloves and goggles
- 50ml 30% Hydrogen Peroxide (H_2O_2).
- Liquid dish soap
- 1gm of KI (potassium iodide)

Procedure

1. On a large sheet of plastic backed absorbent paper, place the 1-liter graduated cylinder on a tray.
2. Add 50 mL 30% Hydrogen Peroxide and 4 drops of soap to the graduated cylinder.
3. Add 1gm of KI.

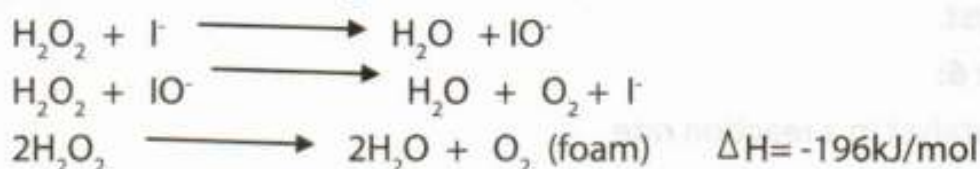
Caution. This reaction generates a lot of heat. Glassware may burn your hand therefore use gloves and goggles.

Observation

- What did you observe?

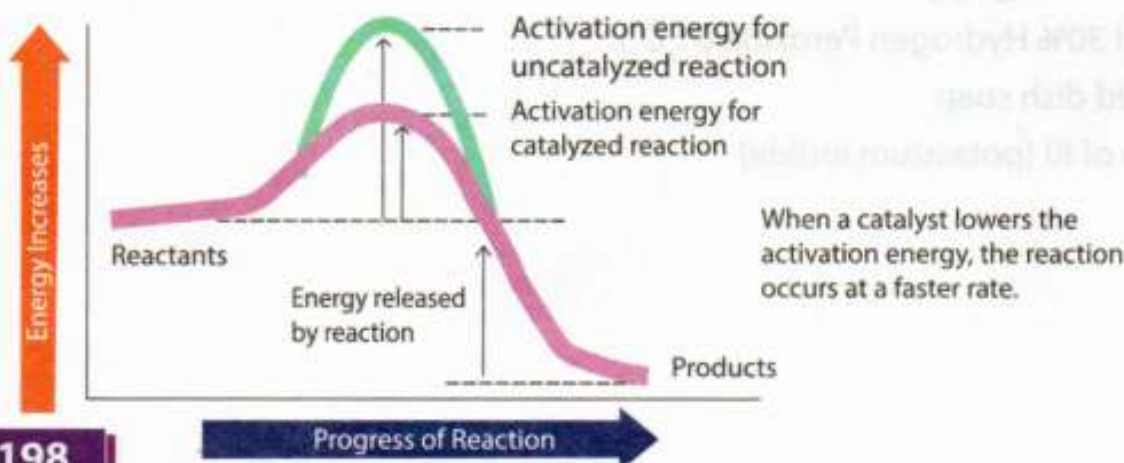
Conclusion

Hydrogen peroxide (H_2O_2) decomposes into water and oxygen gas, which is in the form of **foam**. Under normal conditions, this reaction occurs very slowly, therefore a **catalyst** is added to speed up the reaction, which will result in rapid formation of the foam. The iodide ion from potassium iodide is the catalyst and speeds up the reaction while remaining chemically unchanged in the reaction process. The iodide ion changes the mechanism of the reaction:



A **catalyst** is a substance that takes part in a chemical reaction and speeds up the reaction rate without undergoing a net chemical change itself.

The rate of chemical reaction increases with the increase in the presence of catalyst because a catalyst decreases the activation energy of the reactants molecules.



Activation energy is the minimum amount of energy that is required to activate atoms or molecules to a condition in which they can undergo reaction.



EXERCISE

- Describe the effect of each of the following on the rate of the reaction of: -
 - magnesium metal with a solution of hydrochloric acid
 - the molarity of the hydrochloric acid
 - the temperature of the solution
 - the size of the pieces of magnesium.
- For each of the following situations, state whether the rate of the reaction would increase or decrease, and explain your answer using the collision theory.
 - The concentration of a reactant is doubled.
 - The reaction is moved from a lab bench at room temperature into an ice bath.
 - A solid crystalline reactant is broken down into smaller pieces with a hammer.

6.5 REACTION RATE LAW AND ORDER OF REACTIONS**Lesson objectives**

By the end of the lesson, the learner should be able to:

- Explain the reaction rate law.
- Define and determine the order of reactions.
- Define rate constant.

Rate law is an experimentally determined equation that expresses the rate of chemical reaction in the terms of molecular concentrations of the reactant. It is experimentally observed that the rate of reaction is directly proportional to the concentration of reactants at a constant temperature. Each of the concentrations of expression is raised to power which are determined experimentally. This power is known as order of the reaction.

For the reaction $aA + bB \rightarrow \text{products}$, the rate law is expressed as:

$$\text{Rate} \propto [A]^x[B]^y$$

$$\text{Rate} = K [A]^x[B]^y$$

Where;

x = order of the reaction with respect to A

y = order of the reaction with respect to B

k = rate constant.

Rate constant

Rate constant is defined as the specific rate of reaction when the molar concentrations of the reactants are taken to be unity, i.e.:

$$\text{Rate} = k, \text{ if } [A] = [B] = 1$$

Order of a Reaction

Order of a reaction is the experimental exponent of a reactant in the rate equation. For the rate law expression, $\text{Rate} = K [A]^x[B]^y$

The x and y are known as order of reactions. The Sum of all exponents in the rate law expression is known as overall order of the reactions.

$$x+y$$

6.6 COLLISION THEORY.

Lesson objectives

By the end of the lesson, the learner should be able to:

- Explain the collision theory

The Conditions required for a reaction to occur are:

1. **Collision:** The reactants must collide.
2. **Orientation:** The reactants must align properly to break down and form bonds.
3. **Energy:** The collision must give the energy of activation.

The basic requirement for a reaction to occur is that the reactant species (atoms or molecules) must collide with each other in some way.

Reaction:

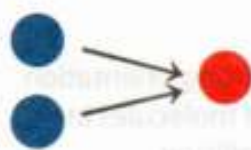


Reactants in
Reaction Flask

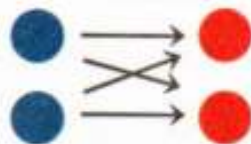
Possible
Collisions



1



2

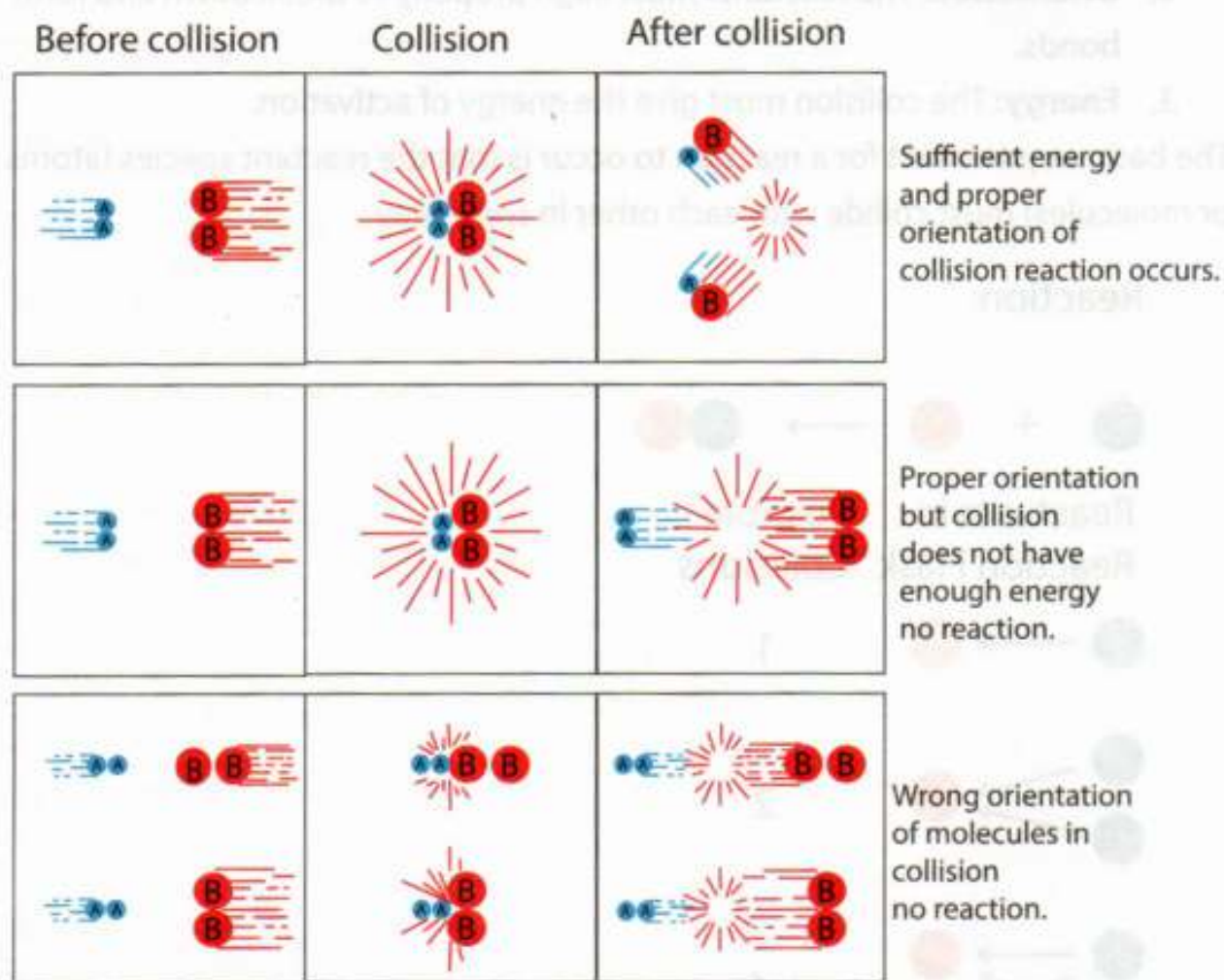


4

Figure 6.4: Increasing the concentration of a reactant increases the number of collisions that are possible.

Collision theory provides an explanation of how particles interact to cause a reaction and the formation of new products.

Collision theory states that the speed or the rate of a chemical reaction is proportional to the number of collisions between reactants. The more often reactant species collide, the more often they combine with one another, and the faster the reaction rate. In fact, only a small number of the collisions are effective collisions. Effective collisions are those which result in a chemical reaction.



In order to have an effective collision, reactant species or particles must possess some minimum amount of energy. This energy used to initiate the reaction, is called the activation energy. For each sample of reactant particles there will be some that possess this amount of energy. The bigger the sample, the higher the number of effective collisions, and the faster the rate of reaction. The number of particles possessing sufficient energy depends on the temperature of the reactants. If reactant particles do not have the required activation energy when they collide, they bounce off each other without reacting.

In some cases, chemical reactions also require that the reactant particles be in a particular orientation to produce an effective collision. Unless the reactant particles have this orientation when they collide, the collision will not be an effective one.



EXERCISE

1. Why is a collision between two particles necessary for a reaction to occur?
2. Explain the reaction of energy graph for an exothermic reaction in terms of collision theory.
3. What is activation energy?
4. Which two factors govern whether a collision between reacting particles will be effective?
5. If two reactants collided with energy but no reaction occurred, what would be the most likely explanation?

6.7 MECHANISM OF REACTION

Lesson objectives

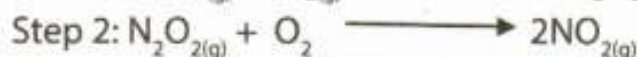
By the end of the lesson, the learner should be able to:

- Describe mechanism for a chemical reaction.

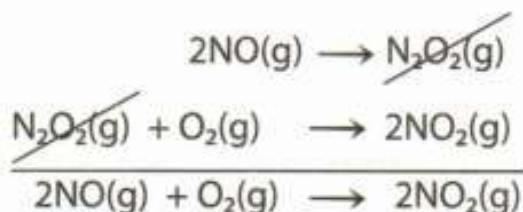
Mechanism of a reaction is the sequence of elementary steps that together make a complete chemical reaction. For example, consider the route that you might take while walking to the grocery store. That route may consist of several turns. The overall reaction determines only the beginning point (your house) and the final destination (the grocery store), with no information about the points in between. Studying the mechanism of a reaction can be very important, because this knowledge often gives us insights into how to improve the reaction, get better products, or make the reaction occur more rapidly. Additionally, many different reactions often have very similar reaction mechanisms. The types of elementary steps that are common to many different chemical reactions allow us to make predictions about the reactivity patterns of unknown compounds. An example of a reaction mechanism concept can be shown by the reaction between Nitrogen monoxide (NO) and Oxygen to form Nitrogen Dioxide (NO_2):



It looks like that this reaction would occur as the result of a collision between two NO molecules and one O_2 molecule. However, in the mechanism of the reaction it has detected the presence of N_2O_2 during the reaction. Thus, the mechanism of this reaction consists of two elementary steps:



In step 1, two molecules of NO collide to form one molecule of N_2O_2 . In step 2, N_2O_2 collides with a molecule of O_2 to form two molecules of NO_2 . The overall of this chemical reaction will be the sum of the two above elementary steps:



N_2O_2 molecule forms in the first elementary step and reacts with O_2 in the second elementary step. N_2O_2 is an intermediate, species that appear in the mechanism of the reaction but not in the overall equation.

**EXERCISE**

1. Do chemical reactions usually occur in a single step?
2. What does the overall balanced equation not tell us?
3. How do overall reactions usually proceed?
4. Explain an intermediate species.
5. Describe the steps that a mechanism of a reaction consists of.

Chapter Review questions

1. Explain the following:
 - a) Reaction Rate
 - b) Collision theory
 - c) Activation energy
 - d) Catalyst
2. In the following reaction:
$$2\text{Ca} + \text{O}_2 \longrightarrow 2\text{CaO}$$
1 g of calcium is oxidized. The reaction takes 10 minutes to reach completion. What is the rate of the formation of CaO ?
3. 10 g of Mg reacts with 0.15 mol.dm^{-3} solution of HCl at a temperature of 25°C .
 - a) Write a balanced chemical equation for the reaction.
 - b) State two ways of increasing the average rate of production of $\text{H}_{2(g)}$.
4. A table of results is given below:

Time elapsed	Volume of $\text{H}_{2(g)}$ cm^3
0	0
0.5	17
1.0	25
1.5	30
2.0	33
2.5	35
3.0	35

5. Consider the following hypothetical reaction:



Increasing the average rate of the reaction, which of the following would you do?

- Decrease the concentration of B.
- Grind A into a fine powder.
- Decrease the temperature

7. The reaction between the peroxydisulfate ion ($S_2O_8^{2-}$) and the iodide ion (I^-) is:



- From the following data (Table below), which were collected at a set temperature, determine the rate law and calculate the specific rate constant.

Experiment	$[S_2O_8^{2-}]$ (M)	$[I^-]$ (M)	Initial Rate (M/s)
1	0.080	0.034	2.2×10^{-4}
2	0.080	0.017	1.1×10^{-4}
3	0.16	0.017	2.2×10^{-4}

- What is the initial rate of reaction if the concentrations of both $S_2O_8^{2-}$ and I^- are 0.5 M?

8. Explain the following:

- Hydrogen peroxide (H_2O_2) decomposes to water and oxygen very slowly. When solid manganese dioxide is added, however, oxygen evolves rapidly.
- People who are lactose intolerant cannot break down milk sugar (lactose).

9. Indicate whether the following changes will increase, decrease, or have no effect on the rate of reaction:
- increasing the temperature
 - increasing the number of reacting molecules
 - adding a catalyst
10. Why does bread grow mold more quickly at room temperature than in the refrigerator?
11. How does a catalyst affect the activation energy?
12. For the decomposition reaction $X \rightarrow Y + Z$, the reaction is first-order with respect to X, and the value of the specific rate constant is 0.0296 s^{-1} . Calculate the initial reaction rate when starting with the following concentrations of X.
- $[X] = 0.410 \text{ M}$
 - $[X] = 0.0223 \text{ M}$
13. In the following reaction, what happens to the number of collisions when more $\text{Br}_{2(g)}$ molecules are added?
- $$\text{H}_{2(g)} + \text{Br}_{2(g)} \longrightarrow 2\text{HBr}_{(g)}$$
14. In the following reaction, what happens to the number of collisions when the temperature of the reaction is decreased?
- $$\text{C}_{(g)} + \text{O}_{2(g)} \longrightarrow \text{CO}_{2(g)}$$
15. How would each of the following change the rate of the reaction shown here?
- $$2\text{SO}_{2(g)} + \text{O}_{2(g)} \longrightarrow 2\text{SO}_{3(g)}$$
- adding some $\text{SO}_{2(g)}$
 - increasing the temperature
 - adding a catalyst
 - removing some $\text{O}_{2(g)}$
16. How would each of the following change the rate of the reaction shown here?
- $$2\text{NO}_{(g)} + 2\text{H}_{2(g)} \longrightarrow \text{N}_{2(g)} + 2\text{H}_2\text{O}_{(g)}$$
- adding some $\text{NO}_{(g)}$
 - decreasing the temperature
 - removing some $\text{H}_{2(g)}$
 - adding a catalyst

17. Classify the following processes as spontaneous or nonspontaneous. Explain your answers in terms of whether energy must be continually supplied to keep the process going:

- a) Water is decomposed into hydrogen and oxygen gas by passing electricity through the liquid.
- b) An explosive detonates after being struck by a falling rock.
- c) A coating of magnesium oxide forms on a clean piece of magnesium exposed to air.
- d) A light bulb emits light when an electric current is passed through it.
- e) A cube of sugar dissolves in a cup of hot coffee.

18 Classify the following processes according to their rates as fast, slow, or very slow:

- a) The souring of milk stored in a refrigerator
- b) The cooking of an egg in boiling water
- c) The ripening of a banana stored at room temperature
- d) The rising of bread dough in a warm room
- e) The melting of butter put into a hot pan
- f) The change of apple juice to cider
- g) The movement of sound from a slammed door to your ear
- h) The ringing of your phone after you step into the shower
- i) The combustion of gasoline in the engine of your car
- j) The perceived passage of time when you are doing something enjoyable

19. Describe the observations or measurements that could be made to allow you to follow the rate of the following processes:

- a) The melting of a block of ice
- b) The setting (hardening) of concrete.
- c) The burning of a candle

20. Consider the following hypothetical reaction:



Calculate the average rate of the reaction on the basis of the following information:

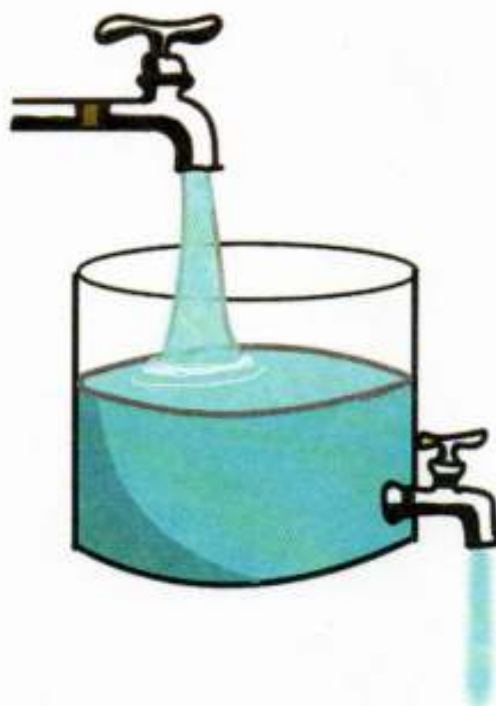
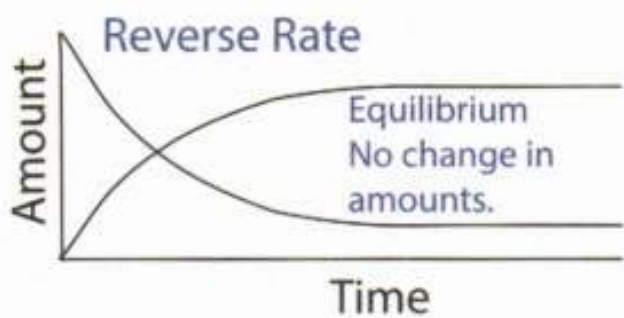
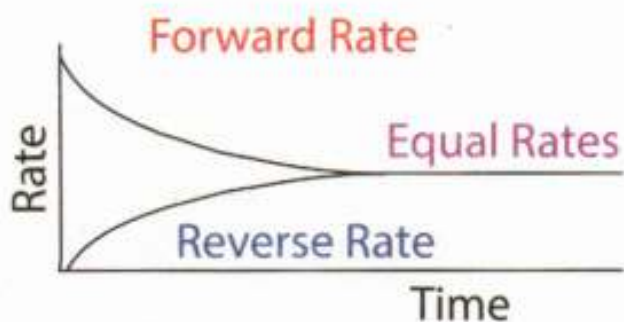
- Pure A and B are mixed, and after 12.0 minutes the measured concentration of C is 0.396 mol/L.
- Pure A, B, and C are mixed together at equal concentrations of 0.300 M. After 8.00 minutes, the concentration of C is found to be 0.455 M.

21. Use energy diagrams to compare catalyzed and uncatalyzed reactions.



CHAPTER 7

CHEMICAL EQUILIBRIUM



CHAPTER CONTENTS

- 7.1 Reversible reactions and equilibrium
- 7.2 Chemical equilibrium.
- 7.3 Types of chemical equilibrium
- 7.4 The equilibrium constant
- 7.5 Calculating equilibrium constants, K_c
- 7.6 Calculating equilibrium concentrations
- 7.7 The reaction quotient
- 7.8 Factors affecting chemical equilibrium
- Chapter review questions

CHAPTER OBJECTIVES

By the end of the chapter, the learner should be able to:

- Explain reversible reaction and how it relates to equilibrium.
- Explain what is meant by chemical equilibrium and how it relates to reaction rates.
- Classify chemical equilibrium on the basis of different phases.
- Define equilibrium constant.
- Write equilibrium constant expressions.
- Manipulate the equilibrium constant to reflect changes in chemical reactions.
- Calculate chemical equilibrium concentrations.
- Predict the direction of a reaction given the equilibrium constant and the concentrations of reactants and products by comparing reaction quotient to equilibrium constant.
- Predict the direction which equilibrium shifts under stress using Le Chatelier's principle.

7.1 REVERSIBLE REACTIONS AND EQUILIBRIUM

Lesson objectives

By the end of the lesson, the learner should be able to:

- Explain the possibilities of the reaction
- Compare reversible and irreversible reactions
- Explain reversible reaction and how it relates to equilibrium



ACTIVITY 1

Single arrow ($\longrightarrow$) is used in some reactions while double arrow ($\rightleftharpoons$) is used for some other reactions. We discuss what each of them indicate in a reaction.

In some reactions, all the reactants are used up thus they are said to have gone to completion. This type of reaction is known as irreversible. In irreversible reaction the reactants react to produce products but the products *cannot* react to produce back to the reactants, however, in some cases reactions do not go to completion but instead attain a state of chemical equilibrium.

This type of reaction is known as reversible reaction. Most reactions are theoretically reversible in a closed system.

A reversible reaction is a reaction that can proceed in both the forward and backward directions. It may be represented in general terms as:



For example:

$H_2(g) + I_2(g) \rightarrow 2HI(g)$ its back ward will be $2HI(g) \rightarrow H_2(g) + I_2(g)$. These two reactions can be expressed as;



The double arrow indicates that the reaction is reversible and can go either forward to produce products, or backward to yield reactants.

This type of reaction can reach a point where the rate of forward reaction is equal to the rate of the reverse reaction and there is no net change in the number of reactant and product molecules. At this point the reaction is said to be in chemical equilibrium.

**EXERCISE**

1. What are the differences between reversible and irreversible reactions?
2. What are the possibilities of the reaction?
3. Name 2 reversible and 2 non-reversible reactions.

7.2 WHAT IS CHEMICAL EQUILIBRIUM?**Lesson objectives**

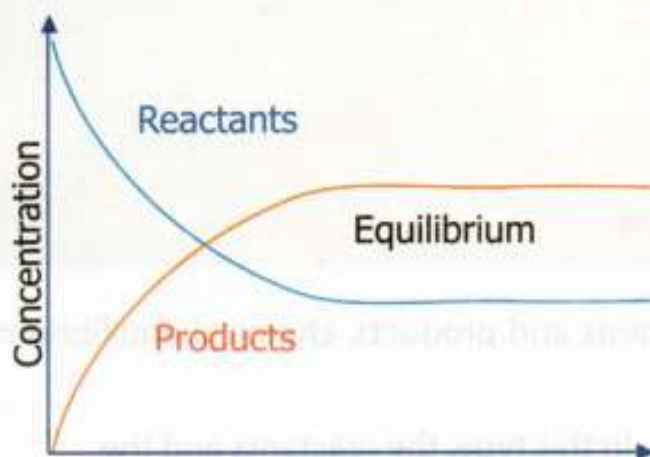
By the end of the lesson, the learner should be able to:

- Explain what is meant by chemical equilibrium and how it relates to reaction rates.

Chemical equilibrium is a state in which the rates of the forward and reverse reactions are equal and the concentrations of the reactants and products remain constant.

Since the conversions of reactants to products and products to reactants are still going on at equilibrium, the chemical equilibrium is also known as **dynamic equilibrium**.

At equilibrium, the number of reactant molecules converting into products and product molecules into reactants are the same.



The conditions necessary for the reaction to attain equilibrium are:

1. The reaction must be in a closed system.
2. The reaction must occur at a constant temperature.
3. The reaction must be reversible.
4. The rate of forward reaction must be equal to the rate of the reverse reaction.



EXERCISE

1. Define chemical equilibrium. Give an example.
2. What are the conditions of chemical equilibrium?
3. Explain what is meant when it is said that chemical equilibrium is dynamic.

7.3 TYPES OF CHEMICAL EQUILIBRIUM

Learning objectives

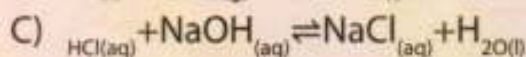
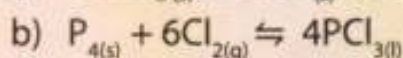
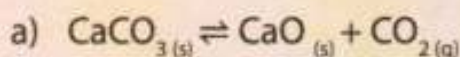
By the end of the lesson, the learner should be able to:

- To Classify chemical equilibrium on the basis of different phases.
- To compare and contrast homogeneous and heterogeneous chemical equilibrium.



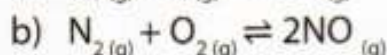
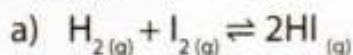
ACTIVITY 2

We identify the physical states of each species in the following chemical equations:

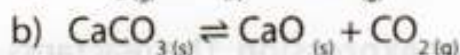
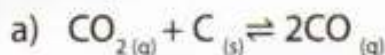


On the basis of the phase of the reactants and products, chemical equilibrium can be classified into two types:

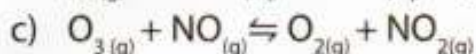
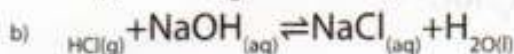
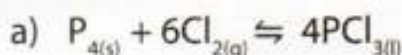
1. Homogeneous chemical Equilibrium: In this type, the reactants and the products of chemical equilibrium are all in the same phase.

For example

2. Heterogeneous chemical Equilibrium: In this type, the reactants and the products of chemical equilibrium are present in different phases.

For example:**EXERCISE**

- What are the differences between homogeneous and heterogeneous equilibrium?
- Classify the following reactions as homogeneous and heterogeneous.

**7.4 THE EQUILIBRIUM CONSTANT****Lesson objectives**

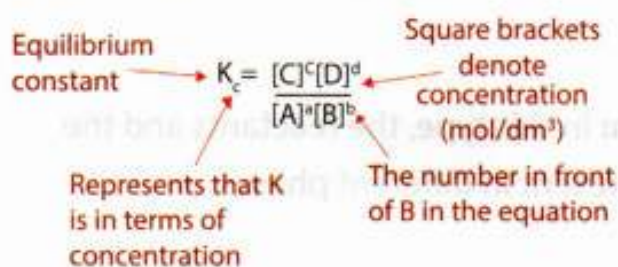
By the end of the lesson, the learner should be able to:

- Define equilibrium constant.
- Write equilibrium constant expressions.
- Manipulate the equilibrium constant to reflect changes in the chemical reaction.

In the mid 1860s, Norwegian scientists C. M. Guldberg and P. Waage noted a peculiar relationship between the amounts of reactants and products in equilibrium. They observed a certain ratio of reactants and products at equilibrium and this ratio is known as the equilibrium constant.

The equilibrium constant K_c is the ratio of the equilibrium concentrations of products over the equilibrium concentrations of reactants each raised to the power of their stoichiometric coefficients.

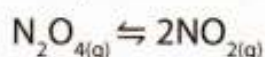
For a reaction: $aA + bB \rightleftharpoons cC + dD$ equilibrium constant is expressed as :



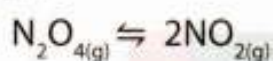
This expression is known as **Equilibrium constant expression**. The reactants or products that are pure solids or pure liquids do not appear in the equilibrium expression because the concentration of pure solid or liquid does not change due to their fixed volume.

Example 1

Write the equilibrium constant expression, K_c , for;



Solution

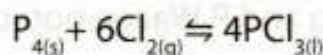


$$K_c = \frac{[\text{product}]}{[\text{reactant}]}$$

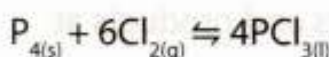
$$= \frac{[NO_2]^2}{[N_2O_4]}$$

Example 2

Write the equilibrium constant expression, K_c , for



Solution

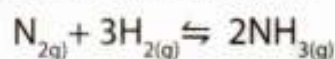
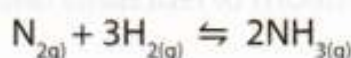


$$K_c = \frac{[\text{product}]}{[\text{reactant}]}$$

$$= \frac{1}{[Cl_2]^6} \quad \text{Write } Cl_2 \text{ using subscript}$$

Example 3

Write the equilibrium constant, K_c , for

**Solution**

$$\begin{aligned} K_c &= \frac{[\text{product}]}{[\text{reactant}]} \\ &= \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} \end{aligned}$$

The equilibrium constant K_c has no unit because it is the **ratio** of the concentration of products and reactants.

The magnitude of K_c gives us some information about the reactant and product concentrations at equilibrium if:

1. The K_c value is greater than 1 ($K_c > 1$), the reaction mixture contains mostly products i.e. products favored.
2. The K_c value is less than 1 ($K_c < 1$), the reaction mixture contains mostly reactants i.e. reactants favored.

**EXERCISE**

1. Why are solids and liquids not included in chemical equilibrium?
2. Write the equilibrium expressions for the following reactions in the space below:
 - a) $\text{O}_{3(g)} + \text{NO}_{(g)} \rightleftharpoons \text{O}_{2(g)} + \text{NO}_{2(g)}$
 - b) $2\text{CO}_{(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{CO}_{2(g)}$
 - c) $\text{NH}_4\text{NO}_{3(s)} \rightleftharpoons \text{N}_2\text{O}_{(g)} + 2\text{H}_2\text{O}_{(l)}$
3. Indicate whether reactants or products are likely to be favored. Justify your decisions.
 - a) $K_{eq} = 1 \times 10^{12}$
 - b) $K_{eq} = 1.5$

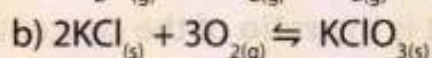
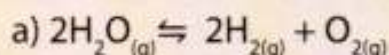
7.5 CALCULATING EQUILIBRIUM CONSTANTS, K_c **Lesson objectives**

By the end of the lesson, the learner should be able to:

- Calculate equilibrium constant, when the concentrations of reactants and products at equilibrium are known,
- Calculate missing equilibrium concentrations.

**ACTIVITY 3**

We write the equilibrium constant expression of the following.



The equilibrium constant (K_{eq}) is a quantitative means of expressing the relationship between the concentrations of reactants and products at equilibrium.

$$K_c = \frac{[\text{products}]}{[\text{reactant}]}$$

If equilibrium concentrations are known, simply substitute the concentrations into the equilibrium constant expression to calculate the equilibrium constants, K_c .

Example 1

For the reaction, $\text{CO}_{(g)} + 3\text{H}_{2(g)} \rightleftharpoons \text{CH}_{4(g)} + \text{H}_2\text{O}_{(g)}$, calculate K_c from the following equilibrium concentrations:

$[\text{CO}] = 0.0613 \text{ M}$

$[\text{CH}_4] = 0.0387 \text{ M}$

$[\text{H}_2] = 0.1839 \text{ M}$

$[\text{H}_2\text{O}] = 0.0387 \text{ M}$

Solution

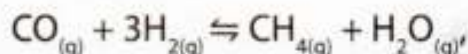
$K_c = ?$

$[\text{CO}] = 0.0613 \text{ M}$

$[\text{H}_2] = 0.1839 \text{ M}$

$[\text{CH}_4] = 0.0387 \text{ M}$

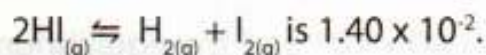
$[\text{H}_2\text{O}] = 0.0387 \text{ M}$



$$\begin{aligned} K_c &= \frac{[\text{CH}_4][\text{H}_2\text{O}]}{[\text{CO}][\text{H}_2]^3} \\ &= \frac{(0.0387)(0.0387)}{(0.0613)(0.1839)^3} \\ &= 3.93 \end{aligned}$$

Example 2

At a given temperature, the equilibrium constant for the reaction stated below is 1.40×10^{-2}



If the concentrations of both H_2 and I_2 at equilibrium are $2.00 \times 10^{-4}\text{M}$, find the concentration of HI .

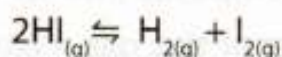
Solution

$$K_{\text{eq}} = 1.40 \times 10^{-2}$$

$$[\text{I}_2] = 2.00 \times 10^{-4}\text{M}$$

$$[\text{H}_2] = 2.00 \times 10^{-4}\text{M}$$

$$[\text{HI}] = ?$$



$$K = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2}$$

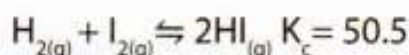
$$1.40 \times 10^{-2} = \frac{(2.00 \times 10^{-4})(2.00 \times 10^{-4})}{[\text{HI}]^2}$$

$$[\text{HI}]^2 = \frac{(2.00 \times 10^{-4})(2.00 \times 10^{-4})}{1.40 \times 10^{-2}}$$

$$\begin{aligned} [\text{HI}] &= \sqrt{2.86 \times 10^{-6}} \\ &= 1.69 \times 10^{-3} \text{ mol/L} \end{aligned}$$

Example 3

Are reactants or products favored in the following reaction?

**Solution**

Since $50.5 > 1$, then $K_c > 1$ therefore product is favored in the reaction.



EXERCISE

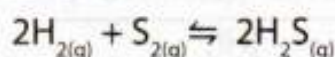
1. What is the equilibrium constant, K_{eq} , if the equilibrium concentrations are as follows: PCl_5 is 0.0096 M, PCl_3 is 0.0247M and Cl_2 is 0.0247 M?



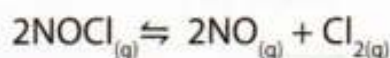
2. Acetic acid dissociates in water. If $K_{eq} = 1.80 \times 10^{-5}$ and the equilibrium concentrations of acetic acid is 0.09986M, what is the concentration of $H^+_{(aq)}$ and $CH_3COO^-_{(aq)}$?



3. Hydrogen sulfide gas decomposes into its elements and establishes equilibrium at 1400°C. One liter of this gas mixture contains 0.18 mol of hydrogen sulfide, 0.014 mol of hydrogen gas and 0.035 mol of sulfur. Calculate the equilibrium constant, K_{eq} , for this reaction. Are products or reactants favored? How do you know?



4. Consider the reaction:



3.00 mol NOCl, 1.00 mol NO, and 2.00 mol Cl_2 are mixed in a 10.0L flask. After the system has reached equilibrium, the concentrations at 35°C are observed to be:

$[Cl_2] = 1.52 \times 10^{-1} M$ $[NO] = 4.00 \times 10^{-3} M$ $[NOCl] = 3.96 \times 10^{-1} M$ Calculate the value of K for this reaction at 35°C

7.6 CALCULATING EQUILIBRIUM CONCENTRATIONS

Lesson objectives

By the end of the lesson, learners should be able to:

- Calculate the concentrations of reaction components at equilibrium given the starting concentrations and the equilibrium constant

If the value of the equilibrium constant and a set of concentrations of reactants and products that are not at equilibrium are known, the concentrations at equilibrium can be calculated by the use of Initial Change equilibrium (ICE) method.

Example 1

If 2.00mol of $\text{H}_{2(g)}$ and 3.00mol of $\text{I}_{2(g)}$ are placed in a 1.00 L vessel at 100°C , what is the concentration of each gas at equilibrium? If k_c is 25.0

**Solution**

Step 1: convert the values into concentrations expressed into molarities

$$\text{Molarity (M)} = \frac{\text{mole}}{\text{volume}}$$

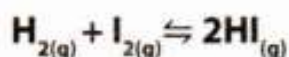
$$[\text{H}_2] = \frac{2.00\text{mol}}{1.00\text{l}} = 2.00\text{mol/L}$$

$$[\text{I}_2] = \frac{3.00\text{mol}}{1.00\text{l}} = 3.00\text{mol/L}$$

Step 2: Set the ICE table to analyze the problem

	Concentrations (mol/L)		
	$[\text{H}_{2(g)}]$	$[\text{I}_{2(g)}]$	$[\text{HI}_{(g)}]$
I	2.00	3.00	0
C	-x	-x	+2x
E	2.00-x	3.00-x	+2x

Step 3: Evaluate the value of "X" by using equilibrium constant expression and quadratic equation.



$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

$$25.0 = \frac{(2x)^2}{(2.00 - x)(3.00 - x)}$$

$$25.0 = \frac{(2x)^2}{(2.00 - x)(3.00 - x)}$$

$$4x^2 = 150 - 125x + 25.0x^2$$

$$21x^2 - 125x + 150 = 0$$

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$X = \frac{-(-125) \pm \sqrt{(-125)^2 - 4(21)(150)}}{2(21)}$$

$$= \frac{125 \pm \sqrt{15,625 - 12,600}}{42}$$

$$= \frac{125 \pm 55}{42}$$

$$X = \frac{125 \pm 55}{42} \quad \text{OR} \quad \frac{125 \pm 55}{42}$$

$$X = 4.29 \text{ OR } X = 1.67$$

Step 4: Choose one of the values of x which doesn't make the concentration negative. In this case we use 1.67 to determine the concentrations of;

$$[\text{H}_2] = 2.00 - X$$

$$= 2.00 - 1.67$$

$$= 0.33 \text{ mol/L}$$

$$[\text{I}_2] = 2.00 - X$$

$$= 3.00 - 1.67$$

$$= 1.33 \text{ mol/L}$$

$$[\text{HI}] = 2X$$

$$= 2(1.67)$$

$$= 3.34 \text{ mol/L}$$

**EXERCISE**

1. For this reaction; $\text{H}_{2(g)} + \text{F}_{2(g)} \rightleftharpoons 2\text{HF}_{(g)}$, $K_c = 1.15 \times 10^2$ at a certain temperature. Suppose 3.000 moles of H_2 and 6.000 moles of F_2 are placed in a 3.000 L container, what are the equilibrium concentrations of H_2 , F_2 and HF ?
2. Consider the reaction; $2\text{NOCl}_{(g)} \rightleftharpoons 2\text{NO}_{(g)} + \text{Cl}_{2(g)}$. At 35°C the equilibrium constant is 1.6×10^{-5} . In an experiment in which 1.0 mol of NOCl is placed in a 2.0 liter flask, what are the equilibrium concentrations?

7.7 THE REACTION QUOTIENT**Lesson objectives**

By the end of the lesson, learners should be able to:

- Define reaction quotient.
- Predict the direction of a reaction given the equilibrium constant and the concentrations of reactants and products.

The Reaction Quotient (Q) is a quantity that provides a measurement of the relative amounts of products and reactants present in a reaction at a given time. We can also define the Reaction Quotient (Q) as the ratio of concentration of products to the concentration of reactant at any given points other than equilibrium. It is expressed as the equilibrium constant (K_c) but it uses the values for concentrations of substances at points other than equilibrium. K is constant while Q is the activities of products and reactants at any given time. Therefore, for a reaction:



The reaction quotient is expressed as:

$$Q = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$



The value of reaction quotient can be compared to the Equilibrium Constant, K , to determine the direction of a reaction shifts in order to reach equilibrium if:

- $Q > K$; the concentration of products is too high; the reaction will try to use more product to produce reactants therefore the reaction will shift to the left to reach equilibrium
- $Q < K$;the concentration of products is too low; the reaction will try to use more reactants to produce products therefore the reaction will shift to the right to reach equilibrium
- $Q = K$; the system is at equilibrium

Example 1

If the $K_c = 1.0$, $[\text{CO}_2] = 2.0 \text{ M}$, $[\text{H}_2] = 2.0 \text{ M}$, $[\text{CO}] = 1.0 \text{ M}$ and $[\text{H}_2\text{O}] = 1.0 \text{ M}$ For the reaction of $\text{CO}_{(g)} + \text{H}_2\text{O}_{(g)} \rightleftharpoons \text{CO}_{2(g)} + \text{H}_{2(g)}$. What is the Q value for this equation and which direction will the reaction shift?

Solution

$Q = ?$

$K_c = 1.0$

$[\text{CO}_2] = 2.0 \text{ M}$ $\text{CO}_{(g)} + \text{H}_2\text{O}_{(g)} \rightleftharpoons \text{CO}_{2(g)} + \text{H}_{2(g)}$

$[\text{H}_2] = 2.0 \text{ M}$ $Q = \frac{[2.0][2.0]}{[1.0][1.0]}$

$[\text{CO}] = 1.0 \text{ M}$ $= 4.0$

$[\text{H}_2\text{O}] = 1.0 \text{ M}$

Since $4.0 > 1.0$, then $Q > K$ therefore the reaction shifts to the left toward the reactants.

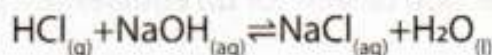
Example 2

Find the value of Q and determine which side will the reaction shift.

If K_c of the reaction; $\text{HCl}_{(g)} + \text{NaOH}_{(aq)} \rightleftharpoons \text{NaCl}_{(aq)} + \text{H}_2\text{O}_{(l)}$ is 0.5 and the concentrations of HCl , NaOH and NaCl are 3.2M, 4.3M and 6M respectively

Solution

$Q = ?$



$K_c = 0.5$

$$Q = \frac{[\text{NaCl}]}{[\text{HCl}][\text{NaOH}]}$$

$[\text{HCl}] = 3.2\text{M}$

$$= \frac{[6.0]}{[3.2][4.3]}$$

$[\text{NaOH}] = 4.3\text{M}$

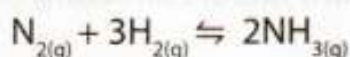
$$= 0.44$$

$[\text{NaCl}] = 6.0\text{M}$

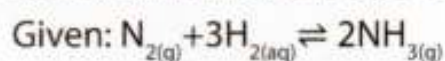
Since $0.44 < 0.5$ then $Q < K$ therefore the reaction shifts to the right toward the product to achieve equilibrium.

**EXERCISE**

- For the synthesis of ammonia, the value of K is 6×10^{-2} at 500°C . In an experiment, 0.50 mol of Nitrogen, 1.0×10^{-2} mol of Hydrogen, and 1.0×10^{-4} mol of Ammonia are mixed at 500°C in a 1.0 L flask. In which direction will the system proceed to reach equilibrium?



- Find the value of reaction quotient and determine which direction of the reaction will shift in order to reach equilibrium.



$$[\text{N}_2] = 0.04\text{M}$$

$$[\text{H}_2] = 0.09\text{M}$$

$$K = 0.040$$

7.8 FACTORS AFFECTING CHEMICAL EQUILIBRIUM**Lesson objectives**

By the end of the lesson, the learner should be able to:

- Define Le Chatelier's principle.
- Predict the direction of shift for an equilibrium under stress.

Chemical equilibria can be shifted by changing the conditions that the system experiences. Any factor that disturbs the equilibrium state is considered as stress. When we stress the equilibrium, the chemical reaction is no longer at equilibrium, and the reaction starts to move back towards equilibrium to decrease the stress.

Le Chatelier was one of the first chemists to investigate the effects of concentration, temperature, and pressure on equilibrium. His investigations led him to the following conclusion:

"If a system at equilibrium is disturbed by a change in temperature, pressure or the concentration of one of the components, the system will shift its equilibrium position so as to reduce the effect of the disturbance".

There are several factors such as temperature, pressure, and concentration of the system which affect equilibrium. Some of the important factors affecting chemical equilibrium which were studied by Le Chatelier are:

1. Change in concentration
2. Change of temperature

7.9.1 The effect of Change in concentration

Adding a reactant or a product, the equilibrium shifts to reduce the effect by consuming part of the added substance. Similarly, removing a reactant or a product, the equilibrium shifts to reduce the effect by replacing part of the removed species.

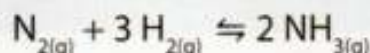
However, the change of solids and liquids does not affect the concentration of the equilibrium system because their concentration does not change due to their fixed volume.

Concentration for both gases (g) and solutions (aq) can be expressed as moles per liter (mole/l) which is equal to molarity (M).

Note: The concentration is indicated by the symbol [].

Example 1

Given this reaction at equilibrium:



In which direction, does the reaction shift if the equilibrium is stressed by:

- Adding $\text{H}_{2(g)}$ to the reaction?
- Adding $\text{NH}_{3(g)}$ to the reaction?

Solution

- If $\text{H}_{2(g)}$ is added, there is now more reactant, so the reaction will shift toward products to reduce the added $\text{H}_{2(g)}$.
- If $\text{NH}_{3(g)}$ is added, there is now more product, so the reaction will shift toward reactants to reduce the added $\text{NH}_{3(g)}$.

Example 2

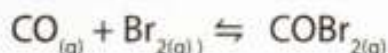
For $\text{H}_{2(g)} + \text{I}_{2(s)} \rightleftharpoons 2\text{HI}_{(g)}$, does the equilibrium shift left or right if we remove I_2 ?

Solution

I_2 is solid, removing solid to the equilibrium system does not effect on equilibrium therefore the equilibrium neither shifts left nor right.

**EXERCISE**

- Given this reaction at equilibrium:



In which direction—toward reactants or toward products—does the reaction shift if the equilibrium is stressed by each change?

- Br_2 is removed.
- COBr_2 is removed.

7.9.2 The effect of Change in Volume or Pressure



ACTIVITY 4

In groups, we discuss the relationship between volume and pressure, then we name the scientist who explained about that relationship.

Changing the pressure by increasing or decreasing the volume of a container will disturb an equilibrium system. However in a heterogeneous chemical equilibrium, the change of pressure in both liquids and solids are being ignored because their effective concentration stay constant throughout the reaction. The changes in their pressure or volume have no effect on the value of equilibrium constant (K).

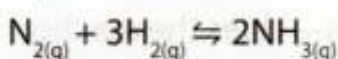
If Pressure increases, the volume decreases then the system shifts to the side with a smaller number of gas molecules and vice verse.

Example 1

For $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$ does the equilibrium shift left or right if the pressure is increased?

Solution

The coefficients of the balanced equation indicate the number of molecules of gases thus the total number of molecules on product side and reactant side are represented as



4 moles

2moles

According to Le Chatelier's principle, if pressure is increased, the volume decreases then the equilibrium shifts to the side with the fewer number of moles of gas. Thus, the equilibrium shifts to right.

Example 2

Consider the following reaction at chemical equilibrium



What is the effect of decreasing the volume of the reaction mixture?

Solution

The coefficients of the balanced equation indicate the number of moles of gases thus the total number of moles on product side or reactant side are represented as



Zero moles 3 moles

According to Le Chatelier's principle, if volume is decreased, the volume decreases then the equilibrium shifts to the side with the fewer number of moles of gas. Thus, the equilibrium shifts to left.

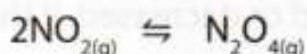
Similarly, if pressure decreases the volume increases, the system shifts to the side with a greater number of moles of gas.

Example 1

For the reaction $2\text{NO}_{2(g)} \rightleftharpoons \text{N}_2\text{O}_{4(g)}$, which direction will the equilibrium shift, if the pressure of the system is decreased?

Solution

The coefficients of the balanced equation indicate the number of moles of gases thus the total number of moles on the product and reactant side are represented as

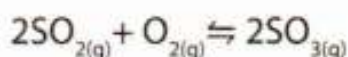


2moles 1mole

According to Le Chatelier's principle, if pressure is decreased, the volume increases then the equilibrium shifts to the side with the greater number of moles of gas. Thus, the equilibrium shifts to left.

Example 2

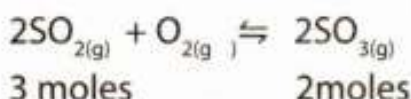
Consider the following reaction at the equilibrium:



What is the effect of increasing the volume of the reaction mixture?

Solution

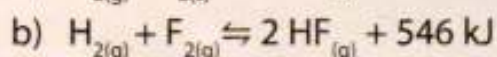
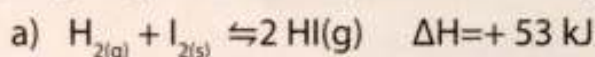
The coefficients of the balanced equation indicate the number of moles of gases thus the total number of moles on product side and reactant side are represented as



According to Le Chatelier's principle, if volume is increased, the pressure decreases then the equilibrium shifts to the side with the greater number of moles of gas. Thus, the equilibrium shifts to left.

7.8.3 The Effect of Changes in Temperature**ACTIVITY 5**

In groups we identify the following as endothermic or exothermic.



The temperature is the only effect that affects the equilibrium constant. A temperature change occurs when temperature is increased or decreased. If the temperature of an equilibrium system is changed, the chemical equilibria shift towards the product side or reactant side, which can be determined by studying whether it is endothermic or exothermic.

Endothermic reaction is the reaction that absorbs energy. The reaction is an endothermic, if the ΔH value is positive or the heat is added to the reactant side.



Exothermic reaction is the reaction that releases energy. The reaction is an exothermic, if the ΔH value is negative or the heat is added to the product side.

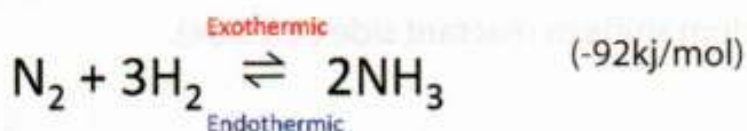


$$\Delta H = -ve$$



The individual reaction in the equilibrium can be either endothermic or exothermic. If the reaction is endothermic in forward then its backward is exothermic.

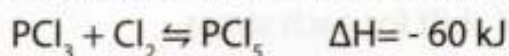
For example:



According to Le Chatelier's principle, when the temperature increases in case of exothermic reactions, the equilibrium shifts towards the reactant side (left side), the product and the equilibrium constant (K_{eq}) decreases, for endothermic reactions the equilibrium shifts towards the product side (right side). The product and the equilibrium constant (K_{eq}) increases with an increase in temperature.

Example 1

Predict the effect of increasing the temperature on this equilibrium.



Solution

The reaction is exothermic, because its ΔH is negative. Therefore, if the temperature is increased, the equilibrium shifts to reactant side (left side).

Example 2

$\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$ is endothermic. What occurs with increasing temperature?

Solution

The reaction is endothermic, if the temperature is increased, the equilibrium shifts to the product side (right side).

On the other hand, when the temperature decreases in case of exothermic reactions, the equilibrium shifts towards the product side (right side), the product and the equilibrium constant (K_{eq}) increases. For endothermic reactions the equilibrium shifts towards the reactant side (left side), the product and the equilibrium constant (K_{eq}) decreases with an decrease in temperature.

Example 1

The following reaction is exothermic:



What is the effect of decreasing the temperature of the reaction?

Solution

The reaction is exothermic, because its ΔH is negative. Therefore, if the temperature is increased, the equilibrium shifts to reactant side (left side).

Example 2

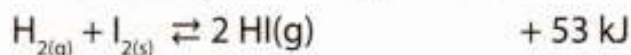
What effect will the decrease in temperature have on the reaction below?

Solution

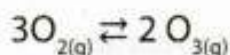
The reaction is exothermic, because its ΔH is negative. Therefore, if the temperature is increased, the equilibrium shifts to reactant side (left side)

**EXERCISE**

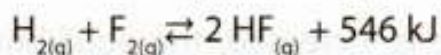
1. Define the Le Chatelier's principle.
2. What is meant by a stress? State how equilibrium can be stressed?
3. Given this equilibrium, predict the direction of shift for each stress.



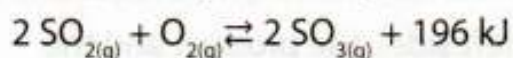
- a. decreased temperature
 - b. increased pressure
 - c. removal of HI
4. What is the effect on this equilibrium if pressure is decreased?



5. Given this equilibrium, predict the direction of shift for each stress.

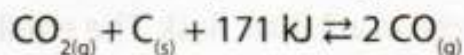


- a. increased temperature
 - b. addition of H_2
 - c. decreased pressure
6. Given this equilibrium, predict the direction of shift for each stress.



- a. removal of SO_3
- b. addition of O_2
- c. decreased temperature

7. Given this equilibrium, predict the direction of shift for each stress listed.



- addition of CO
- increased pressure
- addition of a catalyst

8. $\text{SO}_{24(g)} \rightleftharpoons 2 \text{NO}_{2(g)}$ is endothermic. What occurs with increasing temperature?

9. Predict the effect of decreasing the temperature on this equilibrium.



10. If the temperature is decreased for the reaction:



- Will the equilibrium shift left, or right?
- Does K_c become larger or smaller?

CHAPTER REVIEW QUESTIONS

Answer the following questions

1. Indicate whether reactants or products are likely to be favored. Justify your decisions.

a) $K_{eq} = 1$

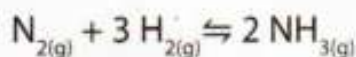
b) $K_{eq} = 5.6 \times 10^{-4}$

2. Calculate the number of moles of chlorine produced at equilibrium in a 10.0 L flask when 2.0 mol of phosphorus pentachloride is heated to 150°C.

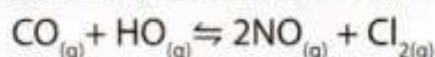
$$(K_c = 4.10 \times 10^{-6})$$



3. What is the effect on this equilibrium if pressure is increased?



4. At 700 K, Carbon Monoxide reacts with water to form Carbon Dioxide and hydrogen:



The equilibrium constant for this reaction is 5.10. Consider an experiment in which 0.500 mol of carbon monoxide and 0.500 mol of water vapor are mixed together in a 0.500-L flask. Calculate the concentrations of all species at equilibrium.



5. For the synthesis of hydrogen fluoride from hydrogen and fluorine, 3.000 mol of hydrogen and 6.000 mol of fluorine are mixed in a 3.000L flask. The equilibrium constant for the reaction at this temperature is 1.15×10^2 . Calculate the equilibrium concentration of each component.

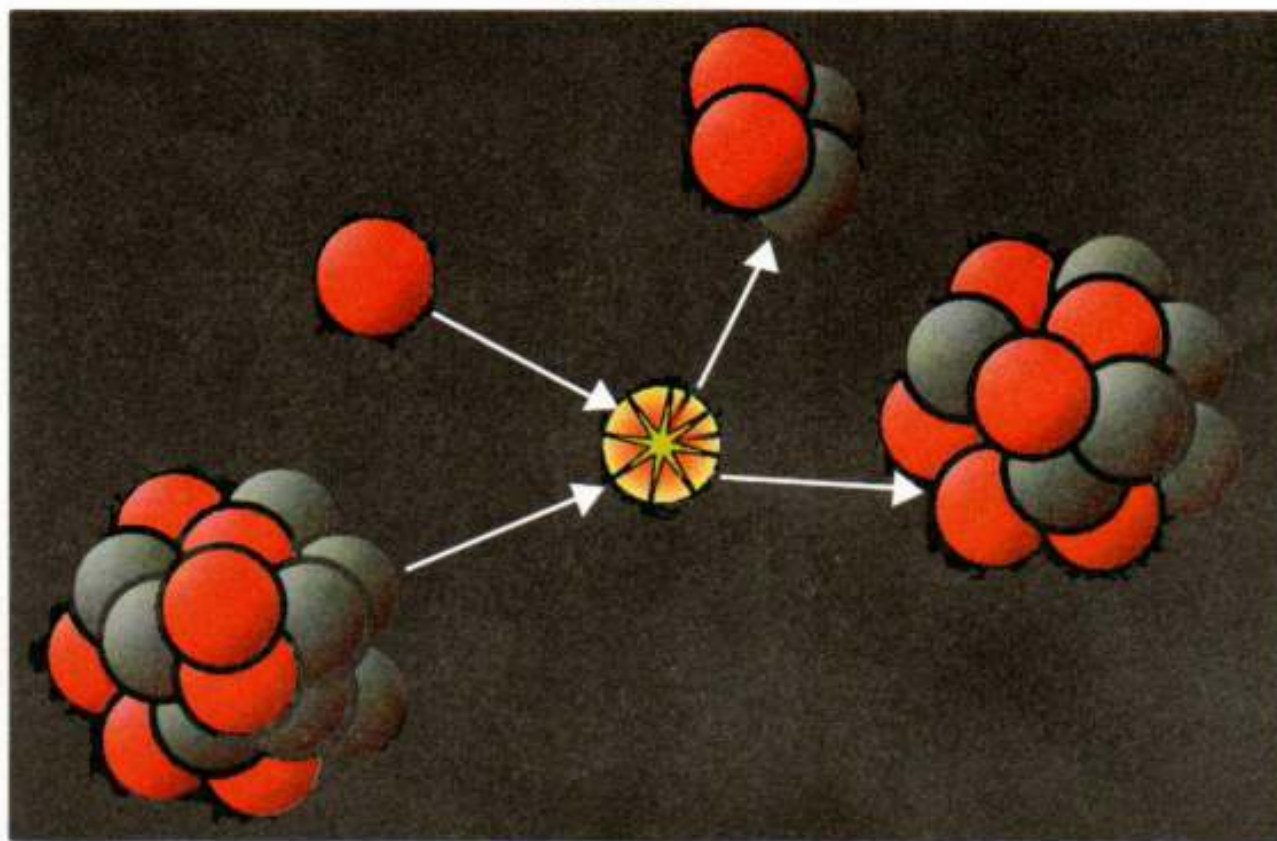
Choose the correct answer

- System of reaction in which no reactant leaves reaction mixture, is termed as
 - open system
 - closed system
 - semi-open system
 - partially closed system
- Reactions in which products cannot turn back into reactants are termed as
 - irreversible reactions
 - reversible reactions
 - equilibrium reactions
 - redox reactions
- On increasing concentration of one or more reactants, more products will be generated until
 - equilibrium
 - closed system
 - open system
 - All of Above
- Position of equilibrium explains amounts of
 - products
 - reactants
 - catalyst
 - both A and B
- Reversible reactions are those which products react together to give original
 - reactants
 - products
 - enzymes
 - All of Above

6. In equilibrium state, backward and forward reactions take place at
- same time
 - simultaneously
 - one time
 - All of Above
7. Le chatelier's principle states that if one or more factors disturbs equilibrium, position of equilibrium will shift where change will be
- reduced
 - induced
 - increased
 - oxidized
8. System will not be in equilibrium when there is an increase in concentration of
- reactants
 - products
 - catalysts
 - activation energy
9. Catalysts speed up reactions but when equilibrium is reached it has
- huge effect
 - no effect
 - slow effect
 - ended

CHAPTER 8

INTRODUCTION TO NUCLEAR CHEMISTRY



CHAPTER CONTENTS

- 8.1 Introduction
- 8.2 The Nucleus
- 8.3 Half-life of radioactive isotopes
- 8.4 Nuclear radiations
- 8.5 Nuclear Reactions
 - 8.5.1 Nuclear fission
 - 8.5.2 Nuclear fusion
 - 8.5.3 Radioactive or nuclear decay
 - 8.5.4 Nuclear transmutation
- 8.6 Applications of Nuclear chemistry
- Chapter Review Questions

CHAPTER OBJECTIVES

By the end of the chapter, the learner should be able to:

- Define nuclear chemistry and compare with Chemical reactions.
- Summarize the events that led to understanding radiation.
- Identify alpha, beta and gamma radiations in terms of their composition and properties.
- Explain why certain nuclei are radioactive.
- Define half-life.
- Write balanced nuclear equations.
- Relate mass and energy.
- Interpret data from a half-life graph.
- Identify uses of nuclear chemistry.

8.1 INTRODUCTION

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define nuclear chemistry.
- Compare nuclear reaction to chemical reaction.

**ACTIVITY 1**

We have been playing soccer as a team and one of my friends injured and felt pain in his leg. He couldn't tell exactly what was causing the pain.

In groups, we will explain how nuclear chemistry will help him to figure out his pain?

Nuclear chemistry is the sub-discipline of chemistry that is concerned with changes in the nucleus of elements. In our earlier lessons we have studied various forms of chemical reactions. In these reactions, atoms can gain, lose, or share valence electrons, leaving the atomic nuclei largely unaffected, and the identity of the atoms does not change. On the other hand, the nuclear chemistry is concerned with the structure of atomic nuclei and the changes they undergo. Whereas chemical reactions involve only small energy changes, nuclear reactions involve much larger energy changes.

In 1895, German physicist Wilhelm Roentgen (1845–1923) found that invisible rays were emitted when electrons bombarded the surface of certain materials. These invisible rays caused photographic plates to darken, and Roentgen named these high-energy emissions X-rays. At that time, French physicist Henri Becquerel (1852–1908) was studying minerals that emit light after being exposed to sunlight, a phenomenon called phosphorescence. Building on Roentgen's work, Becquerel wanted to determine whether phosphorescent minerals also emitted X-rays.

Becquerel discovered by chance that phosphorescent uranium salts produced spontaneous emissions that darkened photographic plates. He observed this phenomenon even when the uranium salts were not exposed to light. Later, chemist Marie Curie (1867–1934) and her husband Pierre Curie (1859–1906) took Becquerel's mineral sample, called pitch-blende, and isolated the components emitting the rays. They concluded that the darkening of the photographic plates was due to rays emitted from the uranium atoms present in the mineral sample. Marie Curie named the process by which materials give off such rays radioactivity; the rays and particles emitted by a radioactive source are called radiation. The work of Marie and Pierre Curie was extremely important in establishing the origin of radioactivity and developing the field of nuclear chemistry. In 1898, the Curies identified two new elements; Polonium and Radium, on the basis of their radioactivity.

Table 8.1: Comparison of Chemical and Nuclear Reactions

Chemical Reactions	Nuclear Reactions
Occur when bonds are broken and formed	Occur when nuclei combine, split, and emit radiation
Involve only valence electrons	Can involve protons, neutrons, and electrons
Associated with small energy changes	Associated with large energy changes
Atoms keep the same identity although they might gain, lose, or share electrons, and form new substances	Atoms of one element are often converted into atoms of another element
Temperature, pressure, concentration, and catalysts affect reaction rates	Temperature, pressure, and catalysts do not normally affect reaction rates



**EXERCISE**

1. Explain the most radical difference between chemical and nuclear reactions
2. Research how nuclear chemistry is different from nuclear physics?
3. In nuclear reactions, Atoms of one element are often converted into atoms of another element. Explain.
4. What is the role of uranium in nuclear science?
5. How was radiation discovered?

8.2 THE NUCLEUS**Lesson objectives**

By the end of the lesson, the learner should be able to:

- State the particles of nucleus.
- Explain nuclear binding energy.
- Explain mass defect and its causes.

As we studied earlier, the atom is made up of protons, neutrons and electrons. Electrons are on the shell while positively charged protons and neutral neutrons are located in the nucleus, and are collectively called nucleons.

In nuclear chemistry, when referring to a single type of nucleus, we often use the term nuclide also known nuclear species, and identified by the number of protons and neutrons in its nucleus. We identify nuclides in two ways;

-When a symbol such as $^{228}_{88}\text{Ra}$ is used, the superscript is the mass number, and the subscript is the atomic number.

-The same nuclide can also be written as radium-228, where the mass number is written following the element name.

Mass Defect:

As a simple example of the energy associated with the strong nuclear force, consider the helium atom composed of two protons, two neutrons, and two electrons. The total mass of these six subatomic particles may be calculated as:

$$(2 \times 1.0073 \text{ amu}) + (2 \times 1.0087 \text{ amu}) + (2 \times 0.00055 \text{ amu}) = 4.0331 \text{ amu}$$

Protons neutrons electrons

However, mass spectrometric measurements reveal that the mass of an atom is 4.0026 amu, less than the combined masses of its six constituent subatomic particles. This difference between the calculated and experimentally measured masses is known as the mass defect of the atom.

In the case of helium, the mass defect indicates a loss in mass of $4.0331 \text{ amu} - 4.0026 \text{ amu} = 0.0305 \text{ amu}$. The loss in mass accompanying the formation of an atom from protons, neutrons, and electrons is due to the conversion of that mass into energy that is evolved as the atom forms. The nuclear binding energy is the energy produced when the atoms' nucleons are bound together; this is also the energy needed to break a nucleus into its constituent protons and neutrons.

**EXERCISE**

1. Define mass defect.
2. How is nuclear stability related to the neutron- proton ratio?
3. Explain how nuclear binding energy produced?

8.3 HALF-LIFE OF RADIOACTIVE ISOTOPES**Lesson objectives:**

By the end of the lesson, the learner should be able to:

- Define half-life of radioactive isotope.
- Determine the amount of radioactive substance remaining after a given number of half- lives.
- Construct a half-life graph.



ACTIVITY 2

Materials

- Shoe box
- 120 coins (say, shillings as a metal radioactive element)
- Stop watch

Instructions

1. Count 120 coins and lay them heads-up in the bottom of the shoe box. In a table like the one in Data and Observations, record this as daughter atoms and 120 parent coin atoms at time 0.
2. Put the lid on the shoe box and, holding the lid on tightly; shake the box with moderate force up and down 20 times while your lab partner times the decay process to the nearest second. Assume that all the subsequent decay trial times are identical.
3. Open the box and count the tails-up coins, which represent the daughter atoms, and remove them from the box. Subtract this number of tails-up from the original number of heads up coins to find the number of remaining parent coin atoms. Record the time taken for the shaking, the number of tails-up atoms removed, and the number of remaining parent atoms.

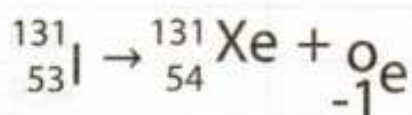


4. Repeat steps 2 and 3 four more times, recording your data after each trial. Simply add on the original shaking time repeatedly to arrive at the total elapsed time.

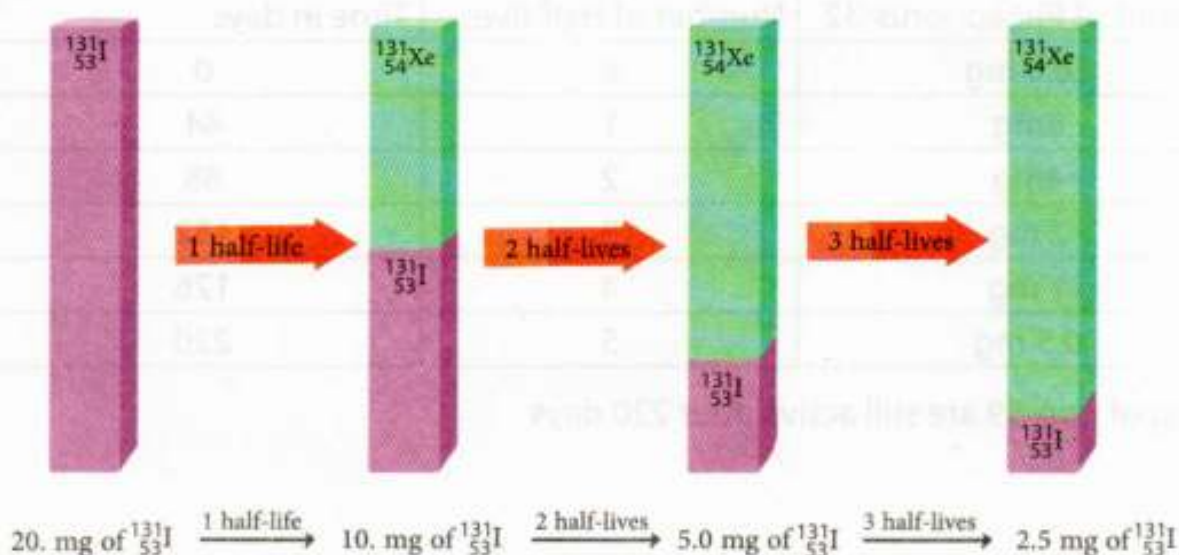
- Construct a graph of your data, plotting the number of remaining parent coins on the vertical axis (y-axis) and the total elapsed time on the horizontal axis (x-axis).
- Write down your data and observations:

Half-life is the time required for half the original sample of nuclei to decay. For example, if a certain radioactive sample contains 500 nuclei at a given time and 250 nuclei (half of the original number) 10 days later, this radioactive nuclide has a half-life of 10 days.

For example, $^{131}_{53}\text{I}$ has a half-life of 8.0 days. As $^{131}_{53}\text{I}$ decays, it produces the non-radioactive isotope $^{131}_{54}\text{Xe}$ and a beta particle.



Suppose we have a sample that initially contains 20. mg of $^{131}_{53}\text{I}$. In 8.0 days, one-half (10. mg) of all the $^{131}_{53}\text{I}$ nuclei in the sample will decay, which leaves 10. mg of $^{131}_{53}\text{I}$. After 16 days (two half-lives), 5.0 mg of the remaining $^{131}_{53}\text{I}$ decays, which leaves 5.0 mg of $^{131}_{53}\text{I}$. After 24 days (three half-lives), 2.5 mg of the remaining $^{131}_{53}\text{I}$ decays, which leaves 2.5 mg of $^{131}_{53}\text{I}$ nuclei still capable of producing radiation.



Example 1

Phosphorus-32, a radioisotope used in the treatment of leukemia, has a half-life of 14.3 days. If a sample contains 8.0 mg of phosphorus-32, how many milligrams of phosphorus-32 remain after 42.9 days?

Solution

Its easiest way is to draw a table showing the amount of Phosphorus-32 as a function of number of half-lives. For each half-life, divide the amount of Phosphorus-32 by 2.

Amount of Phosphorus-32	Number of Half-lives	Time in days
8.0 mg	0	0
4 mg	1	14.3
2mg	2	28.6
1 mg	3	42.9

1 mg of Phosphorus-32 remains after 42.9 days

Example 2

Iron-59 has a half-life of 44 days. If a nuclear laboratory receives a sample of 16.0 mg of iron-59, how many mg of iron-59 are still active after 220 days?

Solution

Amount of Phosphorus-32	Number of Half-lives	Time in days
16.0 mg	0	0
8mg	1	44
4mg	2	88
2 mg	3	132
1 mg	4	176
0.5 mg	5	220

0.5 mg of iron-59 are still active after 220 days

Example 3

The half-life of polonium-218 is 3.04 min. A sample of polonium contains 0.00558 g of polonium-218. What mass of polonium-218 will remain after 18.24 min?

Solution

First, you must determine the number of half-lives that have passed in 18.24 min.

$$\text{Number of half-lives} = \frac{\text{time elapsed}}{\text{half-life}} = \frac{18.24 \text{ min}}{3.04 \text{ min}} = 6.00 \text{ half-lives}$$

Then, to determine the mass of polonium-218 remaining, apply the following relationship:

$$\text{mass remaining} = \text{starting mass} \times \text{fraction remaining}$$

After n half-lives, the fraction of a radioactive nuclide that remains is

$$\left(\frac{1}{2}\right)^n, \text{ or } 2^{-n}.$$

Where n = number of half-lives

$$\text{mass } {}^{218}_{84}\text{Po remaining} = 0.00558 \text{ g} \times \left(\frac{1}{2}\right)^6 = 0.00558 \text{ g} \times \frac{1}{64} = 8.72 \times 10^{-5} \text{ g } {}^{218}_{84}\text{Po}$$

Table 8.2: Half-lives of some radioactive isotopes

Element	Radioisotope	Half-Life	Type of Radiation
Naturally occurring Radioisotopes			
Carbon-14	${}^{14}_6\text{C}$	5730 yr	Beta
Potassium-40	${}^{40}_{19}\text{K}$	1.3×10^9 yr	Beta, gamma
Radium-226	${}^{226}_{88}\text{Ra}$	1600 yr	Alpha
Strontium-90	${}^{90}_{38}\text{Sr}$	38.1 yr	Alpha
Uranium-238	${}^{238}_{92}\text{U}$	4.5×10^9 yr	Alpha
Some Medical Radioisotopes			
Carbon-11	${}^{11}_6\text{C}$	20 min	
Chromium-51	${}^{51}_{24}\text{Cr}$	28 days	Gamma
Iodine-131	${}^{131}_{56}\text{I}$	8.0 days	Gamma
Oxygen-15	${}^{15}_8\text{O}$	2.0 min	Positron
Iron-59	${}^{59}_{26}\text{Fe}$	44 days	Beta, gamma
Radon-222	${}^{222}_{86}\text{Rn}$	3.8 days	Alpha
Technetium-99m	${}^{99\text{m}}_{43}\text{Tc}$	6.0 h	Beta, gamma



EXERCISE

1. Technetium-99m is an ideal radioisotope for scanning organs because it has a half-life of 6.0 h and is a pure gamma emitter. Suppose that 80.0 mg were prepared in the technetium generator this morning. How many milligrams of technetium-99m would remain after each of the following intervals?
 - a) one half-life
 - b) two half-lives
 - c) 18 h
 - d) 24 h
2. For each of the following, indicate if the number of half-lives elapsed is:
 - one half-life
 - two half-lives
 - three half-lives
 - a) a sample of Pd-103 with a half-life of 17 days after 34 days.
 - b) a sample of C-11 with a half-life of 20 min after 20 minutes.
 - c) a sample of At-211 with a half-life of 7 hours after 21 hours.
3. The half-life of **polonium-210** is 138.4 days. How many milligrams of polonium-210 remain after 415.2 days if you start with 2.0 mg of the isotope?
4. Assuming a half-life of 1599 years, how many years will be needed for the decay of 15/16 of a given amount of radium-226?
5. The half-life of radon-222 is 3.824 days. After what time will one-fourth of a given amount of radon remain?
6. The half-life of cobalt-60 is 5.27 years. How many milligrams of cobalt-60 remain after 52.7 years if you start with 10.0 mg?
7. A sample contains 4.0 mg of uranium-238. After 4.46×10^9 years, the sample will contain 2.0 mg of uranium-238. What is the half-life of uranium-238?

8.4 NUCLEAR RADIATIONS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Describe Nuclear radiation.
- State and compare different types of nuclear radiation.

Radiation is energy given off by matter in the form of rays or high-speed particles. The nucleus carries a positive electrical charge, while the electrons carry a negative electrical charge. These forces within the atom work toward a strong, stable balance by getting rid of excess atomic energy (radioactivity). In that process, unstable nuclei may emit a quantity of energy, and this spontaneous emission is what we call radiation.

Physical Forms of Radiation

As previously indicated, matter gives off energy (radiation) in two basic physical forms. One form of radiation is pure energy with no weight. This form of radiation — known as electromagnetic radiation — is like a vibrating or pulsating rays or waves of the electrical and magnetic energy. Familiar types of electromagnetic radiation include sunlight (cosmic radiation), x-rays, radar, and radio waves.

The other form of radiation known as particle radiation is tiny fast-moving particles that have both energy and mass (weight). This less-familiar form of radiation includes alpha particles, beta particles, and gamma rays.

Radioactive Decay

As previously indicated, large unstable atoms become more stable by emitting radiation to get rid of excess atomic energy (radioactivity). This radiation can be emitted in the form of positively charged alpha particles, negatively charged beta particles, gamma rays, or x-rays. Through this process — called **radioactive decay** — radioisotopes lose their radioactivity over time. This gradual loss of radioactivity is measured in half-lives.

Essentially, a half-life of a radioactive material is the time it takes one-half of the atoms of a radioisotope to decay by emitting radiation. When radioisotopes are used in medicine or industry, it is vital to know how rapidly they lose their radioactivity, in order to know the precise amount of radioisotope that is available for the medical procedure or industrial use.



Nuclear Fission

In some elements, the nucleus can split as a result of absorbing an additional neutron, through nuclear fission process. Such elements are called **fissile materials**. One particularly notable fissile material is uranium-235. This is the isotope that is used as fuel in commercial nuclear power plants.

When a nucleus fission occurs, it causes three important events that result in the release of energy. Specifically, these events are:

- ✓ The release of radiation
- ✓ Release of neutrons (usually two or three)
- ✓ Formation of two new nuclei (fission products).

Ionizing Radiation

Radiation can be either ionizing or non-ionizing, depending on how it affects matter. Non-ionizing radiation includes visible light, heat, radar, microwaves, and radio waves. This type of radiation deposits energy in the materials through which it passes, but it does not have sufficient energy to break molecular bonds or remove electrons from atoms.

By contrast, ionizing radiation (such as x-rays and cosmic rays) is more energetic than non-ionizing radiation. Consequently, when ionizing radiation passes through material, it deposits enough energy to break molecular bonds and displace (or remove) electrons from atoms. This electron displacement creates two electrically charged particles (ions), which may cause changes in living cells of plants, animals, and people.

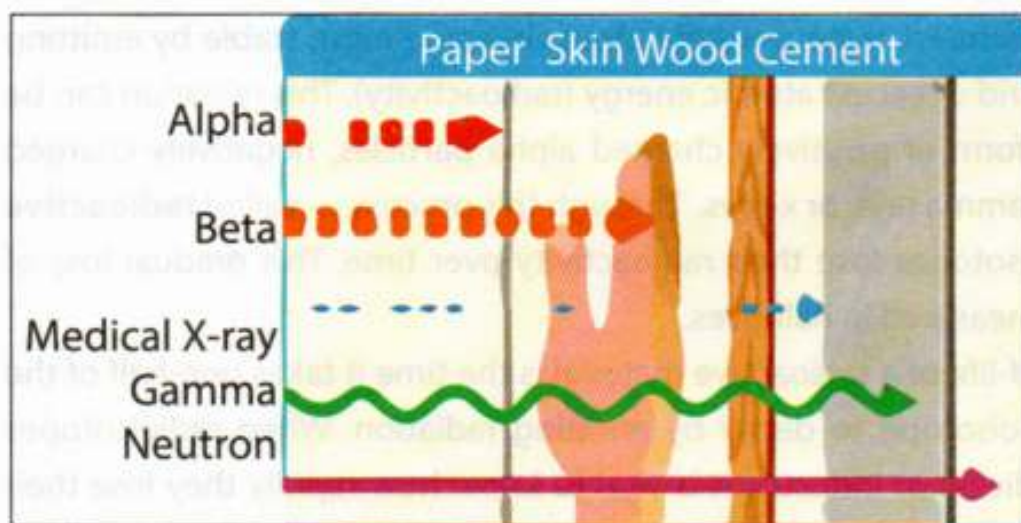


Figure 8.6: Penetrating ability of radioactive emissions.

Relatively large and highly charged alpha particles can be stopped by a sheet of paper. Beta particles penetrate paper, but they are stopped by moderately thick sheets of metals such as lead (0.5 cm 0.2 in.). Gamma radiation has no mass or charge, and it penetrates paper and sheets of metals, but it can be stopped by thick blocks of metals (10 cm =4 in.).

The harmful effects of radiation on living systems come from its ability to break chemical bonds and thereby destroy healthy tissue. But modern practitioners have learned to carefully apply the radiations to unhealthy tissues, but there is still much room for improvement in targeting only the cancerous cells while avoiding the destruction of healthy cells.

The unit most commonly used to express radiation exposure is the **rem**. More than 600 rems in one dose is fatal. The radiation therapy used in fighting cancers gives a *total dose over time* in the 4000 to 7000-rem range.

Sources of radiation are everywhere. Radiation is emitted from rocks, soil, water, and even atoms in us. Although the atmosphere acts as a filter between us and the rest of the universe, we are constantly bombarded by radiation from sources outside the earth's atmosphere.



EXERCISE

1. List alpha, beta, and gamma rays in order of decreasing penetrating power.
2. What is the easiest way you can protect yourself from the harmful effects of natural sources of radiation?
3. Write the pros and cons of using radiation giving concrete examples.



8.5 NUCLEAR REACTIONS

Lesson objectives

By the end of the lesson, the learner should be able to:

- Define nuclear reaction.
- Explain types of nuclear reactions.

Nuclear reactions are processes in which one or more nuclides are produced from the collisions between two atomic nuclei or one atomic nucleus and a sub-atomic particle. The nuclides produced from nuclear reactions are different from the reacting nuclei (commonly referred to as the parent nuclei).

There are four types of Nuclear reactions which are:

1. Nuclear fission reaction.
2. Nuclear fusion reactions
3. Radioactive or nuclear decay
4. Transmutation.

8.5.1 Nuclear Fission reaction

Nuclear fission is the splitting of the nucleus of an atom into parts (Lighter nucleus), often producing free neutrons and smaller nuclei which may eventually produce photons (in the form of gamma rays). Fission of heavy elements is an exothermic reaction which can release large amount of energy both as electromagnetic radiation and as kinetic energy.

Nuclear fission produce energy for nuclear power and to drive the explosion of nuclear weapons.

Bombarding uranium atoms with neutrons result in fission, because changing the ratio of neutrons to protons can make a nucleus less stable. Adding a neutron to a uranium-235 nucleus causes the nucleus to split. Atoms of two different elements are created along with more neutrons. Here is a typical reaction for uranium fission.



Notice that the sums of the mass numbers on the left and right sides are equal, and so are the sums of the atomic numbers. Therefore, the nuclear equation is balanced.

Nuclear Fission Chain Reaction

A chain reaction is a continuing series of reactions in which each produces a product that can react again. Each neutron produced during the fission of one atom of uranium has the potential to cause the fission of another nucleus. The fission reactions can continue in a chain, as shown in Figure 8.7. For a chain reaction to take place, there must be enough fissionable material present for some of the neutrons produced to be able to collide with other nuclei. However, if the chain reaction takes place too quickly, an explosion will result and release an enormous amount of energy all at once. This is what happens when an atomic bomb explodes. If the rate of the fission chain can be controlled to release energy slowly, the energy can be used to heat materials such as water.

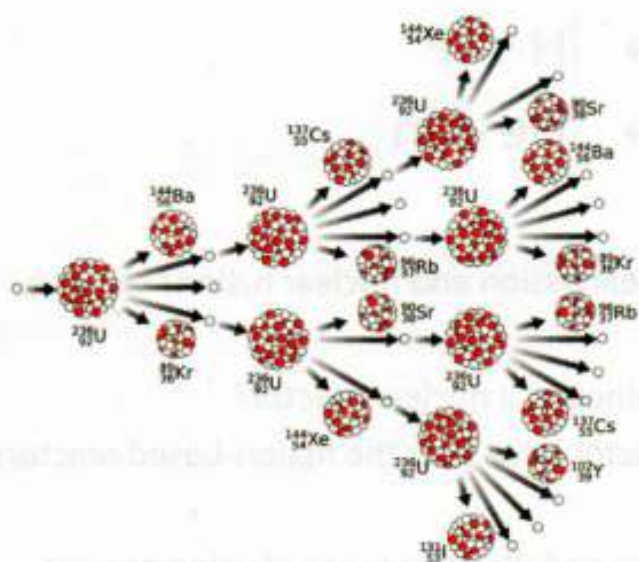
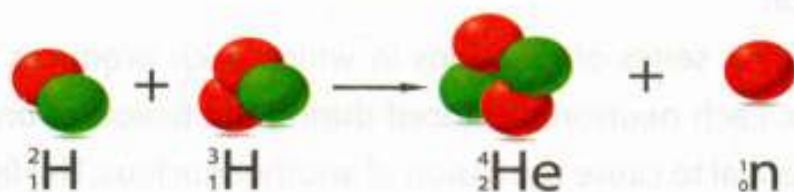


Figure 6.7: Fission Chain Reaction

8.5.2 Nuclear Fusion

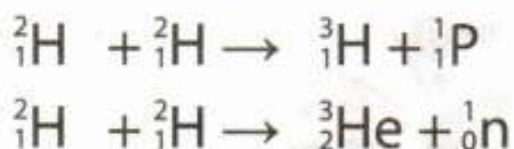
In a nuclear fusion, two or more nuclei combine to form a larger nucleus. Fusion is the nuclear process that produces energy in stars like our sun. In the most typical fusion reaction in the sun, hydrogen nuclei fuse to form helium. The enormous amount of energy that is generated sustains all life on Earth.

Some of the fusion processes in which hydrogen forms helium have been carried out and studied in laboratories. In one common fusion reaction, two different isotopes of hydrogen combine to form helium and a neutron.



Notice that the nuclear equation is balanced. The isotope of hydrogen with a mass number of 2 that is one of the reactants in the equation above is called deuterium (D). The hydrogen isotope with a mass number of 3 is called tritium (T). Nuclei such as these, which have a small mass, can combine to produce heavier, more stable nuclei.

The helium produced has a stable neutron: proton ratio of 1:1. Two deuterium nuclei can also fuse to form either tritium or helium.



EXERCISE

1. What is the difference between nuclear fission and nuclear fusion? List one current use for fission.
2. How is fission controlled and maintained in a nuclear reactor?
3. What advantages would a fusion reactor have over the fission-based reactors currently in use?
4. Compare and contrast the advantages and disadvantages of using energy generated by nuclear fission reactors instead of energy from fossil fuels such as petroleum, natural gas, and coal.

8.5.3 Radioactive or nuclear decay

The spontaneous disintegration of a nucleus into a slightly lighter nucleus is called **radioactive decay**, accompanied by the emission of subatomic particles, electromagnetic radiation or both.

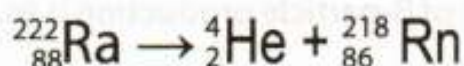
The type of a nuclide and the rate of the decay depend on the nucleon content and energy level of the nucleus.

There are several different types of radioactive decay. One often observed decay process involves emission of an alpha (α) particle, which is a helium nucleus (${}^4_2\text{He}$).

α -particle production is a very common mode of decay for heavy radioactive nuclides.

For example,

${}^{222}_{88}\text{Ra}$, radium 222, decays by α -particle to give radon-218.



Notice in this equation that the mass number is conserved ($222 = 4 + 218$) and the atomic number is conserved ($88 = 2 + 86$).

Another α -particle producer is ${}^{230}_{90}\text{Th}$:

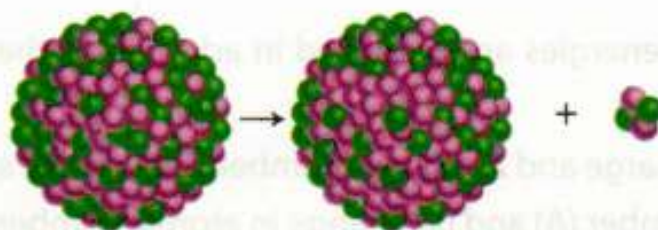


Figure 6.8: uranium-238 undergoes alpha decay

When a nucleus of uranium-238 undergoes alpha decay, a nucleus of thorium-234 and a nucleus of helium-4 are produced. The atomic numbers balance on the two sides of the equation, as do the mass numbers.

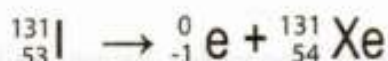
Notice that the production of an α -particle results in a loss of 4 in mass number (A) and a loss of 2 in atomic number (Z). α -particle production is another common decay process.

For example,

The thorium-234 nuclide produces a β -particle as it changes to protactinium-234.



Iodine-131 is also a β -particle producer:

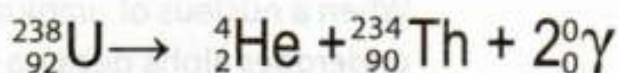


The β -particle is assigned a mass number of 0 because its mass is tiny compared with that of a proton or neutron. The value of Z is -1 for the β -particle, so the atomic number for the new nuclide is greater by 1 than the atomic number for the original nuclide. Therefore, the net effect of β -particle production is to change a neutron to a proton.

A gamma ray, or **γ -ray**, is a high-energy photon of light. A nuclide in an excited nuclear energy state can release excess energy by producing a gamma ray, and γ -ray production often accompanies nuclear decays of various types.

For example,

In the α -particle decay of ${}_{92}^{238}\text{U}$,



Two gamma rays (γ -rays) of different energies are produced in addition to the α -particle (${}^4_2\text{He}$.)

γ -rays are photons of light with zero charge and zero mass number. Emission of a γ -ray results in no change in mass number (A) and no change in atomic number (Z).

Example 1:

Write balanced nuclear equations for each of the following reactions.

- $^{11}_6\text{C}$ produces a positron.
- $^{214}_{83}\text{Bi}$ emits a β -particle.
- $^{237}_{93}\text{Np}$ emits a α -particle.

Solution

- a) We must find the product nuclide represented by ^A_ZX in the following equation:
- $$^{11}_6\text{C} \rightarrow ^0_1\text{e} + ^A_Z\text{X}$$

↑
Positron

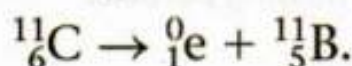
The key to solving this problem is to recognize that both A and Z must be conserved. That is, we can find the identity of ^A_ZX by recognizing that the sums of the Z and A

values must be the same on both sides of the equation.

Thus, for X, Z must be 5 because $Z+1=6$.

A must be 11 because $11+0=11$.

Therefore, ^A_ZX is $^{11}_5\text{B}$ (The fact that Z is 5 tells us that the nuclide is boron. So, the balanced equation is:



CHECK:

Left Side

Right Side

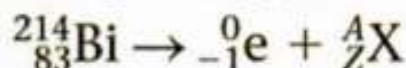
Z = 6

Z = 1 + 5 = 6

A = 11

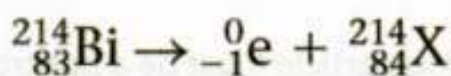
A = 11 + 0 = 11

- b) Knowing that a β -particle is represented by $^0_{-1}\text{e}$, we can write:



Where $Z-1=83$ and $A+0=214$. This means that $Z=84$ and $A=214$.

We can now write:



Using the periodic table, we find that $Z=84$ for the element polonium, so $^{214}_{84}\text{X}$ must be $^{214}_{84}\text{Po}$.

CHECK:

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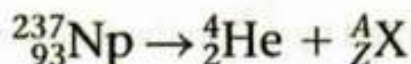
$Z = 83$

$Z = 84 - 1 = 83$

$A = 214$

$A = 214 + 0 = 214$

- c) Because an α -particle is represented by ${}^4_2\text{He}$, we can write:



Where $A + 4 = 237$ or $A = 237 - 4 = 233$, and $Z + 2 = 93$ or $Z = 93 - 2 = 91$. Thus $A = 233$, $Z = 91$, and the balanced equation must be:



CHECK:

Left Side Right Side

$Z = 93$

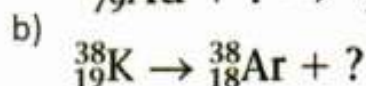
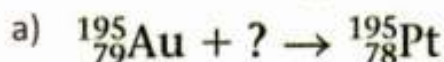
$Z = 2 + 91 = 93$

$A = 237$

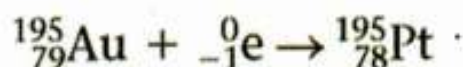
$A = 4 + 233 = 237$

Example 2

In each of the following nuclear reactions, supply the missing particle:

**Solution**

- a) A does not change and Z for Pt is 1 lower than Z for Au, so the missing



CHECK:

Left Side Right Side

$Z = 79 - 1 = 78$

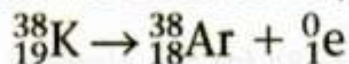
$Z = 78$

$A = 195 + 0 = 195$

$A = 195$

This is an example of electron capture.

- b) For Z and A to be conserved, the missing particle must be a positron.



CHECK:

Left Side Right Side

$Z = 19$

$Z = 18 + 1 = 19$

$A = 38$

$A = 38 + 0 = 38$

Potassium-38 decays by positron emission.

**EXERCISE**

- Define radioactive decay.
- What are the different types of common radioactive decay?
 - List the types of radioactive decay that convert one nuclide into another.
- Write the nuclear symbol and electrical charge of an alpha particle and a beta particle.
- List alpha, beta, and gamma rays in order of decreasing penetrating power.
- Write a nuclear equation for the alpha decay of Ra-224.
- Write a nuclear equation for the beta decay of Bk-249.
- Write a nuclear equation for the positron emission of potassium-40.
- Supply the missing particles in each of the following nuclear equations:
 - ${}^{222}_{86}\text{Rn} \rightarrow {}^{218}_{84}\text{Po} + ?$
 - ${}^{15}_8\text{O} \rightarrow ? + {}^0_1\text{e}$
- What fraction of a given sample of a radioactive nuclide remains after four half-lives?
- Distinguish between natural and artificial radioactive nuclides.
- Write the balanced nuclear equation for the radioactive decay of radium-226 to give radon-222, and determine the type of decay.
- Write a balanced equation for the nuclear reaction in which neon-23 decays to form sodium-23, and determine the type of decay.

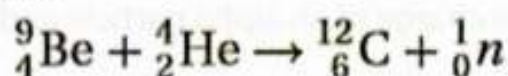


8.5.4 Nuclear transmutation

Unstable nuclei undergo spontaneous changes that change their numbers of protons and neutrons. In this process, they give off large amounts of energy and increase their stability. These changes are a type of nuclear reaction called **Transmutation**. In this process the atomic number changes which results the identity of the element to change. We can define **Transmutation** is a change in the identity of a nucleus as a result of a change in the number of its protons.

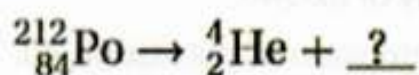
As the other nuclear reactions, balancing nuclear equations the total mass number and atomic number must be equal on both sides of the equation.

Example 1



Example 1

Identify the product that balances the following nuclear reaction:



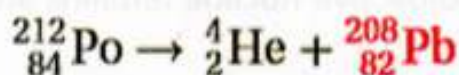
mass number: $212 - 4 = 208$

atomic number: $84 - 2 = 82$

Solution

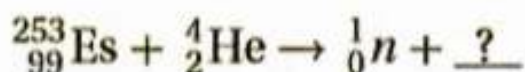
The nuclide has a mass number of 208 and an atomic number of 82, ${}^{208}_{82}\text{Pb}$

The balanced nuclear equation is:



EXERCISE

- 1- Define transmutation,
- 2- Explain how transmutation reaction is different from fission reaction?
- 3- Using 1_0n to represent a neutron and ${}^0_{-1}e$ to represent an electron, complete the following nuclear equations:



8.6 APPLICATIONS OF NUCLEAR CHEMISTRY

Lesson Objectives

By the end of the lesson, the student should be able to:

- Describe the uses of radioactivity and other applications.



ACTIVITY DISCUSSION

In groups of three, we will discuss different ways of using radioactive.

Although, nuclear chemistry has many negative effects on humans depending the amount of exposure, its applications in daily life impacts numerous scientific fields.

For example,

1. In medicine, radioactive sources are used to diagnose, treat disease, and to sterilize medical instruments. A common example is taking X-ray to examine whether bones are broken, etc
2. In Agriculture, radioactive materials are used to improve food crops, preserve food, measure soil moisture content, erosion rate and control insects, etc
3. In industrial development, improving productivity to monitor all steps of industrial process and safety control measures, etc
4. In energy, to produce energy for heat and electrical power and also nuclear weapons, etc

5. In archaeology, we use to determine the age of rocks or archaeological relics, etc
6. In environment, , to detect smoke and to monitor pollutants, etc

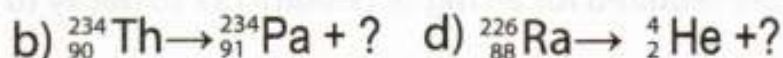
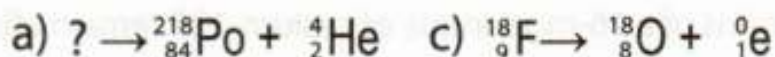
**EXERCISE**

1. In the treatment of abdominal carcinoma, a person is treated with seeds of gold-198, which is a beta emitter. Write the balanced nuclear equation for the beta decay of gold-198.
2. In an experimental treatment, a person is given boron-10, which is taken up by malignant tumors. When bombarded with neutrons, boron-10 decays by emitting alpha particles that destroy the surrounding tumor cells. Write the balanced nuclear equation for the reaction of this experimental procedure.

Chapter Review Questions

1. Determine the number of protons and number of neutrons in the nucleus of each of the following:
 - a) magnesium-26 b) strontium-88 c) iodine-131 d) copper-64
2. Identify each of the following as alpha decay, beta decay, positron emission, or gamma emission:
 - a) ${}_{88}^{226}\text{Ra} \rightarrow {}_{86}^{222}\text{Rn} + {}_2^4\text{He}$
 - b) ${}_{27}^{60}\text{Co} \rightarrow {}_{28}^{60}\text{Ni} + {}_{-1}^0\text{e}$
 - c) ${}_{9}^{18}\text{F} \rightarrow {}_{8}^{18}\text{O} + {}_{+1}^0\text{e}$
3. What part of the atom produces radioactivity?
4. Distinguish between natural and artificial radioactive nuclides.
5. Identify the type of particle or radiation for each of the following:
 - a. 4He b. $0 + 1\text{e}$ c. $0\text{ }0\text{g}$
6. What is ionizing radiation?
7. What is given off by a nucleus undergoing each of the following types of decay?
 - a) Alpha b) Beta c) gamma
8. What types of material will block:
 - a) Alpha particles?
 - b) Beta particles?
 - c) Gamma rays?

9. Balance the following equations, which represent nuclear reactions in the uranium-238 decay series.



10. Write the balanced nuclear equation for each of the following:

a) Th-225 (alpha decay) b) Bi-210 (alpha decay)

c) cesium-137 (beta decay) d) Tin-126 (beta decay)

11. Explain why carbon-14 dating cannot be used to determine how old a dinosaur fossil is.

12. a) What starts the chain reaction that occurs in nuclear fission reactors used today?

b) What keeps it going?

c) How is its rate controlled?

13. What type of nuclear reaction takes place in stars?

14. List three ways that radioisotopes are used in medicine.

15. How is cobalt-60 used to treat cancer?

16. Why are foods irradiated? Does this make them radioactive?

17. What new element is formed if magnesium-24 is bombarded with a neutron and then ejects a proton? Write a balanced nuclear equation for this transmutation.

18. Some types of heart pacemakers contain plutonium-239 as a power source. Write a balanced nuclear equation for the alpha decay of plutonium-239.

19. Write the balanced nuclear equation for each of the following:

a) When two oxygen-16 atoms collide, one of the products is an alpha particle.

b) When californium-249 is bombarded by oxygen-18, a new element, seaborgium-263, and four neutrons are produced.

c) Radon-222 undergoes alpha decay.

d) An atom of strontium-80 emits a positron.

20. Cesium-137, used in cancer treatment, has a half-life of 30 yr.
- Write the balanced nuclear equation for the beta decay of cesium-137.
 - How many milligrams of a 16-mg sample of cesium-137 remain after 90 yr?
 - How many years are required for 28 mg of cesium-137 to decay to 3.5 mg of cesium-137?
21. Scientists created a new synthetic element californium, Cf-246, by bombarding U-238 with C-14. Other than Cf-246, 4 neutrons are produced in the reaction. Write the balanced nuclear equation for the synthesis of californium.
22. Nuclear fission of U-235 can be induced by the bombardment of neutron to give Ba-141, Kr-92, and three neutrons. Write the balanced nuclear equation for the reaction and explain why a chain reaction will occur after the nuclear fission of one U-235.
23. In a paper-making factory, beta radiation is used to check that the paper being produced is the correct thickness. If the paper gets too thin, the reading on the detector increases causing the roller to move apart to make the paper thicker. If the paper gets too thick, the reading on the detector goes down causing the rollers to move closer together. A diagram of this set-up is shown below:

